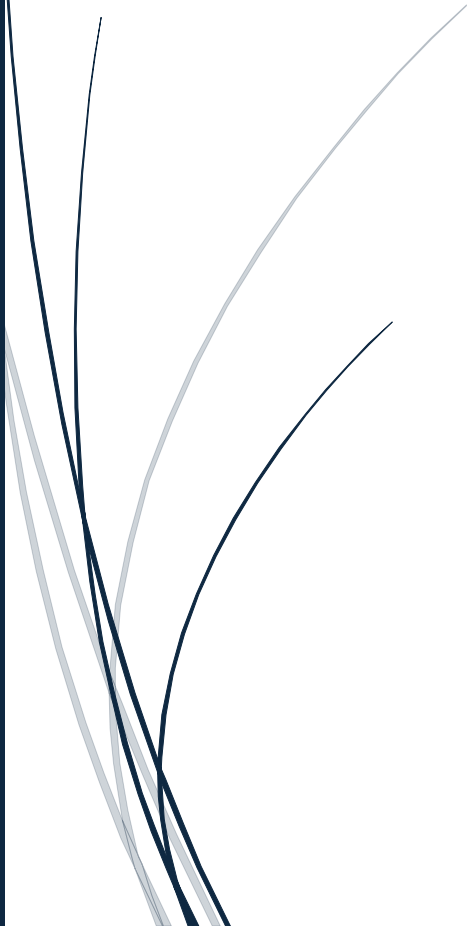


Spring 2025

Course 12101

Quantitative Sustainability



Indholdsfortegnelse

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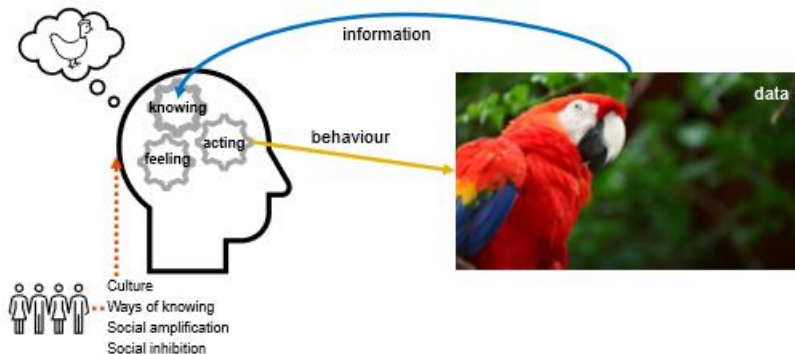
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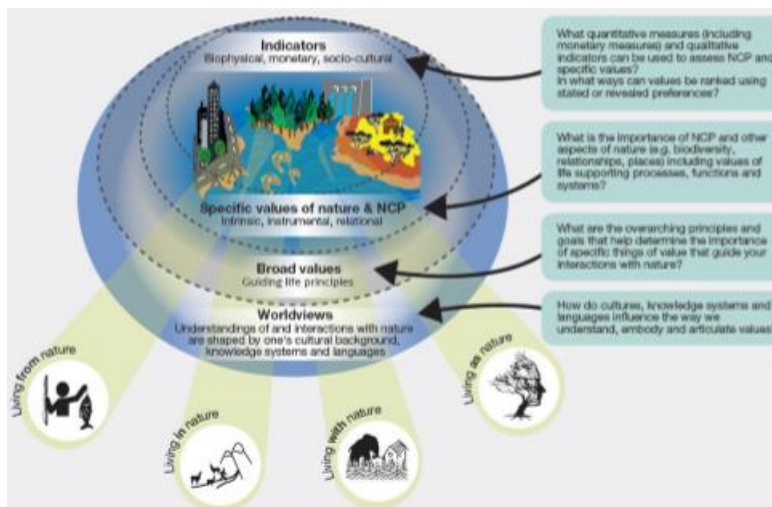
1. What is sustainability

- It is a diverse concept – all have an intrinsic view of this notion but those views differ slightly
- How we understand sustainability depends on our worldviews about the relationship between people and nature
- The challenge: there is not one optimal definition

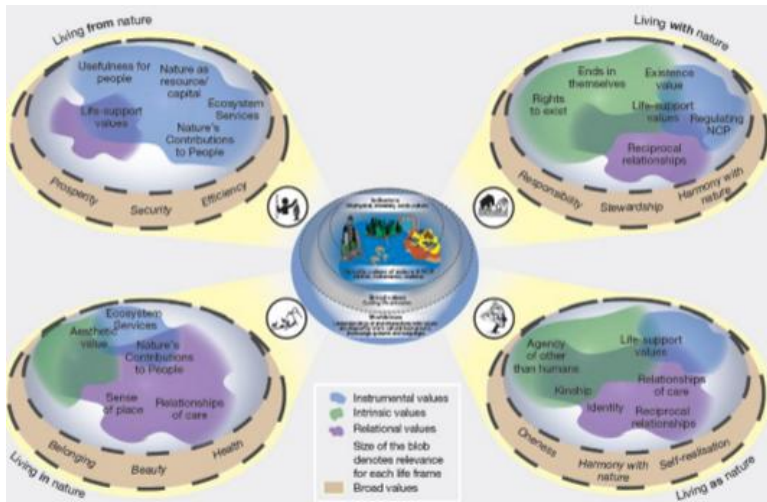
Worldviews – foundations – a model of the world



Worldviews affect values



- Living from values
- Living in nature
- Living with nature
- Living as nature



Global consensus definition of sustainability

- "Human beings are at the centre of concerns for sustainable development. They are entitled to a healthy and productive life in harmony with nature."
- "The right to development must be fulfilled so as to equitably meet developmental and environmental needs of present and future generations."
- "In order to achieve sustainable development, environmental protection shall constitute an integral part of the development process and cannot be considered in isolation from it."

Quantitative sustainability

Rio Declaration 1992

- *"Human beings are at the centre of concerns for sustainable development..."*
- *"...equitably meet developmental and environmental needs of present and future generations."*
- *"... environmental protection shall constitute an integral part of the development process...."*

What is at the origin of nowadays SDGs?

- The modern *sustainability* concept emerged from interactions between economic development & environmentalism

Consensus I: 1972 Stockholm Declaration

- Our activities (particularly industrialization) are having an impact on the global environment
- We need to assess the impact of our activities and manage them

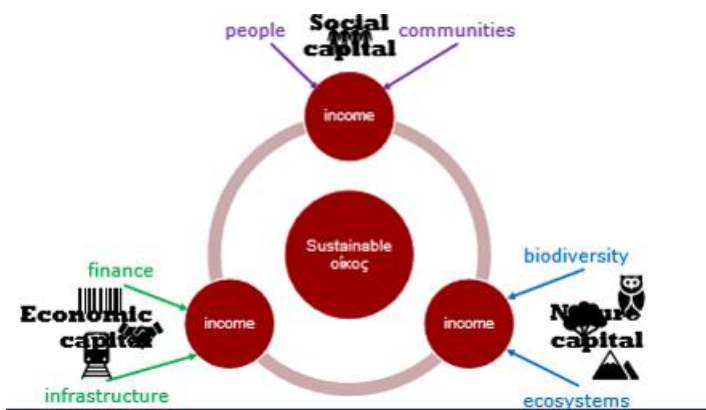
Universal Declaration of Human Rights (1948)

- **Article 1** – All human beings are born free and equal in dignity and rights.
- **Article 2** – Everyone is entitled to all the rights and freedoms set forth in this Declaration, without distinction of any kind.
- **Article 25** – Everyone has the right to a standard of living adequate for the health and well-being of himself and of his family.

What is at the origin of nowadays SDGs?

- The modern *sustainability* concept emerged from interactions between economic development & environmentalism
- **Consensus II: 1992 Rio Declaration**
 - Reconcile Stockholm Declaration with economic development & UDHR
 - Needed for global security
 - Needed to adhere to Universal Human Rights

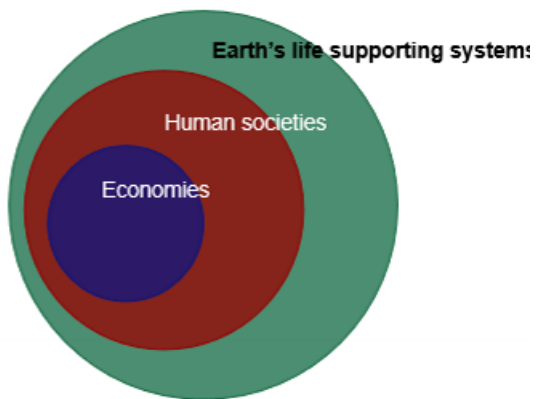
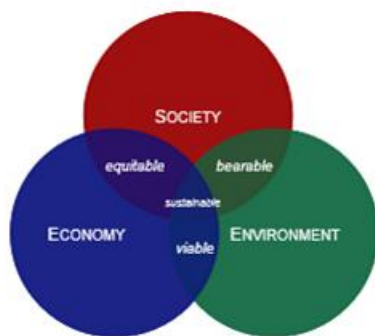
Implementing Rio 1992: Sociological perspective



Alternative formulation



The Three Pillars



Operational definition

- We have a consensus definition
- We have a framework in which human-nature relations can be operationalized across worldviews to meet this definition
- Consensus-building takes a long time

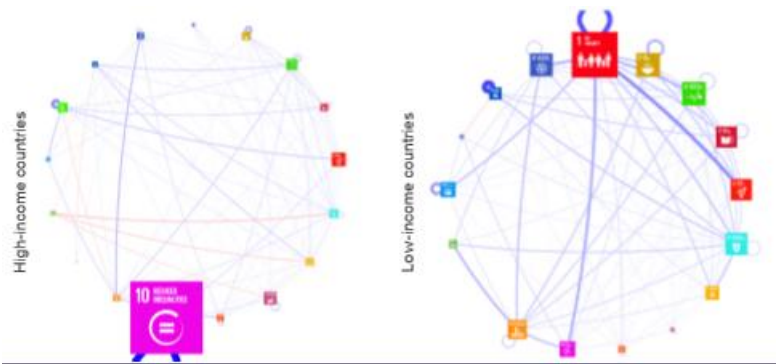
How do we translate it for public & private sectors



Capturing the key dimensions of the problem – 17 goals



The network of SDG interactions



Data availability



Conclusions

- We need S.M.A.R.T. objectives & targets
 - (*Specific, Measurable, Achievable, Relevant & Time-Bound*)
- They need to capture the three dimensions of sustainability
- Data remains sparse 30 years on
- Globally we still confuse ambitions and objectives

Quantitative Sustainability: Influence on companies

United Nation's Sustainable Development goals

17 goals to 'end poverty, protect the planet, and ensure prosperity for all as part of a new sustainable development agenda'



How does this influence companies?

- Probably the strongest external factor industry has yet seen in the sustainability area
- Should affect prioritizations and actions internally in the company

- SDGs are expected to be translated to national programmes containing a targeted mix of legislation, regulation, financing, and price governing mechanisms that companies have to relate to

SDG Compass

- Objective to support companies in aligning their strategies with the SDGs and in measuring and managing their contribution
- Developed by GRI, the UN Global Compact, and the World Business Council for Sustainable Development (WBCSD)



Sustainable alternatives have various advantages.

- It will create direct value by reducing energy costs and reduced waste processing costs, and by avoiding pollution taxes that embrace CSR.
- It will diminish vulnerability regarding future energy costs and other material costs as one would consume less.
- It will be the basis for a sound and positive reputation.
- It might result in a stronger company spirit and a collective company goal improving the company's synergy.
- It will facilitate the acquirement of licenses.
- It will stimulate learning and innovation. New opportunities and emerging markets.
- Last but not least, it will lead to more employee satisfaction. It can fulfill people's desire for good citizenship. Personal motivation of management and other employees can increase, improving social behavior and other business results.

Some academic studies

- **Yu et al., 2015**

- Positive relation between sustainability performance and firm value, after controlling for variables that have been found to affect firm value in the existing literature.
- **Lourenco et al., 2012**
 - Their findings suggest that CSP is positively associated with the financial performance of large and profitable firms which are able to signal their sustainability performance and has a negative association with the performance of large and profitable firms that are not able to signal their sustainability performance.
 - CSP has significant explanatory power for stock prices over the traditional summary accounting measures such as earnings and book value of equity.
- **Lo & Sheu, 2007**
 - A significantly positive relation between corporate sustainability and its market value is found. We also find a strong interaction effect between corporate sustainability and sales growth on firm value.

Different initiatives to engage in

- UN Global Compact
- Science-Based Targets initiative
- EU initiative – UNGPs
- CBK

Global ESG engagement levels (WDI 2022)

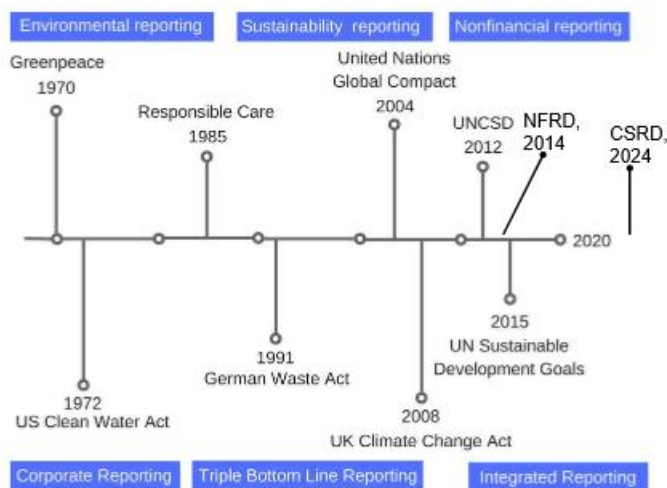
Figure 33: Frequency of prioritized SDGs (2022)



Base: 3,275 N100 companies identify specific SDGs they consider the most relevant for their business.



Reporting



Streamlining diverse approaches

- **EU Requirements:**
 - Corporate Sustainability Reporting Directive
 - Non-Financial Reporting Directive
- **How?**
 - European Sustainability Reporting Standards
 - Technical Screening Criteria

Non-financial reporting requirements

- **EU directive:**

- 500 employees (from 2025 >250 employees; from 2026 all quoted companies)
- Business model and strategy
- Sustainability policies
- **Performance:**
 - Environmental matters
 - Social matters and treatment of employees
 - Respect for human rights
 - Anti-corruption and bribery
 - Diversity on company boards (in terms of age, gender, educational, and professional background)
 -
- **EU reporting standards (ESRS) developed by European Financial Reporting Advisory Group (EFRAG)**

Taxonomy Regulation

- **Environmental objectives:**
 1. Climate change mitigation
 2. Climate change adaptation
 3. Sustainable use and protection of water and marine resources
 4. Transition to a circular economy
 5. Pollution prevention and control
 6. Protection and restoration of biodiversity and ecosystems
- **Social objectives (to come):**
 1. Employees
 2. Customers
 3. Sustainable and inclusive communities

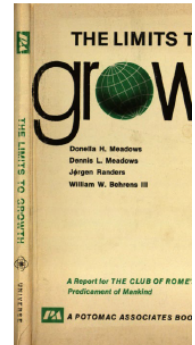
2. Conceptual frameworks

The sustainability challenge

$$I = P \cdot A \cdot T = \text{Pop} \cdot \frac{\text{GDP}}{\text{person}} \cdot \frac{I}{\text{GDP}}$$

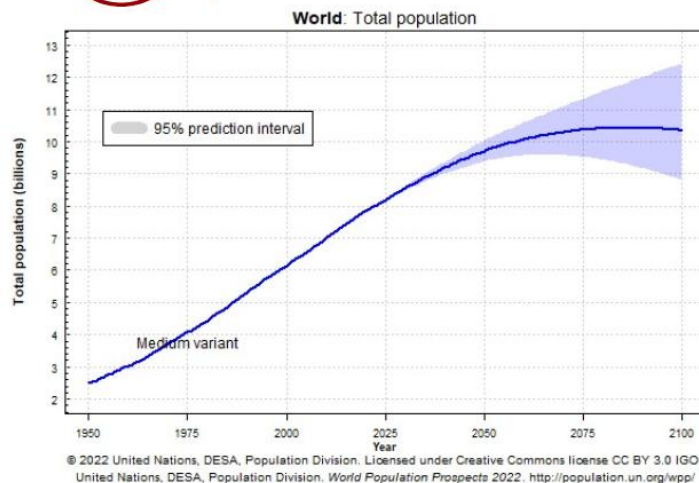
- I is the environmental impact
- Pop is the **global population**
- $\frac{\text{GDP}}{\text{person}}$ is the **Affluence**, the material standard of living
- $\frac{I}{\text{GDP}}$ is the **Technology factor** – environmental impact per created value

Ehrlich P, Holdren J (1971) Impact of population growth. *Science* 171, pp. 1212–1217.
 Commoner B (1972) The environmental cost of economic growth. In Ridker RG (ed.) *Population, Resources and the Environment*, pp. 339–63. U.S. Government Printing Office, Washington, DC.
 Graedel and Allenby (1995) *Industrial ecology*. Prentice Hall, New Jersey.



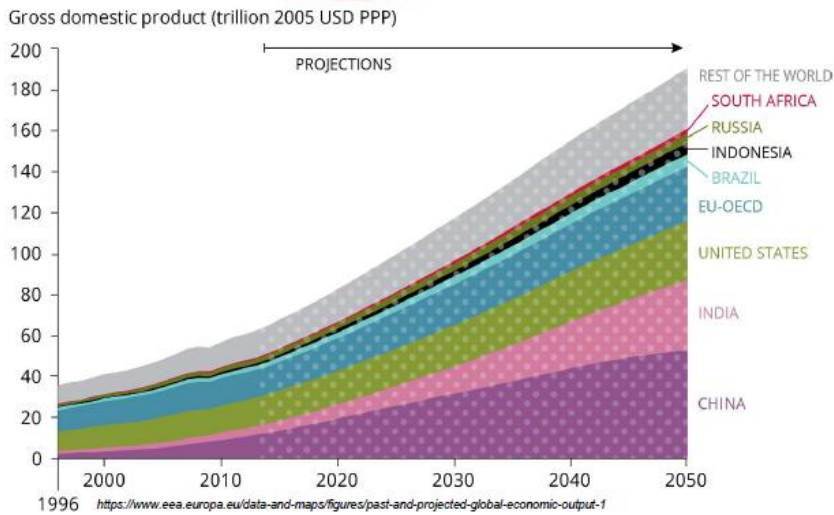
The global population is increasing

$$I = \text{Pop} \cdot \frac{\text{GDP}}{\text{person}} \cdot \frac{I}{\text{GDP}}$$



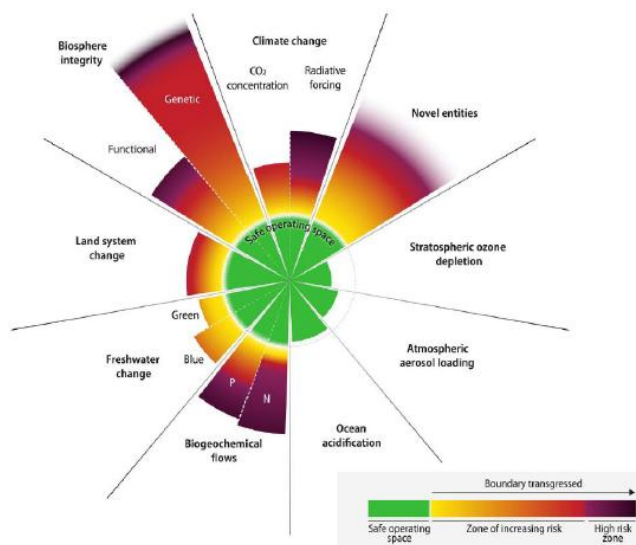
The affluence is increasing

$$I = Pop \cdot \frac{GDP}{person} \cdot \frac{I}{GDP}$$

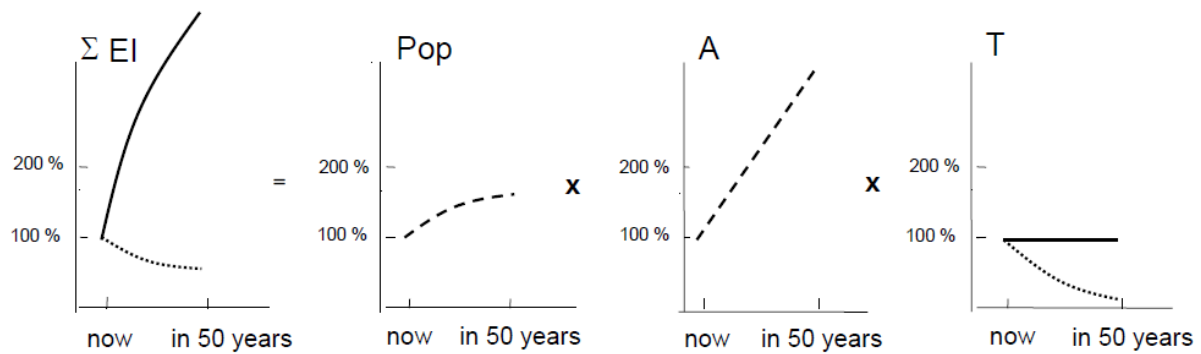


The environmental impacts should be reduced

$$I = Pop \cdot \frac{GDP}{person} \cdot \frac{I}{GDP}$$



How much should the technology factor be reduced?

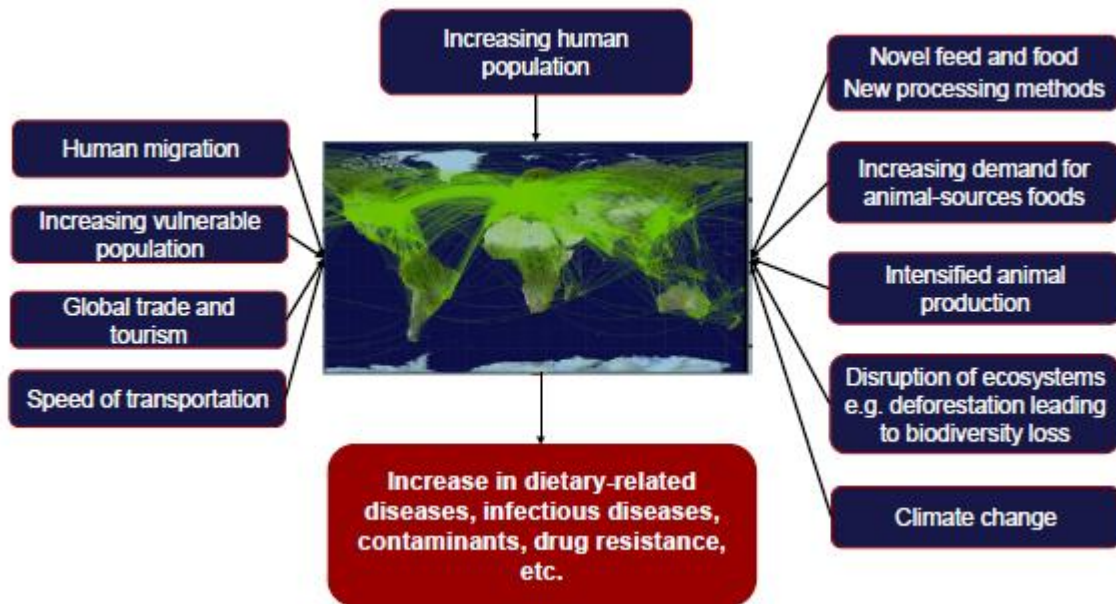


P .A = the total human demand/consumption will grow by at least a factor of 4-5 in 50 years

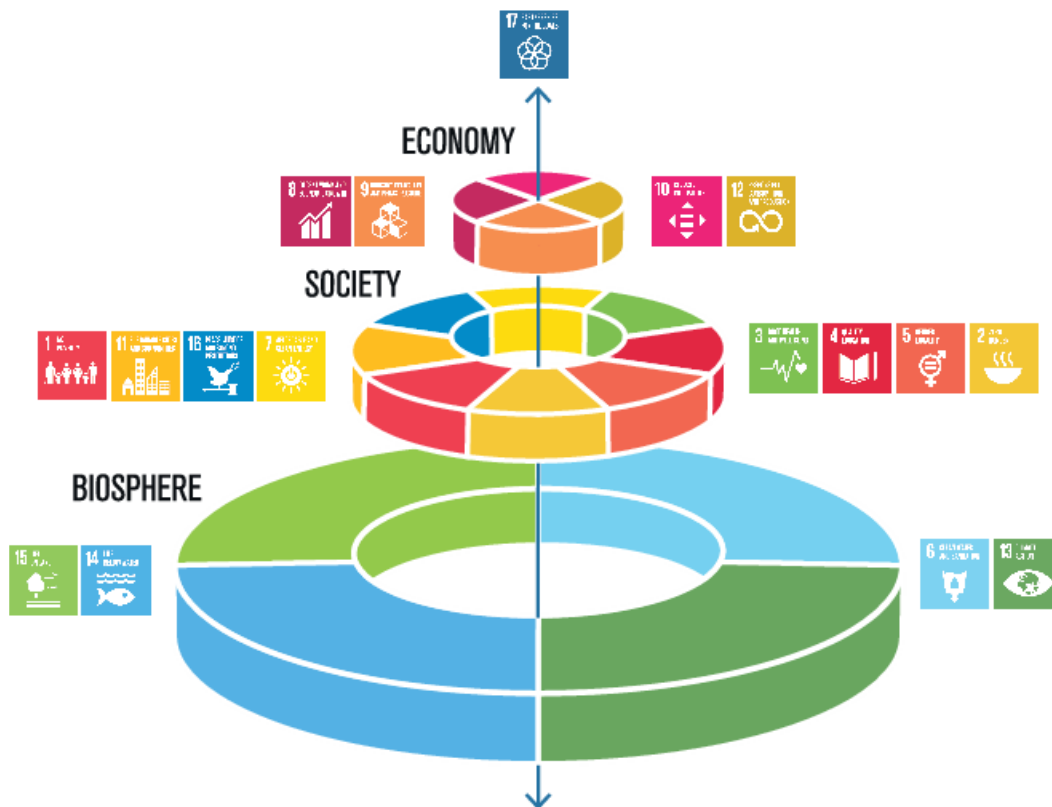
ΣEI = the total human impact on the environment will grow accordingly, if the technology factor remains unchanged

3. Sustainability assessment of foods -A brief intro

Factors influencing the production of healthy, safe and sustainable food.



The wedding cake model of Sustainability Development Goals



Graphics by Dennis Lokrentz/Monte

Earth's supporting functions for all life à Planetary boundaries

Sustainability assessment of food

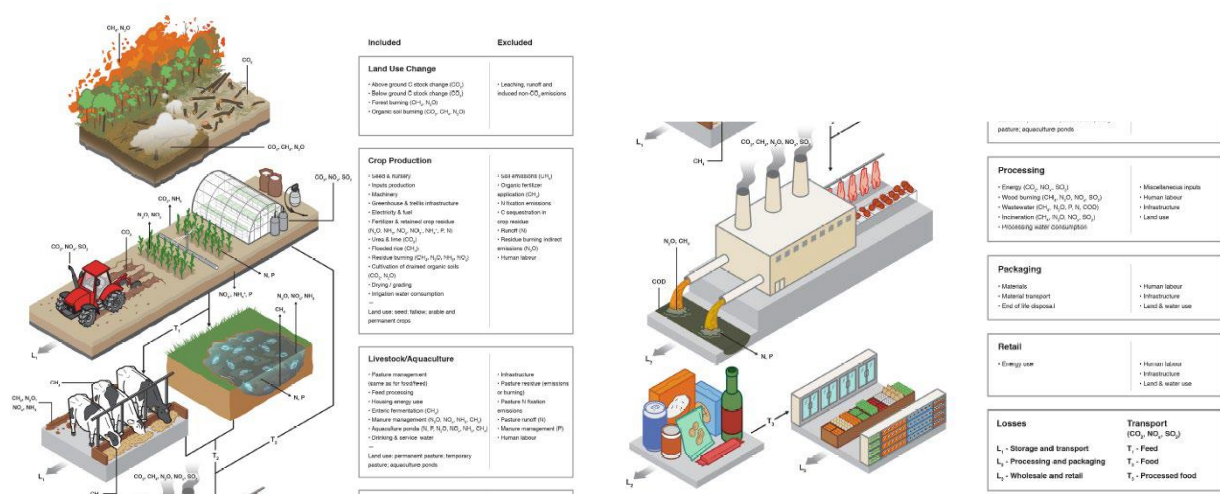
How food is unique compared to other consumer products e.g. EVs

- Biological origin:
 - Food production depends on biological processes
- Perishability:
 - Food has a short lifespan
- High variability:
 - Food production varies based on climate, soil, and farming practices
- Direct link to ecosystems:
 - Agriculture affects biodiversity, water cycles, and soil health in ways that are different from industrial production

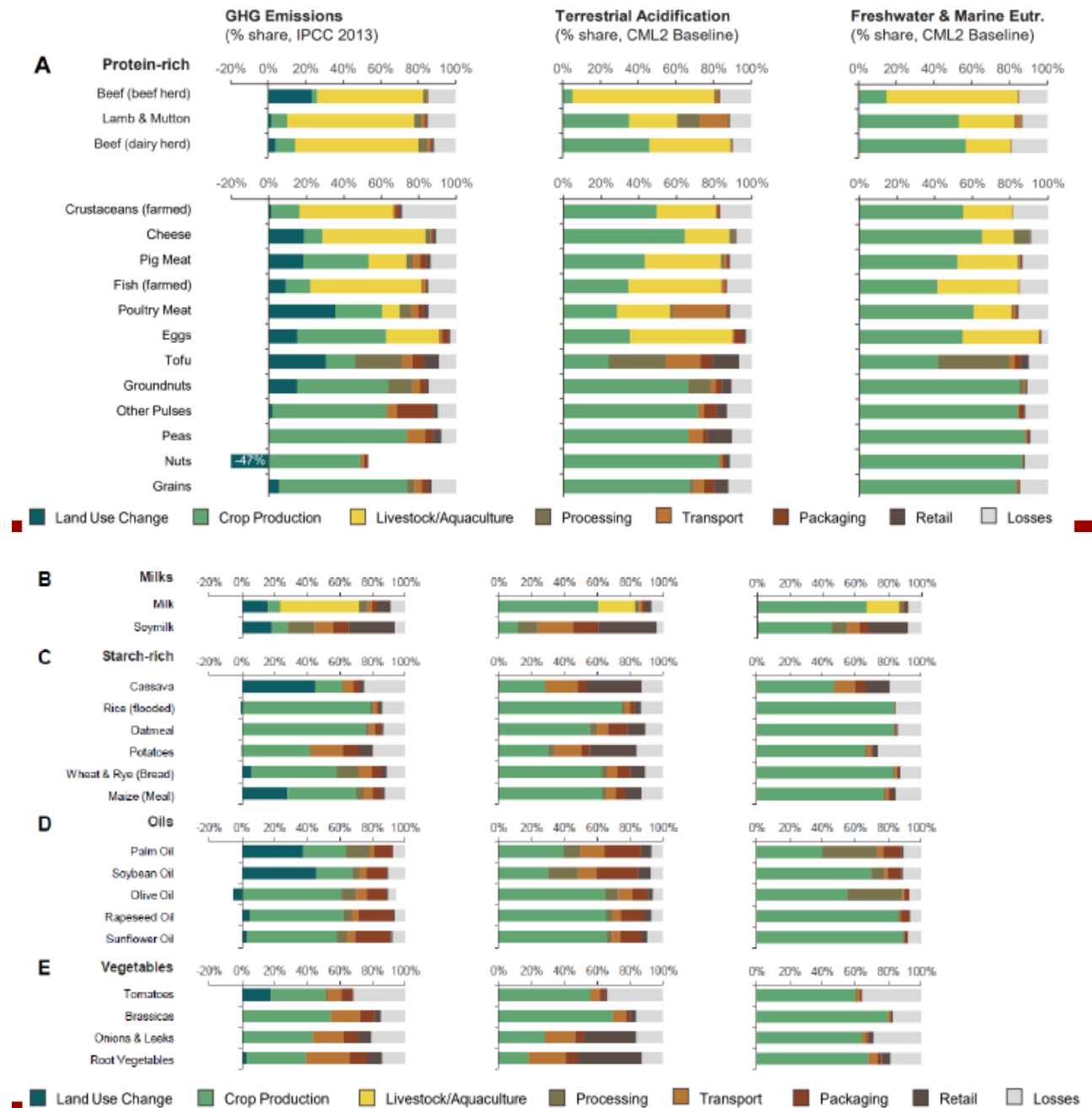
Comparison with EVs:

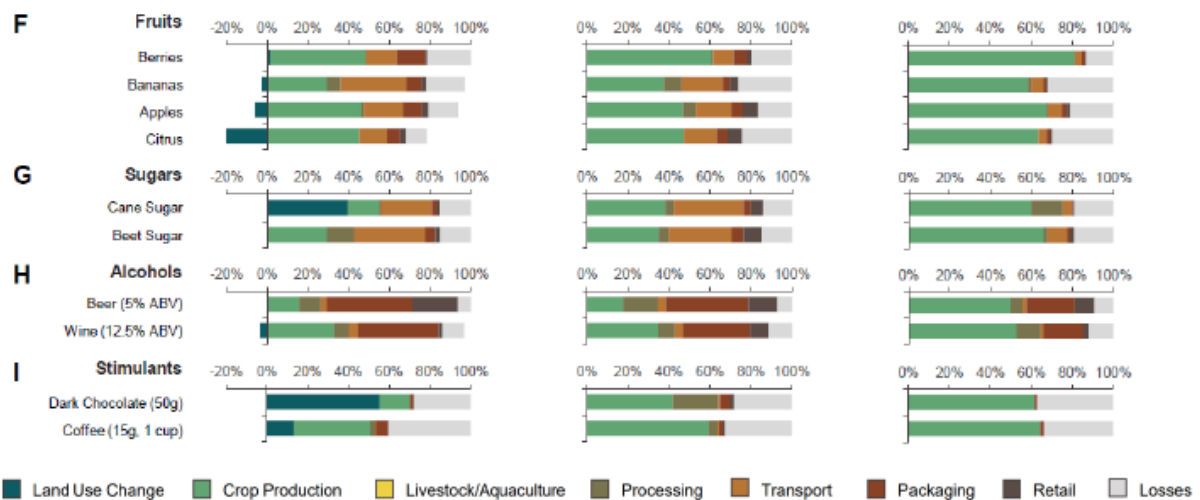
- EV sustainability is primarily about energy efficiency, raw materials (batteries), and end-of-life recycling
- Food sustainability is about resource use (water, land), emissions (methane, nitrous oxide), food loss and waste, and nutrient cycles

Global study of emissions and resource use in food production



Which parts of the life cycle have the highest impact?





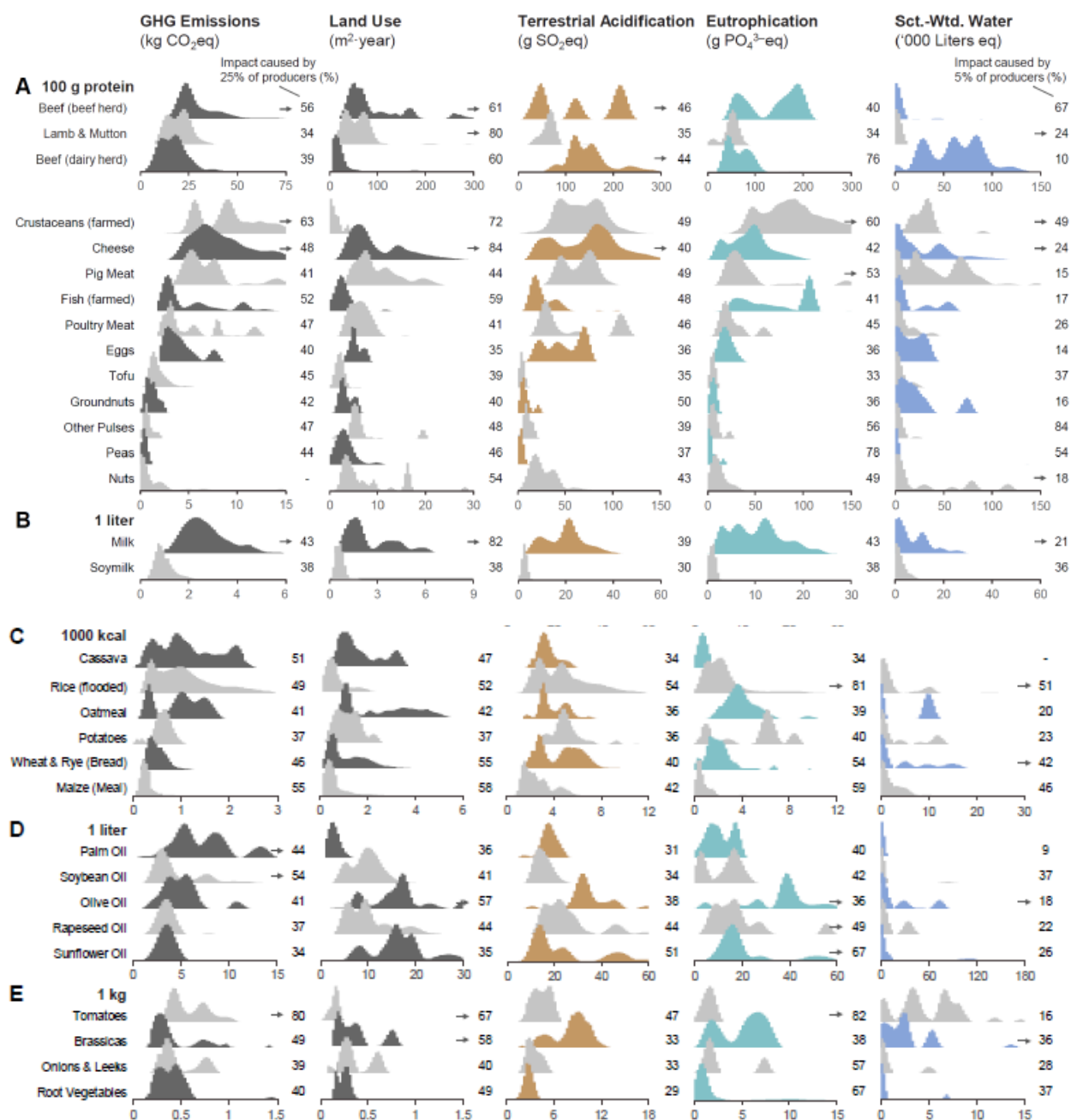
Challenges in defining the Functional Unit in food sustainability assessment

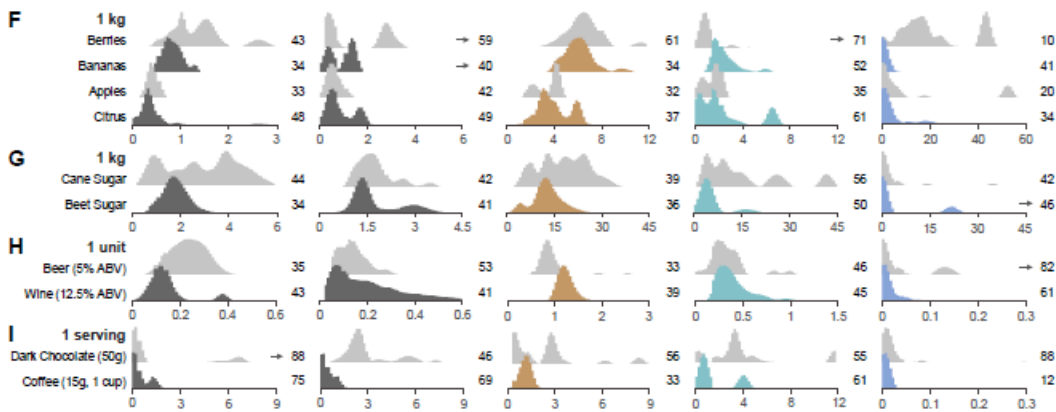
- **Mass-based** (e.g., per kg of food product) Simple but does not account for different nutritional values (e.g., 1 kg of lettuce vs. 1 kg of beef)
- **Nutritional-based** (e.g., per kcal, per g of protein, or micronutrient content) More relevant for human needs but harder to standardize across diverse foods
- **Service-based** (e.g., per meal, per portion of a balanced diet) Captures real-world consumption but requires complex modeling

Comparison with EVs:

- EV functional units are often per kilometer driven, making them easier to standardize
- In food, different functional units lead to very different sustainability rankings of products

LCA – comparison of various foods





Circular Economy in food systems

- Circular economy principles in food: Reduce: Prevent food waste at production, retail, and consumer levels
 - Reuse: Repurpose by-products (e.g., using surplus food for animal feed)
 - Recycle: Composting, biogas production, and nutrient recovery (e.g., phosphorus from wastewater)
- Examples of circular economy applications in food: Food waste valorization: Converting waste into animal feed, bio-based materials, or energy
 - Alternative proteins: Insects, lab-grown meat, or fermentation-based products reduce land and water use
 - Regenerative agriculture: Farming techniques that restore soil health and increase biodiversity

Comparison with EVs:

- EVs focus on material recovery (battery recycling, rare metal reuse), extending product lifespan, and remanufacturing parts
- Unlike food, EV circular economy strategies do not involve biological cycles—they deal mostly with physical materials

Environmental impact hotspots in food production

- Land use and deforestation (e.g., feed for livestock, cattle ranching, palm oil plantations)
- Water footprint (irrigation, washing and cleaning, drinking)
- Greenhouse gas emissions:
 - Methane (CH_4) from livestock digestion
 - Nitrous oxide (N_2O) from fertilizers (pollution from runoff)
 - CO_2 from food transport and processing
- Chemical pollution:
 - Pesticides and fertilizers harming ecosystems (pollution from runoff)

- Antibiotic use leading to antibiotic-resistant bacteria from livestock farming

Policy and consumer behavior in food sustainability

- Key policies influencing food sustainability: Subsidies for sustainable agriculture (e.g., supporting organic or regenerative farming)
 - Food labeling and certification (organic, fair trade, carbon footprint labels)
 - Food waste regulations (bans on food disposal, incentives for redistribution)
- Role of consumer behavior: Dietary choices directly impact sustainability (e.g., plant-based vs. meat-heavy diets)
 - Food waste reduction at the household level is crucial
 - Demand for sustainable packaging and local food affects supply chains

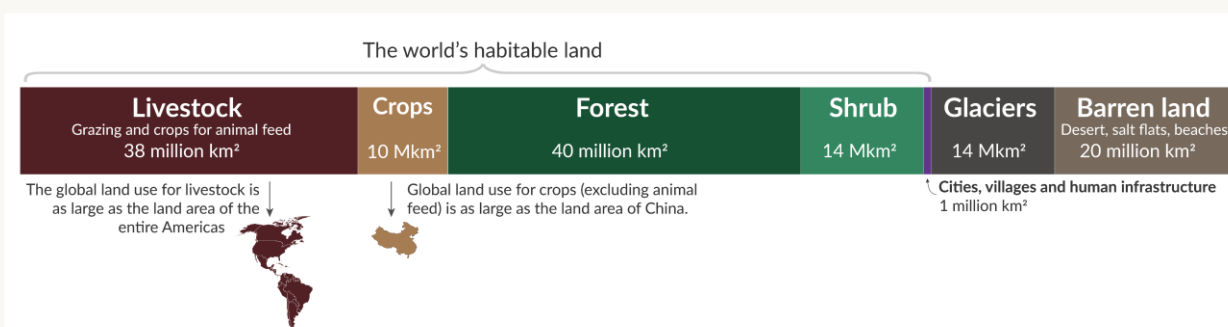
Rethinking the way we produce and consume food

Food systems are responsible for 60% of global biodiversity loss

- 33% of degraded soils
- 24% of the global GHG emissions
- 61% of the depletion of commercial fish
- 20% of overexploitation of the world's aquifers

How is the world's land used?

Our World
in Data

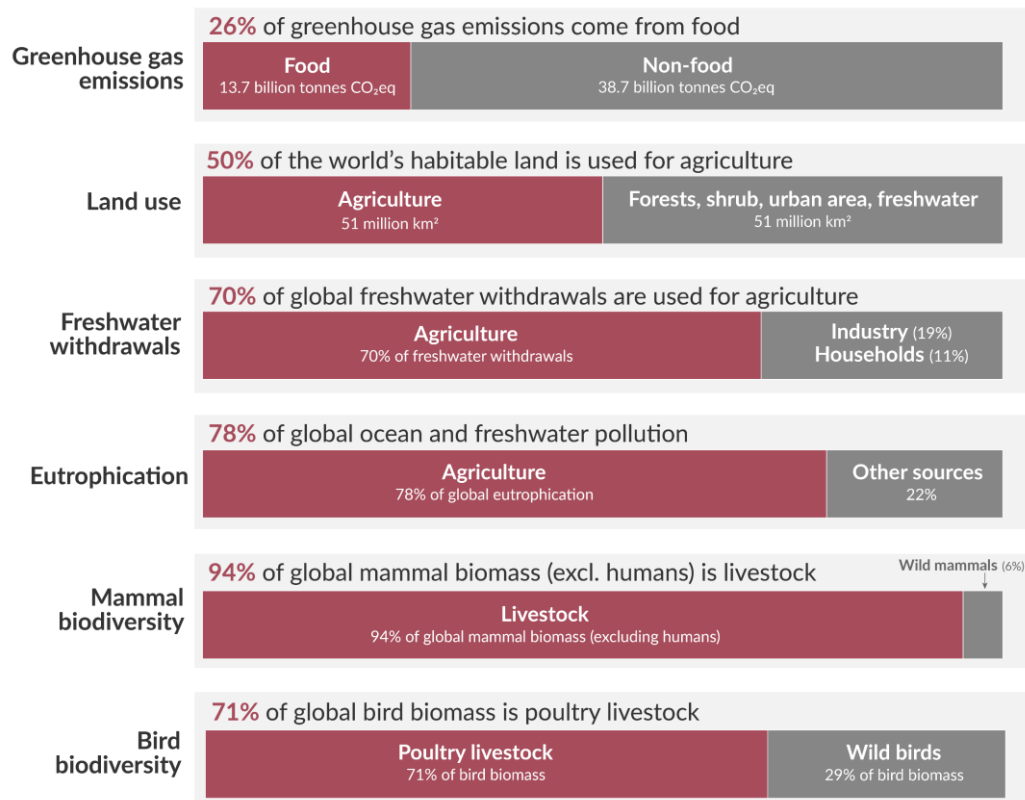


Note: The major uncertainty in these assessments is the allocation of 'rangelands': in some regions it can be difficult to accurately quantify how much of rangelands are used for grazing, and how much is free from human pressure. Despite this uncertainty, most analyses in the scientific literature find very large land use for grazing and livestock production.

Source: Calculated by the authors based on data from the Food and Agriculture Organization of the United Nations and Poore and Nemecek (2018).

OurWorldinData.org — Research and data to make progress against the world's largest problems. Licensed under CC-BY by the authors Max Roser and Hannah Ritchie

The environmental impacts of food and agriculture



Data sources: Poore & Nemecek (2018); UN FAO; UN AQUASTAT; Bar-On et al. (2018).
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Date published: November 2022.

Towards more sustainable and healthy food

- Optimization of resources
 - Produce more with fewer resources
 - Reduce food waste
 - Using side streams (e.g. brewer's spent grain)
 - More recycling/reuse e.g. packaging
- One Health
 - More plant based, less meat
 - Less processed foods
 - New/alternative production methods (e.g. precision fermentation)
 - 'Novel foods' (e.g. alternative protein sources)



- New or reemerging food safety risks??



3. Are plant-based beverages good alternatives for cow's milk?

The Tale of Two Beverages: A Journey from Farm to Glass

In the quiet countryside, two neighbors, Farmer Joe and Millie the Miller, had a friendly rivalry. Joe owned a dairy farm, producing fresh cow's milk, while Millie ran an oat processing facility, crafting creamy oat drinks. Both were proud of their products, but a question lingered in their minds: Which drink was truly better for the planet and human health?

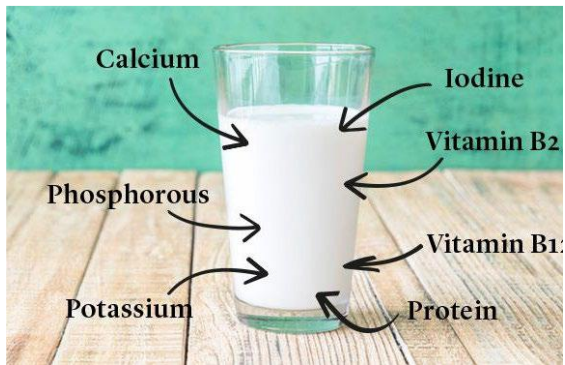
The Footprint of Farming

Joe's dairy farm was bustling with activity—cows grazing, milk trucks rolling in and out. But behind the picturesque scene was an invisible cost. Raising cows required vast pastures, massive amounts of feed and fields to grow them, and plenty of water. The cows also produced methane, a greenhouse gas much more potent than carbon dioxide. Millie's oat fields stretched in golden waves under the sun. Growing oats required less land and water than keeping cows. The processing of oats into a drink took energy, but compared to the emissions from dairy farming, it was less.

Water & Land: Who Drinks More? One morning, Joe and Millie met at the village well, where a conservationist, Dr. Green, was giving a talk on water use. "Did you know," Dr. Green began, "producing one liter of cow's milk takes about **628 liters of water**, while oat drink needs only **48 liters**?" Joe frowned. He knew dairy farming was water-intensive—his cows needed to drink daily, and cleaning the barns used up even more. Millie, feeling triumphant, smiled. Oats only needed rainwater, and their processing used far less than dairy farms.

The Energy Dilemma But Joe had another argument. "My milk is natural," he said. "You just milk the cow, and there it is. But your oat drink? Doesn't it need processing?" Millie nodded. "Yes, but that's still much less than your dairy product. Pasteurization, homogenization, and refrigeration all use a lot of energy, and transportation of fresh milk requires constant cooling. My oat drink can be stored at room temperature before opening." Joe had to admit—keeping milk cold from farm to fridge was an energy drain.

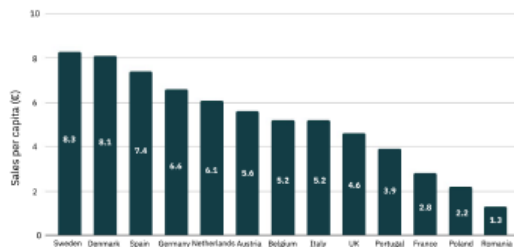
Nutrition: Strength in Different Ways Then came the final question: **Which drink was healthier?** Joe was quick to defend his milk. "Cow's milk is packed with protein, calcium, and vitamin B12. It's a complete protein source, meaning it contains all essential amino acids." Millie countered, "True, but my oat drink is fortified with calcium and B12 too. Plus, it has **more fiber and less saturated fat**, which is better for heart health." Dr. Green stepped in to settle the debate.



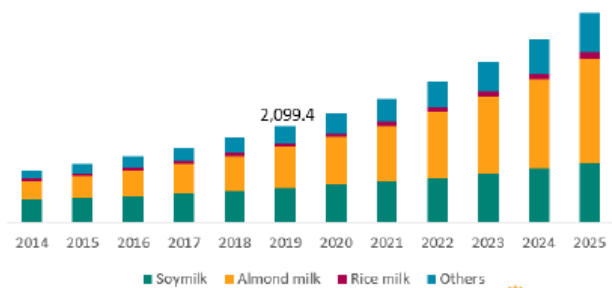
Background

- The number of consumers wanting to make more sustainable diet and lifestyle choices is increasing rapidly
- Denmark has Europe's second highest sales of plant-based milk per inhabitant, surpassed only by Sweden
 - Plant-based drinks make up 7% of Denmark's total market for milk
 - Sales of plant-based milk have increased by 17% between 2020 and 2022

Average plant-based milk spend per country per capita (in euros) 2022



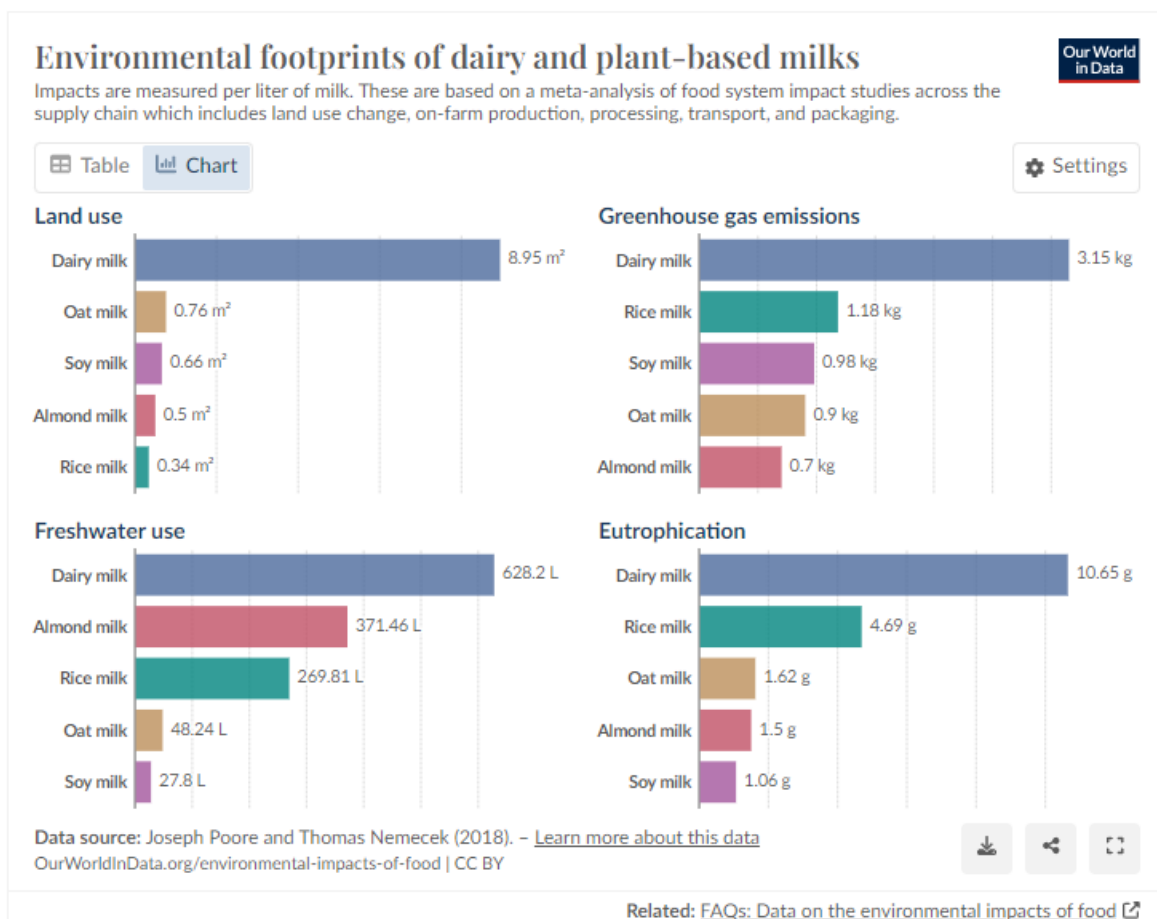
Europe dairy alternatives market estimates and forecasts, by product, 2014 - 2025 (USD Million)



- Cow's milk is an important part of the Danish diet
 - Official dietary guidelines recommends that children from 2 years consume 250-350 mL of low-fat dairy products a day, due to the content of calcium, vitamins B2 and B12, phosphorous, and iodine
 - Sales of animal milk, primarily cow's milk, have decreased by 10% between 2020 and 2022
- In this case study, you will be investigating whether milk alternatives, such as oat and almond beverages are good alternatives for Danish-produced cow's milk

Considerations

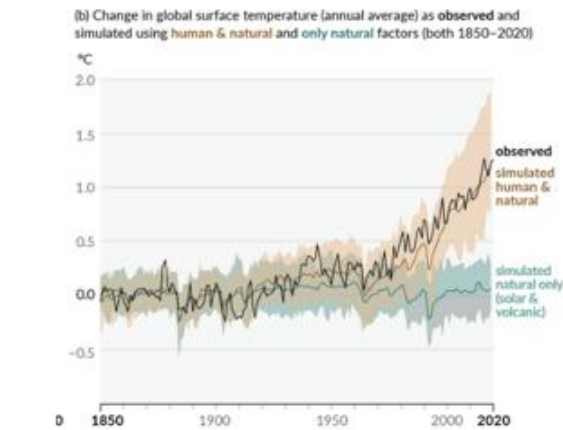
- Oat drink produced in Denmark from Danish raw materials is more environmentally sustainable than milk from cows
 - but is that also the case if the milk is organic?
 - or if the final product or the raw materials (e.g. almonds, soy, rice) are imported from overseas?
- And what about health and economy?
 - Cow's milk and the alternatives are very different products wrt their chemical and physical properties.
 - a comparison necessitate a consideration of the nutritional value and how the products is used
 - cow's milk naturally contains many minerals, vitamins, and protein that the alternatives lack unless supplemented



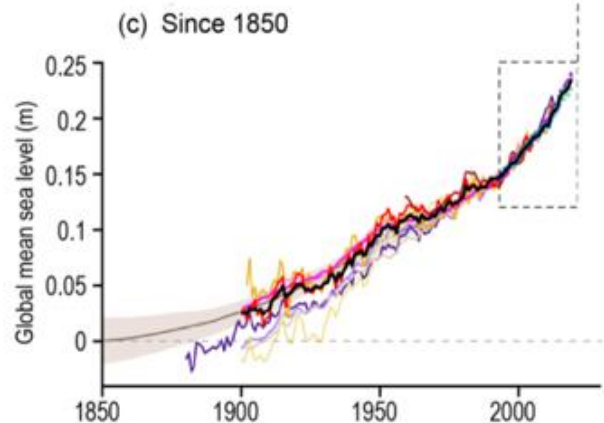
4. Introduction to environmental indicators - Carbon footprint

Global warming – Diagnostic and causes

Global warming relative to 1850-1900 (°C)



CC, October 2019, SPECIAL
PORT: GLOBAL WARMING OF 1.5
Summary for Policymakers")



1. Temperature Increase

- Temperature has increased significantly since the 1950s.
- As of 2020, global temperature has risen by approximately **1.2°C – 1.5°C** above pre-industrial levels (1850-1900).
- The **Intergovernmental Panel on Climate Change (IPCC)** confirms that this increase is largely due to **human activities**.

2. Human vs. Natural Causes

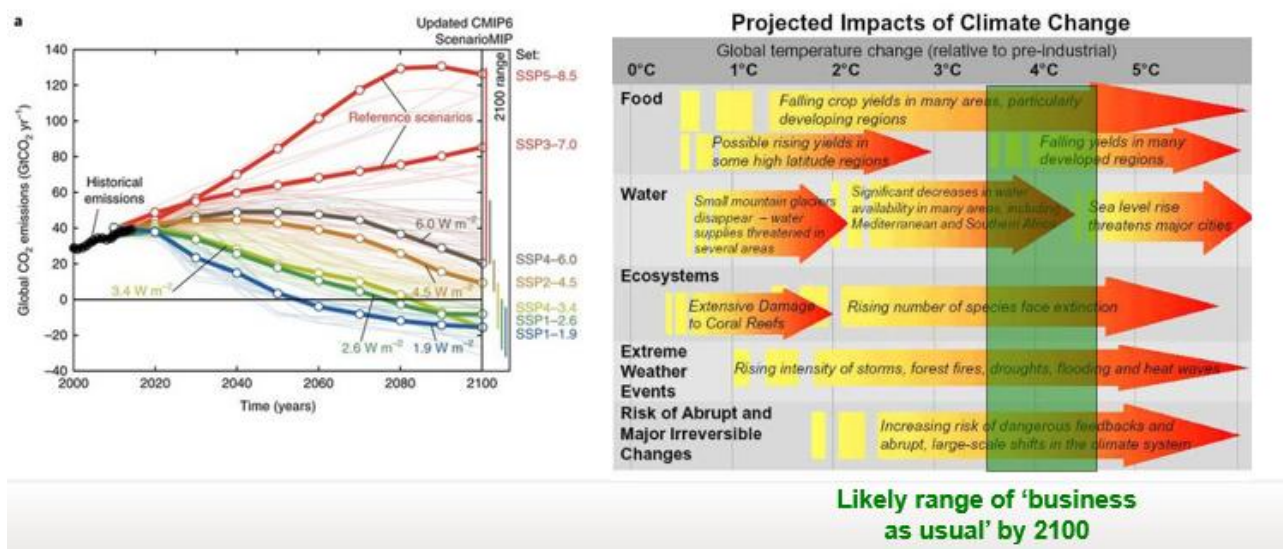
- **Human Influence:** Greenhouse gas emissions from industries, transport, and deforestation have led to a rapid temperature rise.
- **Natural Factors (Solar & Volcanic Activity):** If only natural factors were at play, the global temperature would remain relatively stable.
- Simulations confirm that the **observed warming aligns with human-induced factors**, not just natural variability.

3. Impact on Sea Level Rise

- Temperature increase leads to:

- **Glacier and polar ice melt**
- **Thermal expansion** of ocean water
- Since 1850, the global mean sea level has risen by **~25 cm**.
- Future projections indicate:
 - By 2100: **Up to 1 meter** rise
 - In extreme scenarios: **Up to 7 meters** rise
- Major risk to **coastal cities** like **Copenhagen** and other urban areas located near water bodies.

The emissions projections and related impacts



1. Future Temperature Scenarios

- If **business as usual** continues with fossil fuel use, the global temperature could **increase up to 8.5°C** by 2100 (worst-case scenario, SSP5-8.5).
- A likely outcome is at least a **4°C increase** if no significant action is taken.
- With **strong climate action**, we could limit warming to **1.5°C – 2°C**.

2. Climate Change Impacts by Temperature Increase

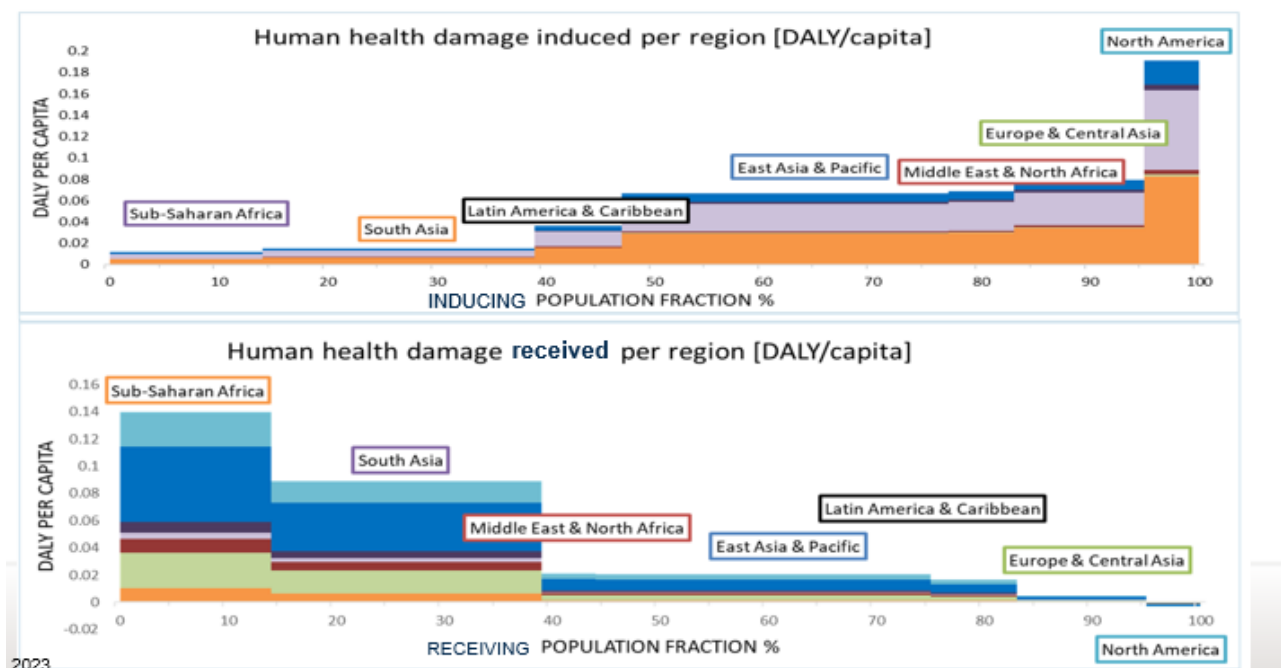
- **Food:**
 - **Falling crop yields** in many areas, especially in developing and later developed regions.
 - Some regions (high latitudes) may initially see **rising yields**, but this is temporary.

- **Water:**
 - **Glacier disappearance** → Threatens water supplies.
 - **Decreased water availability** in Mediterranean and Southern Africa.
 - **Sea level rise** threatens major cities.
- **Ecosystems:**
 - **Coral reef damage** and bleaching.
 - **Mass extinction** of species as temperature increases.
- **Extreme Weather Events:**
 - Increased **storm intensity, forest fires, droughts, flooding, and heatwaves.**
- **Irreversible and Large-Scale Changes:**
 - **Higher risk of climate system tipping points**, such as:
 - Potential **Gulf Stream collapse.**
 - Large-scale shifts in **climate patterns.**

3. The Need for Action

- Some countries are taking action, but **global efforts** are required.
- **Strong climate policies** and emission reductions can limit warming.
- To stay within **1.5°C – 2°C**, drastic and immediate action is needed.

Disparities: Climate change impacts of heat and cold on health



1. Who is Inducing Climate Change?

- **North America (USA & Canada) has the highest carbon footprint.**
 - About **twice** the emissions per capita compared to Europe.
- **Europe** is the second-highest contributor.
- **East Asia & Pacific (China in particular) has rapidly increased emissions** and is now at a similar level to Europe.
- **Sub-Saharan Africa and South Asia have minimal contributions** to climate change.

2. Who is Suffering the Most?

- **Heatwaves, extreme weather, and health damages disproportionately affect:**
 - **Sub-Saharan Africa**
 - **South Asia**
- These regions have contributed **least** to the problem but experience **the most severe health impacts**.

3. Measuring the Impact – Disability-Adjusted Life Years (DALY)

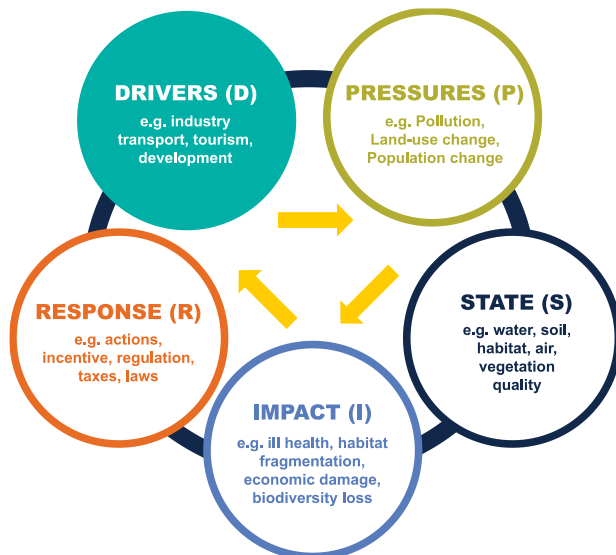
- **DALY per capita** (Years of healthy life lost due to climate-related health issues):
 - **Highest in Sub-Saharan Africa & South Asia.**
 - **Much lower in high-emission regions** like North America and Europe.

- **This creates a strong unfair disparity:**
 - Wealthier, high-emission countries can **adapt better** to climate change.
 - Poorer, low-emission countries suffer **more severe** consequences.

4. Why This Matters

- Climate change **affects ecosystems and human health.**
- **Moral and ethical responsibility:**
 - High-emission countries must **act urgently** to mitigate damage.
 - Priority should be on **helping the most vulnerable regions** adapt.

The DPSIR model: to categorize and causally link indicators

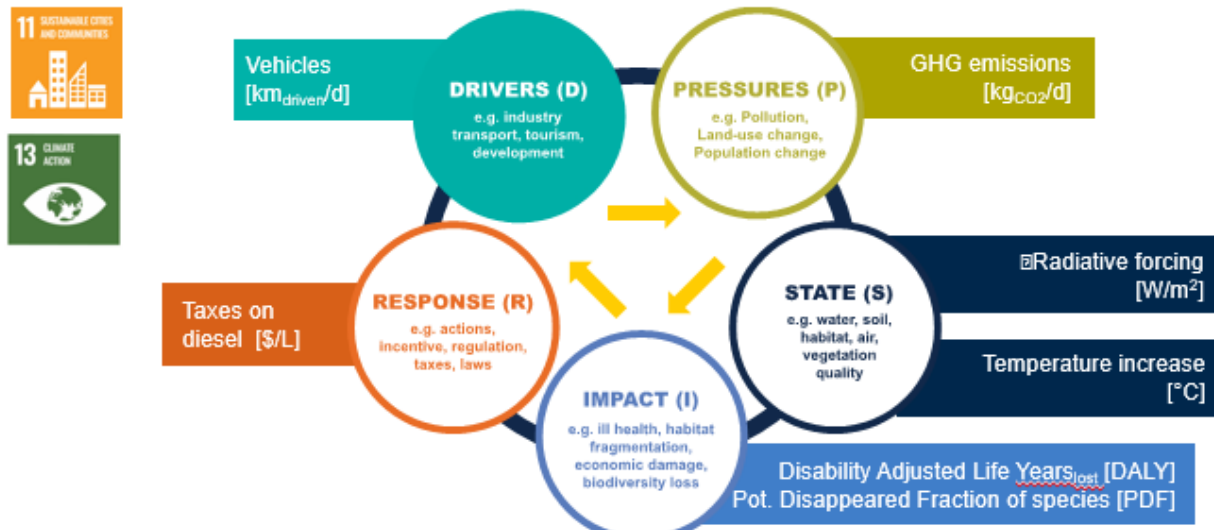


The **DPSIR** (Drivers-Pressures-State-Impact-Response) model is used to **categorize and link** human activities to environmental impacts.

1. Components of the DPSIR Model

- **D – Drivers:** Underlying human activities causing environmental change.
- **P – Pressures:** Emissions or resource use affecting the environment.
- **S – State:** Changes in environmental conditions.
- **I – Impact:** Consequences on ecosystems and human health.
- **R – Response:** Actions taken to mitigate or adapt to impacts.

The DPSIR model: applied to climate change impact of transportation



2. Example: Climate Change Impact of Transportation

- **Drivers (D)**: Transportation sector (e.g., personal cars, trucks).
 - Indicator: **Vehicle kilometers driven (km/driver/day)**.
- **Pressures (P)**: Emissions from transport.
 - Indicator: **Greenhouse gas (GHG) emissions (kg CO₂/day)**.
- **State (S)**: Changes in the climate system.
 - Indicators:
 - **Radiative forcing (W/m²)**
 - **Temperature increase (°C)**
- **Impact (I)**: Consequences of climate change.
 - Indicators:
 - **Human health impact (Disability Adjusted Life Years lost – DALY).**
 - **Biodiversity loss (Potentially Disappeared Fraction of species – PDF).**
- **Response (R)**: Policies and regulations to mitigate impacts.
 - Indicator: **Taxes on diesel/fossil fuels (\$/L).**
 - Encourages reduced fuel use and lower emissions.

3. Importance of DPSIR in Climate Policy

- Helps **quantify and track** climate change impacts.

- Provides a **structured way to design interventions** (e.g., carbon pricing, transportation policies).
- Links **human activities to environmental degradation** and supports sustainable development goals (**SDG 11: Sustainable Cities & SDG 13: Climate Action**).

Carbon Footprint: at personal and product levels

A **carbon footprint** is the weighted sum of greenhouse gas (GHG) emissions caused by an individual, event, organization, service, or product, expressed as carbon dioxide equivalent. GHG emissions are weighted based on their global warming potentials (GWP100, as defined by the Intergovernmental Panel on Climate Change, AR6, Tables 7.15 and 7.SM.7)

Greenhouse Gas	GWP100
Carbon Dioxide (CO ₂)	1 kg _{CO2e} /kg _{CO2}
Methane biogenic (CH ₄)	27 kg _{CO2e} /kg _{CH4}
Methane fossil (CH ₄)	29.8 kg _{CO2e} /kg _{CH4}
Nitrous Oxide (N ₂ O)	273 kg _{CO2e} /kg _{N2O}
Hydrofluorocarbons (HFC-32)	771 kg _{CO2e} /kg _{HFC}
Perfluorocarbons (PFC-14)	7380 kg _{CO2e} /kg _{PFC}
Sulphur Hexafluoride (SF ₆)	24300 kg _{CO2e} /kg _{SF6}

Notes:

A **carbon footprint** is the total greenhouse gas (GHG) emissions caused by an individual, event, organization, service, or product, expressed in **CO₂-equivalents (CO₂e)**.

1. Greenhouse Gas Comparison: Global Warming Potential (GWP100)

- Different GHGs have **different warming effects** on the climate.
- The **Intergovernmental Panel on Climate Change (IPCC)** defines **Global Warming Potential (GWP100)**, measuring how much heat a gas traps over **100 years** relative to CO₂.
- **Example Interpretation:**
 - 1 kg of methane (CH₄, biogenic) = 27 kg of CO₂e
 - 1 kg of SF₆ = 24,300 kg of CO₂e, making it extremely potent.

2. Using GWP100 in Carbon Footprint Calculations

- Convert emissions from different gases to **CO₂-equivalents (CO₂e)**.
- Example:

- If **1 kg of CH₄ is emitted**, it is equivalent to **27 kg CO₂**.
- If **1 kg of SF₆ is emitted**, it is equivalent to **24,300 kg CO₂**.
- This allows comparison of emissions across different activities and gases.

3. Why This Matters

- Helps prioritize **which gases to reduce** (e.g., SF₆ has a much larger impact than CO₂).
- Essential for **climate policies, carbon taxes, and emissions trading**.
- Reducing **high GWP gases** can have a **greater impact** on mitigating climate change.

Average DK carbon footprint per capita

Substance		Unit kg _d	Emissions kg/pers/y	GWP100 kg _{CO2e} /kg _d	GW score kg _{CO2e} /pers/d
Carbon dioxide	fossil	kg CO ₂	8000*	1	8000
Methane	biogenic	kg CH ₄	46.1	27.0	1250
Methane	fossil	kg CH ₄	4.7	29.8	140
Nitrous Oxide*		kg N ₂ O	3.2	273	870
HFCs, PFCs, SF ₆					80
TOTAL kg _{CO2e} /pers/y					10340
TOTAL kg _{CO2e} /pers/d					28.3

*consumption based emissions

2. Calculation Method

1. Multiply **each gas's emissions** by its **GWP100 factor** to convert to CO₂e:
 - Example: **Methane (CH₄, biogenic)**:
 - **46.1 kg CH₄ × 27 GWP = 1250 kgCO₂e**
 - Example: **Nitrous Oxide (N₂O)**:
 - **3.2 kg N₂O × 273 GWP = 870 kgCO₂e**
2. Sum up all contributions to get **total CO₂e emissions per year**.
3. Divide by **365 days** to get **daily emissions per capita**.

3. Key Insights

- **Carbon Dioxide (CO₂) is the largest contributor** (~8000 kg/person/year).

- Methane (CH₄) and Nitrous Oxide (N₂O) have much higher warming effects per kg than CO₂.
- Total carbon footprint per capita in Denmark is ~10.3 tons CO₂e per year (~28 kg CO₂e per day).
- Reducing high-GWP gases (like N₂O & CH₄) is critical for effective climate mitigation.

At which level of the cause effect chain is the carbon footprint?



1. Position of Carbon Footprint in DPSIR Framework

- The carbon footprint (kgCO₂e/day) is measured between the "Pressure" and "State" levels in the DPSIR model.
- It is a **pressure indicator** that quantifies GHG emissions using a state indicator (radiative forcing, W/m²).

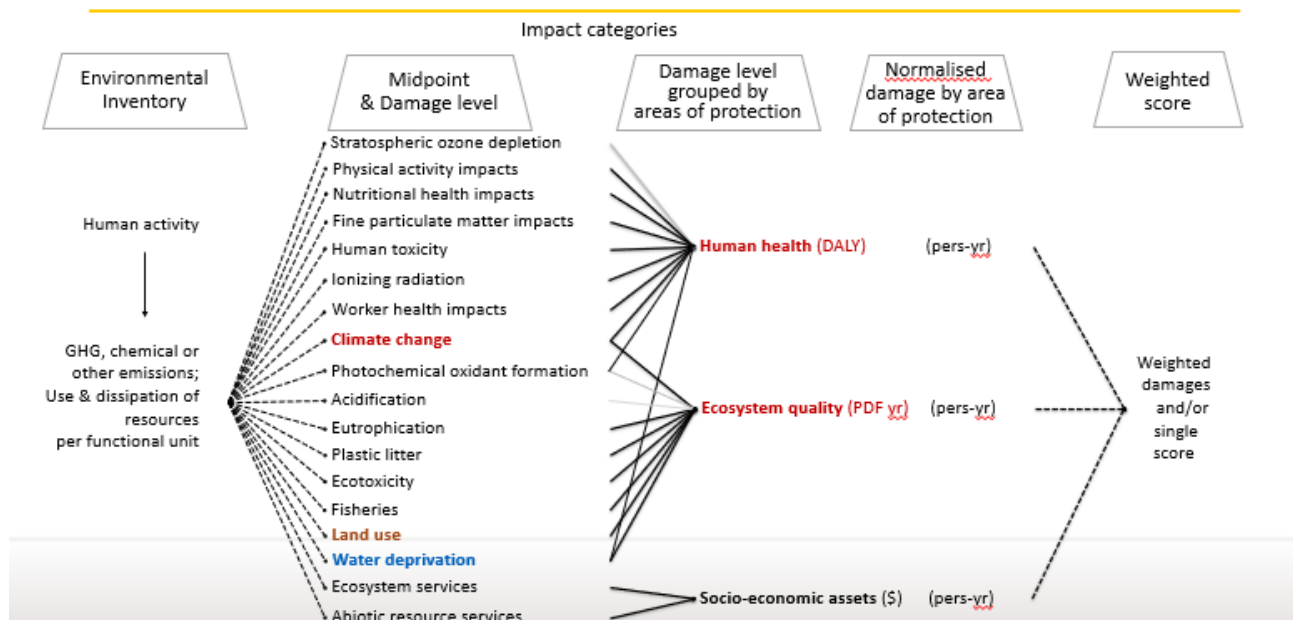
2. Why Measure Carbon Footprint at This Level?

- Measuring at **GHG emissions level (Pressure)** allows **comparison between different greenhouse gases**.
- Using **Global Warming Potential (GWP100)** ensures a **standardized comparison** between gases like CO₂, CH₄, and N₂O.
- **Beyond this stage (State and Impact levels), uncertainty increases:**
 - Temperature increase and health impacts (DALY, biodiversity loss) **are proportional** to emissions.
 - Measuring at **temperature or health impact level** would introduce **higher uncertainties**.

3. Practical Use of Carbon Footprint

- Carbon footprint is **used to track emissions from industries, transportation, and daily activities**.
- It **guides climate policies** by identifying **key sources of emissions**.
- Helps set **targets for emission reductions and sustainability goals** (e.g., SDG 11 & 13).

UNEP - GLAM: the global life cycle impact assessment framework



1. The Role of Carbon Footprint in Impact Assessment

- **Carbon footprint** is just **one** of many environmental impact categories.
 - It primarily falls under **climate change**, which is a **midpoint indicator** in Life Cycle Assessment (LCA).
-

2. The UNEP - GLAM Framework (Life Cycle Impact Assessment)

This framework categorizes **impact pathways** from **human activity** to environmental, health, and economic effects.

A. Environmental Inventory

- **Human activities** (e.g., industries, transportation, agriculture) lead to **GHG emissions, chemical emissions, and resource use**.

B. Midpoint & Damage Levels

- Midpoint categories measure **specific environmental problems**, such as:
 - **Climate change** (CO₂ emissions, GHGs)
 - **Ozone depletion**
 - **Air pollution** (fine particulates, human toxicity)
 - **Eutrophication** (nutrient pollution in water)
 - **Acidification**
 - **Land use & water deprivation**
 - **Biodiversity loss and ecotoxicity**

C. Damage Level by Areas of Protection

- Impacts are aggregated into three main areas:
 1. **Human health (DALY - Disability Adjusted Life Years)**
 2. **Ecosystem quality (PDF·yr - Potentially Disappeared Fraction of species per year)**
 3. **Socioeconomic assets (\$ - Financial losses)**

D. Normalized & Weighted Scores

- Impacts can be normalized and compared.
 - Some methodologies use **a single impact score**, but many prefer to **keep different categories separate** for better decision-making.
-

3. Application to Case Studies

- The goal is to **apply impact measurement methods** to real-world case studies.
 - **First case study: Automobiles & carbon footprint**
 - **Second case study: A custom case study** for further analysis.
-

Key Takeaways

- The **carbon footprint** is a **pressure indicator** that falls within **climate change**.
- LCA considers **multiple environmental impacts**, not just carbon emissions.
- **Measuring impacts early** in the cause-effect chain improves accuracy.
- The **GLAM framework** helps **compare different impact categories** effectively.

4. Primary energy and carbon footprint - Gasoline vs electric vehicles

Building the energy balance and carbon footprint

- 1 **Functional unit (FU):** Identify the functional basis for comparing the different scenarios. Footprints are calculated per functional unit (FU representative of **what service you want, not a ratio**)
- 2 **Collect composition and consumption data:** collect the inputs to product manufacturing and its consumption during operation and calculate each input/FU (what you need to provide the service, in e.g. $\text{kg}_{\text{input}}/\text{FU}$, kWh/FU)
- 3 **Find resource consumption and GHG emission factors per unit input:** Find in databases for each input the $\text{MJ}_{\text{non renewable primary energy}}$ and GHG emission in $\text{kg}_{\text{CO2 equivalent}}$ per unit input
- 4 **Calculate energy balance and carbon footprint per FU:** Multiply 2 and 3 to obtain the contribution of each input to the energy and carbon footprint ($\text{MJ}_{\text{non renewable primary energy}}/\text{FU}$, $\text{kg}_{\text{CO2 equivalent}}/\text{FU}$)
- 5 **Total footprint and interpretation:** Sum up the footprint of each input to calculate the total energy and carbon footprint of the considered service or product

1. Define the Functional Unit (FU)

- The **functional unit** is the basis for comparison.

- Footprints are calculated **per FU**, which represents **the service provided, not just a ratio**.
- Example: "**1 km driven by a car**", "**1 kg of steel produced**", or "**1 kWh of electricity consumed**."

2. Collect Composition and Consumption Data

- Identify all **inputs required** for the product or service.
- Measure **consumption per FU**, such as:
 - **Material input (kg/FU)**
 - **Energy use (kWh/FU, MJ/FU)**
 - **Fuel consumption (liters/FU)**

3. Find Resource Consumption & GHG Emission Factors

- Use **LCA databases** to find:
 - **Primary energy demand (MJnon-renewable/FU)**
 - **Greenhouse gas (GHG) emissions (kgCO₂e/FU)**
- Example sources:
 - **Emission factors for steel, aluminum, gasoline, electricity, etc.**

4. Calculate Energy Balance & Carbon Footprint per FU

- Multiply data from Step 2 (inputs) by Step 3 (GHG emission factors).
- Obtain:
 - **Total energy demand (MJ/FU)**
 - **Total carbon footprint (kgCO₂e/FU)**

5. Total Footprint & Interpretation

- Sum up **all emissions from different inputs** to determine:
 - **Total energy consumption**
 - **Total carbon footprint per FU**
- **Interpret the results:**
 - Compare different scenarios (e.g., gasoline vs. electric car).
 - Identify **hotspots** for emissions reduction.
 - Guide **decision-making** in product design and policy.

Energy and CO₂ balance gasoline: calculate each input/FU

1		2							
Vehicle lifespan (km/vehicle)	150000			Non renewable primary energy			Carbon footprint		
Transportation by Gas Vehicle FU= 1 vehicle ton	Quantity original unit	Quantity per FU	Unit	Energy/Unit [MJ/Unit]	Energy per FU [MJ/FU]	fraction energy	CO ₂ eqs/Unit [kg/unit]	CO ₂ eqs/FU	fraction carbon
Manufacturing (Materials + Processing)	per vehicle over lifespan	per FU							
steel, low-alloyed, hot rolled	1040		kg		0.00E+00	0.0%		0.00E+00	0.0%
sheet rolling, steel	780	5.20E-03	kg	4.06	2.58E-02	8.9%	0.32	1.66E-03	9.8%
aluminum, primary, ingot	55	3.67E-04	kg	218.90	8.03E-02	27.7%	20.97	7.69E-03	45.4%
Copper & other metals (Zn, Ni, Pb)	38	2.53E-04	kg	74.58	1.89E-02	6.5%	5.86	1.49E-03	8.8%
platinum	1.70E-03	1.13E-08	kg	1095329	1.24E-02	4.3%	69212	7.84E-04	4.6%
polyethylene, polypropylene & PVC	175	1.17E-03	kg	79.36	9.26E-02	32.0%	2.32	2.70E-03	15.9%
synthetic rubber	46	3.07E-04	kg	83.60	2.56E-02	8.8%	2.73	8.36E-04	4.9%
glass	32	2.13E-04	kg	11.44	2.44E-03	0.8%	1.01	2.16E-04	1.3%
electronics, for control units	6.1	4.07E-05	kg	448.25	1.82E-02	6.3%	31.50	1.28E-03	7.6%
ethylene, paints, fluids	30	2.00E-04	kg	66.77	1.34E-02	4.6%	1.45	2.91E-04	1.7%
tap water	3400	2.27E-02	kg	0.01	1.19E-04	0.0%	0.00	7.55E-06	0.0%
heat, district or industrial, natural gas	2400		MJ		0.00E+00	0.0%		0.00E+00	0.0%
electricity, medium voltage	2250		kWh		0.00E+00	0.0%		0.00E+00	0.0%
Transportation	per vehicle	per FU							
transport, freight train	556.50		t-km	0.70	0.000	0.00%	0.04	0.00E+00	0.0%
transport, freight, lorry 16-32 metric ton	59.10		t-km		0.000	0.00%		0.00E+00	0.0%
transport, freight, sea, container ship	10400.00		t-km	0.13	0.000	0.00%	0.01	0.00E+00	0.0%
Use phase	per 15 km	per FU							
Supply chain petrol, low-sulfur	0.0019		kg		0.000	0.0%		0.00E+00	0.0%
Direct combustion	0.0019		kg	0.00	0.000	0.0%	3.11	0.00E+00	0.0%
Disposal	per vehicle over lifespan	per FU							
plastic mixture incineration	283.00		kg	0.48	0.00E+00	0.0%	2.35	0.00E+00	0.0%
metals recycles	38.00		kg	1.68	0.00E+00	0.0%	0.23	0.00E+00	0.0%
aluminum recycled	55.00		kg						
iron and mixed metal scrap recycled	1078.00		kg						
Total		per FU							
Total	-	-	-		0.290	100.0%	8.30E-07 [DALY/kgCO ₂ e]	0.017 [DALY/FU]	100.0%

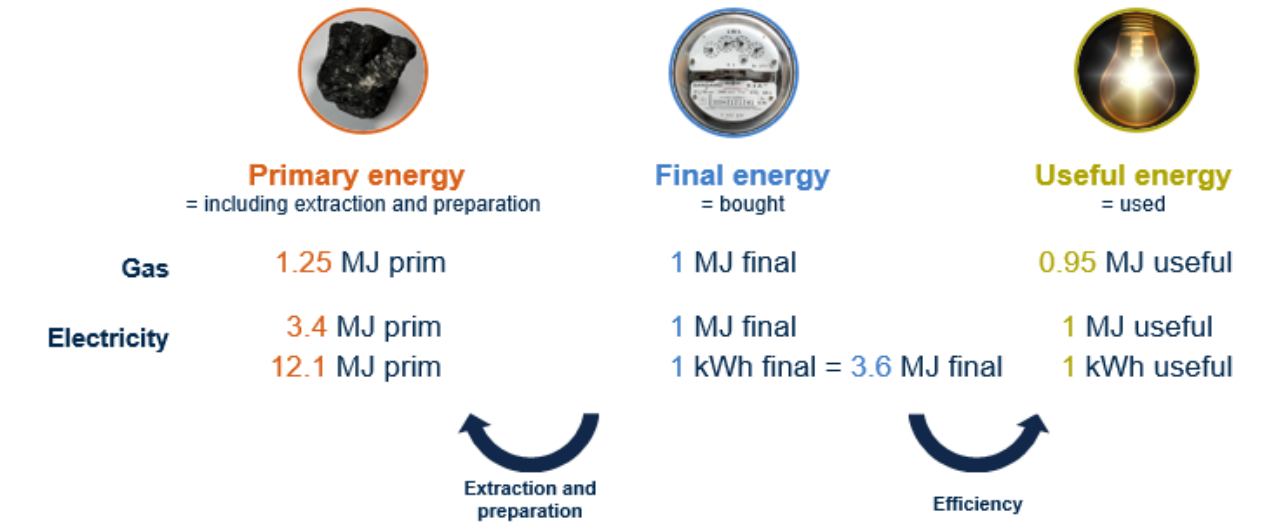
Key Takeaways

- **Aluminum and steel production** are major contributors to emissions.
- **Fuel consumption dominates total emissions during vehicle use.**
- **Material recycling helps reduce end-of-life emissions.**
- **Comprehensive Life Cycle Assessment (LCA) includes all phases** (manufacturing, transport, use, and disposal).

Primary energy definition

- Primary energy is defined as the energy contained in the energy carriers at the point of extraction from the environment, including the upstream energy usage for extraction, preparation, and distribution.
- The final energy is the energy purchased by the consumer and delivered to him.
- The useful energy is the portion of final energy which is actually available after final conversion to the consumer for the respective use.

Energy: definitions



Non-renewable primary energy and CO₂ aggregated inventory flows

		Non-renewable primary energy (MJ per unit)	CO ₂ (kg per unit)	gCO ₂ /MJ ratio
Energy Carriers	1 kWh electricity (Europe)	8.8	0.49	47
	1 kWh electricity (USA)	12.1	0.71	59
	1 kWh electricity (Japan)	11.5	0.53	46
	1 kWh electricity (Switzerland)	7.9	0.11	13
	1 kWh electricity (China)	10.4	0.98	94
	1L gasoline (no combustion)	42.9	0.49	11
	1L gasoline (with combustion)	42.9	2.88	65
	1 kg light oil (42.7 MJ final)	56.8	3.71	65
Transport	1,000 km kg transport <u>16-32 ton</u> Lorry	2.6	0.15	58
	1 pers • km by train (Intercity)	0.98	0.06	58
	1 pers • km by plane (Intercontinental)	3.28	0.19	60
	1 pers • km by car	3.0	0.17	57

Energy Carriers

- The carbon footprint of electricity varies by country based on the energy mix.
- Coal-dependent countries (e.g., China, USA) have higher emissions per kWh.
- Hydropower/nuclear-based countries (e.g., Switzerland, Norway) have lower emissions per kWh.
- 1 liter of gasoline (with combustion) emits 2.88 kg CO₂.
- Diesel and light oil have similar emissions per unit energy.
- Gasoline has higher emissions per energy unit compared to electricity in low-carbon regions.
- Switching from gasoline to electricity significantly reduces emissions, especially in clean electricity grids.

Transport

- Air travel and cars emit similar amounts of CO₂ per passenger per km.
- Trains have the lowest emissions per passenger per km.
- Planes and cars have similar emissions per km per passenger.
- Trains are significantly more efficient in terms of CO₂ emissions.

		Non-renewable primary energy (MJ per unit)	CO ₂ (kg per unit)	gCO ₂ /MJ ratio
Materials	1 m ³ concrete	1,381	257	186
	1 m ³ water	5.55	0.30	54
	1 kg steel, low alloy	27.4	1.63	59
	1 kg copper	31.2	1.86	60
	1 kg primary aluminium	160.4	9.55	59
	1 kg recycled aluminium	22.4	1.32	60
	1 kg gold	269,000	16,500	61
	1 kg paper newsprint	24.3	1.22	50
	1 kg glass	11.5	0.63	55
	1 kg polyethylene HDPEs	76.4	1.56	20
Material processing	1 kg plastic film extrusion	9.25	0.50	54
	1 kg injection moulding	27.1	1.24	46
End of life	1 kg landfilled steel	0.197	0.00657	33
	1 kg landfilled aluminium	0.521	0.02010	39
	1 kg incinerated municipal solid waste	0.43	0.50	1,161
	1 kg incinerated polyethylene	0.22	2.99	13,606

Materials

- Different materials have different carbon intensities (kgCO₂/unit).
- Lightweight materials (e.g., aluminum) can help reduce emissions in transport.
- Some emissions occur later in the lifecycle, such as plastic incineration.

Material processing

- Plastic processing is relatively low in emissions but depends on energy source.

End of life

- Landfilling has a lower carbon impact than incineration.
- Plastic incineration emits ~3 kg CO₂/kg, as stored carbon is released during combustion.
- Steel and aluminum landfilling have very low emissions, but recycling is preferred.

Use of ecoinvent: tips for good use – be careful



ELECTRICITY:

select the mix corresponding to each scenario

Medium voltage for industrial use,
Low voltage for household use



TRANSPORT:

beware the load rate
(includes 40% empty trips for trucks)

Data in energy sector available in three forms:

- Energy data **per fuel unit quantity**,
e.g. per kg oil or liter gasoline: **without** combustion,
- Energy **per MJ final (bought) energy**: **"burned"** = **with** combustion
Use the net calorific values to convert back to fuel quantities
(kg oil, m³ gas or liter gasoline)
- Energy **per MJ useful (delivered) energy**: **"heat"** = **with** combustion

Be sure to include end-of-life process & **only use fossil CO₂**,
not biogenic CO₂. **Avoid to use heavy metal uptake by crops.**

Notes:

- ✓ Check whether combustion emissions are included.
- ✓ Use the correct electricity mix based on usage (industrial vs. household).
- ✓ Transport emissions should account for load rates and empty trips.
- ✓ Ensure fossil CO₂ is used, avoiding biogenic CO₂ in calculations.

Energy and CO₂ balance gasoline: calculate footprint/input

Vehicle lifespan (km/vehicle)		150000				3		3	
Transportation by Gas Vehicle		Quantity original unit	Quantity per FU	Unit	Energy/Unit (MJ/Unit)	Energy per FU (MJ/FU)	Fraction energy	CO ₂ eq/Unit (kg/unit)	CO ₂ eq/FU (kg/FU)
FU= 1 vehicle km									
Manufacturing (Materials + Processing)		per vehicle over lifespan	per FU						
steel, low alloyed, hot rolled		1040		kg		0.00E+00	0.0%	0.00E+00	0.0%
sheet rolling, steel		780	5.20E-03	kg	4.96	2.58E-02	8.9%	0.32	1.66E-03
aluminium, primary, ingot									
Copper & other metals (Zn, Ni, Pb)									
platinum		1.71							
polyethylene, polypropylene & PVC									
synthetic rubber									
glass									
electronics, for control units									
ethylene, paints, fluids									
tap water		3							
heat, district or industrial, natural gas		2							
electricity, medium voltage		2							
Transportation		per							
transport, freight train		55							
transport, freight, lorry 16-32 metric ton		51							
transport, freight, sea, container ship		104							
Use phase		per							
Supply chain petrol, low sulfur		0.1							
Direct combustion		0.1							
Disposal		per vehicle over lifespan	per FU						
plastic, medium incineration		283.00		kg	0.48	0.00E+00	0.0%	2.35	0.00E+00
metals recycled		38.00		kg	1.68	0.00E+00	0.0%	0.23	0.00E+00
aluminium recycled		55.00		kg					
iron and mixed metal scrap recycled		1078.00		kg					
Total			per FU						
Total						0.290	100.0%	8.30E-07	0.017
								DALY/kgCO ₂ e	DALY/FU
								1.41E-08	100.0%

4. Scenario comparison and interpretation - Interpretation Principles

Results Interpretation: look from different angles!



Each level:

after input flows, pollutants inventory, after characterisation and carbon footprint impact evaluation;

Interpret results from multiple angles, for:

Each LCA stage (raw material, manufacturing, transportation, use, disposal), concentrating on those which generate the largest impact and on those which offer the best impact reduction potential;

Each component contribution for a multi-component product.

Each pollutant and processes, analysing their respective contribution to each midpoint impact category.

Each midpoint impact category:

analysing their respective contribution to each damage or weighted final score per scenario

Interpretation question for each scenario



Which are the largest contribution to the environmental impact in terms of production phase, in terms of pollutant?

- The grouping of processes can be arbitrary and depends on how the system is modeled.
- Also consider large sets of individual processes that each appear to have small impacts but lead to substantial impacts when summed

Which life cycle stage has the highest potential to reduce impacts (also depending on feasibility)

- In some win-win cases, both impacts and costs can be reduced at the same time.
- Even a limited low-cost intervention can be extremely efficient in reducing impacts.

Interpretation questions across scenarios: why? Why? WHY?

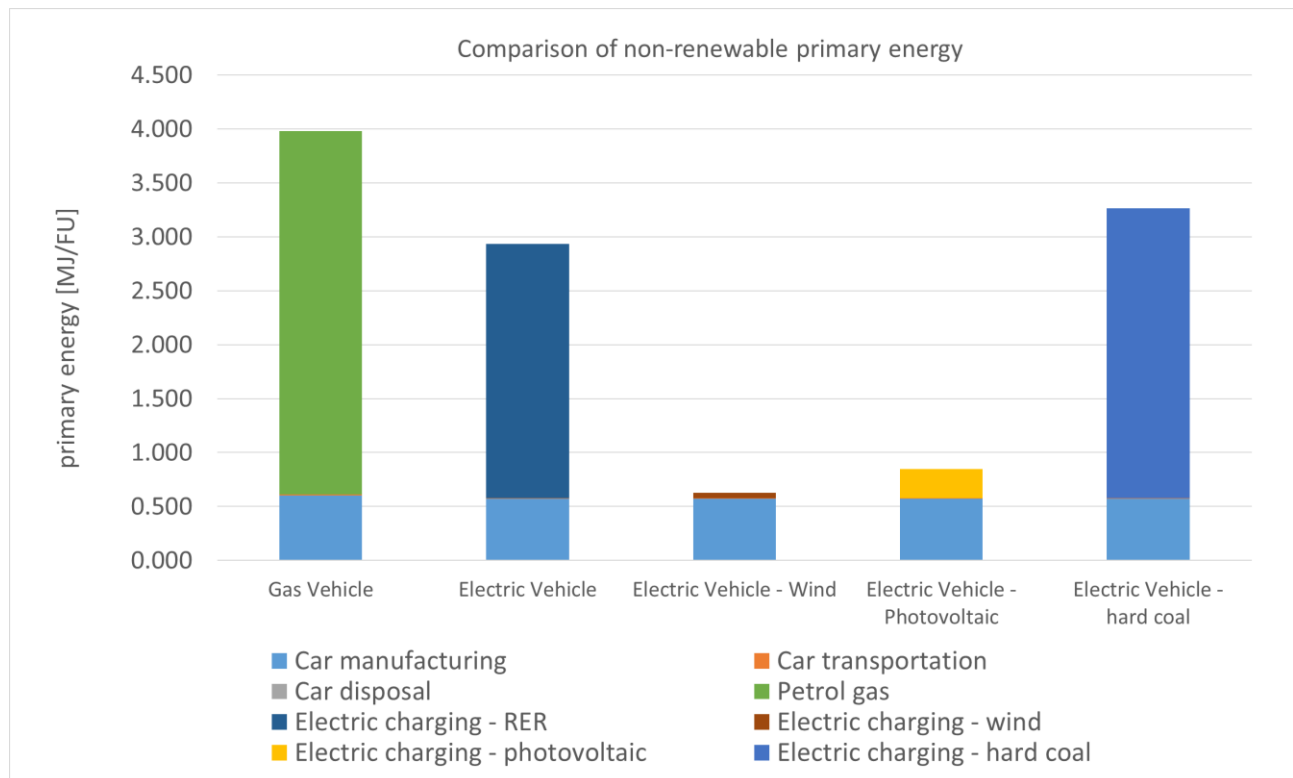
- Are the differences between scenarios significant, relevant?
For which impact category?
- Are differences/conclusions correct and robust: *check check check*
- What is the cause for difference in term of process, inventory flow?
- What are the key parameters affecting the environmental impact?
(*sensitivity studies*)

- **If some results are surprising: WHY?** What are the underlying parameter components and stages responsible for it?

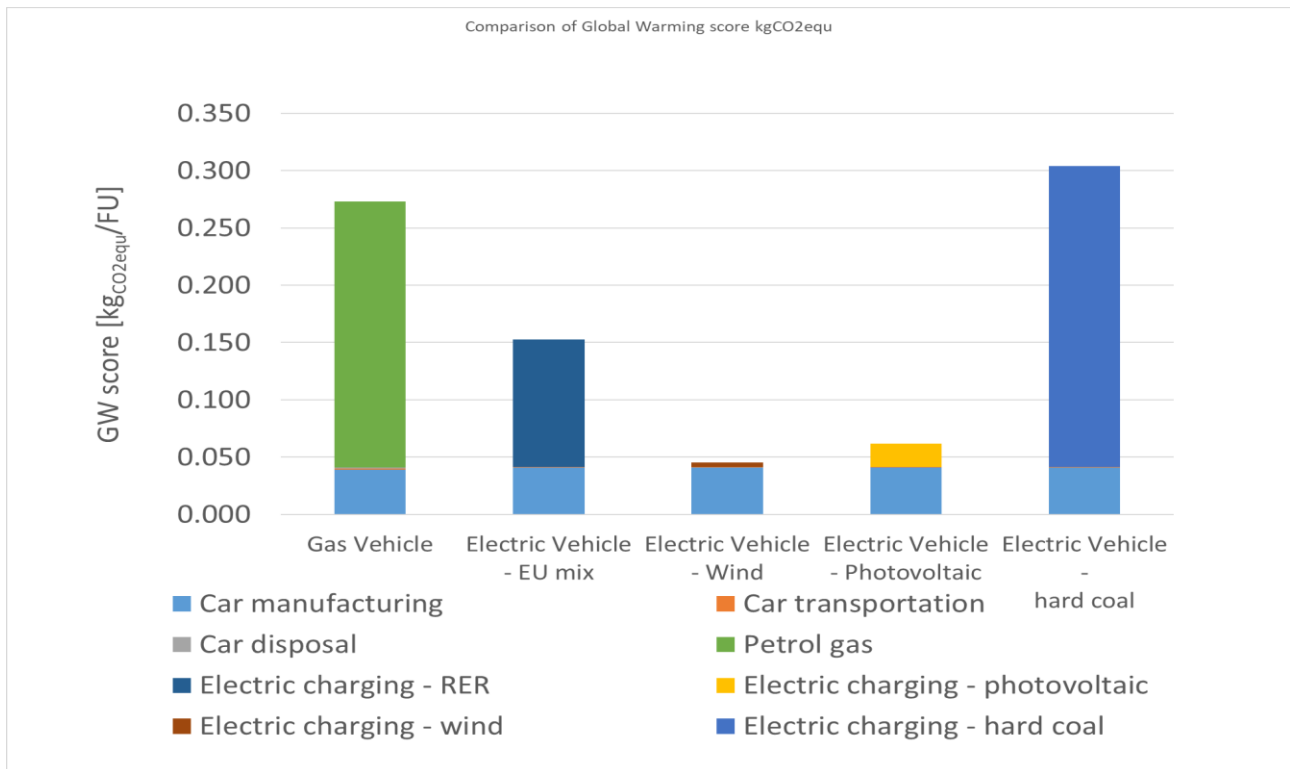
Energy and CO₂ balance gasoline: calculate total and interpret

Vehicle lifespan (km/vehicle)	150000			Non renewable primary energy			Carbon footprint		
Transportation by Gas Vehicle FU= 1 vehicle km	Quantity original unit	Quantity per FU	Unit	Energy/Unit [MJ/unit]	Energy per FU [MJ/FU]	fraction energy	CO ₂ eq/Unit [kg/unit]	CO ₂ eq/FU	fraction carbon
Manufacturing (Materials + Processing)	per vehicle over lifespan	per FU							
steel, low-alloyed, hot rolled	1040		kg	0.00E+00	0.00E+00	0.0%	0.00E+00	0.00E+00	0.0%
sheet rolling, steel	780	5.20E-03	kg	4.96	2.56E-02	8.9%	0.32	1.66E-03	9.8%
aluminum, primary, ingot	55	3.67E-04	kg	218.90	8.03E-02	27.7%	20.97	7.65E-03	45.4%
Copper & other metals (Zn, Ni, Pb)	38	2.53E-04	kg	74.58	1.89E-02	6.5%	5.86	1.49E-03	8.8%
platinum	1.70E-03	1.13E-06	kg	1095329	1.24E-02	4.3%	69212	7.84E-04	4.6%
polyethylene, polypropylene & PVC	175	1.17E-03	kg	79.30	9.26E-02	32.0%	2.32	2.70E-03	15.9%
synthetic rubber	46	3.07E-04	kg	83.60	2.56E-02	8.8%	2.73	8.36E-04	4.9%
glass	32	2.13E-04	kg	11.44	2.44E-03	0.8%	1.01	2.18E-04	1.3%
electronics, for control units	6.1	4.07E-05	kg	448.25	1.62E-02	6.3%	31.50	1.28E-03	7.6%
ethylene, paints, fluids	30	2.00E-04	kg	66.77	1.34E-02	4.6%	1.45	2.91E-04	1.7%
tap water	3400	2.27E-02	kg	0.01	1.19E-04	0.0%	0.00	7.55E-06	0.0%
heat, district or industrial, natural gas	2400		MJ	0.00E+00	0.00E+00	0.0%	0.00E+00	0.00E+00	0.0%
electricity, medium voltage	2250		kWh	0.00E+00	0.00E+00	0.0%	0.00E+00	0.00E+00	0.0%
Transportation	per vehicle	per FU							
transport, freight train	556.50		t.km	0.70	0.00E+00	0.00%	0.04	0.00E+00	0.0%
transport, freight, lorry 16-32 metric ton	59.10		t.km	0.00	0.00E+00	0.00%	0.00	0.00E+00	0.0%
transport, freight, sea, container ship	10400.00		t.km	0.13	0.00E+00	0.00%	0.01	0.00E+00	0.0%
User phase	per vehicle	per FU							
Supply chain petrol, low sulfur	0.0619		kg	0.00E+00	0.00E+00	0.0%	0.00E+00	0.00E+00	0.0%
Direct combustion	0.0619		kg	0.00	0.00E+00	0.0%	3.11	0.00E+00	0.0%
Disposal	per vehicle over lifespan	per FU							
plastic, mixture incineration	283.00		kg	0.48	0.00E+00	0.0%	2.35	0.00E+00	0.0%
metals recycled	38.00		kg	1.68	0.00E+00	0.0%	0.23	0.00E+00	0.0%
aluminum recycled	55.00		kg						
iron and mixed metal scrap recycled	1078.00		kg						
Total		per FU							
Total	-	-	-	0.290	100.0%		8.30E-07	1.41E-06	100.0%
							DALY/kgCO ₂ e	DALY/FU	

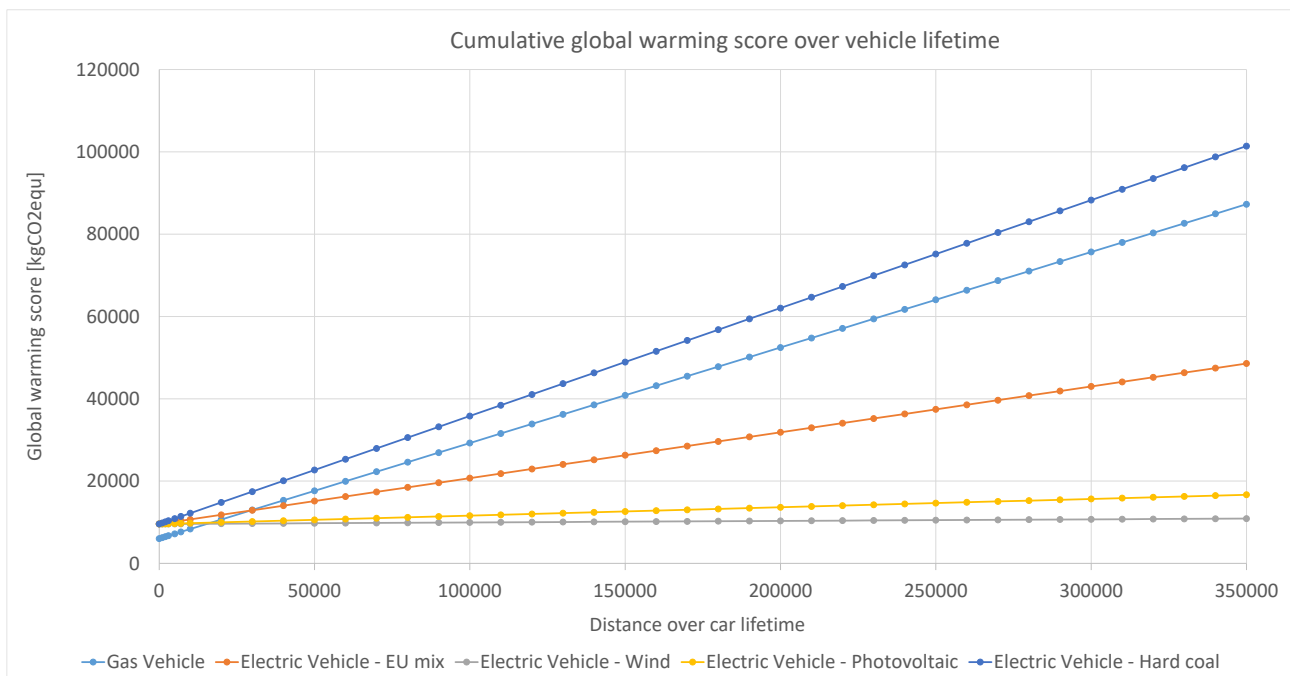
Energy balance: gasoline vs electric vehicle



Carbon footprint: gasoline vs electric vehicle



Sensitivity study: cumulative carbon footprint as a function of vehicle lifetime



1. Understanding the Sensitivity Study

- This study evaluates the **cumulative carbon footprint** of gasoline and electric vehicles over their **entire lifetime**.

- The carbon footprint is measured as the **global warming score (kg CO₂e)** as a function of **distance driven (km)**.
-

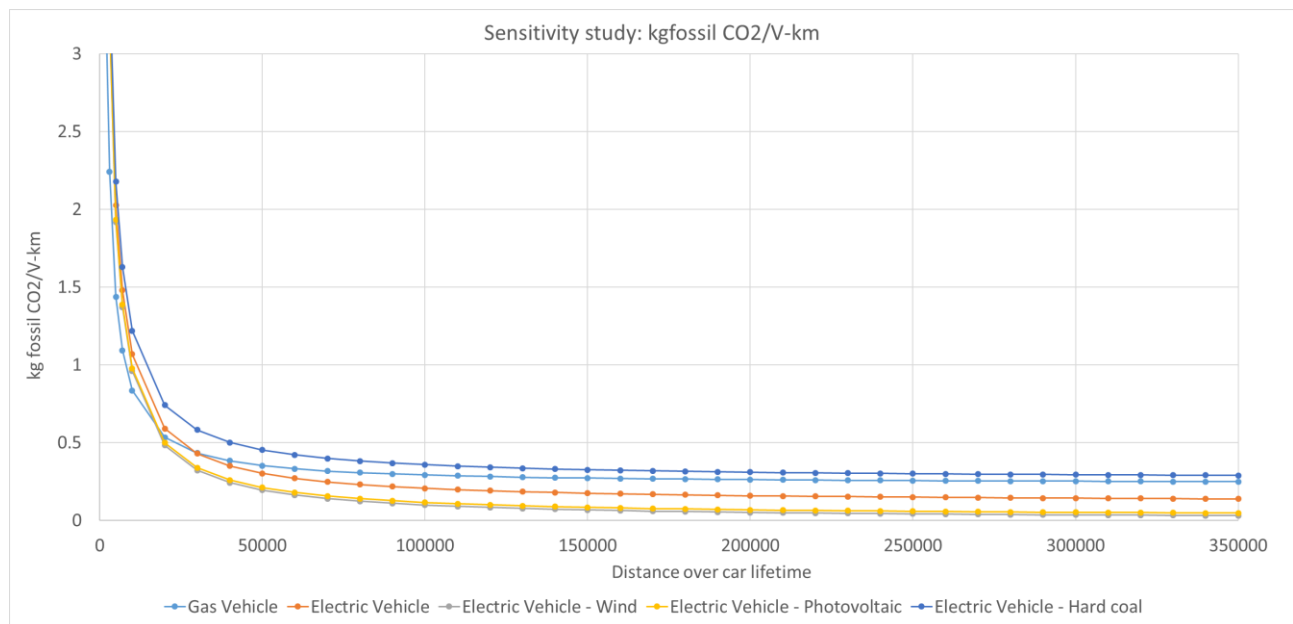
2. Key Observations from the Graph

- **Gasoline Vehicle (blue line):**
 - Starts with a **lower manufacturing footprint**.
 - Has a **higher carbon footprint per km** due to fuel combustion.
 - The total carbon footprint **increases linearly** over time.
 - **Electric Vehicle – EU Mix (orange line):**
 - **Higher emissions at manufacturing** (due to battery production).
 - **Break-even point at ~25,000 km**, where it becomes **better than a gasoline vehicle** in terms of emissions.
 - Over **full lifetime (150,000 - 225,000 km)**, the EV emits about half the CO₂ of a gasoline vehicle.
 - **Electric Vehicle – Wind (gray line) & Photovoltaic (yellow line):**
 - Much **lower lifetime emissions** due to clean electricity.
 - **Break-even point at ~15,000 km**.
 - **Lowest total emissions** over the vehicle's lifespan.
 - **Electric Vehicle – Hard Coal (dark blue line):**
 - **Higher emissions than the EU mix** due to coal-based electricity.
 - Still **performs slightly better than a gasoline vehicle**, but only after ~100,000 km.
-

3. Break-even Analysis

- **Break-even point** = The distance at which an EV becomes **more carbon-efficient** than a gasoline vehicle.
- **Break-even varies based on electricity mix:**
 - **Wind / solar energy** → ~15,000 km
 - **EU average electricity mix** → ~25,000 km
 - **Coal-powered EVs** → ~100,000 km

Sensitivity study: carbon footprint per FU as a function of vehicle lifetime



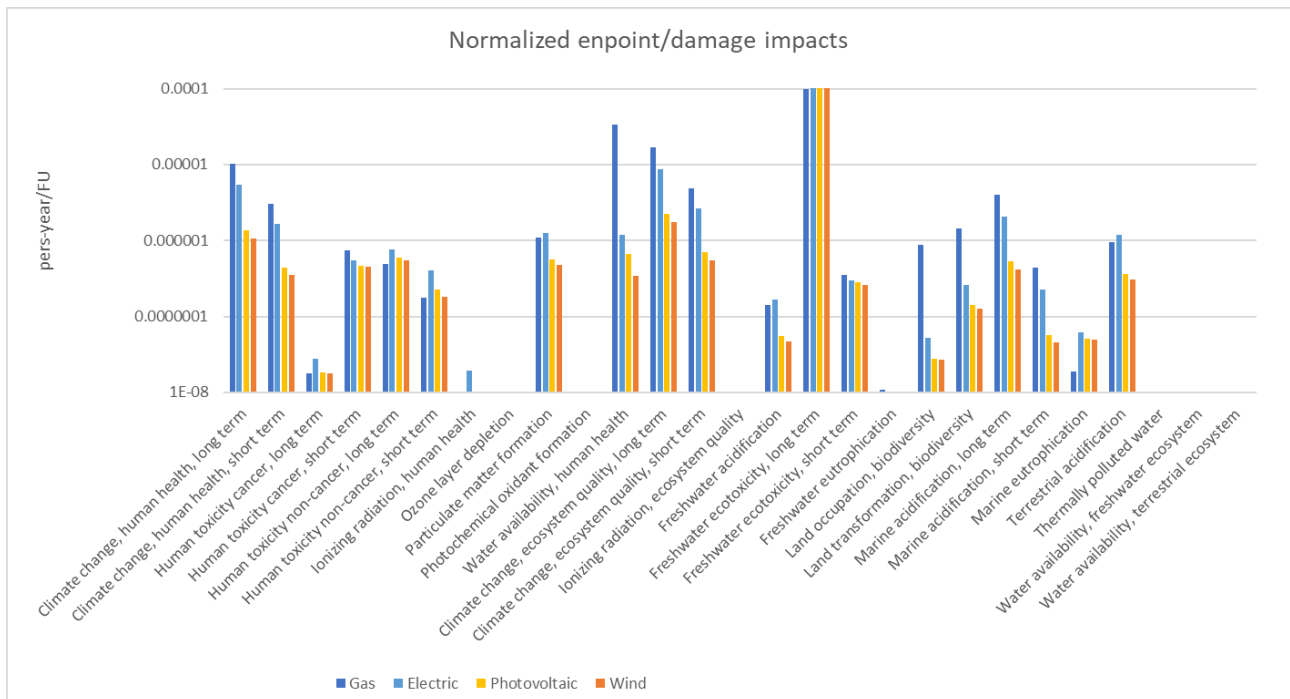
1. Understanding the Sensitivity Study

- Instead of **cumulative emissions**, this graph shows **CO₂ emissions per vehicle-kilometer (kg CO₂/V-km)** over the vehicle's lifetime.
- This is useful to observe the **impact of manufacturing and use phase emissions separately**.

2. Key Observations from the Graph

- **At the start (low mileage):**
 - **Manufacturing emissions dominate**, leading to very high CO₂ per km.
 - **CO₂ per km is initially very high but decreases as the vehicle is driven more.**
- **As distance increases (longer vehicle lifetime):**
 - **Manufacturing emissions become negligible**, and the footprint per km is mostly determined by the energy source used for driving.
 - **Gasoline vehicles stabilize at a higher CO₂/km than electric vehicles.**
 - **Coal-powered electric vehicles have a similar or slightly worse footprint than gasoline cars.**
 - **Renewable-powered EVs (wind, solar) have the lowest CO₂ emissions per km.**

Comparison of normalized damage Impact World+ Life cycle impact Assessment



Key Takeaways

- ✓ Gasoline vehicles have the highest overall damage impact in human health, climate change, and ecosystems.
- ✓ Electric vehicles reduce these impacts significantly, but their effectiveness depends on the electricity mix.
- ✓ Wind and solar-powered EVs are the best options, minimizing most impacts.
- ✓ Photovoltaic EVs have slightly higher water usage impacts but still perform better than gasoline.

5. Resource Management

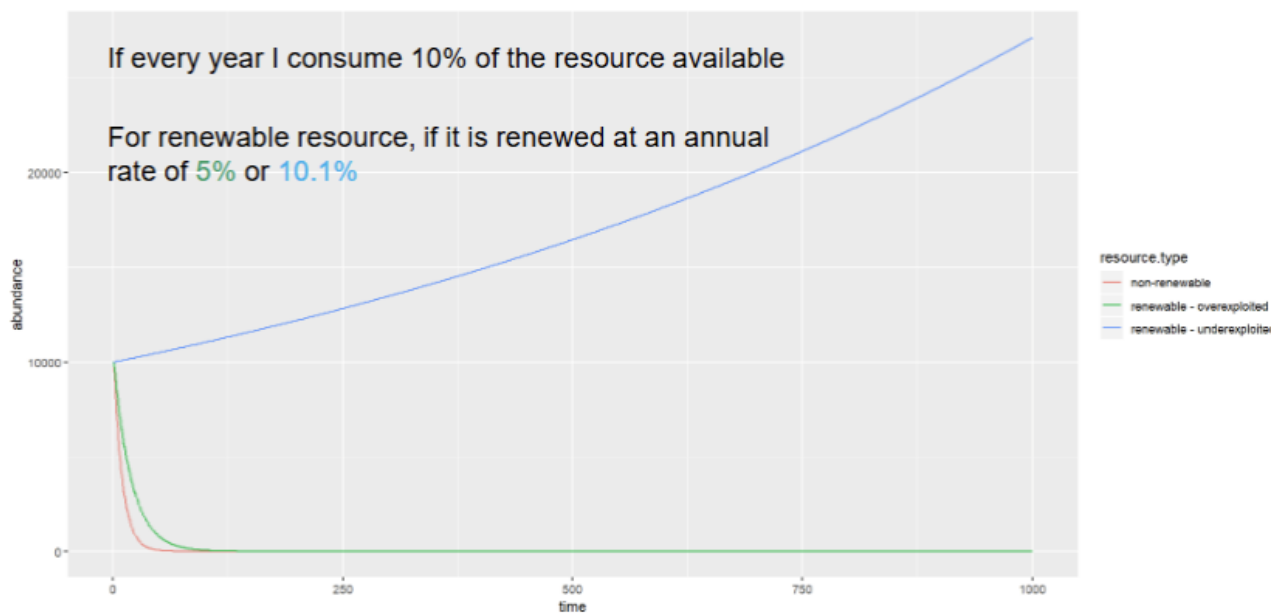
Natural resources

- Can the resource I exploit harness earth systems to make more of itself within a timeframe suitable for my exploitation?

- No: Non-renewable resource
 - Minerals, oil, gas
- Yes: Renewable resource'
 - Soil, plants, animals, water, wind

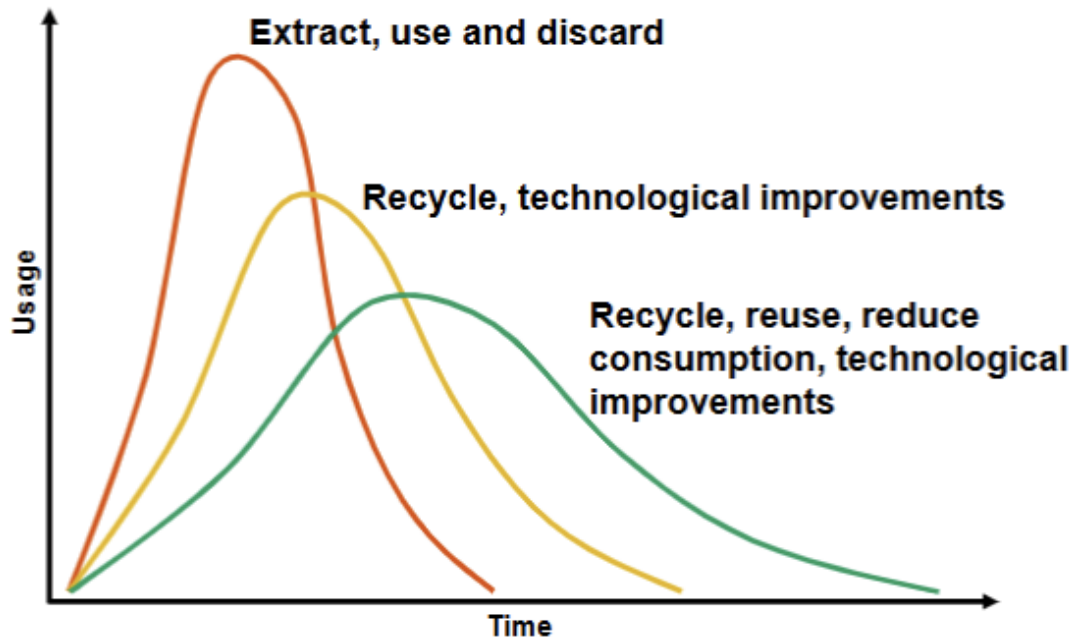
Exploitation types

- Do we extract resources or use products generated by resources
 - Consumptive, non-consumptive
- When using products generated by resources, do we impair resource renewing when doing so?
 - Consumptive?



Non-renewable resource use pattern

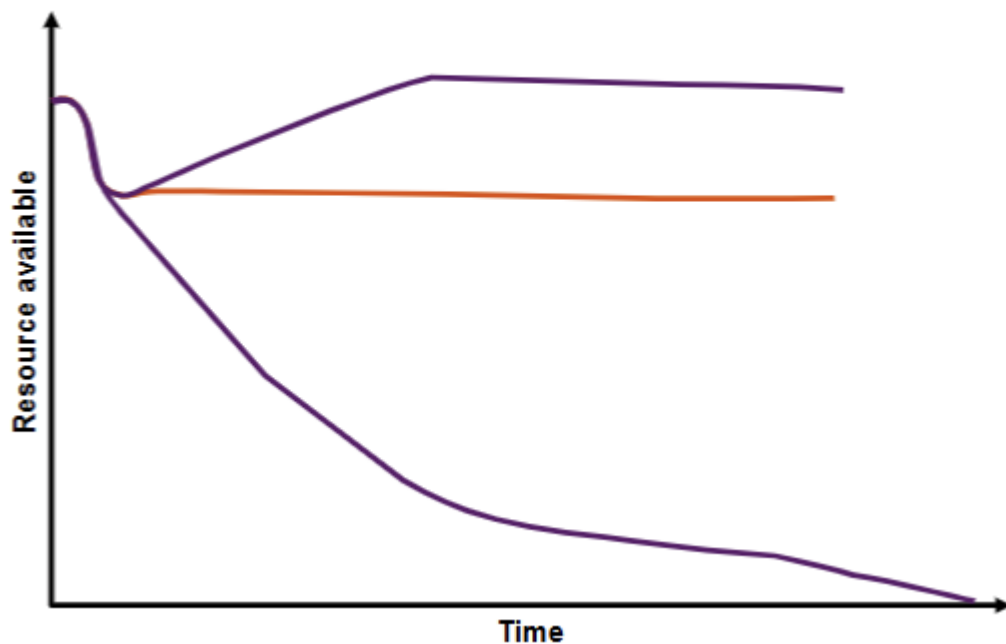
Adoption → demand peak → transition/abandonment



Renewable resource use pattern

Adoption → demand peak → transition/abandonment

Adoption → demand peak → sustained



Sustainability of exploitation

- Ability to manage depends on
 - Whether extractions by one actor will diminish ability to extract by another
 - Actors can be excluded from extraction

Resource types

	Excludable	Non-excludable
Rivalrous	Private good	Common good
Non-rivalrous	Club good	Public good

Private good

- Competitors can be prevented to access the resource (private property rights can be assigned)
- use of that good (consumption) prevent others to use it e.g.
- parking space, bread, a field

Club good

- Property rights can be assigned
- Using that good does not prevent others to use it
- Eg, a cinema, alliance/club/union (services provided to members excluding non members), a private nature park

Public good

- Access cannot be restricted
- The use of the resource does not prevent others from using it
- E.g., knowledge, streetlights, public park

Common good

- Short for common-pool resource
- Access cannot be restricted
- The use of the resource prevents others from using it
- E.g., fish stocks, but also... public park
 - If the use of a natural place degrades it, then some of its use is prevented by the use of others

Tragedy of the commons

- Several countries can exploit a fish stock, if a country has more fishing effort than allotted (e.g., more vessels) the stock can be overexploited
- The benefits of extra vessels are received by the one country increasing its fleet, but the costs are shared among all countries
- Without interventions –recall exclusion interventions are not possible –it will always be beneficial to defect
- Leads to a race to the bottom and resource collapse

ToC game theory: Prisoner's dilemma

Decision consequences

- both confess, both go to jail for five years;
- neither confesses, both go to jail for 1 year (concealed weapons);
- one confesses & the other not, confessor is free (witness) and the silent one is jailed 20 years.
- Nash equilibrium:
- *No player can improve his payoff by changing his strategy from his equilibrium strategy to another strategy provided his opponent keeps his equilibrium strategy."*

ToC - fixes

Assign property rights!

- Then common goods become private [/club] goods (e.g., quotas)
 - European "common policies" approach: common to club goods
- (this is still a failure of common good governance)

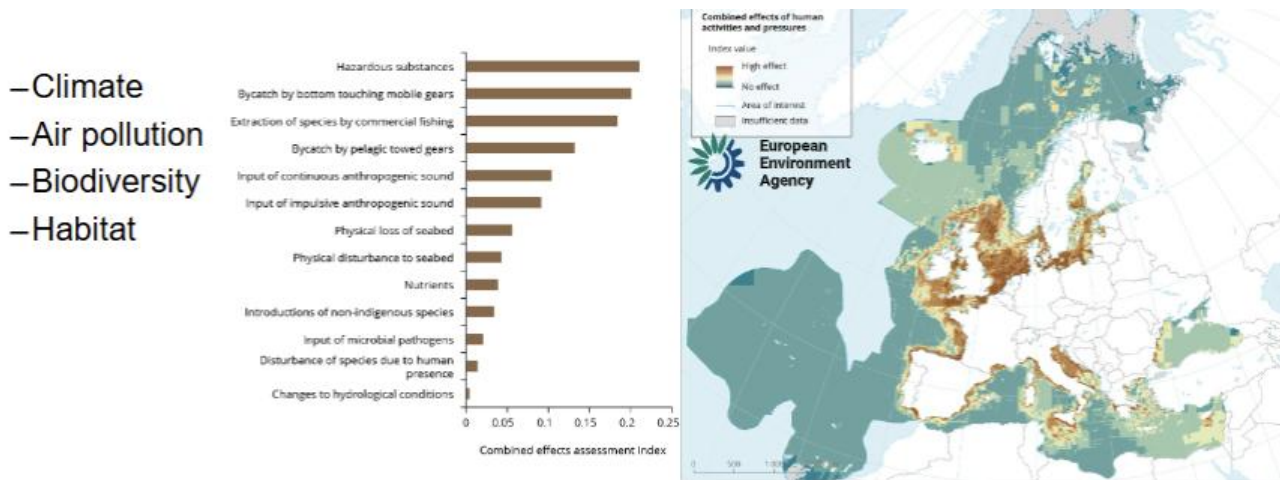
Can we actually sustain common goods exploitation?"

Summary

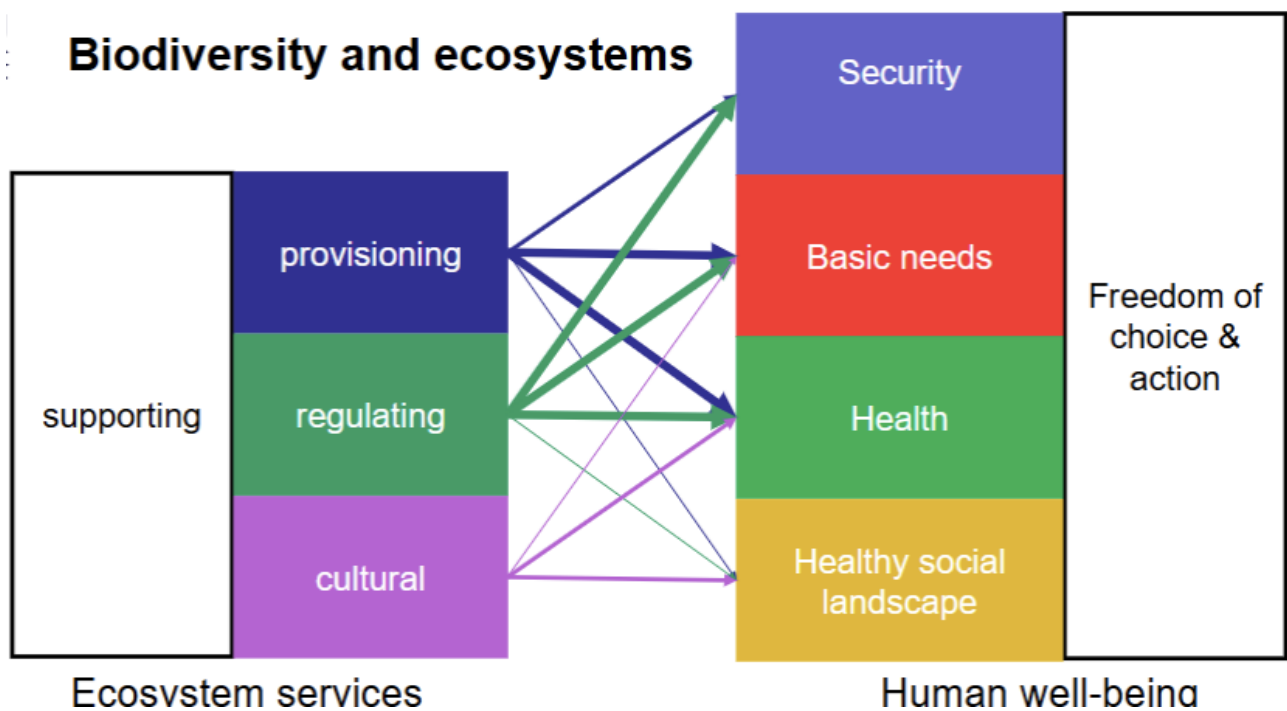
- Renewable natural resources lead to different exploitation patterns because they need not become exhausted
- Our means and abilities to govern resource exploitation depends on whether people can be excluded from its exploitation and whether resource use by some diminish resource use opportunities for others
- Common goods are complex to govern, unfortunately most regulating and cultural ecosystem services are common goods.

Indirect use of resources

- Our activities can impose pressures on resources which are not the source of exploitation

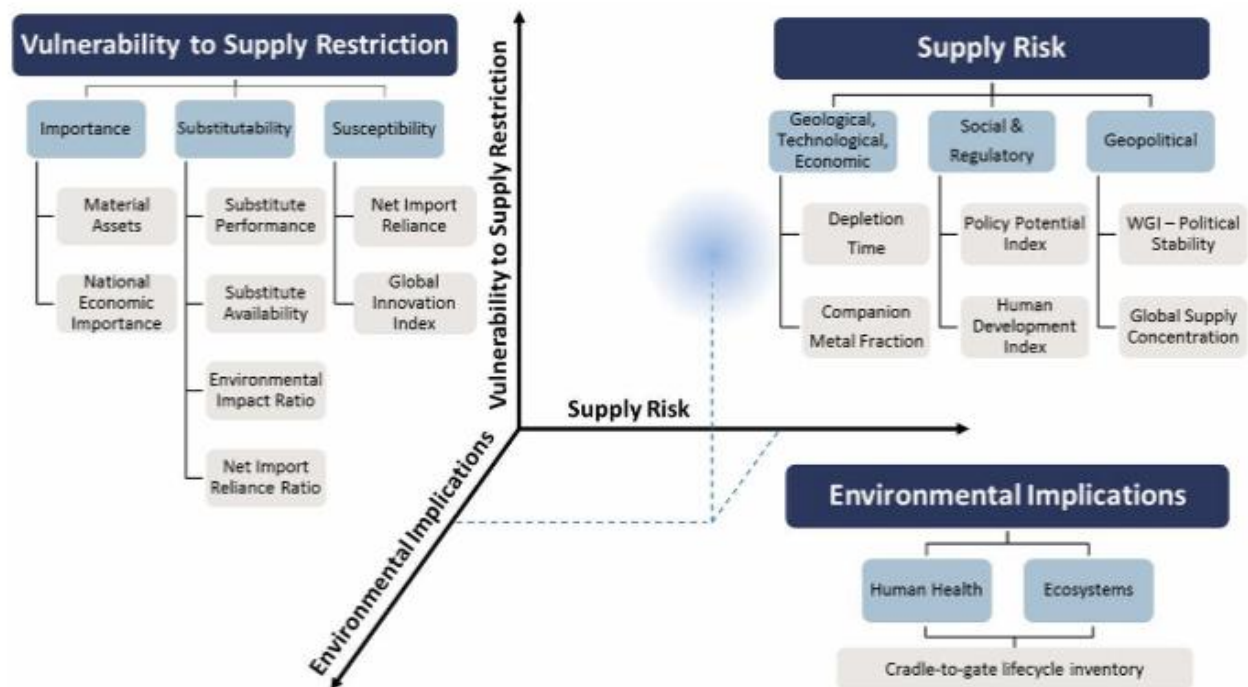


Biodiversity and ecosystem

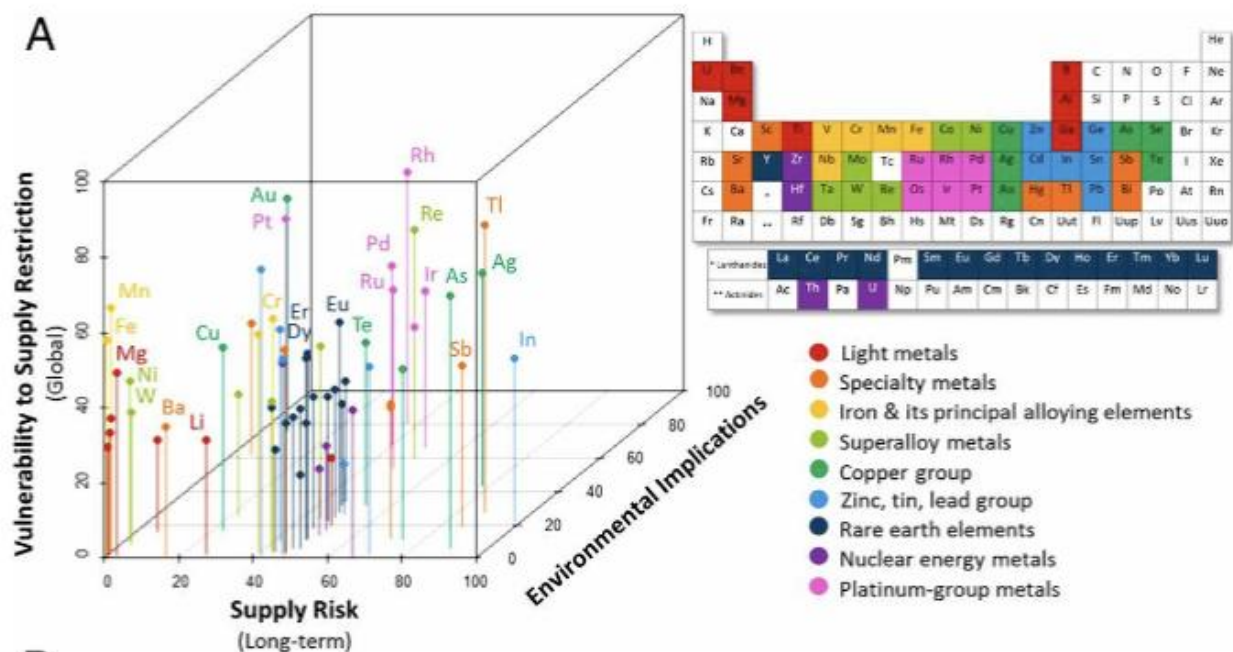


Criticality analysis

Metal criticality: risk appraisal



Criticality outcomes



Measuring maximum sustainable yield

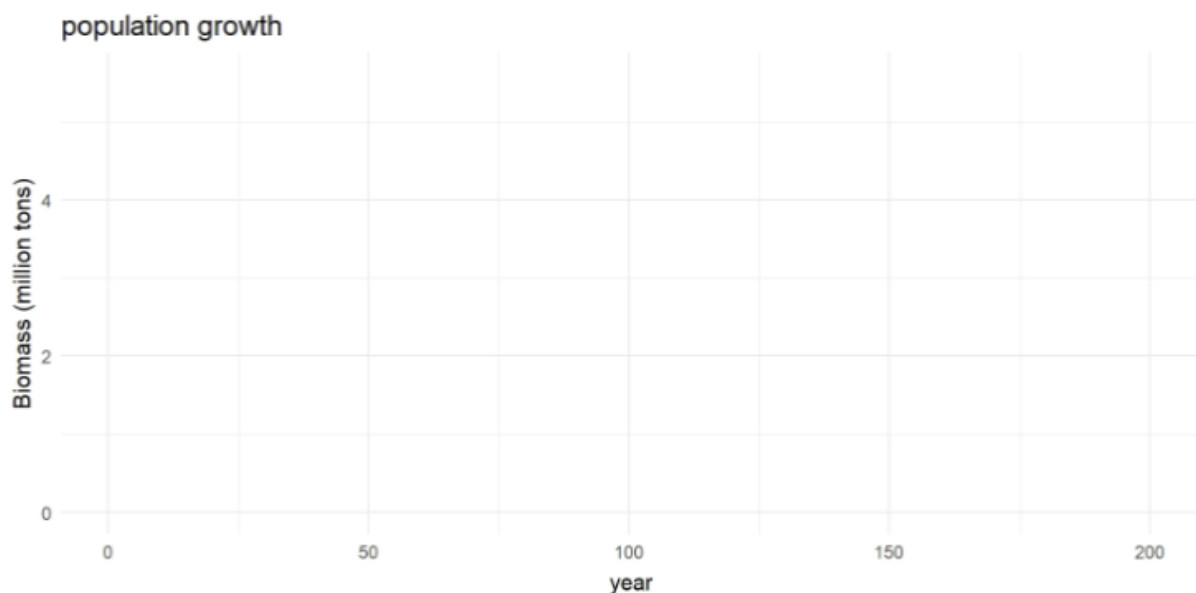
Population model

- Principles described are applicable to all renewable resources
- Let's formulate an equation describing the rate at which a resource renews itself

$$• B_{t+1} = B_t + rB_t \left(1 - \frac{B_t}{K}\right) - C_t$$

- Where B is biomass, r is the intrinsic growth rate, C is catches and K is the carrying capacity (how much of that species the ecosystem can sustain)

K=5.6.10⁶ tons, r=0.3, no catches, starting at 1 ton (cod)

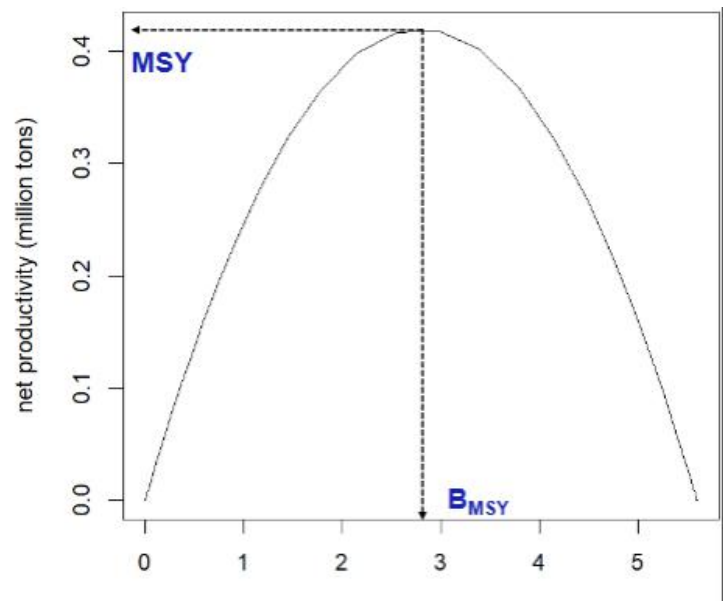


Maximum sustainable yield

$$B_{MSY} = K/2$$

$$MSY = rK/4$$

In the real world:
we don't know r & K ;
we have time series of catches
and how much effort was
needed to achieve them



Catch per unit effort

All things equal, catches should increase at the same rate I increase my effort:

$$C_t = qE_tB_t$$

So the catch per unit effort is related to biomass or abundance:

$$\frac{C_t}{E_t} = qB_t = \text{CPUE}_t$$

This is the Schaefer model; we now know this is a lot more complicated, q (catchability coefficient) is rarely constant, exploited population equilibrium is complicated

at equilibrium, we remove all the growth and growth=0

$$\frac{dB}{dt} = rB_t \left(1 - \frac{B_t}{K}\right) - C_t = 0$$

$$B_t = \frac{CPUE_t}{q}$$

$$C_t = r \frac{CPUE_t}{q} \left(1 - \frac{CPUE_t}{CPUE_{max}}\right)$$

$$\frac{C_t}{CPUE_t} = E_t = \frac{r}{q} - r \frac{CPUE_t}{qCPUE_{max}}$$

$$CPUE_t = CPUE_{max} - \left(\frac{q}{r} CPUE_{max}\right) E_t \text{ and } C_t = CPUE_{max} E_t - \left(\frac{q}{r} CPUE_{max}\right) E_t^2 !$$



One dimension in isolation of others

Bioeconomics of catches (efficiency, supply/demand, quality)

- associated with fuel consumption (and hence C footprint)
- Gordon-Schaefer model

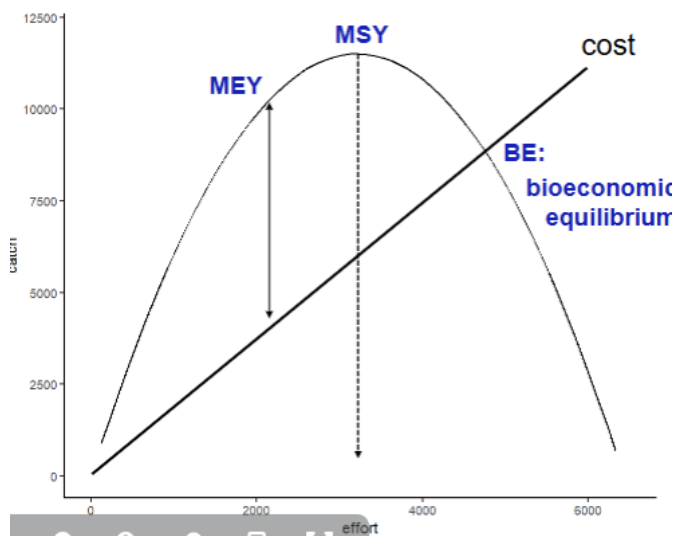
Biodiversity footprint—Incidental catches, habitat deterioration

- Multi-criteria approach

Can I find the effort which will be profitable, have acceptable biodiversity footprint, and yield sufficient catches?

Gordon-Schaefer bioeconomic model

- Revenue = $aC - bE$
- Maximum Economic yield < MSY
- MEY can be > MSY if there are external revenues (subsidies)



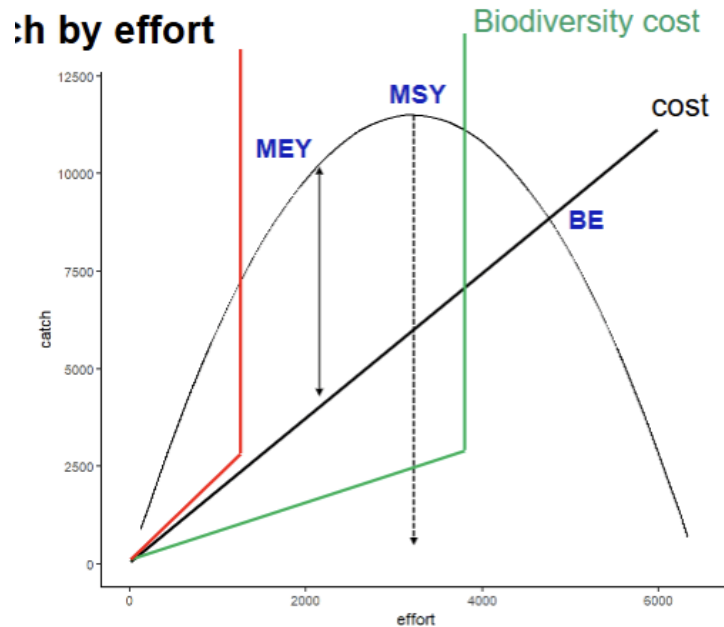
Biodiversity impact: bycatch

- There is a bycatch probability associated with fishing practices for varying sensitive and socially important species
- Total bycatch is not optimised: an independent threshold is set to ensure conservation/restoration objectives are met
- Globally prevailing: Potential Biological Removal (PBR):
- A la Schaefer model:

$$- PBR = N_{\min} \times R_{\text{MSY}} \times F_r$$

Incorporating in catch by effort

- Biodiversity cost is independent of MEY and MSY
- Biodiversity cost can be reduced with mitigation
 - mitigations are species dependent



6. Sustainability tools for society I

Deep-dive on the life-cycle economics for gasoline cars versus EVs for households

- Quantitatively demonstrate economic cost of gasoline vs electric cars across their lifecycle
- Explore in context of different energy sources for electricity
- Explore different ownership models
- Discuss wider societal & equity issues

Notes:

- Focus today: **sustainability of transport modes**, especially **gasoline vs. electric vehicles (EVs)**.
- Use **Life Cycle Cost Analysis (LCCA)** to:
 - Compare long-term economic costs of gasoline vs. EVs.
- In the activity:
 - Assess how costs vary based on **energy source for electricity**.
 - Explore **ownership models** (e.g. leasing vs. buying).
 - Reflect on **societal and equity implications** of vehicle choices.
- Intro to **tools** for evaluating transport sustainability.

Tools

- multi-criteria decision analysis (MCDA),
- material flow analysis,
- life cycle assessment (LCA),
- input–output models,
- sustainability indicators and indices,
- cost–benefit analysis (CBA) and optimisation methods

Notes:

- ☐ Many sustainability tools exist in research, but:
 - Companies often don't know about them or
 - Feel they lack the capacity to use them effectively.
- ☐ These tools can support better environmental, societal, and economic decisions.
- ☐ The paper shown ("Sustainability assessment tools...") gives an overview of key tools.
- ☐ Today's focus: working through a Life Cycle Assessment (LCA).
- ☐ Other tools mentioned:
 - Material Flow Analysis (MFA)
 - Input-Output Models

Material flow analysis

- Analysis of process chains and quantifies the inputs and outputs of materials or substances related to those processes.
- Represents a simple model of the interrelation between the economy and the environment,
 - economy is an embedded subsystem of the environment and dependent on a constant throughput of materials and energy
 - Most countries will provide material economy-wide material flow accounts

Notes:

- ☐ **Material Flow Analysis (MFA)** is commonly used in engineering:
 - It analyzes **process chains** by quantifying **inputs and outputs** of materials.
- ☐ It models the **relationship between economy and environment**:

- Economy is seen as a **subsystem of the environment**, reliant on a **continuous throughput** of materials and energy.
- Example: Apply MFA to **electric vehicles (EVs)**:
 - Assess **costs and environmental impact** of each component **at a single point in time** (production phase).
- Important limitation: MFA is typically **static** (snapshot in time).
- But EVs' impacts extend beyond production:
 - Must consider **use phase, household costs, societal effects**, and **end-of-life** handling.
- So, **MFA is a precursor** to more comprehensive tools like **Life Cycle Assessment (LCA)**.

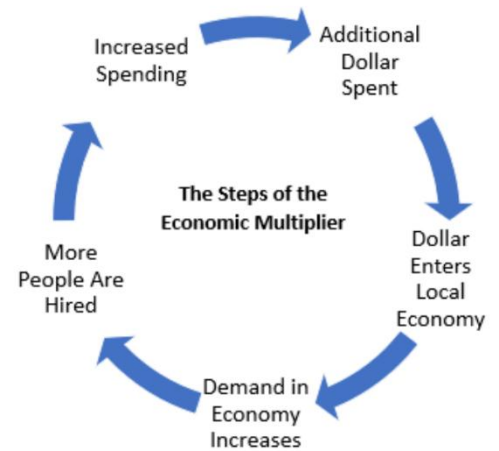
Input–output models

- Describes the interdependencies between different branches of the economy.
- Important (**economic**) quantitative methodology
 - Used to look at the impact of the production and consumption of different products and services on key economic parameters
 - Gross Value Added, employment
 - Used extensively
 - Environmentally extended IO (EEIO) analysis is a method for evaluating the linkages between economic consumption activities and environmental impacts

Notes:

- Goes beyond just buying/producing goods — also includes **direct and indirect impacts** on the economy.
- Offers a **holistic view** of how one industry (like EVs) affects others.
- Useful in understanding fears around the green transition (e.g. **job losses, economic costs**).
- Can **demonstrate positive societal and economic outcomes** to policymakers.
- **Environmental extensions** exist (EEIO), but today's focus is on **economic application only**.

- Like MFA, but flows flows of goods and services in terms of monetary value (underpinned by data on flows of goods and services)
- Calculate for Sector X both the:
- Direct Effect
 - Impact of the expenditures for the sector on the wider economy
- Indirect Effect
 - The impact a sector has on other industries and their subsequent activities to respond to the need for Sector X for the operations of the business.
- Also used to calculate multipliers
- The impact that a change in output in a sector will have on the economy for key parameters GVA, employment
- For example, an 1kr increase in expenditure in Sector X and the potential impact it will have on employment



Notes:

- ☐ Unlike tools like LCA, this model focuses **only on monetary flows**, not emissions or environmental impacts.
- ☐ Especially useful for **sector-specific spending analysis**:
 - E.g., if **€1 or 10 kr** is spent in agriculture, what **total value** (direct + indirect) does it generate?
- ☐ Policymakers use this to understand:
 - Which sectors give **the most value per krone** — in terms of jobs, profit, or economic growth.
- ☐ **Indirect effects are crucial**: Spending in one sector impacts many others through supply chains.
- ☐ **Multipliers** help prioritize funding:
 - “Where do we get the most return?” (e.g., employment vs. profit).
- ☐ But: **Economic value isn’t everything** — we must also consider **environmental and social impacts**, which LCA handles better.

Cost–benefit analysis (CBA)

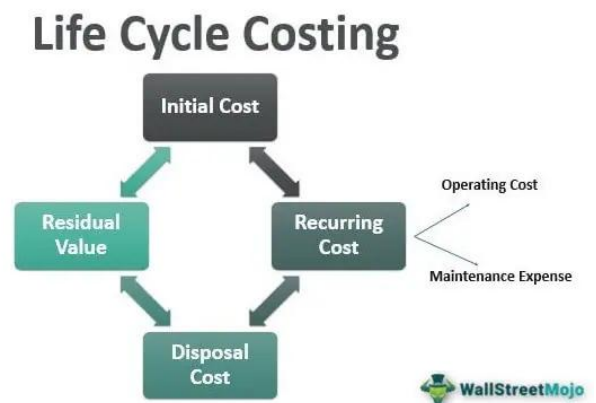
- ... is an economic method that estimates the equivalent monetary value of the benefits and costs of a project.
- Outputs from LCAs and IOs are regularly used within CBAs to quantify costs or benefits

Notes:

- CBA is a **natural next step** after using tools like **LCA** or **Input–Output models**.
- It brings together outputs from **multiple methods** to evaluate **what option is best overall**.
- Can include not just financial costs/benefits, but also **environmental** and **societal impacts**.
- A **basic CBA** might use IO data to assess sector/technology investment choices.
- Often seen as a practical tool for **decision-making**, especially at institutions like **DTU**.
- *“Does exactly what it says on the tin”* — a British saying meaning it’s a straightforward, functional tool.

life cycle cost assessment (LCCA)

- Life cycle costing (LCC) is an approach that assesses the total cost of an asset over its life cycle, including initial capital costs, maintenance costs, operating costs and the asset’s residual value at the end of its life
- Can be used within sustainability analysis to understand the economic cost of different goods or services
- Moving from just environmental focus to an economic focus
- Can be used to aid both producer or consumer decision-making



Notes:

- Builds on **material flow analysis**, but LCCA includes **entire product life** — not just production phase.
- A “sibling” to **Life Cycle Assessment (LCA)**, but with an **economic lens**.
- Helps paint a **full financial picture** — from creation to disposal — not just environmental impact.

- Used by:
 - **Producers** to decide if developing a new product or technology is financially worth it.
 - **Consumers** to compare total lifetime costs of products (e.g., a costly item may be cheaper long-term).
- Practical example: Buying **expensive shoes** that last longer = better life-cycle cost choice.

7. Sustainability tools for society II

Social life cycle assessment (S-LCA)

- As introduced in Module 5:
 - *Life cycle costing (LCC) is an approach that assesses the total cost of an asset over its life cycle, including initial capital costs, maintenance costs, operating costs and the asset's residual value at the end of its life*
- Here we look at:
 - *Social LCA (SLCA) is developed towards evaluating social impacts, such as employment, workplace health and equity*
 - *Assesses the social and socio-economic aspects of products and their potential positive and negative impacts along their life cycle*
 - *Increasingly established*
 - *UNDP - Social Life Cycle Assessment (2020)*

Notes:

- ☐ Part of the same “family” as LCA and LCCA — but **less commonly used historically**.
- ☐ Focuses on **well-being and social justice**, not just money or environment.
- ☐ Looks at **impacts on people**: employment, equity, health (broadly defined).
- ☐ Has become **more prominent in recent years**, thanks in part to experts like *Olivier* (who you’ve met).
- ☐ The **UNDP guide (2020)** is a key reference and highly recommended for deeper understanding.

- Process:
 - *objects: products and services, and*
 - *scope: the entire life cycle.*
- Social and socio- economic aspects assessed in S-LCA are those that may directly affect **stakeholders** positively or negatively during the life cycle of a product.
 - *indirect impacts on stakeholders may also be considered.*
- S-LCA does not have the goal nor pretends to provide information on the question of whether a product should be produced or not.
- Lots of similarities with LCCA
 - *Not least it's data requirements!*

Notes:

- The **impact assessment phase** is central to S-LCA:
 - It calculates and evaluates the **magnitude and significance** of a product's potential social impacts.
- Two main approaches:
 1. **Reference scale approach:** Compares social performance of one product/service to others.
 2. **Pathway (or consequential) approach:** Traces **how a product indirectly impacts society** through multiple channels — like a **ripple effect**.
- Think of it as a **network of social consequences**, not just direct impacts.
- The approach is similar in logic to **input-output models**, but focused on **social effects**.
- Best understood through **real-world examples** (she hinted one was coming).

s-LCA & impact assessment

- Social impact assessment is the phase in S-LCA aimed at calculating, understanding and evaluating the magnitude and significance of the potential social impacts of a product system throughout the life cycle of the product
- 2 approaches
- Reference Scale
 - *social performance or social risk of a product*
- Impact Assessment
 - *consequential social impacts from a product by characterizing the cause-effect chain*

Notes:

□ S-LCA methodology follows **4 key stages**, just like LCA or LCCA:

1. **Define objectives** – What do you want to assess? Which product/service?
2. **Identify stakeholders** – Who is affected by this product or service?
 - Unlike LCA/LCCA (which might focus on a household), S-LCA considers **society at large**.
3. **Understand mechanisms of impact** – How are stakeholders affected? What **social pathways** are involved?
4. **Interpretation** – Making sense of the findings and **what they mean in context**.

□ Importance of stakeholder focus: Helps move beyond individual or business-level impacts to broader **social systems**.

impact pathway approaches

- The main target within the IP S-LCIA is to assess and model relations between the cause (social activities/stressors) that arises from a company's activity and their effect.
- This is usually done by establishing what we call impact pathways.
- Involves tracing the inventory data through the relevant social and socioeconomic mechanism to define the socioeconomic impact
 - *Can be established qualitatively and quantitatively.*
 - *Focus is increasingly on effects on human health measured in terms of DALY's: Disability Adjusted Life Years*

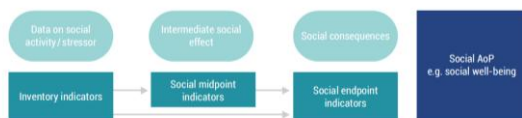


Figure 27: Illustration of the social impact pathway scheme applicable to type II S-LCA. See figure 4 and main text for examples of midpoint and endpoint indicators.

BOX 16. THE LOGIC OF MIDPOINT AND ENDPOINT CATEGORIES FROM E-LCA TO S-LCA (DERIVED FROM NEUGEBAUER, 2016)

Impact categories within life cycle based methods should cover the complete impact pathway by including inventory indicators, midpoint, and endpoint categories (Bare et al., 2000; JRC, 2010). Inventory indicators may be defined as simple variables (e.g. working hours), whereas midpoint impact indicators are seen as parameters in the (social) mechanism network (Bare et al., 2000). Endpoint impact indicators are then understood as measurement endpoints determining damage levels to the Area of Protection (Joliet et al., 2004).

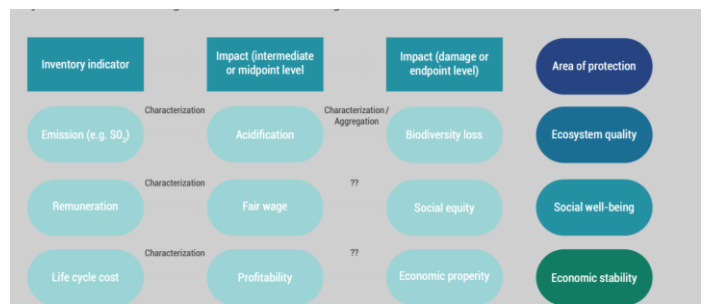


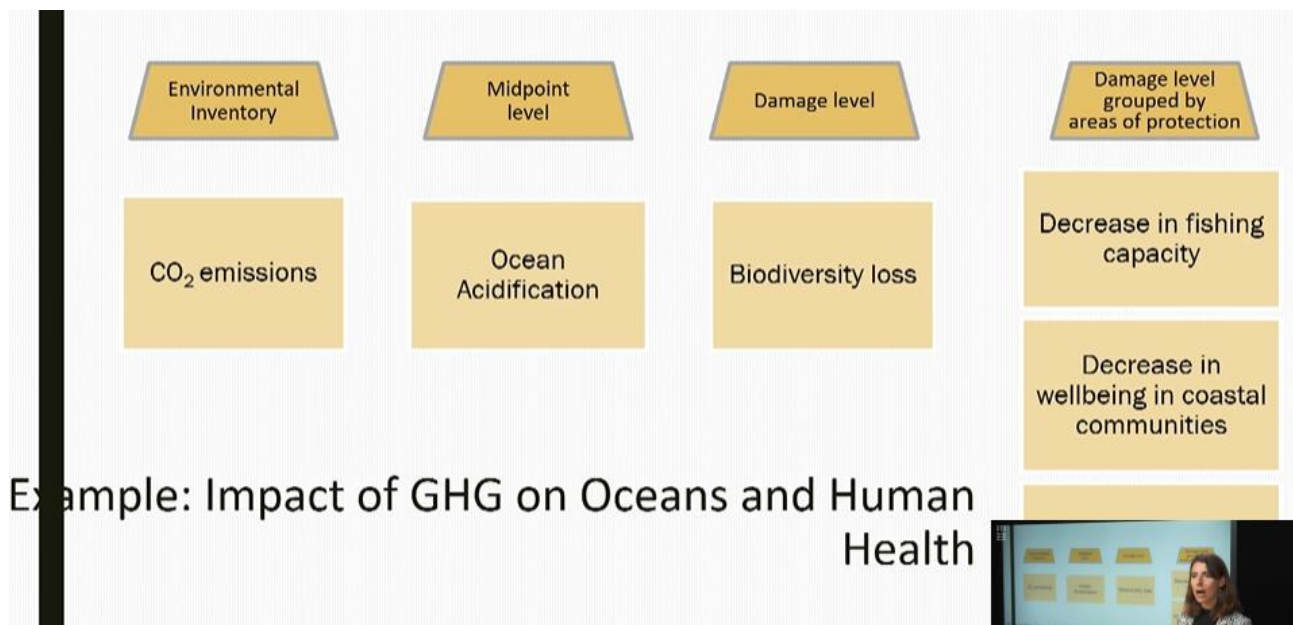
Figure 26: Exemplary impact pathway schemes for LCSA (adapted from Neugebauer 2016).

Development of an IP S-LCIA method, just as for E-LCIA, usually consists of linking inventory data that undergoes a **characterization step**, and which results in **midpoint and/or endpoint impact indicators**.

Notes:

- The focus is on connecting a **company's activity** (the cause) to its **effects on society**, particularly populations.
- “Impact pathways” is the technical term — but essentially, they’re just **pathways of influence**.
- The process is about:
 - Identifying a **product or service**,
 - Determining **who or what it affects**,
 - And mapping **how those effects unfold** socially or economically.
- These pathways can be analyzed in both **qualitative and quantitative** ways.
- The method begins with **inventory data** (e.g. from databases), which is then **translated through a characterization step**.
- This process helps convert raw data into **meaningful social impact indicators**.
- Two types of indicators result:
 - **Midpoint indicators** – reflect parameters along the pathway (e.g., fair wage, working hours).
 - **Endpoint indicators** – reflect **final consequences** (e.g., social well-being, equity).

Exsample: Impact of GHG on Oceans and human health



- This is an example of a **social impact pathway** using **CO₂ emissions** in an **ocean and human health context**.
- **CO₂ emissions** → **ocean acidification** → **biodiversity loss** → human & ecosystem impacts.

- Each link represents a step in the **mechanism or pathway**:
 - Ocean acidification is the **midpoint**.
 - Biodiversity loss is the **damage level**.
- The **real social impacts** come **after** biodiversity loss:
 - **Decrease in fishing capacity** (economic and livelihood effect)
 - **Decrease in well-being** of coastal communities (social effect)
 - Knock-on effects on **ecosystem quality** and **mental well-being**.
- This method helps explain how **environmental stressors** translate into **social and economic harms**.
- Emphasizes that it's not the CO₂ alone, but the **chain reaction** that leads to human and social consequences.
- This framework shows how **all three pillars of sustainability** are connected:
 - **Environmental** (acidification, biodiversity)
 - **Social** (well-being, livelihoods)
 - **Economic** (fishing industry, employment)

Data & databases

- S-LCA case studies,
 - *the Social Hotspot Database (SHDB) and*
 - *the Product Social Impact Life Cycle Assessment (PSILCA) database remain the main sources for generic social*
- Both databases assume pathways exist across value chains and are commonly used to provide country- and sector-specific data on some of the social indicators specified in the UNEP/SETAC

Notes:

- The **available databases for S-LCA** are constantly expanding, just like with traditional LCA.
- The two **main data sources** are:
 1. **Social Hotspots Database (SHDB)**
 2. **Product Social Impact Life Cycle Assessment (PSILCA)**
- Both:
 - Work across the **entire value chain**.
 - Include **country- and sector-specific** social data.

- These tools are **endorsed by UNEP/SETAC** and recommended for practical S-LCA applications.
- They help provide the **underlying data** for modeling social pathways in assessments.

Activity

- Think about the lifecycle stages of an EV
- Stage 1: Resource extraction and processing
- Stage 2: Manufacturing
- Stage 3: Distribution and operation
- Stage 4: Waste and disposal
- For each stage develop a basic impact assessment S-LCIA

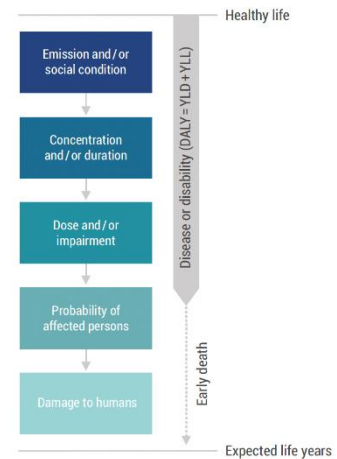
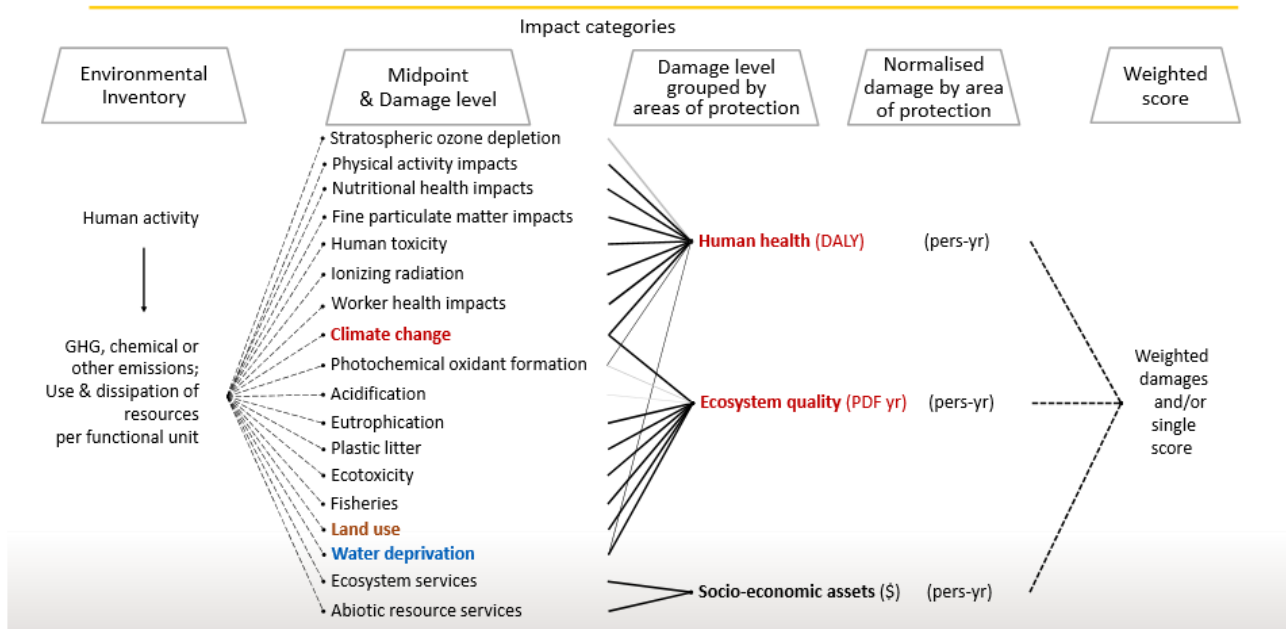


Figure 31: Illustration of a DALY (disability adjusted life years) impact pathway representing the mechanistic approach. YLD = years of life disabled and YLL = years of life lost. Based on figures from Crettaz & Frischknecht et al. (2000) and Golsteijn (N.D.).

Notes:

- **No need to calculate** — this task is **qualitative** only.
- For each EV lifecycle stage, you're asked to **specify an impact pathway**:
 - Identify the **stakeholders** involved.
 - Outline the **mechanisms and consequences** at each stage.
- This involves thinking through:
 - What the **social condition or stressor** is (e.g., mining, labor conditions).
 - What the **midpoint mechanism** could be (e.g., pollution, working hours).
 - How it links to a **social impact** (e.g., community health, employment).
- Most of the challenge lies in **defining the pathway** clearly:
 - What's the **step-by-step connection** between an activity and its **social consequences**?
 - This is where most time would go in **real-world S-LCA** work — e.g., a thesis or consultancy project.
- The **DALY impact pathway** diagram is used here to illustrate **how emissions or social conditions ultimately affect human health** and well-being.

UNEP-GLAM: the global life cycle impact assessment framework

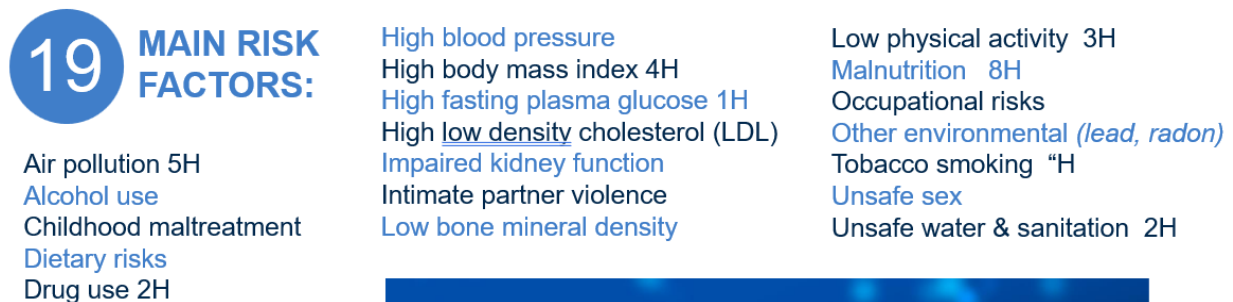


Notes:

- ☐ This diagram shows how you can **map out impact pathways** across **environmental, economic, and social areas**.
- ☐ Start with **human activity** (e.g. GHG emissions), then trace its impacts through:
 - Midpoints (e.g. acidification, toxicity),
 - Damage categories (e.g. human health, ecosystem quality, socio-economic assets).
- ☐ These impacts can be **quantified**:
 - **Human health** → Disability-Adjusted Life Years (DALYs)
 - **Ecosystem quality** → e.g. Potentially Disappeared Fraction of species (PDF·yr)
 - **Socio-economic** → e.g. loss of income, wages, employment
- ☐ These are then **weighted and/or combined** into a single **overall impact score**

7.1. Introduction to global burden of disease - Purpose of the GBD

Global Burden of Disease



Global Burden of Disease: General Approach

The burden of disease can be thought of as a measurement of the gap between current health status and an ideal situation where everyone lives into old age (e.g. the average Japanese woman), free of disease and disability.

Measured in “*Disability Adjusted Life Years*” (DALY)

Measured in “*Disability Adjusted Life Years*”

1 DALY = 1 equivalent year of healthy life lost

DALYs are calculated by taking the sum of 'Years of Life Lost' (YLL) and 'Years Lived with Disability' (YLD)

$$\text{DALY} = \text{YLL} + \text{YLD}$$

Notes:

□ The **burden of disease** is a way to measure the **gap between actual health** and an **ideal health situation**.

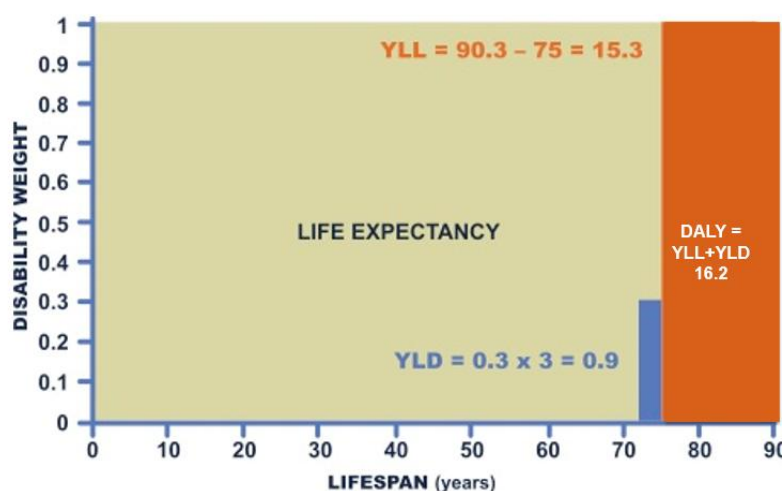
- That ideal is based on **the longest observed life expectancy** for each disease — not just one universal benchmark like the average Japanese woman.
- It's not just about **living longer**, but about **living healthier** (free of disease and disability).
- This is measured using **DALYs (Disability Adjusted Life Years)**:
 - **1 DALY** = 1 year of healthy life lost.
 - DALYs are made up of:
 - **YLL (Years of Life Lost)** – how much shorter a person's life is due to early death.
 - **YLD (Years Lived with Disability)** – time spent living with illness or impairment.
 - **DALY = YLL + YLD**
- This combined measure helps us understand the **true impact of health issues** on a population's life and quality of life

Burden in Disability-Adjusted Life Years (DALY)

DALY = Years Life Disabled (YLD)
+ Years Life Lost (YLL)

Illustration: Person gets a cancer at 72 years old and dies at 75

At age 75: life expectancy is 90.3



Notes:

- **Example:** A person is diagnosed with cancer at age 72 and dies at 75.
- The **ideal life expectancy** at 75 is **90.3 years**, so:
 - **YLL (Years of Life Lost)** = $90.3 - 75 = 15.3$ years
 - Since the **weight** for death is 1, this is **fully counted** as lost years.
- The person **lives 3 years with cancer** before death.
 - WHO assigns a **disability weight** to each condition — cancer is typically **0.3** (30% of a year fully lost).
 - This is based on how much the condition affects daily life (e.g. mobility, eating, self-care).

□ So, for those 3 years:

- **YLD (Years Lived with Disability)** = $3 \times 0.3 = 0.9$

□ **Total DALYs** = YLL + YLD = $15.3 + 0.9 = 16.2$ DALYs

□ The lecturer shared a **personal experience** with leukemia, reinforcing that a condition like cancer may not cause total disability — you might still be able to work or function partially.

GBD standard life expectancy tables

AGE	US LIFE EXPECTANCY	L-STANDARD LIFE EXPECTANCY	EXPECTED AGE AT DEATH
0	78.8	86.6	86.6
5	74.4	81.8	86.8
10	69.4	76.8	86.8
15	64.5	71.9	86.9
20	59.6	66.9	86.9
25	54.9	62.0	87.0
30	50.1	57.0	87.0
35	45.4	52.1	87.1
40	40.7	47.2	87.2
45	36.1	42.4	87.4
50	27.3	37.6	87.6
55	23.2	32.9	87.9
60	19.3	28.3	88.3
65	15.6	23.8	88.8
70	12.2	19.4	89.4
75	9.1	15.3	90.3
80	6.6	11.5	91.5
85	4.6	8.2	93.2
90	3.2	5.5	95.5
95	2.3	3.7	98.7
100		2.5	102.5
105		1.6	106.6
110		1.4	111.4

$$YLL = N \times L$$

N = number of deaths
L = standard life expectancy at age of death in years DALY/death

Constructed based on the lowest estimated age-specific mortality rates from all locations with populations over 5 million in the 2013 iteration of GBD

Notes:

- This table shows the **standard life expectancy** used in **DALY calculations**.
- For someone who dies at **age 75**, the **ideal life expectancy** is **90.3**, meaning **15.3 years of life lost**.
- Interestingly:
 - As you **get older**, your **expected age at death increases** (based on survival).
 - But your **remaining life expectancy decreases** — which is natural.
- The table compares **actual life expectancy** (e.g., in the US = 9.1 years at age 75) to **ideal standards** (15.3 years).
- **YLL (Years of Life Lost)** is calculated as:
 - $N \times L$
 - N = number of deaths
 - L = standard life expectancy at that age

- These standards come from global GBD data based on the **lowest observed age-specific mortality rates** in large populations.

Disability weights: Examples

$$YDL = I \times DW \times L$$

I = number of incident cases

DW = disability weight

L = average duration of the case until remission or death (years)

HEALTH STATE NAME	HEALTH STATE LAY DESCRIPTION	DISABILITY WEIGHT
CANCER controlled phase	has a chronic disease that requires medication every day and causes some worry but minimal interference with daily activities.	0.049 (0.031 – 0.072)
CANCER , diagnosis and primary therapy	has pain, nausea, fatigue, weight loss and high anxiety.	0.288 (0.193 – 0.399)
CANCER metastatic phase	has severe pain, extreme fatigue, weight loss and high anxiety.	0.451 (0.307 – 0.600)
CANCER , terminal phase, with medication	has lost a lot of weight and regularly uses strong medication to avoid constant pain. The person has no appetite, feels nauseous, and needs to spend most of the day in bed.	0.540 (0.377 – 0.687)
MILD PARKINSON DISEASE	has mild tremors and moves a little slowly, but is able to walk and do daily activities without assistance.	0.010 (0.005 – 0.067)
MODERATE PARKINSON DISEASE	has moderate tremors and moves slowly, which causes some difficulty in walking and daily activities. The person has some trouble swallowing, talking, sleeping, and remembering things.	0.267 (0.181 – 0.372)
SEVERE PARKINSON DISEASE	has severe tremors and moves very slowly, which causes great difficulty in walking and daily activities. The person falls easily and has a lot of difficulty talking, swallowing, sleeping and remembering things.	0.575 (0.396 – 0.730)

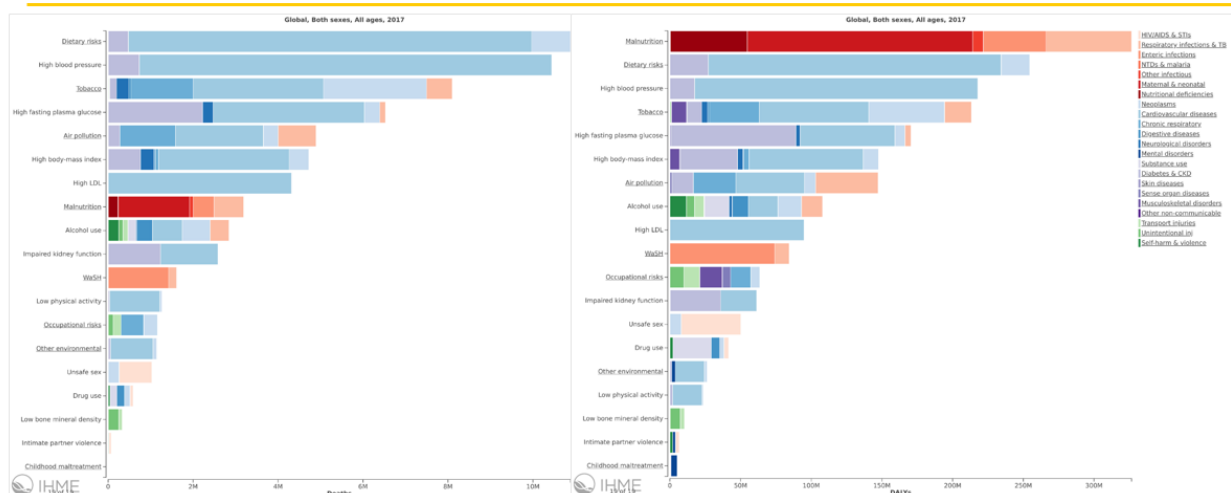
Notes:

- ☐ **Years Lived with Disability (YLD)** is calculated using:
 - The **disability weight (DW)**,
 - Multiplied by **duration of illness (L)**,
 - And the **number of cases (I)**.
- ☐ The disability weight reflects **how much quality of life is lost** due to a condition.
- ☐ **Examples:**
 - **Cancer:**
 - In the **controlled phase**, the weight is low: **0.049** (about 5% of a year lost).
 - In the **metastatic phase**, it increases to **0.45**.
 - In the **terminal phase**, it can go up to **0.54–0.6** — nearly half a year of healthy life lost per year lived.
 - **Parkinson's Disease:**
 - **Mild** = ~0.01 (1% of a year lost).
 - **Moderate** = 0.267
 - **Severe** = up to **0.575** — more than **half a year lost per year lived**.

□ These values highlight how **disease progression increases the burden**, not just in terms of survival but also **quality of life**.

Number of death in the World

Number of DALYs



Institute for Health Metrics and Evaluation (IHME). GBD Compare. Seattle, WA: IHME, University of Washington, 2015. Available from <http://vizhub.healthdata.org/gbd-compare>. (Accessed 08/12/2020)

What is the main difference between death and DALYs ?

Malnutrition is ranked **8th** risk factor of death, but becomes the first risk factor for **DALYs**, since each child death is associated with elevated **YLL**.

Notes:

□ This is where everything gets **combined** into the *Global Burden of Disease* framework.

□ It compares **risk factors** in terms of:

- **Number of deaths**
- **Number of DALYs** (Disability-Adjusted Life Years)

□ Looking at **deaths**:

- **Dietary risks** are the **#1 cause**, leading to around **11 million deaths/year globally** — mostly through **cardiovascular disease**.
- This impact is far greater than, for instance, **COVID**, and it happens **every year**.
- Other major contributors linked to diet include:
 - **High blood pressure**
 - **High fasting plasma glucose** (linked to diabetes)
 - **High BMI**
- **Tobacco** and **air pollution** still contribute significantly (air pollution $\approx$ 5 million deaths/year).

□ Moving to **DALYs**, there's a **notable shift**:

- **Malnutrition** jumps from 8th in deaths to **1st in DALYs**.
 - Why? Because it mainly affects **children**, who have a much **higher life expectancy** — leading to **very high YLL per case**.

□ The **DALY framework** gives a **fuller picture**, emphasizing both:

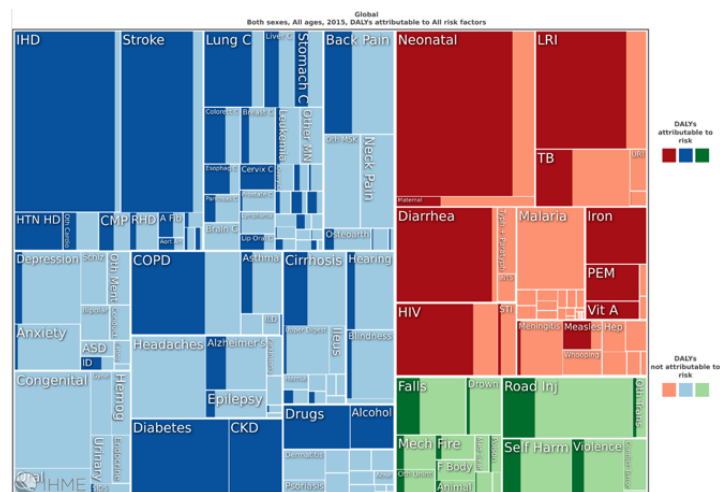
- **How many people die**
- **And how much life is lost in total** — especially among younger populations.

Two main views: Causes and ...

Causes define the illnesses or types of injuries to which **DALYs** are attributed to. On this map, areas are proportional to the magnitude of **DALYs** attributed to each cause.

What are the dominant causes?

Risk factor are potentially modifiable causes of disease and injury.



Institute for Health Metrics and Evaluation (IHME). GBD Compare. Seattle, WA: IHME, University of Washington, 2015. Available from <http://vizhub.healthdata.org/gbd-compare>. (Accessed 08/10/2020)

Notes:

□ The Global Burden of Disease (GBD) doesn't just show **risk factors**, but also the **actual causes** of death and disability.

□ This treemap shows **DALYs by cause** — the **larger the box**, the more **DALYs** are attributed to that illness or injury.

□ Categories:

- **Blue = Non-communicable diseases**, e.g.:
 - Ischemic heart disease
 - Stroke
 - Diabetes
 - COPD
 - Mental health conditions like **depression** and **anxiety** (significant contributors to disability)
- **Red = Communicable, maternal, neonatal, and nutritional diseases**, e.g.:

- Neonatal disorders
- Diarrhea
- HIV/AIDS
- Malaria
- **Green = Injuries**, e.g.:
 - Falls
 - Road injuries
 - Self-harm and violence

□ **Mental health** conditions (especially depression and anxiety) are **major contributors to YLDs** — shown by **light blue color**.

□ Cardiovascular diseases remain **leading causes** of DALYs overall, especially in older populations

Summary

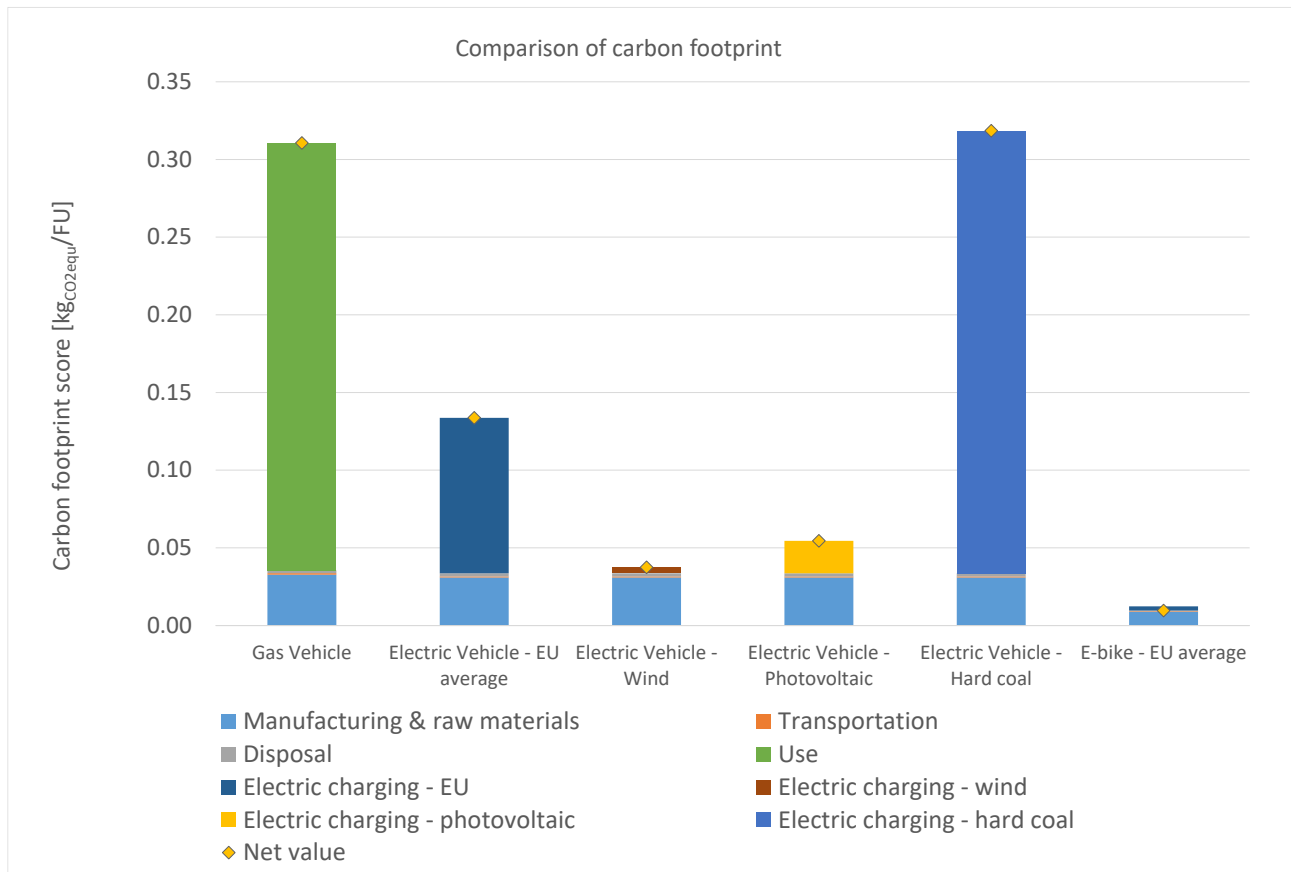
- ▶ DALYs – Disability Adjusted Life Years offer a more comprehensive metrics than death to account for both mortality and morbidity
- ▶ It measures the gap between current health status and an ideal reference situation of living a long and healthy life
- ▶ The Global Burden of Disease project provides worldwide comparison of health status, causes for death and risk factors for 153 countries
- ▶ The website is an amazing source of information, you can mine, relevant to many questions related to human health

Notes:

- **DALYs (Disability-Adjusted Life Years)** provide a **more holistic measure** of population health compared to death counts alone.
 - They combine **mortality (Years of Life Lost, YLL)** and **morbidity (Years Lived with Disability, YLD)** into one indicator.
- DALYs measure the **gap between current health status and an ideal scenario** where everyone lives a full, healthy life.
- The **Global Burden of Disease (GBD)** project enables **global comparisons**:
 - Covers **153 countries**, offering data on **health status, causes of death, and key risk factors**.
- The **GBD interactive website** is a powerful tool for data exploration:
 - You can dive into **country- or region-specific data**, compare across time, and explore causes and risks by **age, sex, or location**.
 - It's a useful resource for research, policymaking, or just general understanding of global health trends.

7.2. Health assessment - Gasoline vs electric vehicles vs bike

Car vs bike carbon footprint



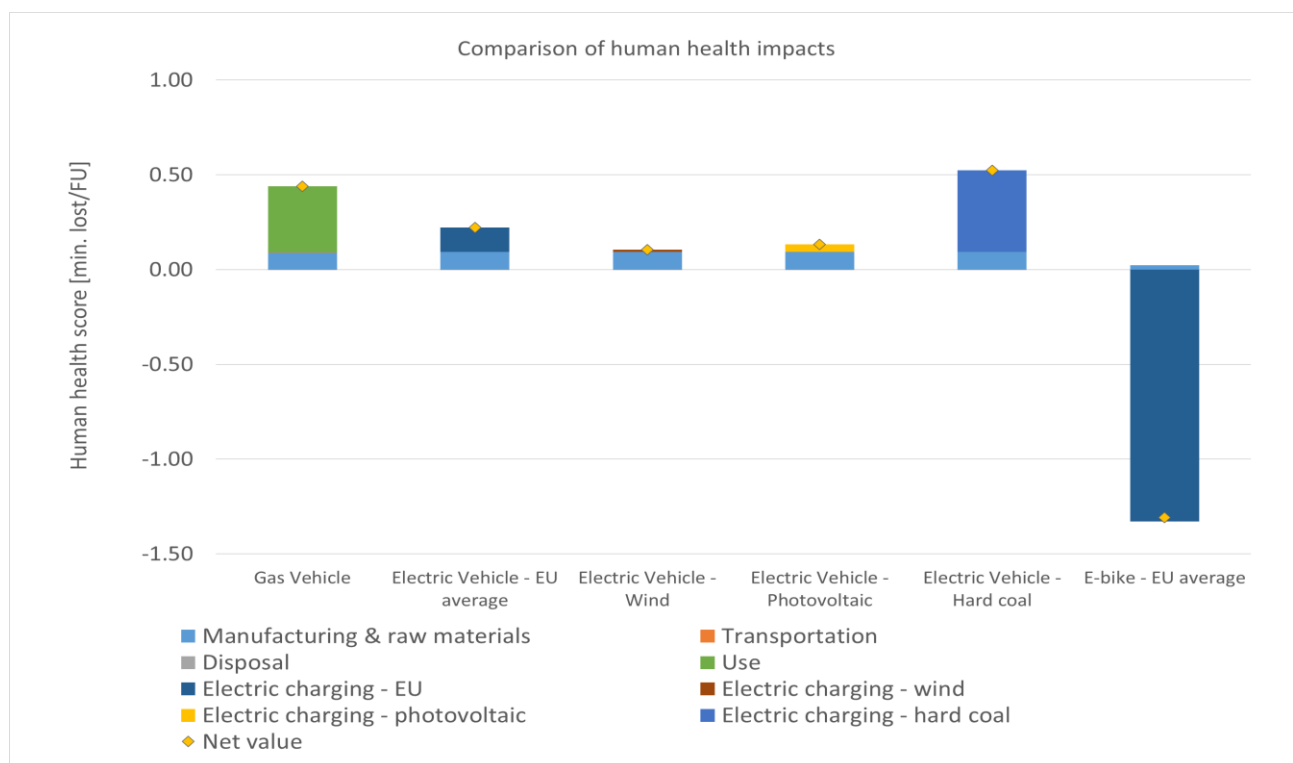
Notes:

- This slide compares the **carbon footprint** of different types of vehicles: gas-powered cars, electric vehicles (EVs) with various charging sources, and e-bikes.
- **Key components** of the carbon footprint shown:
 - **Manufacturing & raw materials**
 - **Use phase emissions**
 - **Disposal**
 - **Electric charging** (broken down by EU mix, wind, photovoltaic, and hard coal)
 - **Transportation-related emissions**

Main Takeaways:

- **Gas vehicles** have the **highest carbon footprint**, mainly driven by emissions during the **use phase** (green bar).
- **Electric vehicles** vary greatly depending on their **electricity source**:
 - EVs charged with **wind or solar** have much lower footprints.
 - EVs charged using **hard coal** emit **more than gas vehicles**, highlighting the importance of clean energy.
- **E-bikes** have a **very low carbon footprint**:
 - Most of their emissions come from **manufacturing**, while their **use phase is minimal**.
 - Charging emissions are also very low on average in the EU context.

Car vs bike human health impacts



Notes:

- This slide compares the **human health impacts** of various transport options, expressed in **minutes of life lost per functional unit (FU)**.
- **Key components** contributing to human health scores:
 - Manufacturing & raw materials

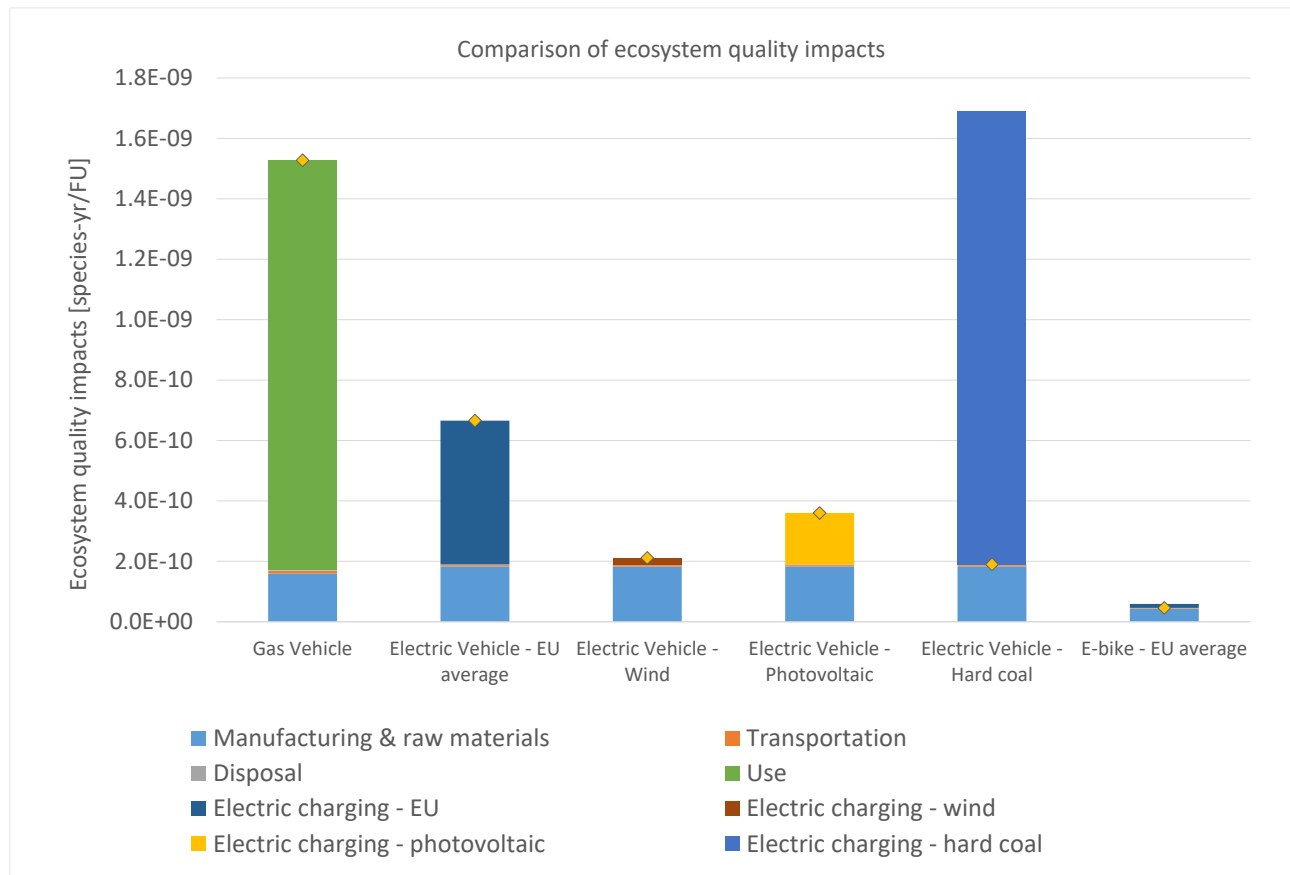
- Disposal
- Electric charging (various energy sources)
- Transportation
- Use phase
- Net value (overall impact)

Main Insights:

- **Gas-powered vehicles** have the **highest negative health impact**, primarily due to the **use phase emissions** (green section), which contribute to **air pollution** and related health issues.
- **Electric vehicles (EVs):**
 - **Health impact varies** depending on how the electricity is generated.
 - Charging with **hard coal** shows a **larger health burden** than wind or photovoltaic.
- **E-bikes** (EU average) show a **net positive health impact** (negative minutes lost = health gain).
 - This is due to the **physical activity** involved, which improves overall health.
 - A **conventional (non-electric) bike** would offer **even greater health benefits**, as all propulsion comes from human effort.

Cycling, especially on a standard bike, is **superior from a public health perspective**. Even with manufacturing and charging, e-bikes offer **net gains** in health due to **increased physical activity**.

Car vs bike ecosystem impacts



Notes:

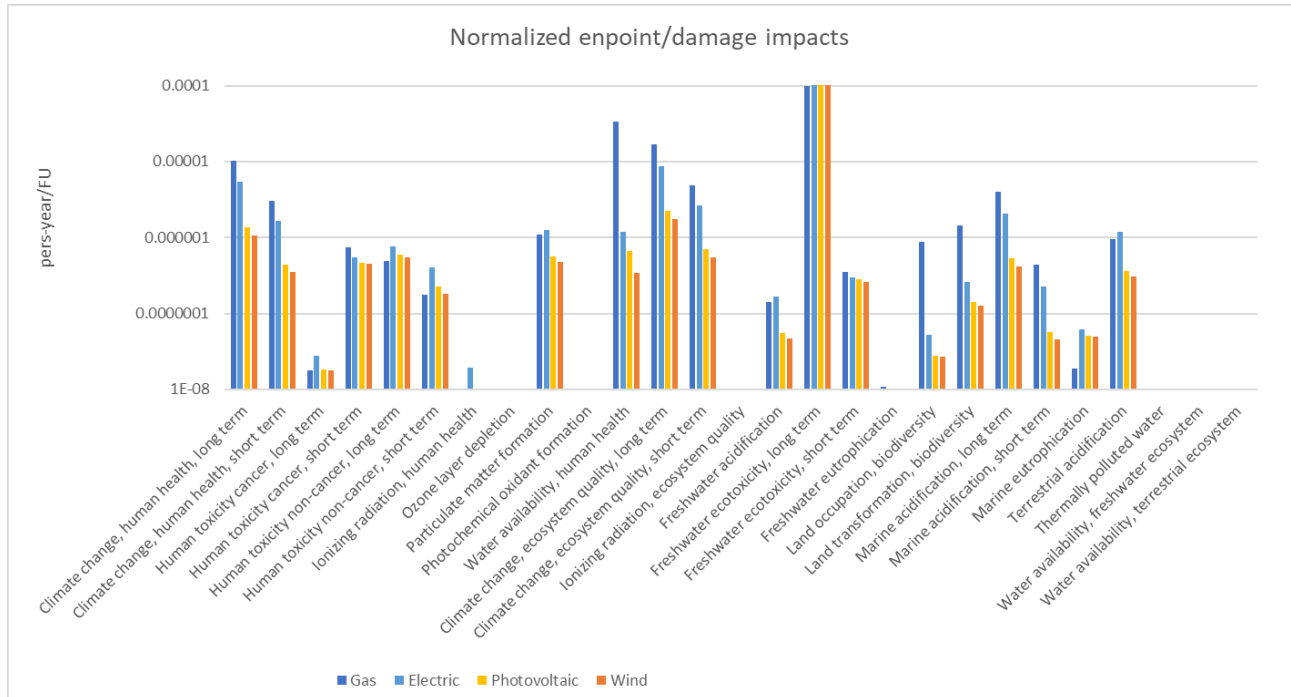
- This graph compares **ecosystem quality impacts** (measured in species·yr/FU) across different vehicle types and charging scenarios.
- The **ecosystem impact** accounts for environmental damage, particularly **biodiversity loss** due to emissions, energy use, and resource extraction.

Key Observations:

- **Gasoline vehicles** have the **highest impact** on ecosystems—primarily from the **use phase** (green bar), which includes emissions harmful to biodiversity.
- **Electric vehicles (EVs):**
 - The **energy source** used for charging plays a major role.
 - EVs charged with **hard coal** have extremely high ecosystem impacts—surpassing even gas vehicles.
 - EVs using **wind or solar** perform significantly better.
- **E-bikes** show the **lowest ecosystem impact**, even when factoring in manufacturing and electricity use.

- Despite including the impact of an “extra sandwich” (i.e., extra food intake for the cyclist), bikes remain more sustainable.

Comparison of normalized damage Impact World+ Life cycle impact Assessment



Notes:

- This chart provides a **more granular breakdown** of environmental and human health impacts across various categories using **normalized endpoint/damage impacts** (in person-years per functional unit).
- Data is shown on a **logarithmic scale** to capture a wide range of values.

Key Takeaways:

- **Consistent Ranking Across Categories:**
In nearly all categories, the relative ranking of the four vehicle types remains consistent:
 - **Gasoline vehicles** generally perform **worse**.
 - **Wind and photovoltaic EVs** consistently show **lower impacts**, especially in **ecosystem-related categories**.
- **Impact Categories Grouped By:**
 - **Human Health** (e.g., climate change, human toxicity, particulate matter, ionizing radiation)
 - **Ecosystem Quality** (e.g., freshwater ecotoxicity, biodiversity, eutrophication)

- **Natural Resources** (e.g., water depletion, land transformation)
- **Minor Categories Have Minimal Influence:**
 - Some impact categories (e.g., **terrestrial acidification, thermal pollution**) are **orders of magnitude lower** (log scale) and don't significantly alter the overall conclusions.

7.3. Consumption, products and health - Exposome and health impacts

Exposome definition (Wild 2012)

Published by Oxford University Press on behalf of the International Epidemiological Association
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International Journal of Epidemiology 2012;41:24-32
doi:10.1093/ije/dyr236

REVIEW

The exposome: from concept to utility

Christopher Paul Wild

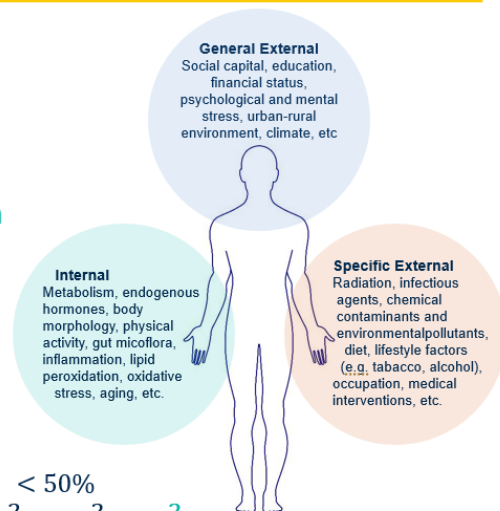
The exposome is composed of every exposure to which an individual is subjected from conception to death.

Therefore, it requires consideration of both the nature of those exposures and their changes over time.

For ease of description, three broad categories of non-genetic exposures may be considered: internal, specific external and general external.

Phenotype = f (Genome, **Exposome**)

Variability: $\sigma_P^2 = \sigma_G^2 + \sigma_E^2$
 < 50%



Inclusion of various impacts on human health

- Climate change
- Fine particulate impacts
- Nutrition impacts
- Physical exercise
- Chemicals in products



These **impact categories go beyond emissions** and incorporate **lifestyle, exposure, and systemic health outcomes** — which is critical when moving toward **integrated health-oriented life cycle assessments** (LCAs or S-LCAs).

Exposome – vision

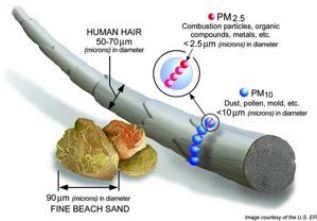


The exposome is composed of **every exposure** (chemical, physical, social, nutritional, etc.) that an individual experiences from **conception to death**.

Air pollution impacts on human health

6.5 mil deaths per year, **65% in Asia**, largely cardiovascular disease

17,800 deaths/day,
equivalent to 60 Malaysian
airplane crashes **per day**



A sunny day in Beijing!



- 6.5 million deaths per year globally
→ 65% of those deaths occur in Asia
→ Mostly due to cardiovascular diseases
- That's equivalent to:
 - 17,800 deaths per day
 - Like 60 Malaysian airplane crashes every single day
- *"A sunny day in Beijing"* ironically shows the severity of smog, even on days considered clear.
- Air pollution is not just a local issue.
Many of the goods we consume are manufactured in regions suffering from poor air quality, making us co-responsible through global supply chains.
- The health toll is far greater than many global crises like COVID-19 — yet it's ongoing and often overlooked.

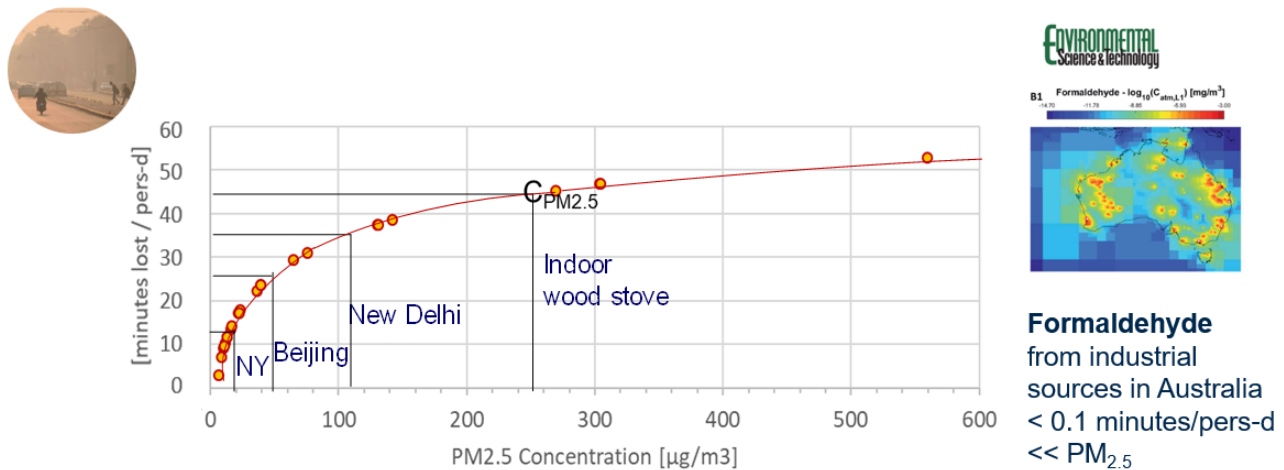
What can be done?

- Solutions like filters on industrial stacks are technically simple and well understood.
- Policy, global cooperation, and awareness are key barriers — not the lack of solutions.

2. Indoor and outdoor PM_{2.5} burden of disease

The graph shows a clear relationship:

As **PM_{2.5} concentration (µg/m³)** increases, so does the **daily loss of healthy life time**.

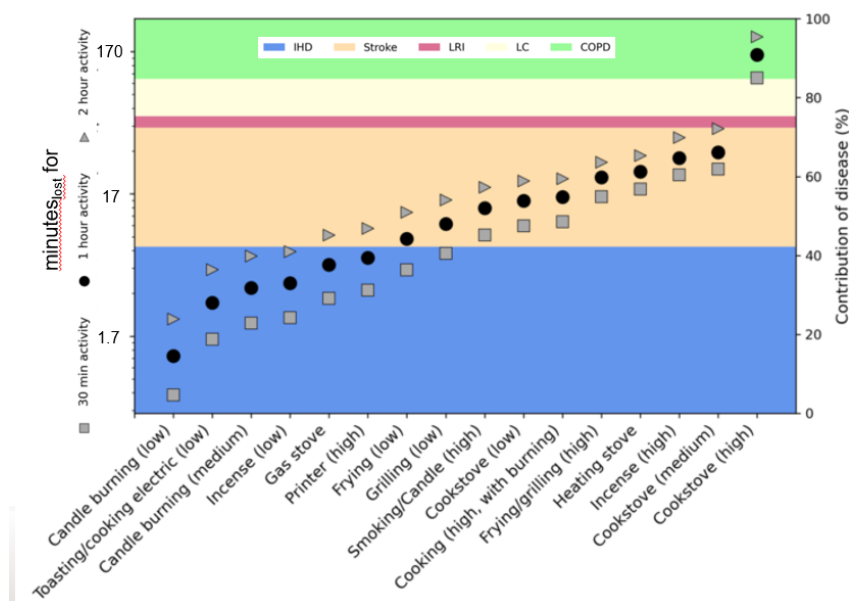


Location / Source	PM _{2.5} Level (µg/m ³)	Life Lost (min/person/day)
New York City	~10–20	~12 minutes
Beijing	~50	~25 minutes
New Delhi	~100	~35 minutes
Indoor wood stove 🔥	~250	~45 minutes

Even clean, cozy wood-burning stoves — often used in rural or alpine homes — have severe health implications if ventilation is poor.

- ❑ PM_{2.5} is a **silent killer** — it causes **measurable daily harm**, especially in cities or poorly ventilated indoor settings.
- ❑ The impact is **not only industrial**; common lifestyle choices (e.g., wood-burning heat) can pose serious risks.

Impact of indoor sources of fine particulates



1 – 17 min.(medium 3.5)
of healthy life lost
for 1 hour indoor
“huggely” candle,

Burning a candle indoors (especially one that smokes a lot) might seem cozy — “hygge” style — but it can cost you **1 to 17 minutes of healthy life per hour** depending on the emission level.

Indoor Source	Life Lost (approximate range)
Candle burning (low)	~1–2 min
Toasting/cooking electric (low)	~1.5 min
Candle burning (medium)	~3.5 min (⚠️ noted on the slide)
Incense (low)	~4–5 min
Gas stove	~5–7 min
Printer use	~6–8 min
Frying (low)	~6–9 min
Grilling (low)	~7–10 min
Smoking/Candle (high)	~10–15 min
Cookstove (low/medium)	~12–17+ min
Incense (high)	~18+ min

▲ The higher up the activity on the chart, the more minutes of life lost per hour it causes.

□ Even seemingly harmless **indoor rituals** like candle use or incense can significantly impact health.

- Activities like **grilling indoors** or **burning incense** have much higher effects than most people expect.
- Ventilation and low-emission alternatives are **critical** for reducing PM_{2.5}-related health impacts indoors.

USEtox framework



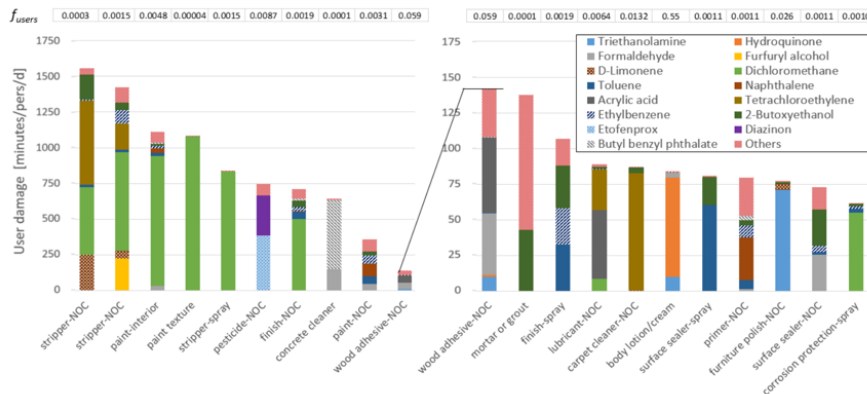
- **Inventory Flow:**
Measured in **kg of chemical in products per functional unit (kg_chemical/FU)**
- **Characterization Factor Output:**
Health impact measured as **DALY per kg of chemical in products**
(i.e., how harmful a certain amount of chemical is when present in a product)

Category	Examples / Relevance
Building materials	Volatile organic compounds (VOCs), flame retardants
Food contact materials	Packaging leaching, phthalates, BPA
Environmental emissions	PFAS, dioxins, air/water/soil contamination
Pesticides	Agricultural residue exposure
Paints	Heavy metals, solvents
Personal care products	Absorption through skin, e.g. parabens, microplastics
Toys (including mouthing)	Young children's exposure via ingestion or skin contact

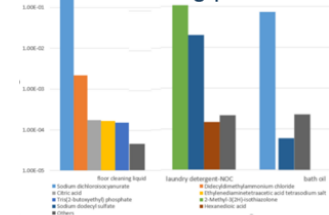
“Don’t put on your skin anything you wouldn’t drink.”

– Because **up to 60%** of substances applied topically can be **absorbed into the body**, contributing to the chemical burden and potential health risks.

Chemicals in household products: Human Health impacts on users



Impacts on freshwater ecosystems: Cleaning products!



Several home maintenance, cosmetics & cleaning product with substantial impacts on **users**, 20-1500 minutes lost per user per day → product-chemical combinations to further study or replace in priority

Left chart:

Measured in **minutes lost/person/day**:

- **Paint strippers & VOCs**: Up to **1500 minutes/day**!
- **Paints (without VOCs)**: Still significant health burden
- **Spray applications**: e.g., pesticide or concrete cleaner sprays also pose major risks

→ **Protective recommendation:**

Use proper **ventilation**, **protective gloves**, and **VOC-filtering masks** – not surgical or COVID masks. These chemicals are **inhaled or absorbed** through skin.

Middel chart:

- **Body lotions, lipsticks, creams**:
Containing **hydroquinone**, **butyl benzyl phthalate**, and other high-risk compounds
→ Up to **75 minutes/day** of healthy life lost if used regularly

Right chart

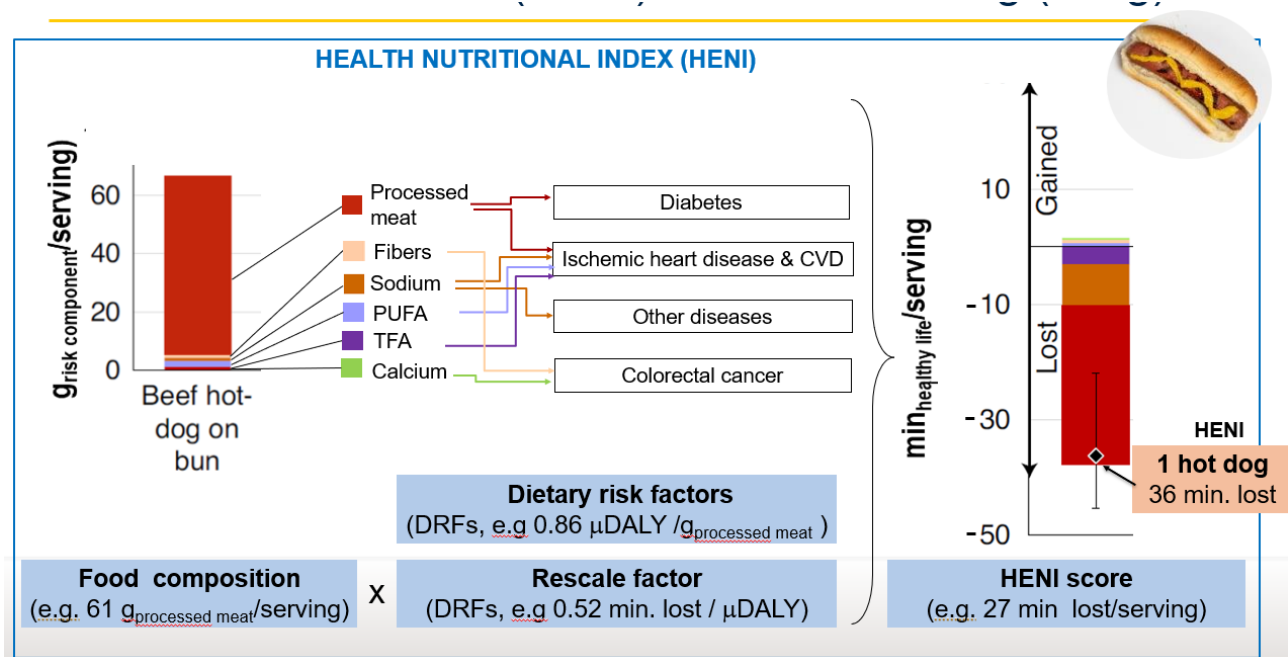
Cleaning products dominate in terms of **freshwater ecosystem damage**

► **Laundry detergents, floor cleaning fluids, toilet cleaners** contribute disproportionately to aquatic toxicity

What You Can Do:

- Minimize exposure: Choose low-VOC and eco-certified products
- Protect yourself: Use appropriate PPE during maintenance tasks
- Rethink product choices: Especially for items in prolonged contact with skin or frequent use

Health Nutritional Index (HENI) for a beef hot dog (140g)



Case Study: Beef Hot Dog (on bun)

Breakdown of Nutritional Components (per serving):

- **Processed meat:** 61g (largest negative contributor)
- **Sodium:** 1g
- **Fibers:** ~1g
- **PUFA (polyunsaturated fats), TFA (trans fats), Calcium:** trace amounts (PUFA & calcium are beneficial, TFA is harmful)

Health Risks & Disease Links:

- **Processed meat** → Diabetes, colorectal cancer, heart disease
- **Sodium** → Heart disease

- **Trans fat (TFA)** → Multiple chronic conditions

✓ **Small gains from:**

- **PUFA, Calcium, Fiber** → Improve cardiovascular and gut health

12 34 Calculation Using HENI Framework:

- Formula:
Food composition × Dietary Risk Factors × Rescale factor → Minutes of healthy life lost
- Example for processed meat:
 $61\text{g} \times 0.86 \mu\text{DALY/g} \times 0.52 \text{ min}/\mu\text{DALY} \approx 27 \text{ min lost}$

🕒 Net HENI Score for 1 Hot Dog:

- **36 minutes of life lost (net)**
- **Range (uncertainty):** ~20 to 45 minutes lost per serving

⚠️ Conclusion:

A single hot dog costs you about **half an hour of healthy life**, due to its processed meat content and other negative nutritional factors.

Health Nutrient Index HENI: Stylianou et al., 2021 Nature food



Small targeted dietary changes can yield substantial gains for human and environmental health

Katerina S. Stylianou¹, Victor L. Fulgoni III² and Olivier Jolliet¹

To identify environmentally sustainable foods that promote health, we combined nutritional health-based and 18 environmental indicators to evaluate, classify and prioritize individual foods. Specifically for nutrition, we developed the Health Nutritional Index to quantify marginal health effects in minutes of healthy life gained or lost of 5,853 foods in the US diet, ranging from 74 min lost to 80 min gained per serving. Environmental impacts showed large variations and were found to be correlated with global warming, except those related to water use. Our analysis also indicated that substituting only 10% of daily caloric intake from beef and processed meat for fruits, vegetables, nuts, legumes and selected seafood could offer substantial health improvements of 48 min gained per person per day and a 33% reduction in dietary carbon footprint.

Minutes of healthy life <https://rdcu.be/cuVht>
Gained or lost for 5800+ foods



120,000 likes on CNN Instagram

High impact research >1000 news media with **potential reach of 1.3 billion people**

HENI Study Summary:

- Title: “*Small targeted dietary changes can yield substantial gains for human and environmental health*”
- Published in: **Nature Food**, 2021
- Authors: **Stylianou, Fulgoni, Jolliet**

Purpose:

To evaluate 5,800+ foods in the U.S. diet for:

- **Health impact** (in minutes of healthy life lost/gained per serving)
- **Environmental footprint**

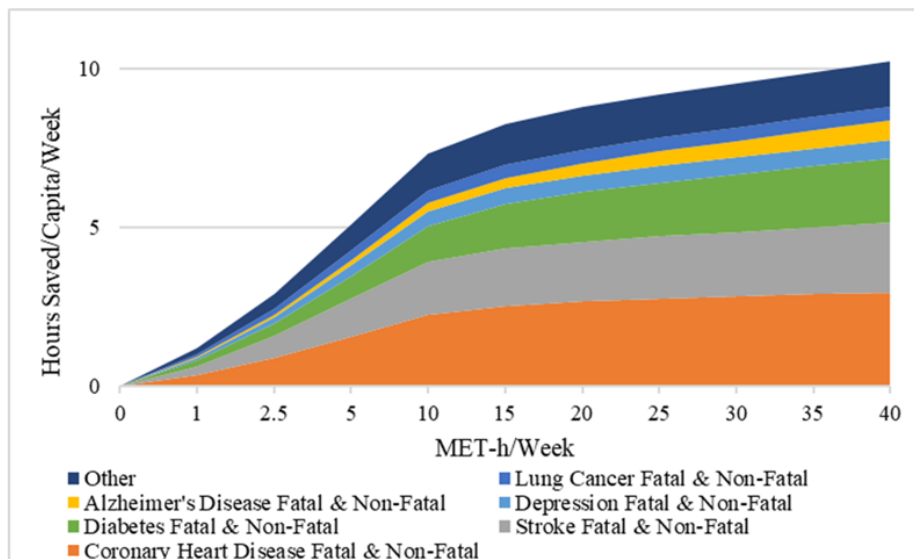
Method:

- Combined nutritional risk factors (based on Global Burden of Disease)
 - Converted to DALYs → then into **minutes of healthy life per serving**
-

The Viral Hot Dog Example:

- **One beef hot dog (140g) = 36 minutes of life lost**
- Became instantly viral:
 -  Picked up by over **1,000 news outlets**
 -  Featured on **CNN’s Instagram** → 120,000 likes
 -  Estimated **1.3 billion people reached** worldwide
-  Even translated and shared globally (including Persian, as noted in the story)

Physical exercise: benefits of 825 physical activities



2011 Compendium of Physical Activities

CODE	METS	MAJOR HEADING	SPECIFIC ACTIVITIES
01015	7.5	bicycling	bicycling, general
21050	4.3	volunteer activities	walking, 3.5 mph, brisk speed, not carrying anything

✓ Physical Exercise Saves Lives:

- The y-axis shows **hours of life saved per capita per week**.
- The x-axis represents **total MET-hours per week**.
- Health benefits increase with more physical activity, especially up to around 20–25 MET-h/week.

🚲 Examples of Physical Activities:

From the **2011 Compendium of Physical Activities**:

Activity METs Description

Bicycling 7.5 General cycling

Walking 4.3 Brisk walk at 3.5 mph, not carrying anything

So:

- **2 hours of bicycling/week** = 15 MET-h/week
- **5 hours of walking/week** = ~21.5 MET-h/week

Health Impacts Prevented:

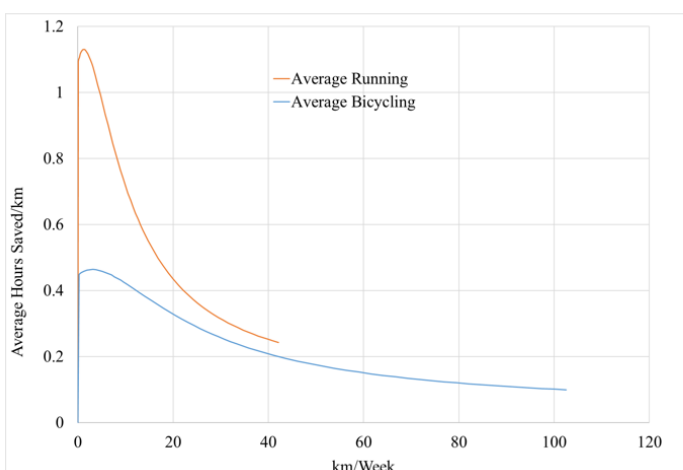
Stacked areas in the graph show time saved from avoiding or reducing:

- **Coronary heart disease**
- **Stroke**
- **Diabetes**
- **Depression**
- **Cancer (lung, etc.)**
- **Alzheimer's disease**
- **Other causes**

DK Special Note on Denmark:

- In Denmark, many people commute by bike, easily reaching **20 MET-h/week**.
- This can translate to saving **~8 hours of healthy life per week**, or over **1 hour per day** — a major gain in public health.

Physical activity gains per km walking/running and cycling: hours of healthy life gained per km



2011 Compendium of Physical Activities

Duration of trip compensate for the MET
→ $\frac{\text{min}_{\text{gained}}}{\text{km}_{\text{activity}}}$ very comparable
Across walking or running, but depends
on overall weekly physical activity

As long as not as not in New Delhi or
forest fire area, much more beneficial
to bike

- **At lower weekly distances:**
 - Running saves **more life per km** (~1.1 hours/km when doing very little total km).

- Bicycling saves less, but still significant (~0.4 hours/km).
 - **As you increase km/week:**
 - Both benefits **decline per km**, due to diminishing returns (you're already "healthy" from earlier km).
 - But total benefit still **adds up** over the week.
-

✓ Practical Example:

If you **bike 20 km per week**:

- You save about **0.3 hours/km** → **6 hours gained/week** (close to a full healthy day!).

If you run the same:

- Slightly higher per km, but it takes more time and physical effort.
-

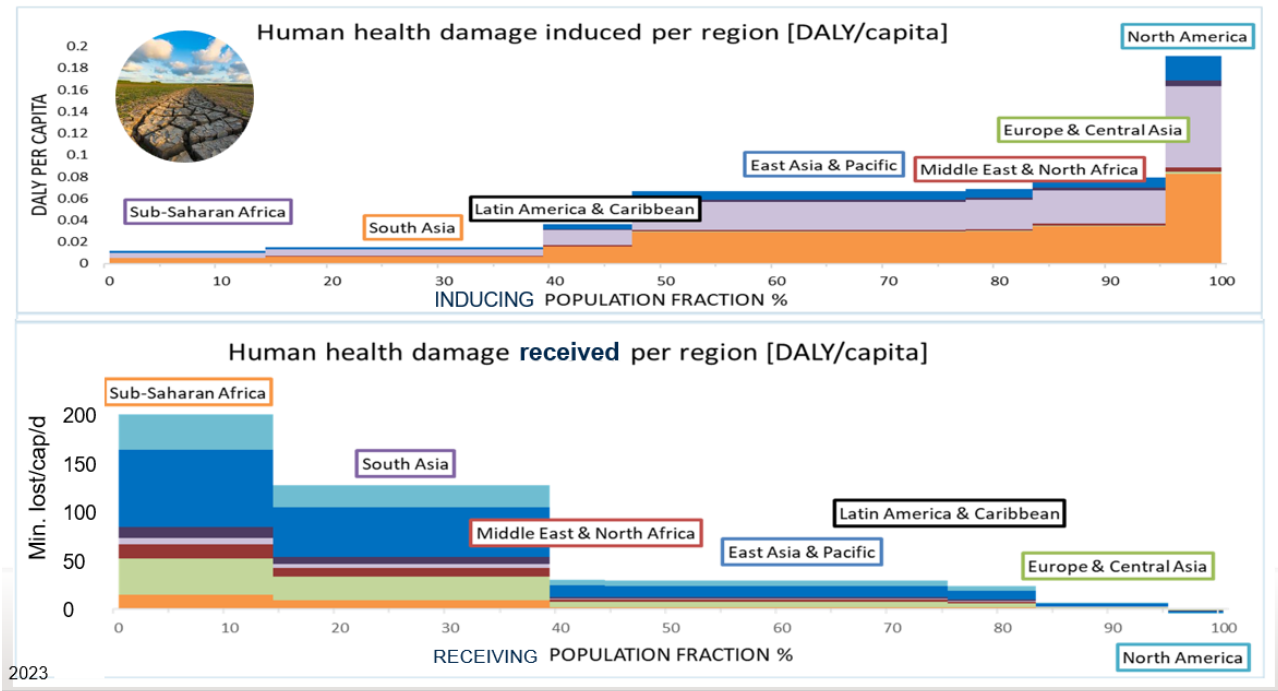
⚠ Caveats:

- The **air quality matters**:
 - In **polluted cities (like New Delhi)** or **forest fire zones**, outdoor exercise might reduce or even reverse the benefit.
 - Always check **PM2.5** levels if exercising outdoors.
-

💡 Bottom Line:

- **Biking is time-efficient and very beneficial**, especially when done moderately (e.g. commuting).
- **Walking and running are great**, but take longer per km for the same benefit.
- Combine activity with clean environments for **maximum health gains**.

Disparities: Climate change impacts of heat and cold on health



- This shows **which regions cause the most damage** through their emissions.
- **North America** is the top inducer of health damage per capita, followed by **Europe & Central Asia**.
- These regions have the highest per-capita carbon footprints, mainly due to high consumption and fossil fuel use.

Bottom: Health Damage *Received* (Minutes lost/capita/day)

- This shows **who is suffering the most** from climate-related health damage.
- **Sub-Saharan Africa**: The most affected, with **~200 minutes of healthy life lost per person per day** due to climate impacts.
- **South Asia** also suffers significantly (over **100 minutes lost/day**).
- Ironically, these regions contribute **very little** to climate change emissions.

Key Message: Extreme Inequality

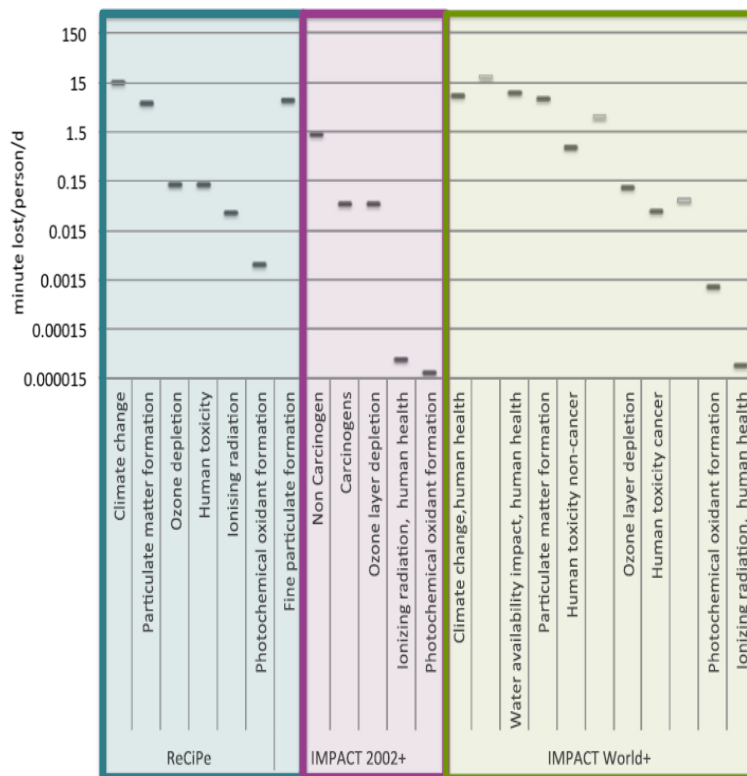
- **High emitters** (North America, Europe) → induce harm
- **Low emitters** (Africa, South Asia) → receive harm

- This is a textbook example of **climate injustice**.

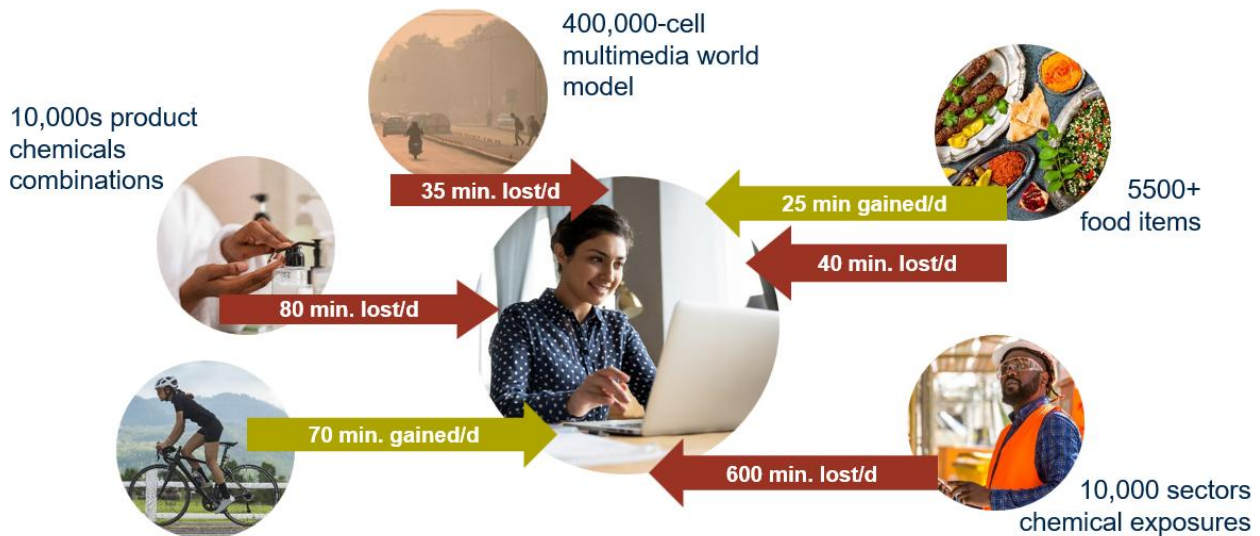
💡 What Can Be Done?

- If you're in a high-emission region, you **have more power and responsibility** to:
 - Reduce carbon-intensive consumption (travel, energy, meat, etc.)
 - Support policies that **invest in adaptation** in vulnerable regions
 - Advocate for **global equity** in climate financing and loss/damage support
 -

Worldwide environmental impacts on human health



Quantitative screening of impacts per pers per day



Each component in the circle around the person reflects a major domain of exposure or behavior that can be quantified in **minutes of healthy life gained or lost per day**.

Healthy Gains

- 🍌 **Food (e.g., nuts, fruits, vegetables)**
→ **+25 min/day**
Good dietary choices add to your lifespan.
- 🚴 **Physical Activity (e.g., biking)**
→ **+70 min/day**
Commuting or moving actively greatly benefits health.

⚠️ Health Losses

- 🧴 **Consumer Product Chemicals (e.g., skincare, cleaners)**
→ **-80 min/day**
Many substances penetrate the skin and accumulate over time.
- 🌫️ **Air Pollution (PM2.5)**
→ **-35 min/day**
Largely invisible, yet one of the most deadly exposures.
- 🏭 **Occupational Chemical Exposure**
→ **-600 min/day**
Massive health toll in certain sectors; highlights global inequalities.

- 🍔 **Unhealthy Food (e.g., processed meat, sugary drinks)**
→ **-40 min/day**
-

🔍 **Behind the Numbers**

This analysis is powered by:

- 📦 **10,000+ product-chemical combinations**
- 🧪 **10,000+ industrial sectors of chemical exposure**
- 🥗 **5,500+ food items**
- 🌐 **400,000-cell multimedia exposure model**

It's a **life cycle perspective**, measuring both **mortality** and **morbidity (disability)** across multiple health domains.

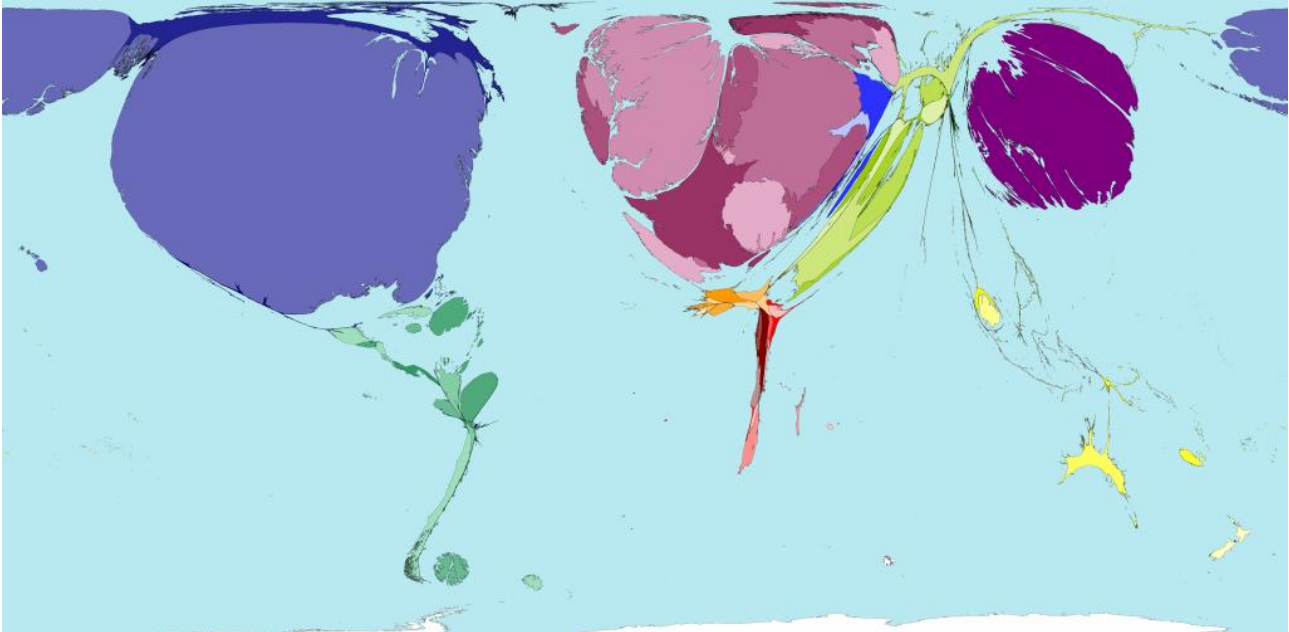
💡 **Your Role**

The takeaway is empowering:

Small changes in what you eat, breathe, apply to your skin, or how you move can compound into big impacts on your health.

7.4. Global trade - Part A

Clothing Imports



- 📦 **U.S. & Western Europe**
These regions dominate the global clothing import market. The U.S. appears **massively bloated**, reflecting its huge consumer market. Western Europe, particularly countries like Germany, the UK, and France, also show high import volumes.
- 🇯🇵 **Japan**
Another significant importer, Japan shows up prominently, though smaller than the U.S. and Europe.
- 🌍 **Africa & South America**
Much smaller in this visualization, indicating **relatively low import volumes**—likely due to smaller economies, local textile markets, or reduced per capita consumption.
- 🌏 **Asia (excluding Japan)**
Often **export hubs** rather than importers. Countries like China, Bangladesh, and India are **major producers** but not significant importers of clothing, so they appear smaller here.

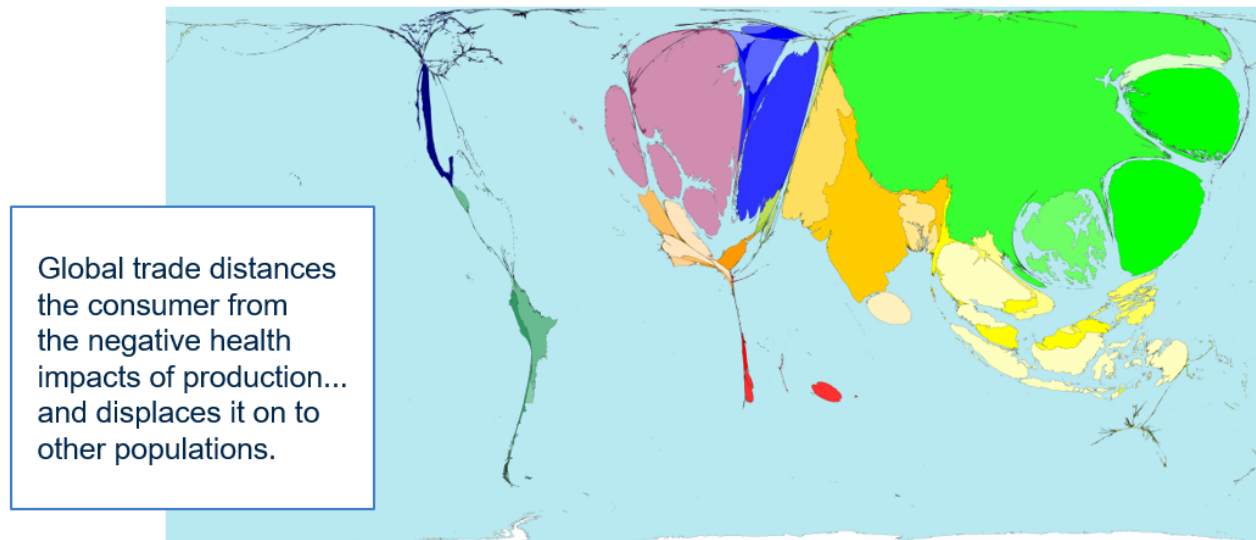
💬 Why It Matters

This map:

- Reveals **economic disparities** in global consumption.

- Highlights where **consumer responsibility** for clothing waste and environmental impact lies.
- Supports the case for **sustainable fashion** and **rethinking fast fashion habits** in high-consuming regions.

Clothing Exports



- 🌍 **Asia Dominates Production**
 - **China** is by far the largest clothing exporter globally, with **India**, **Bangladesh**, **Vietnam**, and other Southeast Asian nations also showing significant export volume.
 - These countries appear disproportionately **enlarged**, reflecting their heavy involvement in textile and garment manufacturing.
- 🌍 **Africa & Latin America**
Some production also happens in countries like **Mexico**, **Turkey**, and parts of **Africa**, though to a lesser extent than Asia.
- 🌍 **U.S. & Western Europe**
Shrink significantly in this map. Despite being major consumers (as seen in the import map), they contribute **very little to clothing production**.

📦 Trade Disparity Insight

The note on the map says it clearly:

"Global trade distances the consumer from the negative health impacts of production... and displaces it on to other populations."

What does that mean?

- **Consumers in wealthier nations** benefit from affordable fashion.
- **Workers in exporting countries** bear the **health, environmental, and social** burdens—exposure to dyes, poor working conditions, low wages, etc.
- This creates a **hidden cost** of fashion—one we don't see on a price tag.

Reflection

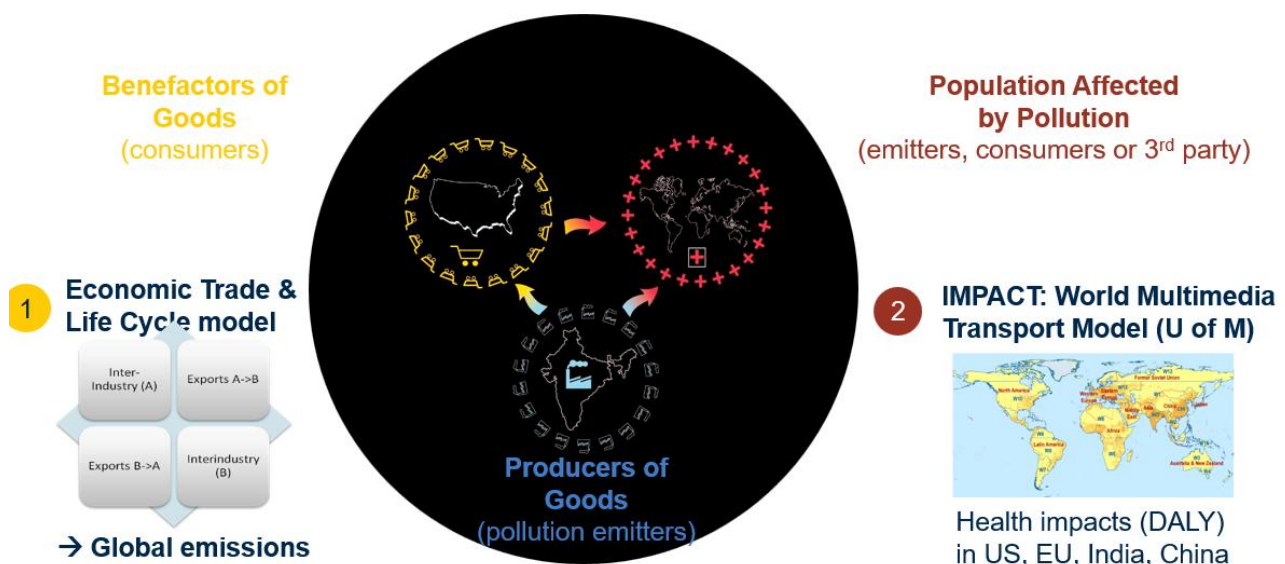
Together, the import and export maps visualize a **global imbalance**:

- **Who wears it vs. who makes it**
- **Who gains convenience vs. who bears the cost**

Would you like to go deeper into:

- The **health/environmental impacts** in textile-producing regions?
- The concept of “**slow fashion**” or **sustainable purchasing behavior**?

Combining models to calculate health impacts of global trade



Two Integrated Models:

1. Economic Trade & Life Cycle Model

- Tracks the **flow of goods** and services between countries and industries.
- Identifies:
 - **Consumers** (benefactors of goods – e.g., U.S., Europe)
 - **Producers** (pollution emitters – e.g., India, China)
- Calculates **global emissions** from producing these goods.

2. IMPACT: World Multimedia Transport Model (University of Michigan)

- Models how **pollutants travel** through air, water, and soil.
- Calculates **health impacts** (in **DALYs** – Disability-Adjusted Life Years).
- Shows that **pollution may affect**:
 - People **in the country of production**
 - People **in other countries** (e.g., pollutants transported back to the consumers or third parties)

Key Takeaways:

-  **Trade-driven emissions** are not just a local issue—they cause **global health consequences**.
-  **Consumers in wealthier countries** (e.g., U.S., EU) enjoy goods made elsewhere, while the **health burden** of pollution is mostly borne by **producers** (e.g., India, China).
-  **Pollution doesn't respect borders**. Airborne particulates and other pollutants can travel across continents, affecting distant populations.

Setup of our M-cubed project



🌍 Project Highlights:

- **Project Origin:** Launched through the **M-cubed program** at U-M, which funds interdisciplinary research.
- **Core Requirement:** 3 faculty members from **3 different schools or disciplines** must come together.
- 🟡 The University of Michigan (the yellow "M" on the map) is the hub.
- ✈️ One member was on a **plane**, another in **China**, and a third on **sabbatical in Australia**—true global coordination!

🤝 Global Collaboration:

The map shows red lines from Michigan to collaborators in:

- **China** CN
- **India** IN
- **Cyprus** CY
- **Australia** AU

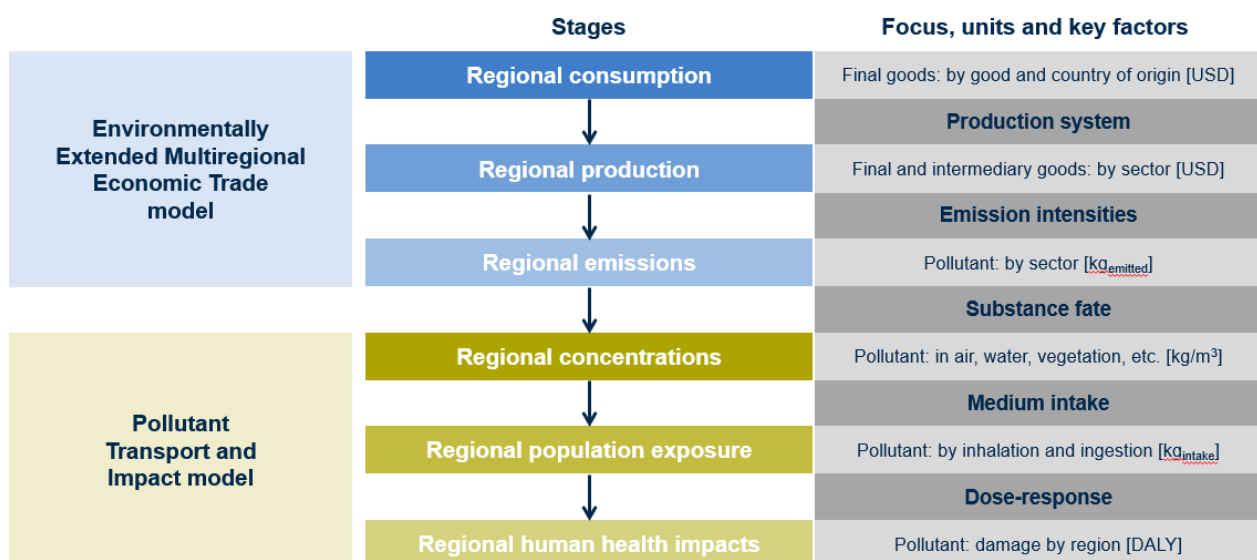
These partners supported work across:

- Health impact modeling
- Global emissions
- Data integration for trade, pollution, and DALYs

💰 Outcome:

- Secured **\$60,000 in seed funding**
- Enabled a **world-scale research initiative**
- Combined expertise across continents in **environmental health, economics, and transport modeling**

Add global economic trade model



1. Regional Consumption

- **What:** Where goods are consumed.
- **Focus:** Final goods, by country of origin and product type.
- **Unit:** USD



2. Regional Production

- **What:** Where goods are produced and how.
- **Focus:** Sectoral breakdown of both final and intermediate goods.

- **Unit:** USD



3. Regional Emissions

- **What:** Emissions from the production process.
- **Focus:** Sector-based emission intensities.
- **Unit:** kg of pollutant emitted



(Now we shift to the environmental model)

4. Regional Concentrations

- **What:** Resulting pollution concentrations.
- **Focus:** Pollutants in air, water, soil, vegetation.
- **Unit:** kg/m³
- *Determined by substance fate modeling*



5. Regional Population Exposure

- **What:** Human exposure to pollutants.
- **Focus:** Inhalation or ingestion pathways.
- **Unit:** kg of pollutant intake per person



6. Regional Human Health Impacts

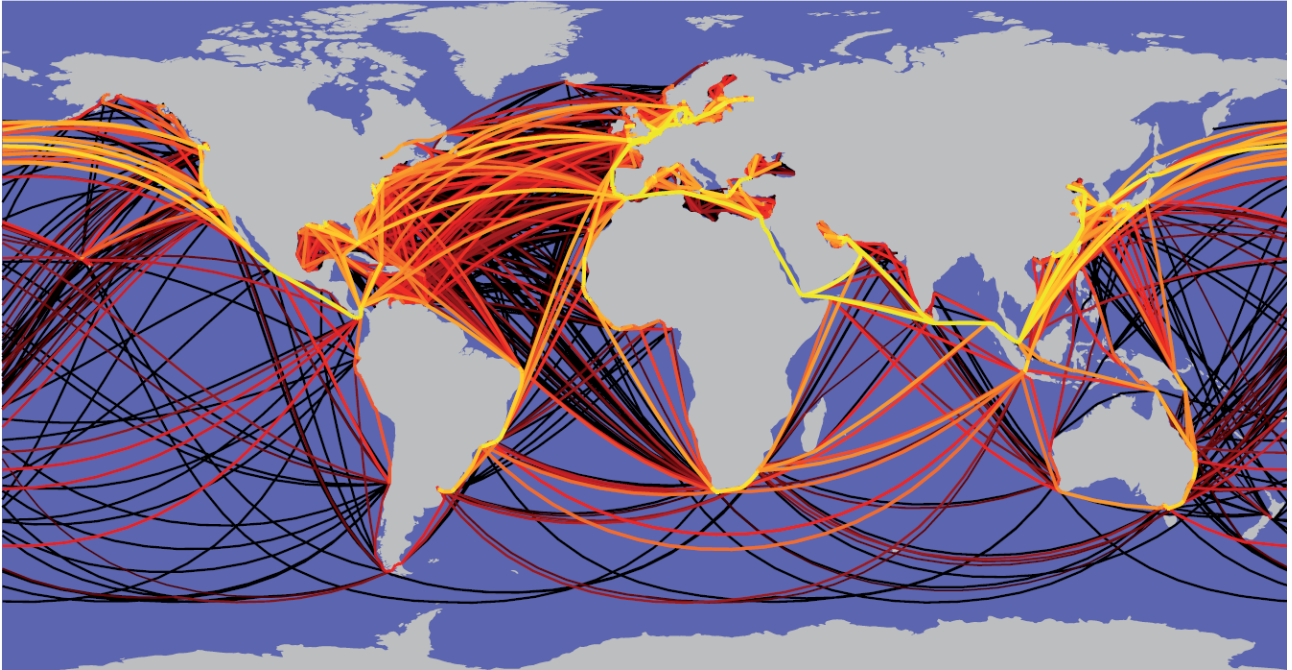
- **What:** Health damage due to exposure.
- **Focus:** Disease burden per region.
- **Unit:** DALYs (Disability-Adjusted Life Years)

This approach combines:

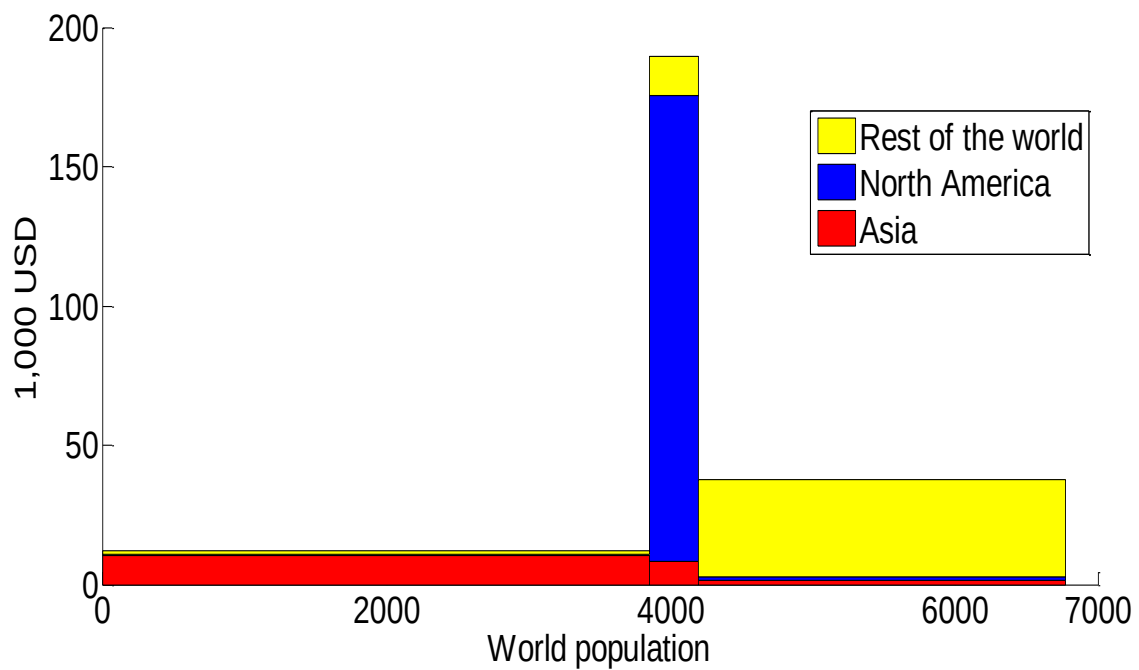
- **Economic modeling** (Environmentally Extended Multiregional Input-Output, EEMRIO)
- **Pollutant fate and transport modeling**

- Health impact assessment (dose-response)

Full global trade network exchange between 15,000 country-sectors



World Production per capita



- **X-axis:** World population (in millions), broken down roughly into:
 - **Asia** (red section, left)
 - **North America** (blue section, centered around 400 million)
 - **Rest of the World** (yellow section, to the right)
 - **Y-axis:** Annual production per capita (in USD thousands)
-

Key Insights from the Commentary:

1. North America Leads in Consumption:

- The average North American **induces \$180,000/year** in economic activity.
- Despite a **smaller population (~300–400 million)**, its **per capita consumption** is extremely high.
- North America is mostly a **closed market**, meaning much of the production supports domestic consumption — but not entirely.

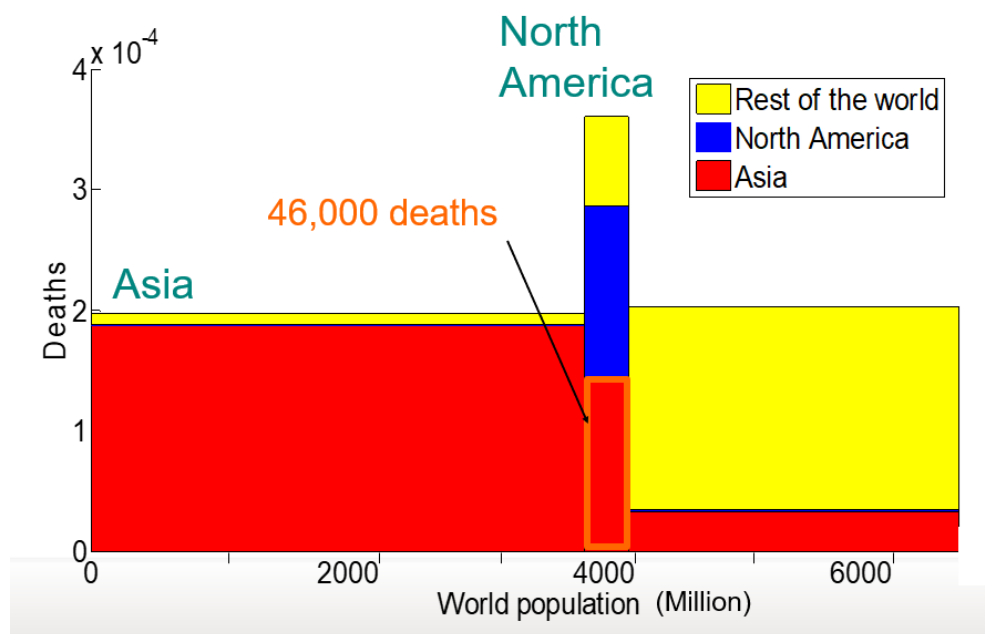
2. Asia's Role as a Producer:

- Asia (with billions of people) **produces far less per capita**, but in total volume it's massive due to population size.
- Interestingly, **Asian production isn't just for Asian consumers** — a large share supports **external demand** (especially from North America).

3. Global Imbalance:

- There's a **mismatch between who consumes and who produces**.
- This disconnect often leads to **outsourced environmental and health impacts**, as seen in earlier slides — where **pollution and labor burdens are carried by producing regions**, often without enjoying the economic benefits to the same degree.

Particulate Matter related deaths induced by consumption per capita



- **X-axis:** Global population (in millions), grouped by region (Asia, North America, Rest of the World).
- **Y-axis:** PM-related deaths (per capita).
- Bars represent the **deaths per capita induced by consumption per region**, with:
 - **Red:** Asia
 - **Blue:** North America
 - **Yellow:** Rest of the world

🧠 Key Takeaways from the Commentary:

1. Asia Bears the Brunt (65% of deaths):

- Particulate matter emissions per dollar spent are **significantly higher in Asia** due to intensive industrial production.
- This results in **460,000 PM-related deaths per year in Asia** — a **large share of which are linked to goods consumed elsewhere**, particularly in North America and Europe.

2. Responsibility of Consumption:

- Roughly **1/3 of the health impact** of a US consumer's lifestyle is **exported to Asia**, meaning products consumed in the US cause pollution and death in Asia.
- European and global consumers also contribute, **adding ~60,000 deaths** in Asia due to imports.

3. Implications for Equity and Policy:

- **Global trade distances** consumers from the health and environmental impacts of their consumption.
- The solution isn't only to reduce local emissions — but also to **support cleaner production abroad**, reduce unnecessary imports, or **invest in cleaner supply chains**.

For Presentation or Summary:

“Our consumption has consequences far beyond our borders. One-third of particulate matter–related deaths in Asia—about 460,000 lives per year—are tied to goods consumed in the US alone. This underscores the urgent need to align global trade with environmental justice.”

Particulate Matter Control Technology is Effective

Best available control technology available with 99-99.9% removal efficiency



“Particulate Matter Control Technology is Effective”

- **99–99.9% removal efficiency** is already achievable using **best available control technologies (BACT)** such as:
 - Electrostatic precipitators
 - Fabric filters

- Scrubbers
 - Cyclones
-

Core Takeaways from the Text:

1. Technology is not the bottleneck:

- PM pollution can be cut by a factor of 100 to 1000 using current solutions.
- These are **widely known and commercially available**.

2. The human cost of inaction is staggering:

- **6 million premature deaths per year** are caused by PM globally.
- That's **more than the annual death toll of COVID-19 at its peak**.

3. Health cost outweighs implementation cost:

- Installing PM controls on **exhaust pipes and industrial stacks** is **cost-effective**, especially when compared to the **economic burden of disease and death**.

4. Urgency & Opportunity:

- This is a **global health opportunity** with **immediate benefits**.
 - Clear **call to action**: implement filters **everywhere**, especially where current emissions are unregulated.
-

For a Strong Summary Slide:

“With 6 million deaths annually from particulate matter pollution, and solutions that can reduce emissions by 99.9%, the question is no longer if we can act—but whether we will. The technology exists. The cost is low. The impact is massive. Let’s filter the air—and save lives.”

Choices

It’s a choice.

We either act—or accept that others (or we ourselves) will continue living in harmful conditions, like the one shown.

Global burden of disease 2019 – in class activity

Take 15 minutes to rank these risk factors from highest to lowest for global number of deaths.

Estimate the total annual number of death in the world associated with the highest and lowest factors.

Without using Internet (just give your best guess!)

19

MAIN RISK FACTORS:

Air pollution
Alcohol use
Childhood maltreatment
Dietary risks
Drug use

High blood pressure
High body mass index
High fasting plasma glucose
High low density cholesterol (LDL)
Impaired kidney function
Intimate partner violence
Low bone mineral density

Low physical activity
Malnutrition
Occupational risks
Other environmental (*lead, radon*)
Tobacco smoking
Unsafe sex
Unsafe water & sanitation

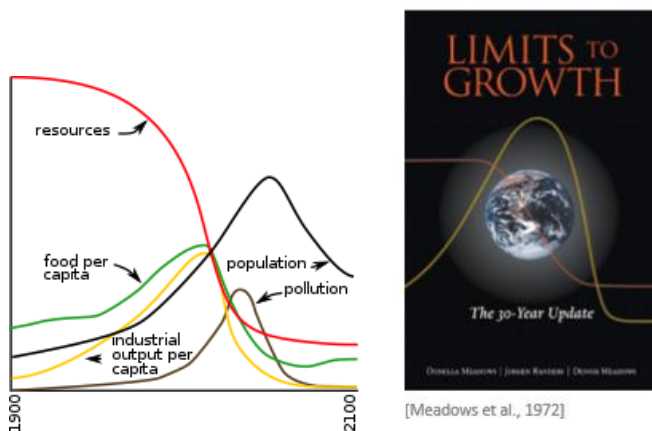
Each of you takes a photo of the group ranking

8. Applying tools (case study)

9. Circular Economy

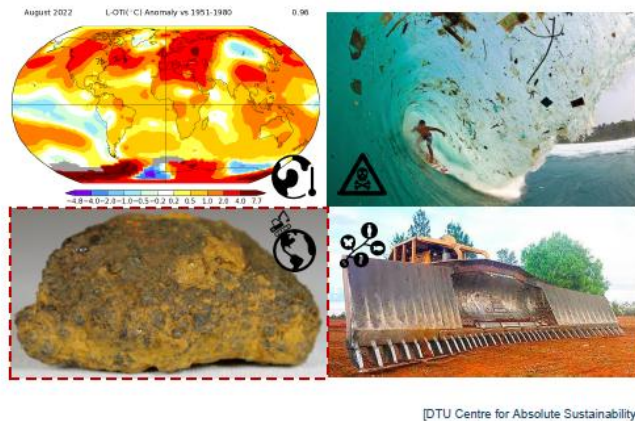
1. Whats the problem

Urgent need for radically different measures to address anthropocentric sustainability



- ◆ **Main message:** There is an urgent need to rethink sustainability from a human-centered perspective.
- ◆ **Reference:** *Limits to Growth* by Meadows et al. (1972)
 - A groundbreaking report warning that unlimited growth is unsustainable on a finite planet.
- ◆ **Key points:**
 - Earth's **resources are finite**.
 - **Population growth**, **industrial output**, and **pollution** are all interconnected.
 - **Food per capita** and **industrial output** rise temporarily, then decline.
 - **Pollution** and **resource depletion** become major risks if current trends continue.
- ◆ **Visual:** The system dynamics model predicts a collapse scenario if no corrective actions are taken.
- ◆ **Takeaway:** This warning was made over 50 years ago, yet it remains highly relevant. Radical changes are necessary to avoid surpassing planetary boundaries.

Urgent needs to boost environmental sustainability



[DTU Centre for Absolute Sustainability]

- ◆ **Main message:** Multiple environmental crises demand urgent action to improve sustainability.

- ◆ **Key environmental challenges highlighted:**

1. Climate crisis

- Global temperature anomalies (e.g., NASA graph) show a clear **warming trend** since the 1970s.
- This contributes to more extreme weather, sea level rise, and disruption of ecosystems.

2. Toxicity crisis

- The image of a surfer surrounded by plastic waste (Bali) illustrates the **global plastic pollution problem**.
- Plastic affects aquatic ecosystems and enters the food chain, posing health risks to humans and wildlife.

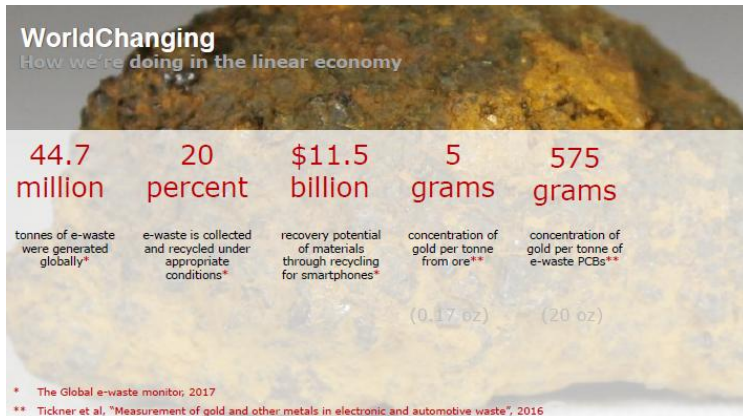
3. Biodiversity crisis

- We are entering what is often called the **sixth mass extinction**, this time driven by human activity.
- Image of deforestation (e.g., Amazon) shows habitat destruction, a major cause of species loss.

4. Resource crisis

- Reaffirms the *Limits to Growth* message from 1972: **natural resources are finite**.
- The image of a raw material (e.g., bauxite or iron ore) and mining equipment symbolizes overexploitation.

Word changing



◆ Main message:

The current linear economy leads to massive waste of valuable materials. We're discarding resources that could be recovered—especially from electronic waste.

◆ Key facts and figures (from 2017 data):

- **44.7 million tonnes** of e-waste are generated annually.
- Only **20%** is recycled properly — meaning **80%** is mismanaged or lost.
- The **monetary value** of recovering materials from smartphones alone is estimated at **\$11.5 billion**.

◆ Focus on gold recovery:

- **Gold ore** yields just **5 grams of gold per tonne**.
- In contrast, **e-waste PCBs** (printed circuit boards) yield **575 grams of gold per tonne** — over **100x more**.
- This highlights the irrationality of mining virgin materials when valuable resources are available in our waste streams.

◆ Takeaway:

We are throwing away massive economic and environmental opportunities by not transitioning to a circular economy — one where we design for reuse, recycling, and resource recovery.

The problem with the linear economy



◆ Main message:

The linear economy model — *Take* → *Make* → *Use* → *Dispose* — is outdated and unsustainable.

◆ Explanation:

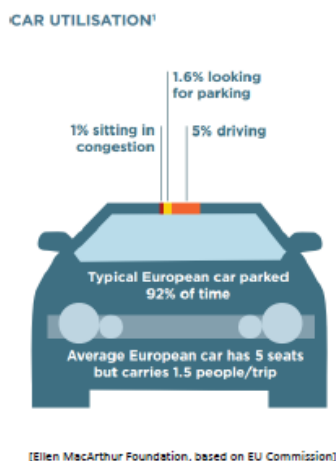
- This model has dominated industrial development for decades.
- While it has brought economic growth and convenience, it has **not been responsible** in terms of environmental or resource use.
- It leads to overconsumption of finite resources and creates massive waste.

◆ Problem:

By following a linear model, we contribute directly to the **resource crisis**, **pollution**, and **unsustainable extraction** of materials.

◆ Takeaway:

To ensure long-term sustainability, we need to **move away from the linear model** and rethink how we produce and consume.



◆ Main message:

Our current transportation system creates **structural waste**, not just material waste.

◆ Key figures (EU Commission / Ellen MacArthur Foundation):

- A typical **European car is parked 92%** of the time.
- Only **5%** of the time is spent **driving**.
- **1.6%** of the time is spent **looking for parking**.
- **1%** is spent **sitting in congestion**.
- **Average occupancy:** 1.5 people per trip in a 5-seat car.

◆ **What is structural waste?**

- Inefficiencies built into how we **use and design systems**, like underused vehicles.
- Even when a car is used, most of the energy (in combustion engines) is **lost before it reaches the wheels** — up to **86% energy loss**.
 - Causes: heat, friction, resistance, auxiliary systems, poor aerodynamics.

◆ **Implication:**

- The linear mindset leads to ownership-heavy, underutilised systems.
- We must rethink product **design, sharing models**, and **engineering efficiency**.

◆ **Potential solutions:**

- Improve **vehicle efficiency** through **lightweighting** (e.g., more plastics, less metal).
- Rethink **mobility systems**: car-sharing, electrification, better public transport.

Plastics as a proxy for our *uncircularity ...or why we desperately*

◆ **Visual evidence:**

- Marine animals entangled in plastic nets (e.g., turtles).
- Fish and whales accidentally ingesting plastic bags and microplastics.
- Birds dying from plastic-filled stomachs.
- Ocean floor covered in dumped **tires**, originally placed as artificial reefs.

◆ **Key issues:**

- Plastics don't biodegrade; instead, they **break down into microplastics**, releasing harmful toxins.

- If current trends continue, **by 2050 the ocean may contain more plastic than fish by weight** (Ellen MacArthur Foundation estimate).
- Plastic waste affects **marine ecosystems, biodiversity**, and ultimately **human health**.

◆ Rebound effect:

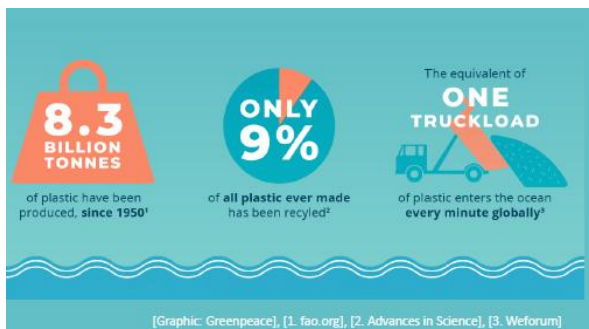
- Example: Tires were dumped in oceans to create coral reefs.
→ This caused more harm than good due to chemical leaching and failed reef structures.
→ A **well-meaning intervention** led to **long-term environmental damage**.

◆ Takeaway:

- There is **no such thing as "away"** when it comes to waste.
- We must rethink our **entire approach to resource use**, especially plastics.
- The solution lies in **closing loops**, designing for **reuse, recycling**, and minimizing **unintended consequences**.

Rebound effects

The plastic problem



◆ Main message:

Plastic waste illustrates the severity of our global resource mismanagement — and it's a call to action.

◆ Key facts:

- Since **1950, 8.3 billion tonnes** of plastic have been produced.
- Only **9% of all plastic ever made** has been recycled.

- Every **minute**, the equivalent of **one truckload of plastic enters the ocean** globally.
-

◆ **Implications:**

- The plastics problem reflects deeper **systemic issues**: overproduction, poor design for end-of-life, and lack of circular solutions.
 - It impacts not just the **resource challenge**, but also **climate change**, **toxicity**, and **biodiversity**.
 - These challenges are **interlinked** — solving one can positively impact others.
-

◆ **Takeaway:**

- While these numbers are alarming, they also point to **huge opportunities for change**.
- By rethinking how we consume, produce, and take responsibility for our resources, we can drive progress across all sustainability areas.
- We **designed** the current system — which means we can also **design the solutions**.

2. The promise of circularity

- Fastest growing business strategy in Europe
- Closing the loops
- Decoupling value creation from resource consumption
- Requires a systemic approach

The circular economy is a growing concept aimed at transforming how we produce, use, and recover resources — replacing the wasteful linear model.

◆ **Key points:**

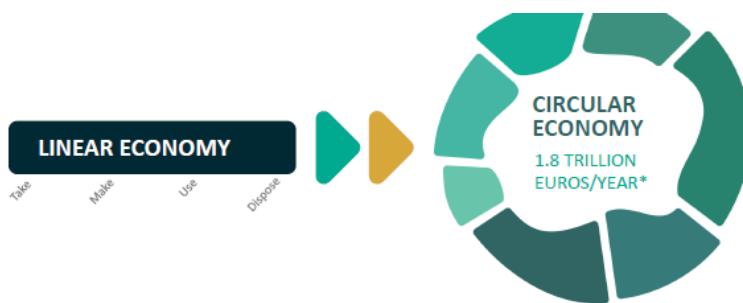
- **Fastest growing business strategy** in Europe (and gaining global political, corporate, and societal interest).
- It's about **closing resource loops** — keeping materials in use for as long as possible, rather than discarding them.
- Focuses on **eliminating the concept of “waste”** — everything becomes a resource.

- **Decoupling value creation from resource consumption:**
 - Delivering value (products, services, satisfaction) without necessarily consuming more finite resources.
 - Examples: product-as-a-service models, reuse systems, digital platforms, modular design.
- **Requires a systemic approach:**
 - Rethinking not only **products** but entire **systems** — from production to end-of-life.
 - Links directly with sustainability goals and needs **interdisciplinary thinking**.

◆ **Takeaway:**

Circular economy thinking is at the core of solving environmental challenges — from **resource scarcity** to **climate**, **toxicity**, and **biodiversity** — by designing smarter, more regenerative systems.

The promise of the circular economy



◆ **Main message:**

The **circular economy** transforms the wasteful "take-make-use-dispose" model into a **regenerative loop**, offering major environmental and economic benefits.

◆ **From linear to circular:**

- Instead of discarding products at end-of-life, circular economy thinking aims to:
 - **Reuse** entire products
 - **Repurpose** components
 - **Recycle** materials
 - **Cascade** materials into other uses if direct reuse is not possible

◆ **Massive potential:**

- According to McKinsey & the Ellen MacArthur Foundation:
 - The circular economy could generate up to **€1.8 trillion per year in new business opportunities** within the EU alone.
 - Even if this estimate is optimistic, the economic potential is **huge**.

◆ **Key insight:**

- Circular thinking removes the concept of "waste" entirely by treating all outputs as inputs for new processes.
- It's not only about **sustainability**, but also about **innovation and value creation**.

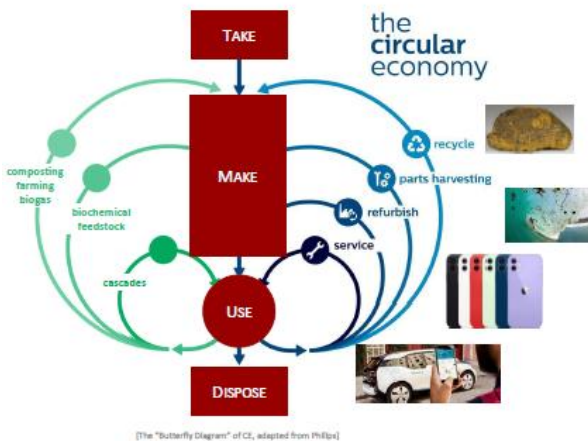
◆ **Takeaway:**

The circular economy offers a **win-win**: better use of resources, reduced environmental impact, and vast business potential. It's not just a necessity — it's a **strategic opportunity**.

The definition of a circular economy

“an economy that provides multiple value creation mechanisms, which are decoupled from consumption of finite resources”

... decoupling value creation from resource consumption!



◆ **Main message:**

The **butterfly diagram** illustrates how the circular economy can keep materials and value circulating — instead of being lost through disposal.

◆ **Linear vs. Circular:**

- The traditional **linear economy** follows: **Take → Make → Use → Dispose**.

- The **circular economy** aims to replace this with **cycles** — keeping materials in use for as long as possible and extracting value at every stage.

◆ **Right side = Technical cycle:**

- For **non-biodegradable materials** (e.g., metals, plastics, electronics).
 - Prioritizes:
 - **Service models** (e.g., car sharing)
 - **Refurbishing** (e.g., phones, electronics)
 - **Parts harvesting** (reuse of components)
 - **Recycling** (materials recovery)
 - Key principle: **The smaller the loop, the better** (e.g., reusing a product is better than recycling it into raw materials)
-

◆ **Left side = Biological cycle:**

- For **organic materials** that can return safely to nature.
 - Includes:
 - **Composting**
 - **Farming**
 - **Biogas**
 - **Biochemical feedstock**
 - Mimics **natural systems** like nutrient cycles in ecosystems.
-

◆ **Why this matters:**

- Helps reduce waste **before it occurs**.
 - Allows us to **create value** again and again from the same resources.
 - Shows how nature can inspire sustainable design and material management.
 - Supports **decoupling** value creation from resource consumption.
-

◆ **Takeaway:** To achieve a circular economy, we must think in **loops** rather than lines — and design systems that allow both **technical** and **biological** materials to flow in regenerative cycles.

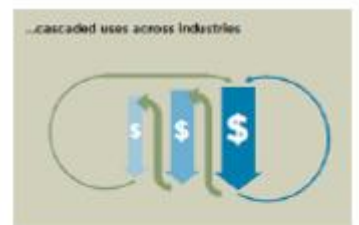
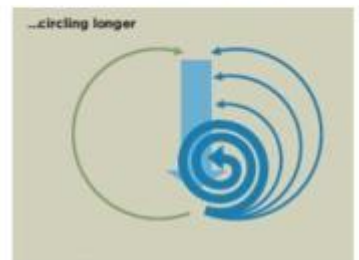
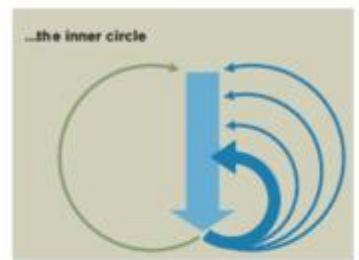
Value Creation

In circular economy thinking, **value is created** by designing products and systems that **stay in use longer** and **require fewer resources to maintain**.

- Inner Circle –Minimising comparative materials use, through re-use. The tighter the circle, the less it has to be changed to be returned to use (with higher savings)
- Circling Longer –Maximising the number of consecutive cycles of reuse, to avoid production of a new component

Cascading–Diversified re-use across the value chain, substituting previously used virgin materials with existing materials (including symbiosis);

- Pure inputs –Avoidance of contaminated materials to increase collection and re-use efficiency whilst maintaining quality.



Value creation			
1. Inner Circle	2. Circling Longer	3. Cascading	4. Pure Inputs
• Aim: Minimize material use by designing for reuse. • The tighter the loop, the less the product needs to be changed to be returned to use. • Examples: ○ Reusing products or components without disassembly ○ Using a product-as-a-service model instead of selling it ○ Keeping items in their most intact, high-value form ✓ The tighter the loop, the greater the material and energy savings.	• Aim: Maximize the number of reuse cycles before replacement is needed. • Products should be durable, repairable, and maintainable. • Focus on longevity to delay or avoid the production of new items. ✓ More loops = more value extracted from the same resources.	□ Definition: Diversified reuse of materials across different value chains. □ Goal: When a material can no longer serve its original purpose, repurpose it elsewhere instead of downcycling or discarding. □ Examples: • Textile scraps used as insulation • Waste heat used in industrial symbiosis □ Key idea: Maintain material value by extending its usefulness — even if it shifts industries or applications.	• Definition: Use non-toxic, easily separable, and uncontaminated materials to enable more efficient reuse and high-quality recycling. • Goal: Design from the start for material health and easy separation. • Example philosophy: Cradle to Cradle — ensures products are safe for both humans and the environment, and easily recycled or composted. • Why it matters: Pure materials are easier to: ○ Disassemble ○ Sort ○ Reuse without quality loss or harm

◆ Takeaway:

Together, the four principles —

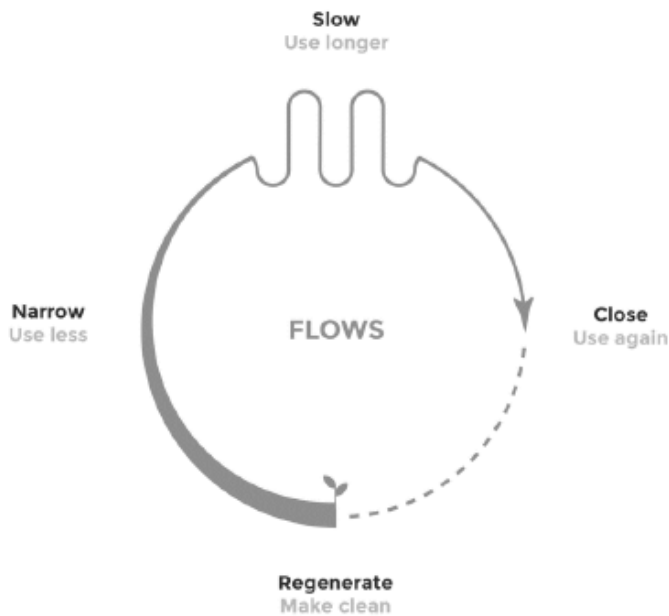
1. **Inner Circle**,
2. **Circling Longer**,
3. **Cascading**, and
4. **Pure Inputs** —

form a strategy for how we can **design out waste, optimize material flows, and create long-term value** in a circular economy.

Four Strategies for Circular Flows

◆ Main message:

To transition from a **linear** to a **circular** economy, we need to manage material flows differently — these four principles are key.



1. Narrow – Use less

- Reduce overall **material input** through efficiency and smarter design.
- Example: lightweighting products, designing with fewer components.

2. Slow – Use longer

- Extend **product lifetimes** through durability, repair, and maintenance.
- Encourages business models like product-as-a-service or resale.

3. Close – Use again

- **Recover and reuse materials** in circular loops.
- Includes reuse, refurbishment, remanufacturing, and recycling.

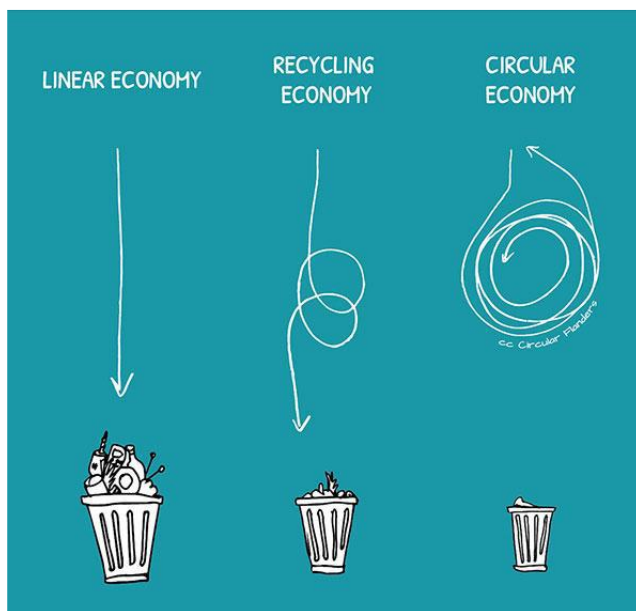
4. Regenerate – Make clean

- Restore natural systems by using **non-toxic, biodegradable, or renewable inputs**.
- Links to **pure input design** and **biological cycles** in the butterfly diagram.

Flow princip	Guiding Idea	Key Strategy	Example
Narrow	Use less materials	Design with minimal material input; focus on lightweighting and efficiency	<i>Riversimple hydrogen car</i> : lightweight composite design to minimize materials

Slow	Use products longer	Extend lifetime and increase intensity of use	<i>Car sharing systems:</i> increase lifetime usage and reduce idle time
Close	Use products/materials again	Encourage reuse through sharing, rental, or subscription models	<i>Clothing subscriptions:</i> repeated use of garments via access instead of ownership
regerate	Make clean/regenerative solutions	Use biodegradable, renewable, or non-toxic materials	<i>Paboco paper bottle:</i> biodegradable beverage packaging replacing traditional glass/plastic

3. Deploying CE strategies



◆ **Main message:**

This diagram (from Circular Flanders) illustrates the **three economic paradigms**:

1. **Linear Economy** – take, make, dispose
2. **Recycling Economy** – slightly reduces waste by reprocessing materials
3. **Circular Economy** – closes the loop to **eliminate waste entirely**

◆ **Why it's important:**

- It **visualizes the amount of waste** going down as we move from linear → recycling → circular.
 - It reminds us that **recycling is not enough** – it's only one part of a much broader circular strategy.
 - True circularity requires **rethinking entire systems** – not just end-of-life material processing.
-

◆ **Key insight:**

- To achieve a **zero waste society**, we need to **go beyond recycling** and implement strategies like:
 - Product-as-a-service
 - Design for reuse and repair
 - Regenerative inputs
 - Sharing systems and business model innovation
-

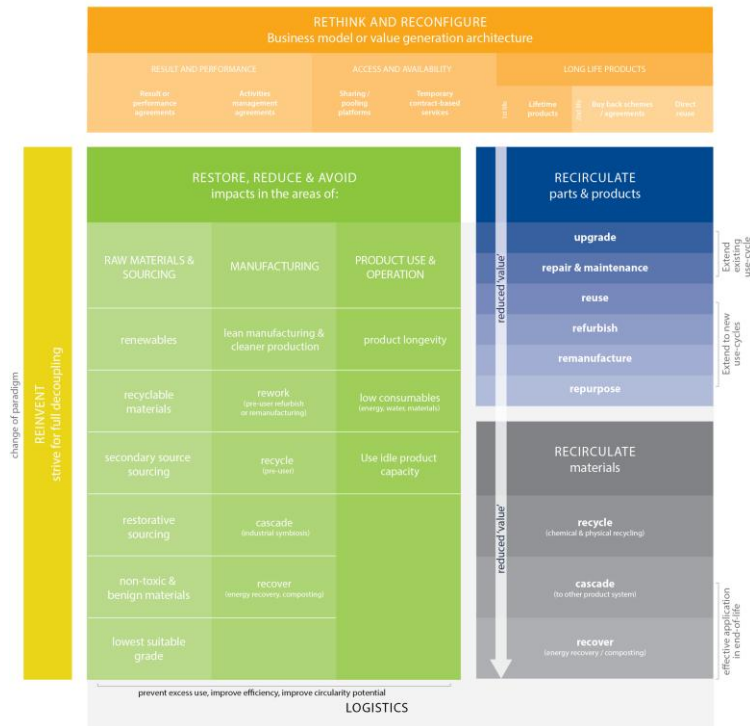
◆ **Takeaway:**

The circular economy is not just a vision — it's a **strategic shift**.

This slide shows how we **move from a mindset of waste to a mindset of opportunity** — but the challenge is in turning that mindset into **real, applied change**.

Circular Strategies Scanner: 32 opportunities to go circular

This framework maps out **32 different circular economy strategies** companies can use — across all stages of the value chain, from sourcing to end-of-life and business model innovation.



Green section – Internal product & production strategy

- Focuses on **product development, manufacturing, and usage improvements**.
- Aims to **restore, reduce, and avoid** resource impacts through:
 - **Raw materials & sourcing:** renewables, recyclable materials, secondary sourcing
 - **Manufacturing:** lean production, rework, recycling, cascading
 - **Product use & operation:** product longevity, low consumables, use of idle capacity

Blue section – Parts & products recirculation

- Deals with **end-of-use strategies** for entire products and components.
- These strategies **retain more value** than material-level recycling:
 - **Upgrade, repair & maintenance, reuse, refurbish, remanufacture, repurpose**

Gray section – Material recirculation

- Lower in the value hierarchy (less value retained), but still important:
 - **Chemical/physical recycling**
 - **Cascade** to other uses
 - **Recover** (e.g. energy recovery, composting)
-

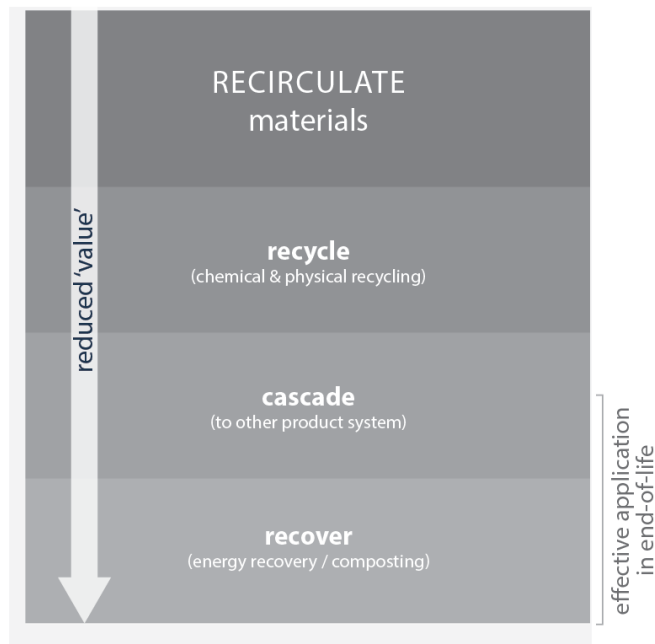
Orange section – Business model innovation

- Encourages **rethinking how value is delivered**, such as:
 - **Product-as-a-service**
 - **Sharing platforms**
 - **Performance or result-based contracts**
 - **Buy-back and lifetime product agreements**
-

Yellow section – Paradigm shift

- A call for **complete transformation**:
 - Reinvent the system
 - Uncouple value from resource use
 - Shift to regenerative and circular thinking

Recirculate materials



- **Recycle**

Zone: Gray (Recirculate Materials)

Stage: End-of-life

- ◆ **Key Idea:**

- Recover **raw materials** from used products, especially **plastics or metals**, and reintegrate them into new production.

- ◆ **Example:**

- **HP (Hewlett-Packard):**
 - 83% of plastic in their products is recycled content.
 - 100% recyclable packaging.
 - Reuse of **returned inkjet cartridges** as feedstock.
 - Shifted product colors (e.g., from white to dark green) to **accommodate recycled plastic** better.
 - Darker colors are easier to produce from mixed/recycled polymers.
 - Lighter colors typically require **bleached, virgin plastic**, which is less sustainable.

- ◆ **Why it matters:**

- Recycling helps close the material loop and reduce demand for virgin materials.

- It can drive **design changes** (e.g. color, material types) to support sustainability goals.
- A widely used and recognizable circular strategy — but still **lower in value** than reuse or repair.

- **Cascade**

Zone: Gray (Recirculate Materials)

Stage: End-of-life (not just end-of-use)

- ◆ **Key Idea:**

- Reuse materials that can **no longer meet their original mechanical or functional requirements**, by “**cascading**” them into new uses with **less demanding properties**.

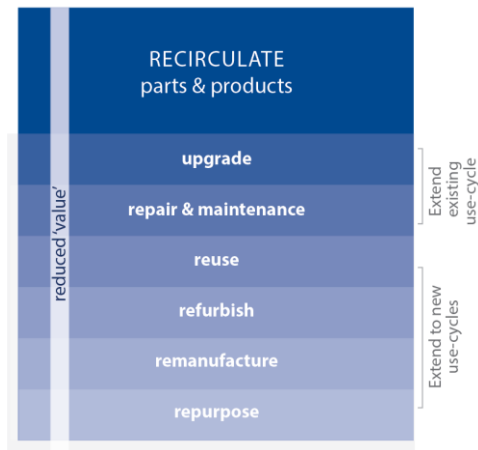
- ◆ **Example:**

- A real business making **new products from mixed materials** that were originally designed for high-performance use.
- These new products may have **lower aesthetic or strength standards**, but they remain functional and useful.
- Cascading is ideal when **direct reuse or refurbishment is no longer viable**, but value can still be recovered.

- ◆ **Why it matters:**

- Keeps materials in use **even after high-end function is lost**.
- Avoids landfilling or incineration.
- A realistic and often overlooked form of material circularity.

Recirculate parts products



- **Reuse**

Zone: Blue (Recirculate Parts & Products)

Stage: End-of-use (before end-of-life)

Value level: Higher than recycling — **more value retained**

- ◆ **Key Idea:**

Reuse means **keeping components or entire products in circulation** by using them again in their existing form — **without reprocessing them into raw materials**.

- ◆ **Example:**

Natura (Brazilian cosmetics company)

- One of the **largest cosmetic manufacturers** globally (they also own The Body Shop).
- Focuses heavily on **reuse**, both for:
 - The **cosmetic substances** inside the product.
 - The **primary packaging** — the part the user touches.
- **Refillable packaging system:**
 - Encourages users to **reuse containers** instead of disposing of them.
 - Product design supports this behavior (easy to refill, durable packaging).

- ◆ **Why it matters:**

- Reuse **extends product life** and **avoids new production**, reducing resource consumption.
- Especially important in **mass-produced sectors** like cosmetics, where packaging is a major waste contributor.
- Reuse becomes **even more valuable** for complex or high-tech products (e.g. electronics, furniture, machinery), where the savings from reuse are even greater.

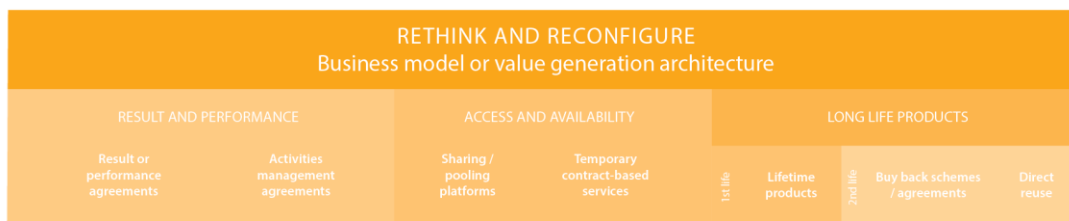
Restore, reduce & avoid impacts in the areas of:

RESTORE, REDUCE & AVOID Impacts in the areas of:		
RAW MATERIALS & SOURCING	MANUFACTURING	PRODUCT USE & OPERATION
renewables	lean manufacturing & cleaner production	product longevity
recyclable materials	rework (pre-user refurbish or remanufacturing)	low consumables (energy, water, materials)
secondary source sourcing	recycle (pre-user)	Use idle product capacity
restorative sourcing	cascade (industrial symbiosis)	
non-toxic & benign materials	recover (energy recovery, composting)	
lowest suitable grade		

Strategy	Focus Area	Description	Example
Recycling (pre-user)	Manufacturing	Recycling production scrap (pre-consumer) into new products.	Novo Nordisk: Used insulin pens turned into chair shells – 120 pens per chair.
Renewables	Raw Material Sourcing	Using renewable, bio-based materials instead of fossil-based or non-renewable inputs.	Paper-mâché beer bottles: From renewable, biodegradable sources – considered if alternatives are worse.
Recyclable Materials	Raw Material Sourcing	Using materials that are easier to recycle, less toxic, and designed for disassembly.	Steelcase (furniture): Designed for easy disassembly, low-toxicity, and full recyclability.
Restorative Sourcing	Raw Material Sourcing	Recovering materials from waste streams, especially	The Ocean Cleanup Project: Plastic collected from oceans

		environmental cleanups.	used as new raw material.
Non-toxic & Benign Materials	Raw Material Sourcing	Selecting materials that are safe for human health and the environment, with minimal toxicity.	Gabriel (DK): Cradle to Cradle–certified textiles – safe, non-toxic at industrial scale.

Rethink and reconfigure – business model or value generation architecture



Level: Business Model / Value Generation Architecture

Strategy: Result or Performance Agreements

Company/Example: Interface (carpet and flooring solutions)

Explanation:

Interface shifted from selling carpets as a one-time product to offering **flooring as a service**. Under this model, customers no longer purchase carpets outright but pay for performance over time. If damage occurs—like wear from office chairs—Interface replaces the flooring as part of the agreement.

This change led Interface to redesign its carpets into **modular tiles** instead of large rolls, allowing for easier maintenance and replacement. The invisible joints between tiles ensure a seamless look, maintaining customer satisfaction.

This model reduces material waste, encourages product longevity, and enables **decoupling of value creation from raw material consumption**, making it a strong example of a circular business strategy.

Reinvent – strive for full decoupling



Title: Ogoori – Plastics as a Service

Level: Paradigm Shift

Strategy: Reinvent – Strive for full decoupling

Company/Example: Ogoori (spun out from collaboration between Tesla/Best and Offecct/Opra)

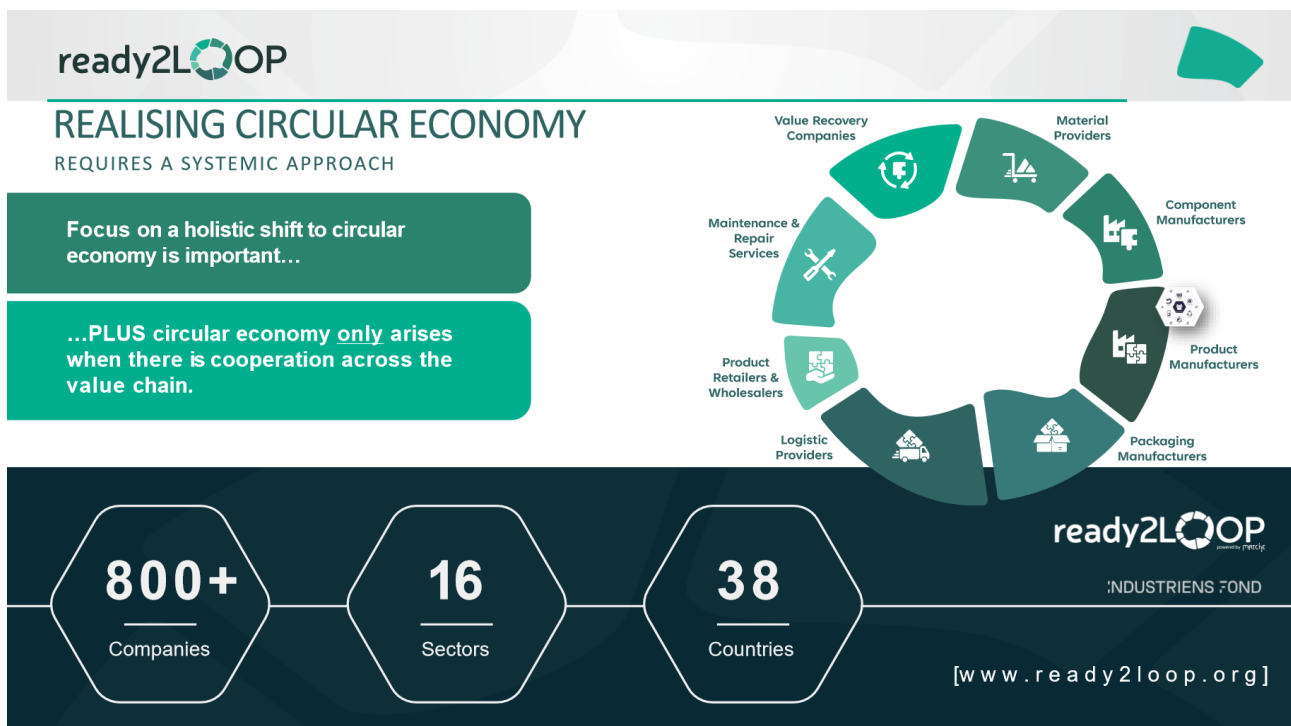
Explanation:

Ogoori is a company born out of the need to rethink how we treat waste plastics—particularly **ocean plastics**—as a valuable resource. The traditional value chain wasn't prepared to handle large-scale use of such reclaimed materials, so Tesla (a Swedish furniture company, not the car manufacturer) and Opra (Norwegian office modules manufacturer) created **Ogoori** to focus on securing and managing a **safe supply of recirculated plastics**.

Their model is based on "**materials as a service**", where companies don't just buy plastics but subscribe to the material stream. Ogoori guarantees plastic supply for companies like Uber and Tesla, while also committing to **take back and recirculate** those plastics after product use.

This approach **completely redefines business logic**, shifting from selling products to ensuring circularity of resources. It's a bold move toward decoupling value creation from raw material consumption—a full **reinvention of how we think business**.

4. CE readiness



Title: Realising Circular Economy – A Systemic Approach

Level: System-Level Collaboration

Strategy: Cooperate Across the Value Chain

Project/Example: Ready2LOOP (funded by Industriens Fond, Denmark)

Explanation:

A truly circular economy cannot be achieved by focusing only on internal efforts. **Circularity requires collaboration across the entire value chain**, from **material providers** to **value recovery companies**. The Ready2LOOP project emphasizes that **only 1 in 4 companies** currently collaborate with their value chain partners regarding circular economy.

The project maps out **nine key roles** in the value chain, each with unique responsibilities:

- **Material Providers:** Supply the foundation for circular design and reuse.
- **Component Manufacturers:** Create complex parts, which outnumber final products, and hold great circular potential.
- **Product Manufacturers (OEMs):** Orchestrate the design, assembly, and user delivery – the core players with leverage across the chain.
- **Packaging Manufacturers:** Crucial for circularity due to the volume of packaging in delivery systems.
- **Logistics Providers:** The connective "glue" that links all parts of the value chain.
- **Product Retailers & Wholesalers:** Influence product placement, return systems, and circular offerings.
- **Maintenance & Repair Services:** Enable lifespan extension and reuse.
- **Value Recovery Companies:** Handle recycling, reuse, and repurposing at the end-of-life stage.

With **over 800 companies**, **16 sectors**, and **38 countries** included, Ready2LOOP provides a **readiness assessment tool** helping each company identify how prepared they are to transition toward circularity. This approach supports a **holistic shift**—not just redesigning products, but redesigning systems.

8 dimensions for the CE transition



Title: Ready2LOOP – Dimensions of Circular Readiness

Level: Product Manufacturers

Strategy: Strategy & Business Model Innovation

Project/Example: Ready2LOOP's 8 Dimensions of the CE Transition

Explanation:

As part of the **Ready2LOOP project**, researchers identified **8 key dimensions** that companies—especially product manufacturers—must work with to **successfully transition to a circular economy**. At the core of these is the **organisation** itself, which must coordinate actions across the following 7 surrounding areas:

1. **Strategy & Business Model** – This has proven to be the most engaging dimension for many companies. It's about asking: *How ready are we to create a long-term business offering that supports circularity?* It involves exploring:
 - Circular value propositions to customers
 - Alternative revenue models (e.g. leasing, product-as-a-service)
 - Economic sustainability (profitable offerings)
 - Strategic vision alignment across departments

This dimension is not only about **sustainability from an environmental point of view**, but also about whether companies can make the **business case** for circularity. The assessment helps companies reflect on whether they are **strategically aligned** with the demands and opportunities of the circular economy.

Strategy & Business Model Innovation

Title: Mobike – Bike Sharing Scheme

Level: Strategy & Business Model Innovation

Strategy: Product-as-a-Service / Sharing Platform

Project/Example: Mobike – Circular Startup

Explanation:

Mobike represents a typical **bike-sharing system**, designed from the ground up as a **circular business model**. Since it was created as a new company, it could establish its own **rules, infrastructure, and goals**, all aligned with **access over ownership**. Mobike doesn't sell bikes but offers **mobility as a service**, encouraging maximum utilization of each product unit.

This is an example of how circular business models can flourish in **newly created ventures**, which are not bound by legacy systems. However, this dimension (strategy and business model) becomes **more complex** for **established companies**—those producing trucks, airplanes, or buses—who need to **rethink their traditional linear models** and assess their **readiness to transform** toward circularity.

The key challenge this dimension poses is:

“How ready are we, as a company, to move from a product-based linear economy to a service-based circular economy?”

Product & Service Innovation

Title: RePack – Reusable Packaging System

Level: Product & Service Innovation

Strategy: Design for Reusability / Circular Packaging

Project/Example: RePack

Explanation:

This dimension focuses on how companies are **designing and developing products and services** that are suitable for a **Circular Economy (CE)**. One such innovation is **RePack**, a company that produces **reusable, returnable packaging**. Rather than using traditional cardboard boxes or disposable envelopes that are discarded after one use, RePack offers a **durable alternative** that operates on a **deposit-return system**.

This innovation enables **multiple reuse cycles**—potentially **5, 10, or even 15 deliveries** with the same package—significantly **reducing raw material consumption** and **packaging waste**. Although the environmental benefit may not be immediately visible in **one-to-one lifecycle assessments**, it becomes highly favorable when the full **functionality over time** is considered.

RePack’s model illustrates the **product development and service design potential** in a CE context, showing how **new value chains** and **user-provider collaboration** can be leveraged to support circular flows.

The key question for this dimension is:

“How ready are we to develop and implement products and services that are circular by design?”

Manufacturing & Value Chain

Title: Green Fibre Bottle – Collaborative Circular Manufacturing

Level: Operations (Manufacturing & Value Chain)

Strategy: Circular Manufacturing & Value Chain Collaboration

Project/Example: Carlsberg’s Green Fibre Bottle

Explanation:

This dimension looks at how **ready production companies are to manufacture in a circular way**, and more importantly, how well they can **collaborate across the value chain** to support that. A key example is the **Green Fibre Bottle** developed by Carlsberg. This innovation requires a **coordinated effort** across multiple stakeholders—not only Carlsberg but also:

- **The bottle manufacturer,**
- **The pulp and paper supplier, and**
- **Transport and logistics partners, all working closely together.**

To truly enable circularity, each player must be **geographically and logistically aligned** to avoid counterproductive emissions and inefficiencies. In addition, systems like **deposit return schemes**

and **consumer education** play a critical role in ensuring the packaging is used and collected in a circular loop.

This case highlights that circular operations aren't just about **in-house manufacturing**, but about **systemic collaboration** across the **entire value chain**.

The key question for this dimension is:

“How ready are we to manufacture products and collaborate with others in a way that supports a circular economy?”

Technology & Data

Title: WasteIQ – Smart Waste Management

Level: Technology & Data

Strategy: Data-Driven Circular Systems

Project/Example: WasteIQ in Bærum, Norway

Explanation:

This dimension evaluates how well a company (or municipality) uses **technology and data** to support and **enhance circular economy efforts**. Technologies such as **IoT sensors, smart meters, apps, GPS, and cloud platforms** are becoming increasingly essential in sharing systems, resource tracking, and efficient operations.

A great example is **WasteIQ**—a smart waste management system used in Bærum, Norway.

WasteIQ uses RFID and other sensor technologies to:

- **Track how much waste is being disposed,**
- **Measure how well it has been sorted,** and
- **Reward residents** based on proper waste behavior (e.g., sorting paper vs. plastics).

Beyond tracking usage, it allows waste containers to signal when they are full (even underground), optimizing **collection logistics** and reducing unnecessary emissions.

This case illustrates how **data collection and sharing** can lead to smarter, more efficient, and circular systems—not just in business but also **public services**.

The key question for companies is:

“How ready are we to collect and leverage data to improve value creation and circularity?”

Use, Support & Maintenance

Title: Konecranes – Predictive Maintenance

Level: Use, Support & Maintenance

Strategy: Extend Use Phase through Service

Project/Example: Konecranes, Nordic CIRCuIT project

Explanation:

This dimension asks companies: “**How ready are we to extend the useful life of our products?**”

It’s about boosting circularity through **maintenance, repair, spare parts availability**, and especially **predictive maintenance**.

Konecranes—active in heavy lifting and industrial cranes—exemplifies this approach. Instead of only selling a product, they are:

- Offering **ongoing support and predictive maintenance** based on usage data,
- **Extending product life** significantly by avoiding breakdowns and downtime,
- Building stronger **value propositions** for clients like mining companies and shipyards.

By turning **maintenance into a service**, they not only create new business opportunities, but also reduce resource use and increase product longevity—central ideas in a circular economy.

This model shifts from “product ownership” to “product performance over time,” tying nicely into **circular business models** and creating long-term value for both company and customer.

Takeback & End-of- Life Strategies

Title: Refurb – Circular IT Equipment

Level: Takeback & End-of-Life Strategies

Strategy: Reuse, Refurbish, and Remanufacture

Project/Example: Refurb (Danish company refurbishing IT equipment)

Explanation:

This dimension evaluates how ready companies are to **prevent products from “falling off the cliff” into waste** and instead **recirculate value** through intentional end-of-use strategies.

Refurb, based in Denmark, has built a thriving business on this principle. They:

- **Collect used IT equipment** (computers, phones, tablets) from companies, municipalities, or individuals,
- **Refurbish and resell them** via a webshop with clear product grading (A, B, C),
- **Extend product life** while reducing waste and creating value from what others may see as obsolete.

They’ve reportedly managed to **recirculate 5,000 units per month**, proving that **end-of-life doesn’t mean end-of-value**.

The takeaway is mindset: seeing waste as **a future resource** — a key principle of the circular economy.

Policy & Market

The next area here — it's the one around the side — often gets missed because it's the biggest one enshrouding the rest. This dimension is about policy and market, and here we're asking companies to reflect on how *ready* they are to understand the **policy instruments** in place, identify potential **barriers**, but also recognize the **openings** that could allow them to grow or even establish entirely new business opportunities. It's about seeing where regulation can either support or hinder your transition toward circularity — and how to work with or influence it.

Example: Copenhagen Village

This example comes from Copenhagen Village — a student housing concept started by two entrepreneurs roughly 7–8 years ago. What they did was create affordable student housing using **old shipping containers**, turning otherwise waste materials into homes. But what's particularly interesting here is not just the reuse of materials — it's the **policy innovation**.

They approached the Municipality of Copenhagen with a problem: there are many areas in the city not suitable for normal family housing due to missing infrastructure like schools or waste collection. However, these areas could work well for temporary student housing. They proposed placing temporary, modular container homes in these locations. To make that happen, they needed **changes to existing regulations**.

After engaging with both the city and national parliament, the law was changed — allowing temporary housing on land previously unavailable for such use. This is a clear case of **using policy as a lever** for circular business, and it shows that companies not only need to understand legislation but can also **help shape it**.

Organisation

This dimension focuses on the internal readiness of a company to support circular economy initiatives. No matter how great a pilot project or idea is, if the organization isn't aligned behind it, it likely won't last. That's why companies are encouraged to evaluate how ready and engaged different departments and teams are in supporting circular strategies.

Example: Bose – Circular Readiness Across the Whole Company

A strong example here comes from Bose, the American audio equipment company. They undertook a broad internal assessment not just of their sustainability maturity, but also their circularity maturity — mapping how well each part of the business was positioned to contribute to circular transformation.

This included everything from:

- Top management and strategy teams
- Product designers and production engineers

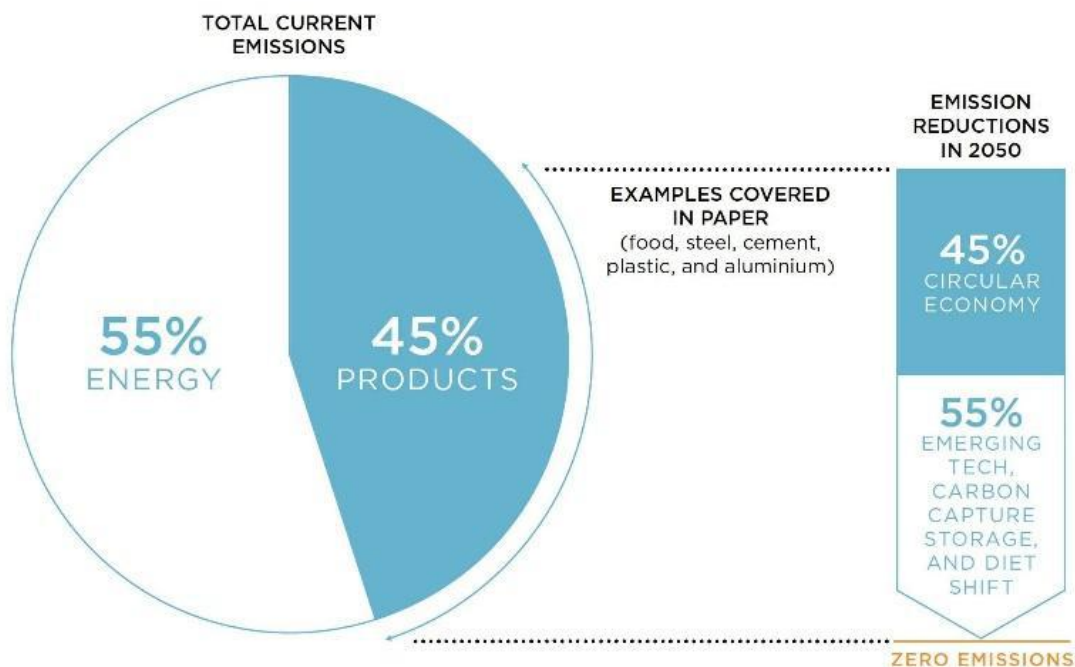
- Logistics and procurement departments
- To people working on the factory floor

The focus was on building a shared mindset and commitment across the whole organization to understand materials and products in circular terms — seeing closed-loop systems as a business priority.

The lesson? Circular economy is not just a technical shift — it's an organizational one. Change sticks when the whole company is on board.

Circularity increasingly recognised as a means to net-zero

COMPLETING THE PICTURE: TACKLING THE OVERLOOKED EMISSIONS



Title: Circular Economy as a Pathway to Net-Zero

Level: Systemic Impact / Climate Mitigation

Strategy: Circularity for Carbon Reduction

Topic: Linking Resource Circularity to Climate Goals

Explanation:

As we conclude, it's important to highlight how circular economy isn't just about better resource use — it's also deeply connected to climate action. Recent estimates show that nearly half of the total emissions reductions needed by 2050 to reach net-zero can come from circular strategies.

A 2023 study shows:

- 55% of emissions reductions can come from energy transition, tech, and diet shifts.

- 45% can come from how we design, use, reuse, and recover products and materials — the domain of the circular economy.

That means circularity is not just a side-topic — it's central to climate solutions.

If society — both companies and citizens — transitions to a circular mindset, it can help “bend the emissions curve” away from the current trajectory and toward a sustainable future.

Takeaway:

Circularity is no longer just about waste. It’s a strategic lever for climate action, making it a core part of the net-zero transition.

Summary

- Consider what type of circularity pattern(s) your project might best be suited to follow: narrowing, slowing, closing, regenerating..?
- Which of the 32 circular strategies could/should you follow? Make a consideration and find a workbook to aid you! www.circitnord.com
- How ready do you think you/your case company is to transition to a circular economy? Try it out! www.ready2loop.org
- Read the appended articles for deeper theoretical understanding and guidance

Title: Summary & Call to Action

Level: Project Guidance & Strategic Planning

Strategy: Evaluate, Select, and Apply Circular Strategies

Focus: Project Alignment & Tools for Action

Explanation:

As you conclude your exploration of circular economy strategies, here are key takeaways and next steps:

1. **Identify your circularity pattern**

Ask: Is your project best suited to narrowing resource loops (e.g. efficiency), slowing flows (e.g. reuse), closing loops (e.g. recycling), or regenerating systems (e.g. bio-based)?

2. **Use the 32 strategies**

Refer to the CIRCItnord resources and **workbooks** to explore the full range of practical circular strategies and align your project accordingly.

👉 www.circitnord.com

3. **Test readiness for circularity**

Assess how ready your case company is to transition using the **Ready2Loop** platform –

available for DTU students and external companies.

 www.ready2loop.org

4. Read, apply, and prototype

Dive into the **appendices and theoretical tools** provided alongside the lectures. These offer depth and practical value to support your circular transition journey.

Lecture 10 - Planetary boundaries and absolute sustainability

1. From relative to absolute sustainability

The sustainability challenge

$$I = P * A * T = Pop * \frac{GDP}{person} * \frac{I}{GDP}$$

- I is the environmental impact
- Pop is the global population
- $\frac{GDP}{Person}$ is the affluence, the material standard of living
- $\frac{I}{GDP}$ is the Technology factor – environmental impact

Notes:

1. What is the IPAT Equation?

- **IPAT** is a formula used to analyze **environmental impact (I)** based on three key drivers:
 - **P** = Population
 - **A** = Affluence (typically measured as GDP per person)
 - **T** = Technology (environmental intensity of producing affluence)
- **Formula:**

Impact (I)=Population (P)×Affluence (A)×Technology (T)
 $\text{Impact (I)} = \text{Population (P)} \times \text{Affluence (A)} \times \text{Technology (T)}$

2. Understanding the Variables

- **Affluence (A):**
 - Often represented by **Gross Domestic Product (GDP) per capita**.

- Commonly used to measure **wealth** across nations.
- **Technology (T):**
 - Interpreted as **environmental impact per unit of GDP**.
 - Reflects the **environmental efficiency** of the technology used to create wealth.

3. How Changes Affect Impact

- If **population** increases by 50% (with A and T constant), **environmental impact** also increases by 50%.
- If **affluence** increases by 20% (with P and T constant), **impact** increases by 20%.
- Each factor contributes **multiplicatively** to the total environmental burden.

4. Population Trends

- Current global population: **8.2 billion**
- Forecasted to **level off around 10 billion** by **mid-century**.

$$I = Pop \cdot \frac{GDP}{person} \cdot \frac{I}{GDP}$$

- The global population may level off around 10 billion
- Material standard of living will grow strongly in newly industrialised countries
- The environmental impact already exceeds sustainable levels in many areas
- So what is the challenge?

18 4

Notes:

Components:

- Population: Expected to level off around 10 billion.
- GDP per person: Rising, especially in newly industrialized regions (e.g., Southeast Asia, South America, Africa).
- Impact per GDP: Must be reduced significantly to meet sustainability targets.

Growth & Consequences:

- Material standard of living is increasing, which raises GDP per capita.
- This leads to a higher environmental impact, which is already above sustainable levels.

Climate Change:

- According to the Paris Agreement, temperature rise must be:
 - Strive for +1.5°C, and
 - Stay well below +2°C (relative to pre-industrial levels).
- To meet this goal, we must:
 - Cut greenhouse gas emissions by ~50% between 2020–2030.

Biodiversity Crisis:

- Current species extinction rates are unprecedented:
 - Comparable to the extinction event that wiped out dinosaurs ~55 million years ago.

What Needs to Happen:

- Environmental impact per GDP unit must sharply decline.
- Technology's environmental intensity needs major improvement.
- "Impact per created value" (i.e., how much damage we do per unit of economic output) must be drastically reduced.
- For climate alone, a factor 3 reduction in impact is needed from 2020 to 2030.

ECO-Efficiency

Eco-efficiency can be defined as the ratio between the functional output and the environmental impacts caused by an activity

$$Eco - efficiency = \frac{Delivered\ service}{Enviromental\ impact} = \frac{1}{T}$$

Eco-efficiency is the reciprocal of the technology factor in the IPAT equation

Eco-efficiency is determined with Life cycle assessment, LCA

Eco-efficiency has to grow strongly to achieve the needed reduction in impact with growing population and affluence

Notes:

Definition: $\text{Eco-efficiency} = \text{Delivered service} / \text{Environmental impact}$

- Eco-efficiency measures how much useful service (output) we get from a process per unit of environmental impact.
 - Example: Transporting a person from A to B using fewer resources and emissions = higher eco-efficiency.
- It's a more practical way of describing the technology factor from the IPAT equation, especially in real-world discussions.
- Eco-efficiency is determined using Life Cycle Assessment (LCA):
 - LCA accounts for all environmental impacts throughout a product or service's life cycle.
 - You've been introduced to LCA earlier in the course.

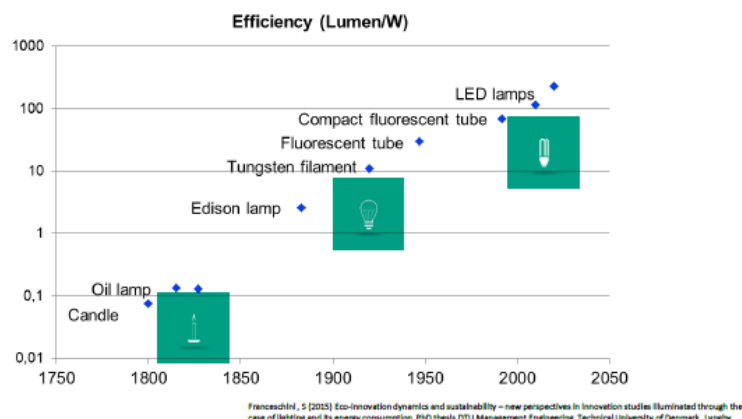
Why It Matters:

- As population and GDP per person increase, **eco-efficiency must increase dramatically** to reduce total environmental impact.
- Current environmental pressures (climate change, biodiversity loss) demand **strong growth** in eco-efficiency.
- The goal is not just small improvements—we need **major changes** to technology and systems to make this realistic.

Developments in efficiency

Example: **Lighting Technology Over 250 Years**

The example of lighting technology



Efficiency Progress (Measured in Lumen/Watt):

- **Historical timeline** of lighting:
 -  Candle →  Oil lamp →  Edison bulb →  Tungsten filament →  Fluorescent tubes →  Compact fluorescent →  LED lamps
- **Efficiency rose from ~0.1 lumen/W to several hundred lumens/W**
- This is plotted on a **semi-logarithmic scale**, showing that:
 - What appears linear is actually an **exponential increase** in energy efficiency over time.

Key Insights:

- **Energy efficiency** is not the same as **eco-efficiency**, but they are **strongly correlated**.
- Lighting technologies have improved **over three orders of magnitude** (~1000×).
- This kind of progress shows what's **technically possible** when driven by innovation and need.

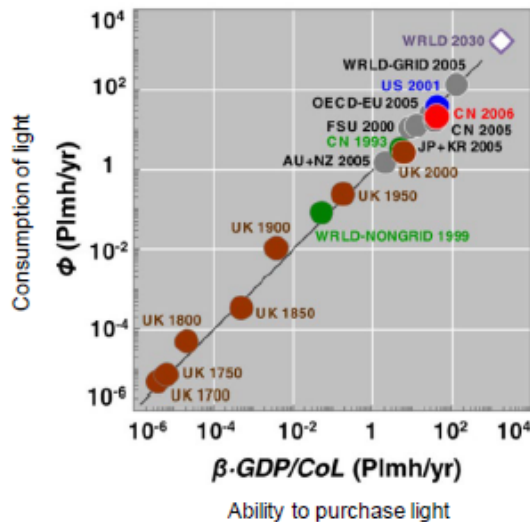
Climate Context:

- To meet **Paris Agreement** goals, we need a **factor 3 reduction in climate impact** within a decade.
- The historical development of lighting technologies shows that **rapid and exponential improvements are achievable**.
- This gives **hope and direction** for making similar progress in other areas (e.g., energy systems, transport, electronics).

Moore's Law (Parallel Example):

- Originally proposed in the **mid-1960s** by Gordon Moore.
- States that the **density of transistors in integrated circuits doubles every ~2 years**.
- Has held true for **4–6 decades**, showing **exponential tech development** is possible and sustainable over time.

Development in consumption



“Over the past three centuries, and even now, the world spends about **0.72%** of its GDP on light and **0.54%** of its GDP on the consumption of energy associated with light”

Tsao JY, Saunders HD, Creighton JR, Coltrin ME, Simmons JA (2010) Solid-state lighting: an energy-economics perspective. *J. Phys.D: Appl. Phys.* 43, 354001 (17p)

💡 Development in Light Consumption

📊 Graph Explanation:

- **X-axis:** Ability to purchase light (based on GDP and cost of light).
- **Y-axis:** Actual **consumption of light** (Plmh/year).
- Each point represents a **region at a given time**, from **UK 1700** → **World 2030**.
- The trend is **remarkably linear**, showing a consistent rise over **>300 years**.

💬 Key Observation:

“Over the past three centuries, and even now, the world spends about **0.72%** of its **GDP on light** and **0.54%** of its **GDP on the energy associated with light**.”

— *Tsao et al., 2010*

🧠 Interpretation and Insights:

- Despite massive increases in **energy efficiency** (up to **3 orders of magnitude**), **spending on light hasn't decreased**.
- Why? Because **GDP has also grown** significantly — so spending on light remains a **fixed % of GDP**.
- This shows the **rebound effect**:
→ More efficient tech → Lower costs → **More usage**.

Rebound Effect:

- Efficiency gains don't always mean lower impact.
 - **Better tech can lead to increased consumption**, offsetting environmental benefits.
 - In IPAT terms:
 - If **T (technology factor)** improves but causes **A (affluence)** to grow, **I (impact)** may not drop as expected.
 - **T, A, and P are not always independent** — this limits the equation's real-world predictive value.
-

Final Takeaway:

- Improving **eco-efficiency** (T) is essential, but **not sufficient** on its own.
- A deeper understanding (e.g. via **Life Cycle Assessment**) is needed to grasp real sustainability impacts.

Relative and absolute sustainability

LCA supports **relative assessments of environmental sustainability** (“*more sustainable than...*”)?

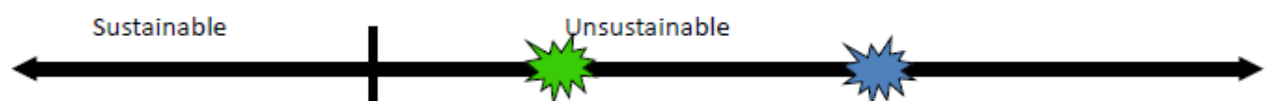
- Same or higher functionality with less environmental impact



Absolute sustainability (“*sustain-able*”)?

- Where is the boundary beyond which the activity becomes unsustainable?

- What is sustainable in absolute terms?



Relative Sustainability

- Life Cycle Assessment (LCA) supports **comparative assessments**:
 - “Is this **more sustainable** than that?”

- Focuses on:
 - **Same or better functionality** with **lower environmental impact**.
 - Example: Riding a bike vs. driving a car:
 - Same transport function, but **bike has lower environmental impact**.
 - Outcome: Helps identify "**greener**" options, but **not whether they are sustainable in absolute terms**.
-

Absolute Sustainability

- Asks:
 - Where is the **limit or boundary** between sustainable and **unsustainable**?
 - What level of environmental impact is actually **sustainable long-term**?
 - Important because:
 - Many things are called “sustainable” just because they’re **better than others**.
 - But that doesn’t mean they’re actually **sustainable enough to be maintained indefinitely**.
 - We need to shift focus from “**better than before**” to “**good enough for the planet to cope**”.
-

Key Takeaways:

- LCA is **strong for relative comparisons**, but **not enough to judge absolute sustainability**.
- Current use of “sustainable” is often misleading if we don’t define **what’s truly sustainable**.
- It’s essential to establish **clear thresholds** of environmental impact beyond which activities become **unsustainable**.

A sustainable level of impact?

Sustainability: Fulfilment of needs

- Today and in the future
- Which needs?
- How to fulfil them?
- For how many?

Planetary boundaries

- Top-down prescription of sustainable level of impact

Notes:

Sustainability = Fulfilment of Needs

Definition (Brundtland Commission):

“Meeting the needs of the present without compromising the ability of future generations to meet their own needs.”

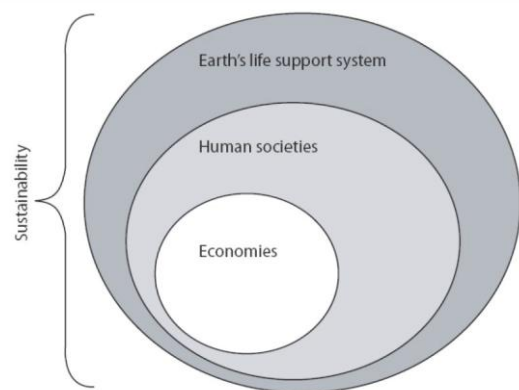
But this raises important questions:

- **Which needs?** (e.g., food, shelter, education, belonging, creativity)
- **How to fulfill them?**
- **For how many people?**
- **At what level?** (basic vs. psychological vs. self-fulfillment needs)

▲ *Maslow's hierarchy of needs* shows that:

- Basic physiological and safety needs are essential for survival.
- Higher levels (belonging, esteem, self-actualization) are also part of a good life, but require more resources and infrastructure.

needs



Absolute Sustainability & Planetary Boundaries

- We need to **go beyond vague goals** and define **concrete limits**.
- **Top-down approach:**
Start by identifying the **biophysical limits of the Earth** (planetary boundaries) → then define what is possible within those.

🌀 Human societies and economies are **nested inside the Earth's life support system**:

- The environment isn't just a resource—it is **foundational**.
- Human systems must **operate within ecological limits** (e.g., climate regulation, ecosystems, biodiversity).

Summarizing

- **Eco-efficiency** is the ratio between created value and caused environmental impact
- Sustainability requires **strong improvements in eco-efficiency**
- Our current focus on eco-efficiency must be informed by an **absolute sustainability perspective** to ensure that solutions are also **eco-effective**

Notes:

Eco-efficiency:

- Defined as the **ratio between created value and environmental impact**.
- It reflects how **efficiently resources are turned into valuable services or products** with minimal harm.

🚀 Why Eco-efficiency Matters:

- **Major improvements** are needed to reduce environmental impacts to sustainable levels.
- It's not just about being *better* — it's about being *enough* to protect ecosystems and future generations.

🔍 Beyond Efficiency: The Need for a Broader View

- **Efficiency** alone isn't enough — we must ask:

“How far do we need to improve before it becomes truly sustainable?”

- That's where the **absolute sustainability perspective** comes in.

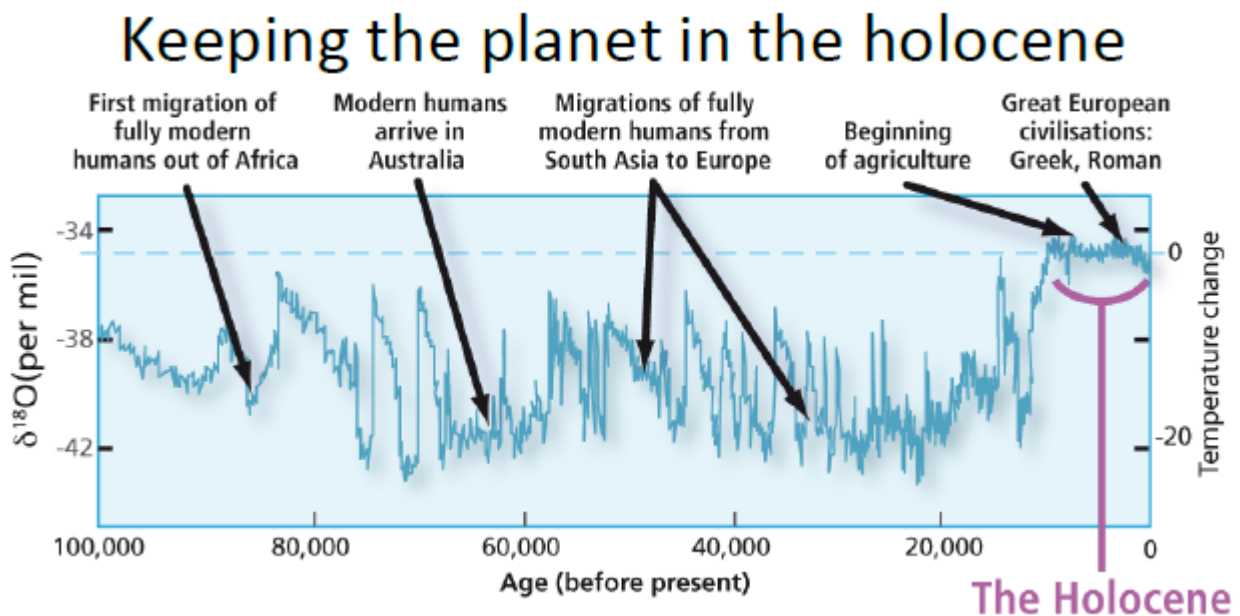
✅ From Eco-efficiency to Eco-effectiveness:

- **Eco-effectiveness** = doing the *right things*, not just doing *things right*.
- We need to design solutions that are not only efficient but also fall within **planetary boundaries**.

- Engineers and designers should aim for **solutions that are both optimized and truly sustainable**.

2. Planetary boundaries and absolute sustainability

The sustainability challenge



Notes:

Goal: Keep the planet in the Holocene

What the Graph Shows:

- **X-axis:** Time (in years before present) — goes back 100,000 years.
- **Y-axes:**
 - **Left Y-axis:** $\delta^{18}\text{O}$ (oxygen-18 isotope ratio), a **proxy for past temperature**.
 - **Right Y-axis:** Interpreted **temperature change**.
- Data is reconstructed from **ice cores** by measuring oxygen isotopes in ancient trapped air.

Temperature History Insights:

- Over the past 90,000 years (before ~12,000 years ago):
 - **Wild temperature fluctuations**, sometimes $\pm 15\text{--}20^\circ\text{C}$ in a century.
 - Extremely harsh and unstable climate → challenging for human survival.
 - **The Holocene** (last ~11,700 years):
 - Exceptionally **stable climate**, with fluctuations of only about $\pm 1.5^\circ\text{C}$.
 - Stability has **enabled agriculture, civilization**, and the growth of human populations from ~100,000 to **8.2 billion** today.
-

Human Evolution and Climate:

- Key human migration and civilization milestones align with this timeline:
 - Out of Africa (~85,000–70,000 years ago)
 - Agriculture (~10,000 years ago)
 - Major civilizations (Greek, Roman, etc.)
 - These developments were only possible due to **climate stability during the Holocene**.
-

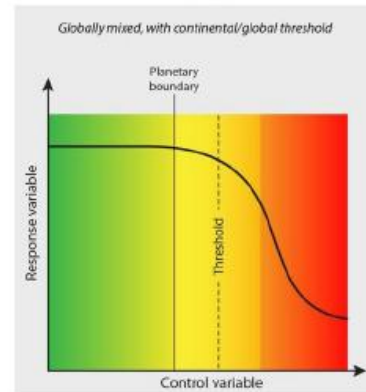
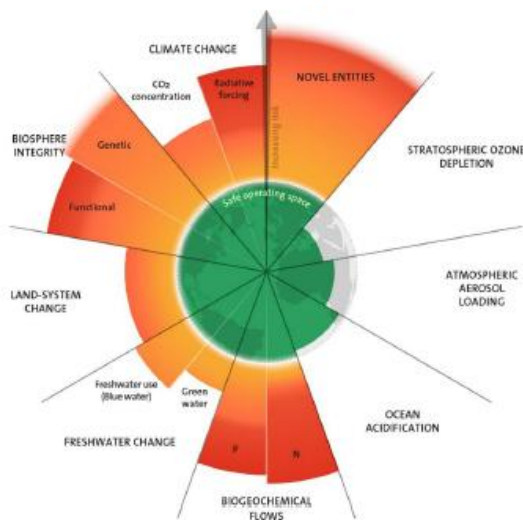
Why It Matters:

- The **Holocene is an outlier** — past climate was much more unstable.
- The **planetary boundary framework** is based on this insight:

"We must understand and protect the Earth systems that have allowed this climate stability."

- If we **disturb the systems** that maintain the Holocene-like conditions, we risk **returning to instability** — a scenario humans are **not adapted to survive** in large numbers.

Planetary boundaries



Steffen W,
Richardson K,
Rockström J et al.
(2015) Planetary
boundaries.
Guiding human
development on a
changing planet.
Science 347(6223),
736-746

Richardson K, Steffen W, Lucht W et al.
(2023) Earth beyond six of nine planetary
boundaries. Science Advances 9, eadh245.

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Planetary Boundaries

Key Concept:

Planetary boundaries define the **safe operating space** for humanity on Earth. They identify environmental limits we should not cross to **avoid destabilizing the planet's life-support systems**.

The 9 Earth System Processes Identified:

1. **Climate Change**
2. **Biosphere Integrity** (genetic and functional diversity)
3. **Land-System Change**
4. **Freshwater Use** (blue and green water)
5. **Biogeochemical Flows** (nitrogen and phosphorus cycles)
6. **Ocean Acidification**
7. **Atmospheric Aerosol Loading**
8. **Stratospheric Ozone Depletion**
9. **Novel Entities** (man-made pollutants like chemicals, plastics, etc.)

The Control-Response Model (Right Figure):

- **X-axis:** Control variable (human pressure on the Earth system)

- **Y-axis:** Environmental response
- **Green Zone:** Safe operating space
- **Yellow Zone:** Uncertainty zone (early signs of damage, risk of tipping points)
- **Red Zone:** Clear boundary transgression — irreversible or catastrophic changes

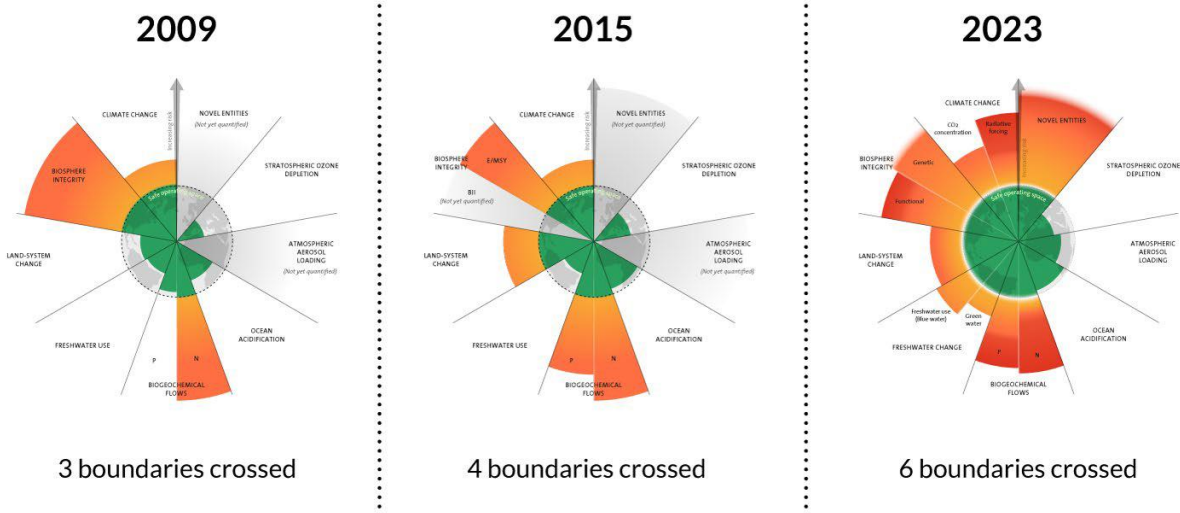
⚠ **The boundary is set at the lower edge of the uncertainty zone** to stay safely below potential tipping points.

🔥 **Example: Climate Change**

- **Control variable:** Atmospheric CO₂ concentration
 - Pre-industrial level: ~280 ppm
 - Current level: ~420 ppm
 - Already entering the **yellow/red zone** → signs of climate instability
 - **Tipping point risk:** Melting **permafrost** could trigger methane release (28× stronger than CO₂ over 100 years), reinforcing global warming in a **positive feedback loop**.
-

🔔 **What This Means:**

- We've **already crossed several boundaries** (e.g., climate change, biogeochemical flows, biosphere integrity).
- We don't always know **exactly where the tipping point is**, so precaution is critical.
- This framework urges **staying well below the risky thresholds**, not just avoiding collapse.



● Crossing Planetary Boundaries: 2009 → 2023

Timeline Overview:

Year	Boundaries Crossed	Notes
2009	3	First publication. Several boundaries were not yet quantifiable .
2015	4	More clarity on boundaries; biosphere integrity added as critical.
2023	6	Latest assessment: clear worsening , with some systems deeply in the red zone .

What It Means:

- **Crossing a boundary $\neq$ immediate collapse**
But: it **increases risk** of irreversible and dangerous system change.
- **Deeper transgression = greater risk:**
 - Some boundaries (e.g., **biosphere integrity**, **biogeochemical flows**) are **severely exceeded**.
 - These "**red zones**" signal urgent concern.
- Planetary systems are **interconnected**:

Breaching one boundary can **destabilize others**.

- E.g., land use change and biosphere loss impact carbon cycles and climate.
- This **amplifies risk** of entering a new, **unstable climate regime**.

Final Concern:

The Earth might still "work" with fewer people under unstable conditions, but **not for 8.2 billion humans** — who depend on **climate and ecosystem stability** for food, health, and security.

Absolute sustainability

Meeting the needs of present and future generations

within the biophysical boundaries of our climate and ecosystems — within the safe operating space

Notes:

Meeting the needs of present and future generations within the biophysical boundaries of our climate and ecosystems — within the safe operating space

What Is the Safe Operating Space?

- Concept defined by the **Planetary Boundary Framework**.
- Refers to the **green zone** — the area where:
 - Human activities stay **within Earth's capacity to cope**.
 - There is **low risk** of triggering environmental tipping points.

Why It Matters:

- Sustainability is **not just about being better** than before — it's about staying within limits that keep the planet stable.
- **Absolute sustainability** emphasizes:
 - Respecting **Earth's life-support systems**.
 - Defining sustainability in **physical, measurable terms**, not just relative improvements.

Absolute boundaries for social sustainability?

- Social science and natural science
- What is for negotiation?

Notes:

From Environmental to Social Sustainability:

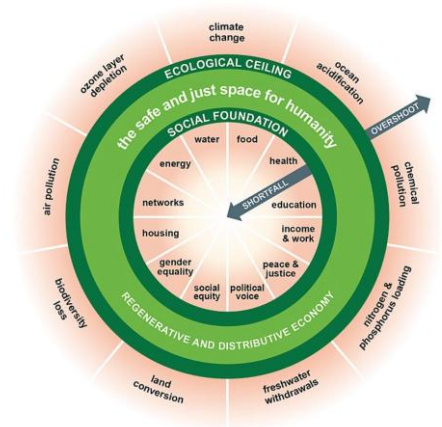
Sustainability has **three pillars**:

Environmental

Social

Economic

Much of the course has focused on **environmental boundaries** — now we consider the **social dimension**.



■ The Doughnut Model (by Kate Raworth):

Visual model combining:

Outer limit: The **Ecological Ceiling** → defined by **planetary boundaries**.

Inner limit: The **Social Foundation** → defined by **minimum social standards**.

● Safe and Just Space for Humanity:

Humanity must live **between these two limits**:

Not exceed environmental thresholds

Not fall short on basic social needs

The green “doughnut” ring is this **safe operating space**.

✿ Key Social Dimensions (Inner Ring):

Examples include:

Water

Food

Health

Education

Gender equality

Income & work

Housing

Political voice

Social equity

Peace & justice

Energy and networks (connectivity)

These often align with the **UN Sustainable Development Goals (SDGs)**.

What Is for Negotiation?

Ecological boundaries (outer ring) are **non-negotiable**:

They are **biophysical facts** with real tipping points.

Social boundaries (inner ring) are partly **normative and political**:

What counts as "enough" access, education, equity, etc. is **debated and context-dependent**.

Summarizing

- **Planetary boundaries** are limits for our pressure on earth system processes that **ensure climate stability**
- They cover **many types of environmental impact** but at the core are climate change and biosphere integrity
- Absolute sustainability is **meeting the needs** of present and future generations while **respecting the Planet's biophysical boundaries**

Planetary Boundaries:

- Define **limits on human pressure** on Earth's systems to maintain **Holocene-like climate stability**.
- Aim to **preserve the stable environmental conditions** we've experienced over the last ~12,000 years.
- Not limited to climate — they **include multiple environmental domains**, such as:

- **Climate change**
 - **Biosphere integrity** (biodiversity)
 - **Biogeochemical flows**, land use, water, pollution, etc.
-

Core Focus:

- The most critical boundaries are:
 - **Climate change**
 - **Biosphere integrity**
 - These are considered **foundational for Earth's overall system stability**.
-

Absolute Sustainability:

**Meeting the needs of present and future generations
while respecting the planet's biophysical boundaries.**

- It's not only about reducing impact, but ensuring that:
 - We **stay within Earth's limits**
 - We **avoid destabilizing feedback loops** and irreversible change
 - Future generations can also **meet their basic and social needs**

3. AESA for assessing absolute environmental sustainability

Performing AESA

- Identify biophysical boundaries and determine safe operating spaces
 - PBs or LCIA

Notes:

Step 1: Identify the Biophysical Boundaries

- **Goal:** Find out **which environmental limits** (planetary thresholds) we must stay within to avoid destabilizing Earth's systems.
- This defines our **safe operating space**:

How much impact is acceptable — and how much must be reduced?

🔧 Two Main Approaches:

1. PBs (Planetary Boundaries)

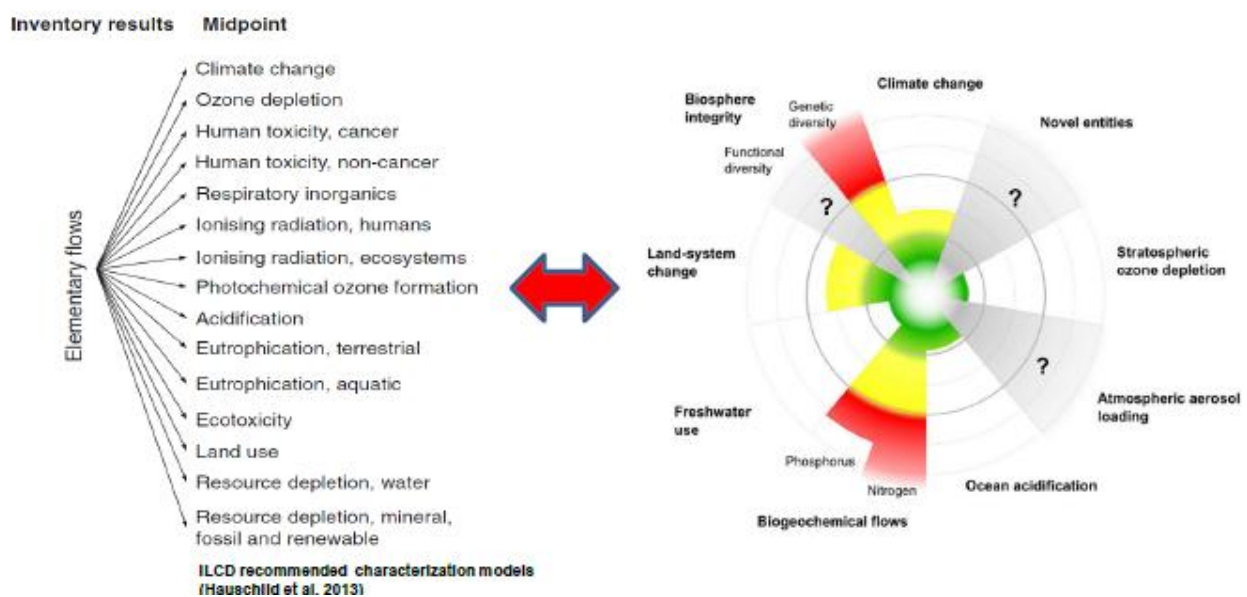
- Top-down approach
- Focus on global system limits (e.g. climate change, nitrogen flow, biosphere integrity)
- Developed by earth system scientists

2. LCIA (Life Cycle Impact Assessment)

- Bottom-up approach
- Focuses on environmental impacts of products/processes over their life cycle
- Developed in LCA communities

✚ These methods **evolved in parallel**, often **independently** — but now converge in AESA to define what is environmentally “safe enough.”

Where is the Boundary ?



Notes:

Life Cycle Impact Assessment (LCIA):

- LCIA assesses environmental impacts from products and processes.
- Starts with **inventory results**:
 - Also called **elementary flows** (e.g. emissions, resource extractions).

- These flows are translated into **impact categories** at the **midpoint level**, such as:
 - Climate change
 - Ozone depletion
 - Human and ecosystem toxicity
 - Acidification, eutrophication, land use, etc.
-

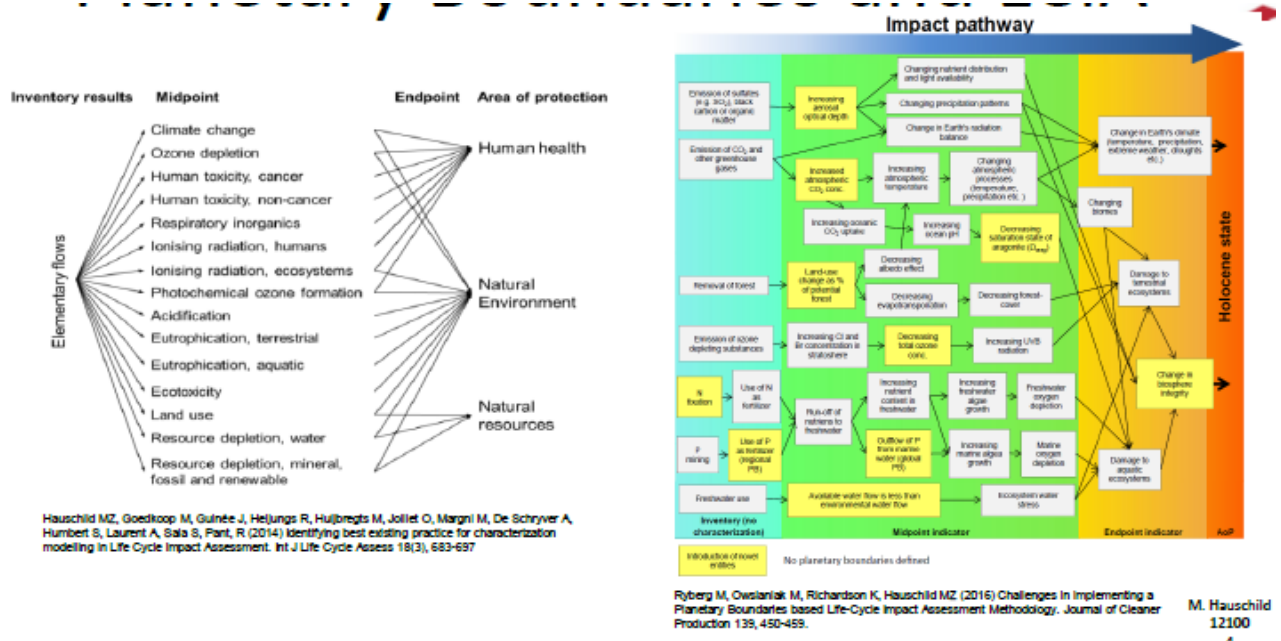
Planetary Boundaries Framework (PBs):

- PBs focus on 9 **critical Earth system processes** (e.g., climate, land use, biosphere integrity).
 - Aim to define **absolute thresholds** humanity must not exceed to maintain global stability.
-

The Integration Challenge:

- LCIA and PBs originate from **different scientific communities**.
- But both approaches aim to assess **human pressure on the environment**.
- Two possible paths:
 1. **Integrate planetary boundary thinking into traditional LCIA.**
 2. **Use PBs as the foundation to create new LCIA models.**
 - This involves **setting boundaries for LCIA impact categories** (e.g. “how much land use is too much?”).
 - Some PBs already match LCIA midpoint categories well (e.g., land-system change, climate change), others still don’t.

Planetary Boundaries and LCIA



LCIA Structure:

Life Cycle Impact Assessment translates environmental emissions into potential damages using three stages:

Stage	Example
Inventory Results (Inputs/Outputs)	Emissions like CO ₂ , SO ₂ , heavy metals, etc.
Midpoints	Impact categories (climate change, acidification, toxicity)
Endpoints / Areas of Protection	Human health, ecosystem quality, resource availability

LCIA provides a systematic and methodical chain from emissions to final damages.

Planetary Boundaries Viewpoint:

- PBs define **control variables** for 9 critical Earth system processes.
- These are **less structured** than LCIA:
 - PBs often **do not link directly** from inventory to final damages.
 - Their focus is **thresholds and systemic stability**.
- Yellow boxes** in the impact pathway diagram show where PBs define control variables — not always in consistent or aligned places.

Common Ground:

- Many PBs are ultimately about preserving **biosphere integrity**.
 - Example: Excess nitrogen/phosphorus → eutrophication → ecosystem damage.
 - Example: Land use → biodiversity loss.
 - The link from midpoints (e.g. acidification, eutrophication) to PBs often centers on their **effect on ecosystems**.
-

Key Insight:

PBs may not follow LCIA's rigor, but they offer **essential boundary conditions** to build on.

- LCIA = **Detailed chain of cause-effect**.
- PB = **System-level safety thresholds**.

Performing AESA

- Identify biophysical boundaries and determine safe operating spaces
 - PBs or LCIA
- Allocate the safe operating space to different activities
- Calculate the impact profile of the activity using LCA
- Compare impact and allocated space

Notes:

Once we have:

- Identified **biophysical boundaries** (via **Planetary Boundaries** or **LCIA**),
 - Defined the **total safe operating space (SOS)** for Earth's systems,
-

We must then decide:

How do we **share this SOS** across different human activities and needs?

Why it matters:

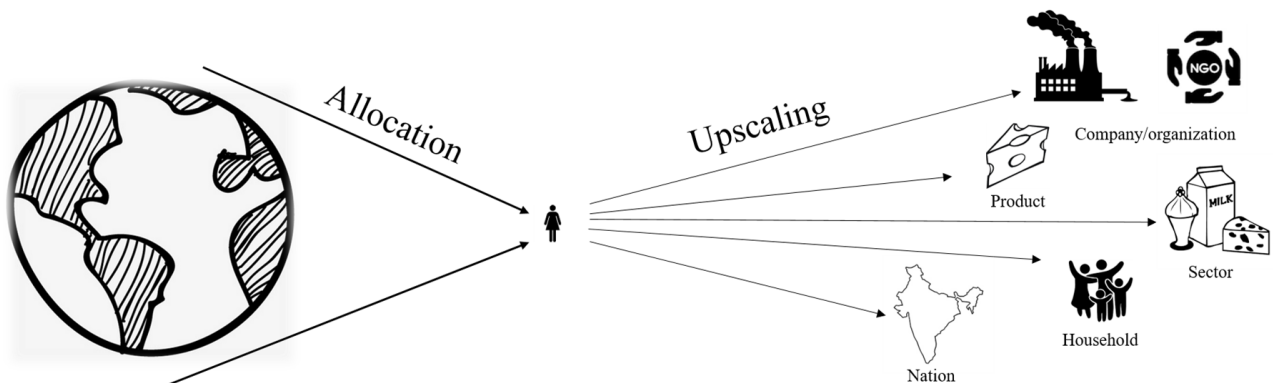
- We need SOS for:
 - 🍲 Food
 - 🚗 Transportation
 - 🏠 Housing
 - 🏥 Healthcare
 - 🌐 Infrastructure
 ...and more — for **all 8.2 billion people** on Earth.
- **Sustainability** requires:
 - Not exceeding **total planetary boundaries**
 - Ensuring **fair and responsible use** across sectors and generations
 - **No prioritization** of one group over another — present or future

🌍 Guiding Principle (from Brundtland):

“Meeting the needs of the present without compromising the ability of future generations to meet their own needs.”

This step in AESA ensures **equity, intergenerational justice, and functional sustainability**.

Allocating the operating space



Notes:

Allocating the Operating Space

After identifying the safe operating space (SOS) at the **global level**, the next question is:

👉 “How do we share it?”

🔄 Step 1: Allocation

- The total SOS must be **divided** (allocated) between individuals, sectors, and systems.
- Example: “How much of the climate budget is **my share**?”

⬆️ Step 2: Upscaling

- Once an individual share is determined, we **scale it up** to:
 - 🏢 Companies / organizations
 - 📦 Products / sectors (e.g., food, clothing)
 - 🌍 Nations
 - 👨👩👧👦 Households

Allocation Principles for Operating Space - Interpretation

Name	Description	Ethical norm (tentative)
Equal per capita (EPC)	Assigned share is the same for all individuals	Egalitarian
Capability to reduce (CR)	Assigned share is negatively correlated with a region's gross domestic product per capita	Prioritarian
Historical debt (HD)	Assigned share is negatively correlated with a region's cumulative environmental impact per capita	Prioritarian

Name	Description	Ethical Norm
Equal per capita (EPC)	Each person receives the same share of the global safe operating space	Egalitarian

Capability to reduce (CR)	Regions with higher GDP/capita (more means) get a smaller share (more reduction duty)	Prioritarian
Historical debt (HD)	Regions with greater cumulative impacts get a smaller share (they "owe" more)	Prioritarian

Notes:

Interpretation

- **EPC** is simple and fair on paper – everyone gets the same – but doesn't reflect infrastructure or opportunity differences.
- **CR** assumes wealthier regions or sectors are more able to make changes, so they carry more responsibility.
- **HD** considers **who caused the problem** – those who emitted more in the past get less space in the future.

Economically-Based Allocation Principles

Name	Description	Ethical norm (tentative)
Grandfathering (GF)	Assigned share is proportional with environmental impact in a reference year	Inegalitarian
Economic value added (EVA)	Assigned share is proportional to economic value added	Utilitarian
Cost efficiency (CE)	Assigned share is inversely proportional with the cost of reducing environmental impact of production	Utilitarian
Final consumption expenditure (FCE)	Assigned share is proportional with final consumption expenditure	Utilitarian

Björn A, Chandrasekhar C, Souley A-M, Doka G, Fang K, Gondran N, Hauschild MZ, Kerkhof A, King H, Margni M, McLaren S, Mueller C, Owsianicki M, Peters G, Rooz S, Sala S, Sandin G, Sim S, Vargas-Gonzalez M, Ryberg M (2020) Review of life-cycle based methods for absolute environmental sustainability assessment. Environ. Res. Lett. 15, 083001

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Name	Description	Ethical Norm
Grandfathering (GF)	Assigned share is proportional to past (reference-year) environmental impact	Inegalitarian

Economic value added (EVA)	Assigned share is proportional to value created (GDP contribution)	Utilitarian
Cost efficiency (CE)	Assigned share is inversely proportional to mitigation cost	Utilitarian
Final consumption expenditure (FCE)	Assigned share is proportional to consumer spending (demand-based)	Utilitarian

Notes:

Key Points & Limitations

- Grandfathering is simple but unfair: past big emitters keep a large share even in a reduced space.
- EVA and FCE focus on money spent or generated – but this can undervalue essentials like food (which is cheap but vital).
- CE favors actions where environmental improvements are cheap – but this may neglect hard-to-decarbonize sectors.

Performing AESA – Step-by-Step Process

- Identify biophysical boundaries and determine safe operating spaces
 - PBs or LCIA
- Allocate the safe operating space to different activities
- Calculate the impact profile of the activity using LCA
- Compare impact and allocated space

☐ Identify biophysical boundaries and determine safe operating spaces

- Use **Planetary Boundaries (PBs)** or **Life Cycle Impact Assessment (LCIA)** frameworks.
- Define what the planet can handle (e.g., emissions, land use) without tipping into instability.

☐ Allocate the safe operating space to different activities

- Decide *who* or *what* gets a share of the total safe space.
- Use ethical allocation principles like:
 - Equal Per Capita (EPC)

- Historical Debt (HD)
- Economic Value Added (EVA)
- Grandfathering (GF)
- And others...

□ **Calculate the impact profile of the activity using LCA**

- Use **Life Cycle Assessment** to assess the actual environmental burden of a product, process, or organization.

□ **Compare calculated impact to the allocated space**

- If **impact < allocated space**, the activity is **absolutely sustainable**.
- If **impact > allocated space**, it is **not sustainable**.
- The **sustainability ratio = impact / allocated space** (must be ≤ 1 to be sustainable).

Planetary Boundaries-based LCIA

Science of the Total Environment 634 (2018) 1406–1416

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Science of the Total Environment

How to bring absolute sustainability into decision-making: An industry case study using a Planetary Boundary-based methodology

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HIGHLIGHTS

- Applied novel Planetary Boundaries based LCIA methodology in case study
- Characterized impacts in metrics of the Planetary Boundaries
- Assessed absolute sustainability of activity relative to Planetary Boundaries
- Assignment of safe operating space found to be largest source of uncertainty
- Allowed for quantifying impact reductions required to be absolutely sustainable

GRAPHICAL ABSTRACT

Laundry washing

Assigned share of safe operating space

Impact categories in metrics of Planetary Boundaries

Unsustainable (Priority for reduction)

Sustainable

Notes:

Case Study: Laundry Washing in Europe

Objective:

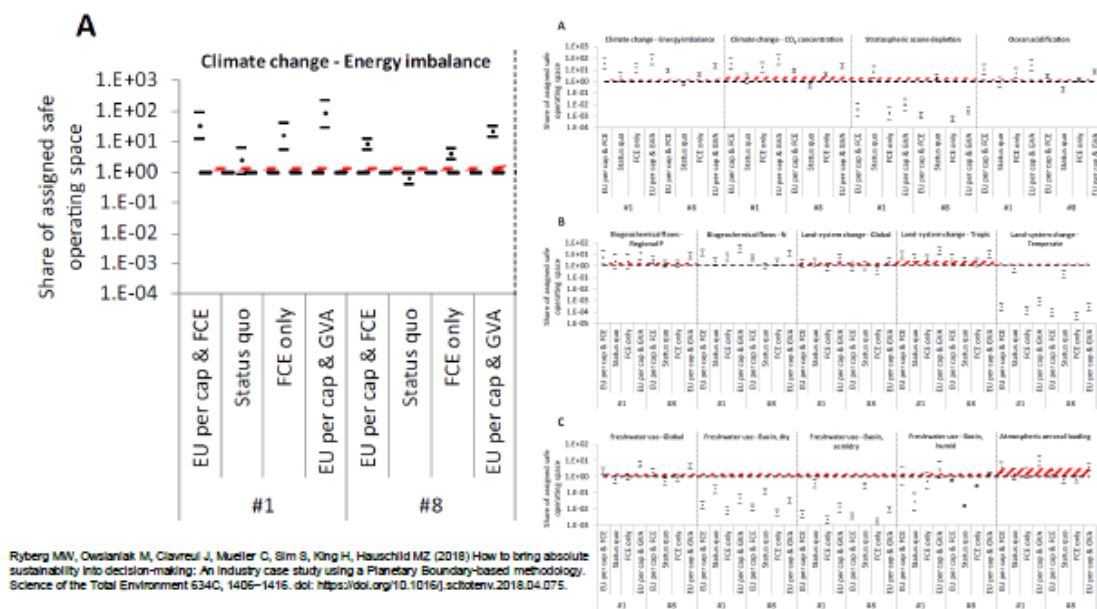
Assess whether current laundry washing practices are sustainable **in absolute terms** using Planetary Boundaries as reference points.

Method:

1. **Model impacts** of laundry washing across multiple **Planetary Boundary categories** (e.g., climate change, land-system change).
2. **Allocate a share** of the global safe operating space to the activity (laundry washing).
3. **Compare actual impacts** with allocated share to assess:
 - o  **Sustainable** if within the space
 - o  **Unsustainable** if exceeding the boundary (priority for reduction)

Improvement Scenarios Tested:

- Lower washing temperature
- More efficient electricity supply
- Better palm oil sourcing
- Lower detergent dosage
- Use of concentrated detergents
- Switching to renewable energy



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Is laundry washing in Europe absolutely sustainable?

And if not, can eco-design changes make it so?

Method Summary:

1. Impact Category Example:

- **Climate Change (measured by energy imbalance)** is shown here as a sample of one Planetary Boundary category.
- Total impacts from laundry are **compared to assigned shares of the global safe operating space (SOS)**.

2. Sustainability Ratio (Y-axis):

- Ratio = **Impact / Assigned Share of SOS**
 - If **> 1**, laundry washing **exceeds** its "safe share" → **unsustainable**
 - If **< 1**, activity is **within limits** → **sustainable**
 - Logarithmic scale helps visualize large differences.
-

Two Scenarios Compared:

Scenario #1 – Current Practice (left side):

- All four allocation methods result in **sustainability ratios > 1** → **not sustainable**
- Allocation methods tested:
 - EU per capita & FCE (Final Consumption Expenditure)
 - Status quo (Grandfathering)
 - FCE only (global)
 - EU per capita & GVA (Gross Value Added)

Scenario #8 – Eco-design Improvements (right side):

- Combines all 6 identified design improvements:
 - Green electricity
 - Reduced detergent dose
 - Renewable-sourced ingredients
 - Lower washing temperatures, etc.
- Shows **significant improvement**

- For **some allocation approaches** (e.g. grandfathering), it **becomes sustainable (ratio < 1)**
-

Key Insights:

- **Absolute sustainability** depends heavily on:
 - How the safe space is **allocated** (e.g. by capita, spending, value)
 - **What improvements** are implemented
- **Scenario 8 proves** that combining eco-design strategies can bring some allocation scenarios below the threshold.
- However, **true sustainability requires improvements across all impact categories**, not just one.

Summarizing

- Absolute environmental sustainability assessment, AESA, builds on LCA
- AESA relates environmental impactsto the assigned operating space to measure whether an activity is sustainable or not
- AESA covers all relevant environmental impacts

Notes:

What is AESA?

AESA (Absolute Environmental Sustainability Assessment) is a framework that:

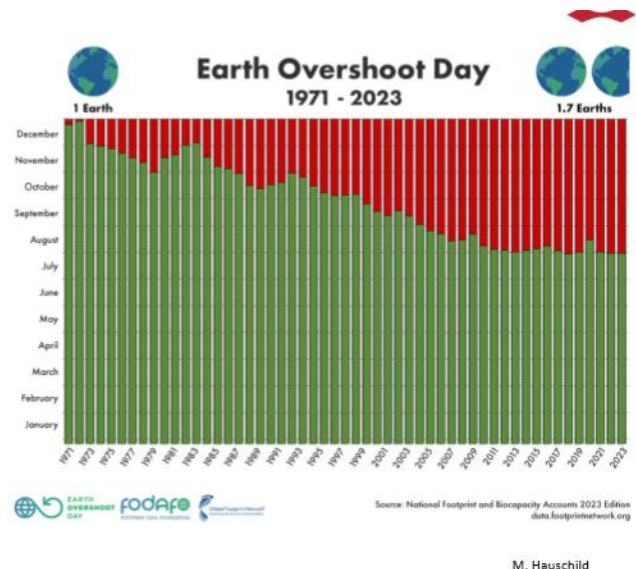
1. **Builds on LCA (Life Cycle Assessment)**
 - LCA is the *analytical engine* of AESA. It quantifies environmental impacts across a product or activity's life cycle.
2. **Adds an Absolute Perspective**
 - AESA doesn't just compare alternatives (as LCA does).
 - It checks whether impacts stay within the **planet's biophysical limits** (safe operating space).
3. **Measures Sustainability in Absolute Terms**
 - It answers: **Is this good enough for the planet?**
 - Not just: *Is this better than another option?*
4. **Covers All Relevant Environmental Impacts**

- AESA includes **multiple planetary boundaries**, not only climate change.
- Provides a full picture of environmental pressure from a product/activity.

4. Examples of other absolute sustainability metrics

Ecological footprint

- Focus on land use and CO₂ emissions
- Life cycle perspective
- Product footprints and national footprints
- Earth overshoot day
 - 2015: 7 August
 - 2016: 9 August
 - 2017: 5 August
 - 2018: 1 August
 - 2019: 3 August
 - 2020: 16 August!
 - 2021: 3 August
 - 2022: 1 August
 - 2023: 2 August



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Notes:

Ecological Footprint – Core Idea

The **ecological footprint** is a measure of how much biologically productive land and water area a population requires to produce the resources it consumes and absorb the waste it generates.

It includes:

1. **Actual land use** (e.g. food, biofuels, materials).
2. **Hypothetical land use** – the forest area required to absorb **CO₂ emissions** from fossil fuel combustion and other energy uses.

For a country like Denmark:

- ~50% of the footprint is **actual land use**.
- ~50% is the **hypothetical land** needed to offset CO₂ emissions.
- Much of this CO₂ is emitted **abroad** (from imported goods), so it uses a **life cycle perspective**.

Earth Overshoot Day marks the date when humanity's demand for ecological resources in a given year exceeds what Earth can regenerate in that year.

- **Formula:**

$$\text{Earth Overshoot Day} = \frac{\text{Biocapacity of Earth}}{\text{Ecological Footprint}} \times 365$$

A clear example of absolute sustainability:

→ We are using more than 1 Earth per year (1.7 in 2023), meaning we are in ecological overshoot.

Trends (1971–2023)

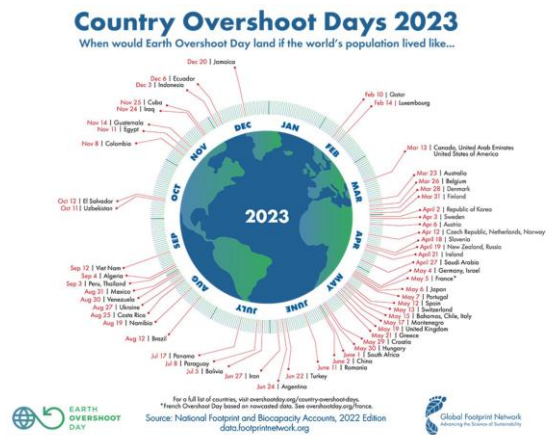
- 1971: First year humanity entered **overshoot**.
- Since then: Overshoot day has crept **earlier each year**.
- 2020: A rare reversal (16 August) due to **COVID-19 lockdowns**.
- 2023: Back to **2 August**, continuing the overshoot pattern.

Country overshoot days 2023

What is Country Overshoot Day?

Country Overshoot Day represents the hypothetical date when the world would reach Earth Overshoot Day if everyone lived like the average person in a given country.

- It's calculated based on the **ecological footprint** of that country (land use + CO₂ emissions).
- The earlier the date, the **more biocapacity** that lifestyle consumes annually.



DK Example: Denmark

- Denmark's overshoot day in 2023 is around **March 28**.
- If everyone lived like Danes, we'd need **~4 planets** to sustain that lifestyle.
- This highlights the **unsustainability of high-consumption societies**.

Global Inequity in Resource Use

- **Qatar, Luxembourg, and the U.S.:** Overshoot day falls as early as **February/March**.
- **Indonesia, Ecuador, Jamaica:** Reach overshoot **in December**, suggesting lower ecological impact lifestyles.

However, **environmental sustainability** $\neq$ **social sustainability**:

- Countries with late overshoot dates often have **less access to healthcare, education, and infrastructure**.
- This brings up a tension: **ecologically sustainable** $\neq$ **socially desirable** if basic needs are unmet

The Science Based Targets Initiative (SBTi)

- **SBTi established in 2015**



- ***"...in line with what the latest climate science says is necessary to meet the goals of the Paris Agreement—to limit global warming to **well-below 2°C** above pre-industrial levels and pursue efforts to limit warming to **1.5°C**."***

<https://sciencebasedtargets.org/>

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What is the SBTi?

The **Science Based Targets initiative (SBTi)** is a global body enabling businesses to set **greenhouse gas (GHG) emissions reduction targets** that are in line with the **latest climate science**.

Established in 2015

Founded by:

CDP

United Nations Global Compact

World Resources Institute (WRI)

Key Purpose

To ensure that corporate climate targets are consistent with the **goals of the Paris Agreement**:

*“Limit global warming to **well below 2°C** above pre-industrial levels and pursue efforts to limit warming to **1.5°C**.”*

Why It Matters for Companies

Companies that want to claim climate leadership must **commit to and validate science-based targets**.

Targets must reflect a **fair share of global emissions reductions** needed to meet climate goals.

SBTi is an example of an **absolute sustainability metric**—meaning it's based on **real global limits**, not just relative improvement.

Five steps



Five Steps in the SBTi Process

1. COMMIT

Submit a letter of intent

► Formally declare that your company will set science-based emissions reduction targets.

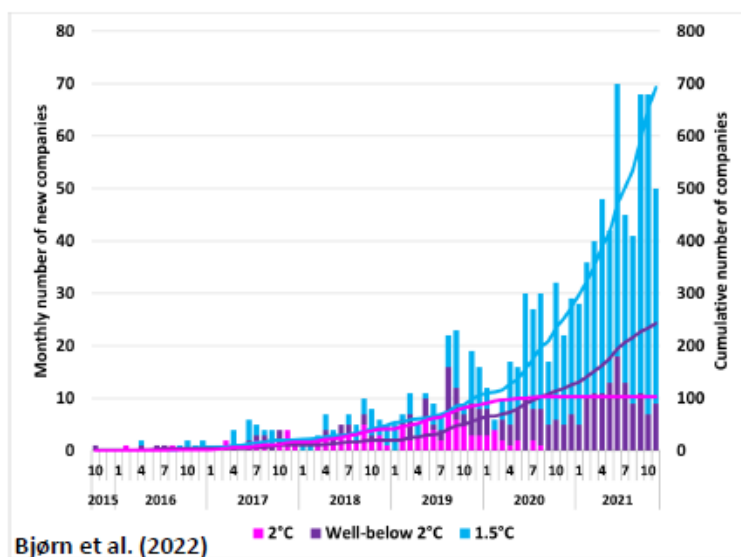
2. DEVELOP

Set emission reduction targets

► Define targets in line with SBTi criteria, using a **baseline year** and **Paris-aligned pathways** (e.g., well-below 2°C or 1.5°C).

- Often applies a **grandfathering logic**: reductions are based on your historical (baseline) emissions.
- 3. **SUBMIT**
Get your targets validated
 - Submit your target to SBTi for **official review and approval**.
- 4. **COMMUNICATE**
Share your commitment publicly
 - Announce your science-based target and position your company as part of the climate solution.
- 5. **DISCLOSE**
Track and report progress
 - Provide annual updates on **emissions and target progress** through platforms like CDP.

Rapidly growing



As of April 19th 2023:

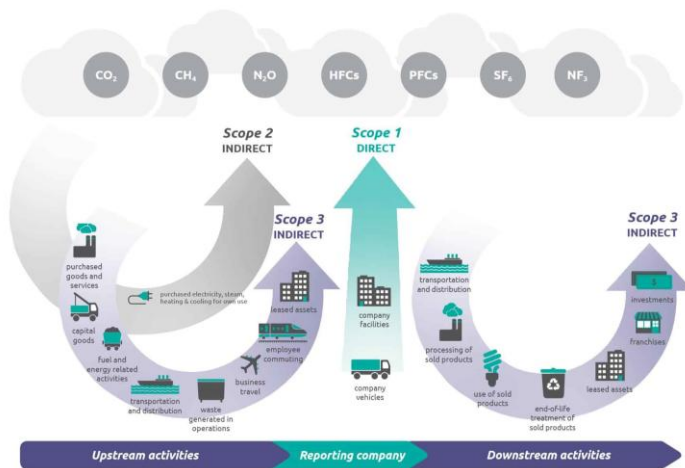


Exponential Growth of SBTi

- As of April 19th, 2023:
 - 4872 companies were taking action.
 - 2538 of those had officially **validated science-based targets**.
 - 1779 had made **net-zero commitments**.
- The bar chart (from Bjørn et al., 2022) displays:

- **Monthly new company sign-ups** (left axis).
- **Cumulative total** (right axis), color-coded by ambition level:
 - Light purple = 2°C
 - Blue = well-below 2°C
 - Dark purple = 1.5°C (the Paris Agreement's most ambitious goal)
- **In Denmark**, over **200 companies** have already joined at different stages—whether through committing, submitting targets, or fully disclosing progress.

The three scopes



◆ Scope 1 – Direct emissions

- Emissions from sources **owned or controlled by the company**.
- Examples:
 - Emissions from company facilities.
 - Company vehicles.

◆ Scope 2 – Indirect emissions from purchased energy

- Emissions from **the generation of purchased electricity, heat, or steam**.
- These emissions **occur at the energy producer's facilities** but are attributed to your company's energy use.

◆ Scope 3 – All other indirect emissions

- Emissions that occur **upstream and downstream** in the value chain.

- Examples:
 - Upstream: purchased goods and services, capital goods, employee commuting.
 - Downstream: product use, end-of-life treatment, investments, leased assets.
- Often accounts for **the majority of a company's total footprint**.

Setting targets

- **Science-based, i.e. in accordance with the Paris agreement's 1.5 or well below 2.0 degree scenario**
 - Sectoral approach
 - Absolute contraction approach (grandfathering)
 - 95% of scope 1 and 2
 - Min. 67% of scope 3 (if it is more than 40% of total emissions)
- **Absolute or intensity depending on sector**
- **Short term (5-10 years)**
 - Min. 4.2% linear annual contraction
- **and long term (2050 or sooner)**
 - 90% reduction

✓ Science-based alignment

- Targets must align with **1.5°C or well below 2°C** global warming scenarios.
- These are modeled pathways consistent with the **Paris Agreement**.

✂ Target-setting approaches

- **Sectoral Approach:** Each sector has different reduction trajectories based on technological feasibility.
- **Absolute Contraction (Grandfathering):** All companies reduce emissions **by the same percentage**, based on their **baseline year emissions**.

📦 Coverage requirements

- **Scope 1 & 2:** Must cover at least **95%** of direct and energy-related emissions.
- **Scope 3:** If it represents **more than 40%** of total emissions, then **at least 67%** of Scope 3 must be included in the target.

📅 Short-term targets (5–10 years)

- Must include a **minimum 4.2% annual linear reduction**, which supports a 50% reduction by 2030.

Long-term targets (by 2050 or sooner)

- Must aim for at least **90% absolute emission reduction**, i.e., **near net-zero**.

Absolute vs. intensity targets

- **Absolute targets:** Total emissions must be reduced regardless of production volume.
- **Intensity targets:** Emissions per product/service (e.g., kg CO₂ per item) – can hide overall growth in emissions if not paired with absolute constraints.

Why absolute targets matter

- If a company grows (e.g., sells more products), only reducing intensity may still lead to an **overall increase in emissions**.
- Absolute targets ensure total impact is limited — essential to remain within a **shrinking global carbon budget**.

Two examples

Coca-Cola European Partners commits to reduce absolute scope 1, 2 and 3 GHG emissions 30% by 2030 from a 2019 base year. Within that target, Coca-Cola European Partners commits to reduce absolute scope 1 and 2 GHG emissions 47% by 2030 from a 2019 base year and reduce absolute scope 3 GHG emissions 29% by 2030 from a 2019 base year.

HeidelbergCement commits to reduce scope 1 GHG emissions 22% per ton of cementitious materials by 2030 from a 2016 base year. HeidelbergCement also commits to reduce scope 2 GHG emissions 66% per ton of cementitious materials within the same timeframe.* *The target boundary includes biogenic emissions and removals associated with the use of bioenergy.

Differences in:

- Base year
- Target year
- Targeted reduction
- Absolute/intensity
- Emission scope

Example 1: Coca-Cola European Partners

- **Target:** Reduce absolute GHG emissions (Scopes 1, 2, 3) by **30%** by **2030** from a **2019** base year.
- Scope 1 & 2 emissions reduced by **47%**, Scope 3 by **29%**.
- **Target type:** **Absolute** reductions.
- **Covers:** All scopes (1, 2, and 3).

Example 2: HeidelbergCement

- **Target:** Reduce Scope 1 GHG emissions by **22% per ton of cementitious material** by **2030** from a **2016 base year**.
- Scope 2 reduction: **65% per ton** over the same timeframe.
- **Target type:** **Intensity-based** (emissions per product unit).
- **Covers:** Scope 1 & 2; no explicit mention of Scope 3.

Key Differences Highlighted

Factor	Coca-Cola	HeidelbergCement
Base Year	2019	2016
Target Year	2030	2030
Reduction Type	Absolute (%)	Intensity (% per unit)
Scopes Covered	1, 2, and 3	1 and 2
Reduction Level	30–47% absolute	22–65% per product

These variations are allowed within SBTi, but **future scrutiny** will likely focus more on:

- **Whether Scope 3 is included** (especially if it's a major share),
- **Whether targets are absolute**, and
- **The ambition level relative to 1.5°C scenarios.**

Summarizing

- Relate performance to a target that is based on environmental limits (e.g., stability of the climate system, available area of land).
- Inform about the size of improvements that are needed to achieve the target.
- Shift the focus from efficiency to effectiveness — from “better” in relative assessments
- ... to “good enough” in absolute assessments.

Notes:

What are Absolute Sustainability Metrics?

They:

- **Relate environmental performance** to scientifically defined **planetary limits**.
 - Example: Limits on **CO₂ emissions** (climate stability) or **land use** (ecological footprint).
 - Are **not comparative** (i.e., better vs. worse), but rather **threshold-based** (i.e., is it "good enough"?).
-

Purpose

They inform:

- **How far we are** from what's **environmentally sustainable**.
 - **How much improvement** is needed to stay within our "safe operating space."
-

Efficiency vs. Effectiveness

- **Efficiency** = doing something better (e.g., fewer emissions per product).
- **Effectiveness** = doing the right thing (e.g., staying within planetary boundaries).
- This shift is vital: **"better" is not enough if it's still unsustainable**.

Lecutre 11 - Decarbonizing energy systems

Outline of Decarbonizing energy systems

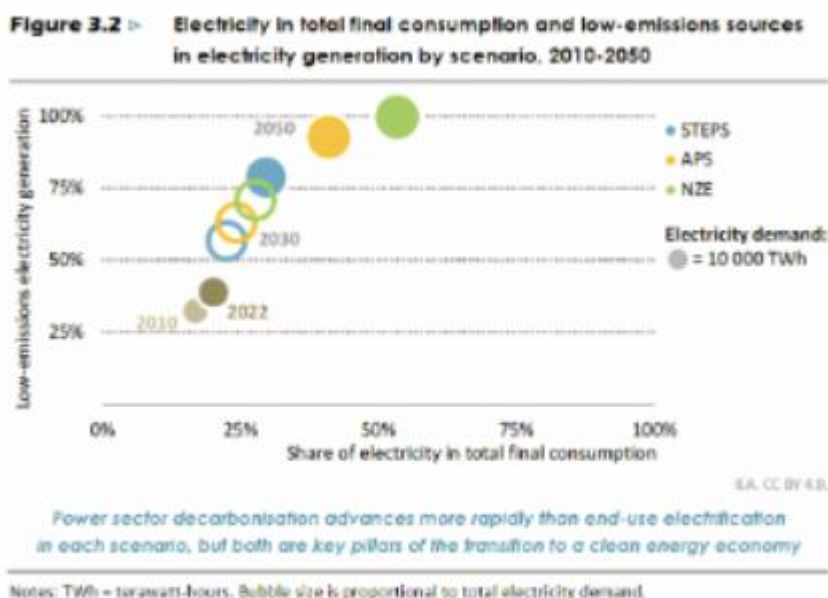
The intention of the module “Decarbonizing the energy systems” in the course Quantitative Sustainability is to enable the student to calculate the resulting Global Warming Potential emission from electricity production given a specific technology mix of a country or region. By scaling up the

emission to the equivalent global emission one can compare the CO₂ emission to the global planetary boundary as dictated by the Paris agreement of a 1.5 degree global warming limit. The students will then be able to determine if a future electricity mix will violate the global planetary boundary of the Global Warming Potential.

Since there is a trend towards electrification of the energy system, then the focus of the module is on

the electricity part of the energy system and especially the renewable energy technologie

Electricity fraction of energy consumption



Source: IEA Energy Outlook 2023

- **Trend over time (2010–2050):**
 - Electricity is expected to cover a growing share of the planet’s total energy consumption.

- Transition from ~20% (2010) to around 50% (by 2050 or earlier, depending on scenario).
- **Electrification & Low-Emission Technologies:**
 - There's a dual trend:
 1. Increased **electrification** of energy use (more sectors relying on electricity).
 2. Higher use of **low-emission technologies** for electricity generation.
 - By **2050**, in the **Net Zero Emissions (NZE)** scenario, nearly **100%** of electricity is expected to be from **low-emissions sources**.
- **Scenarios shown in the chart:**
 - **STEPS** = Stated Policies Scenario
 - **APS** = Announced Pledges Scenario
 - **NZE** = Net Zero Emissions Scenario
 - Each scenario projects different levels of electricity demand and decarbonization.
- **Rationale for Lecture Focus:**
 - Because electricity will play a **central role in clean energy transition**, the course/lecture focuses on the **electricity system** specifically.

Technologies for electricity production:

Fuel based

- Coal
- Gas
- Biomass
- Nuclear

Renewables

- Wind
- Solar
- Hydro
- Tidal
- Geothermal



Main Argument:

- **100% of electricity** in future plans should come from **low-emission technologies**.
 - This slide introduces the **key electricity generation technologies**—both **fuel-based** and **renewable**—to explain why the lecture focuses on the electricity system.
-

Types of Technologies:

1. Fuel-Based (Conventional) Sources:

- **Coal**
- **Gas**
- **Biomass**
- **Nuclear**
 - Nuclear is listed under fuel-based for the sake of this classification, but it could also be considered low-emission.
- **Garbage/Waste incineration**
 - Produces electricity and district heating.
 - Shown as a hybrid case (fuel-based but with partial waste management benefits).

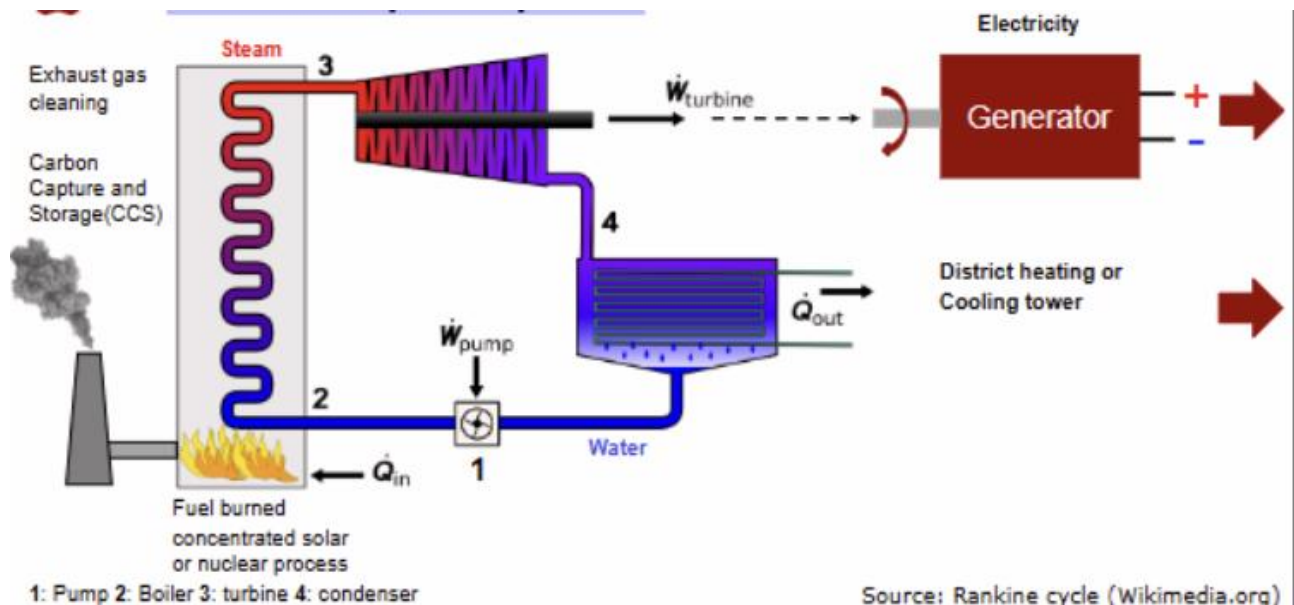
2. Renewable Sources:

- **Wind**
 - **Offshore wind**
 - **Onshore wind**
 - **Solar**
 - **Solar PV** (photovoltaics)
 - Other types mentioned (not pictured):
 - **Hydro**
 - **Tidal**
 - **Geothermal**
-

Note on Classification:

- **Nuclear** could arguably be placed under renewables due to its **low emissions**, but it's included in **fuel-based** here to match the typical operational setup.
- The placement of technologies on the slide reflects how **plants are actually operating**, not just how they're classified theoretically.

Fuel based power plants



Main Principle:

- Follows the **Rankine Cycle**, which is common in both **fossil fuel** and **nuclear** power plants.

Steps in the Process:

1. **Pump (1):**
 - Circulates water through the system.
2. **Boiler (2):**
 - **Fuel (e.g., coal, gas, biomass, or nuclear heat)** is used to heat water.
 - Water turns into **steam** in a heat exchanger.
3. **Turbine (3):**
 - Steam drives a **turbine**, which converts **thermal energy to mechanical energy**.
4. **Generator:**
 - Turbine turns a **generator**, producing **electricity** for the grid.

5. Condenser (4):

- Steam is **cooled and condensed** back to water.
- Cooling can be used for **district heating** or removed via a **cooling tower**.

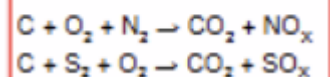
Additional Notes:

- **Nuclear power plants** follow the **same principle**, using nuclear reactions instead of burning fuel to generate heat.
- **Exhaust gases** from fossil fuels can be:
 - Released into the atmosphere via a chimney.
 - Treated with **Carbon Capture and Storage (CCS)** to remove **CO₂ emissions**.
- The process is often referred to as a **Rankine cycle**.

Burning fuels causes CO₂ emissions

Coal	C	+ O ₂	→	CO ₂	+ Heat +
Methane	CH ₄	+ O ₂	→	CO ₂	+ H ₂ O + Heat +
Oil	C _x H _y	+ O ₂	→	CO ₂	+ H ₂ O + Heat +
Wood	C ₆ H ₁₂ O ₆	+ O ₂	→	CO ₂	+ H ₂ O + Heat +
Plastic (PET)	C ₁₀ H ₈ O ₄	+ O ₂	→	CO ₂	+ H ₂ O + Heat +
Hydrogen	H ₂	+ O ₂	→		H ₂ O + Heat +
Ammonia	NH ₃	+ O ₂	→	NO _x	+ H ₂ O + Heat +
Fission	²³⁸ U		→	²³⁴ U + ...	+ Neutrons + Heat
Fusion	² ₁ D + ³ ₁ T		→	⁴ ₂ He	+ Neutron + Heat

Remember that burning fuels in air will also include nitrogen and some sulfur, whereby NO_x and SO_x can result



Main Message:

- Burning fuels causes **CO₂ emissions**, but also other harmful byproducts such as **NO_x** (nitrogen oxides) and **SO_x** (sulfur oxides), depending on the fuel composition.

Environmental Impacts Beyond CO₂:

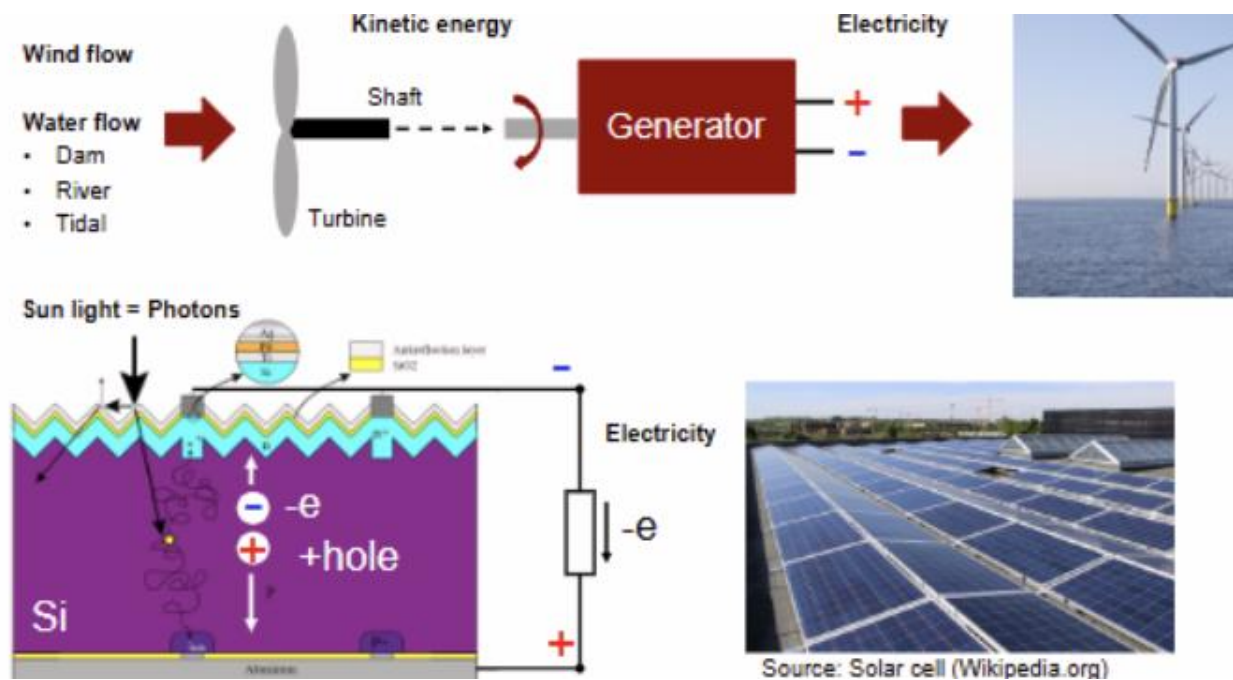
- **Nitrogen in air** and **sulfur in fuels** can lead to:
 - **NO_x**: nitrogen oxides (air pollutants)

- **SO_x**: sulfur oxides (also harmful)
- **Hydrogen fuel**: Clean — only water and heat produced.
- **Ammonia**: CO₂-free but may emit NO_x.
- **Fusion and fission**: No direct CO₂, but produce **neutrons** and radioactive byproducts (fission).

Key Takeaway:

- While CO₂ is the primary concern, **other emissions** (NO_x, SO_x) must be considered when evaluating environmental impacts of fuel-based power.

Renewable based power plants



Main Principle:

Renewable power plants convert **natural energy flows** (wind, water, sunlight) into **electricity**—often **without emissions** and **with minimal moving parts** (in the case of solar).

1. Wind and Water-Based Systems (Kinetic Energy Conversion)

Examples:

- **Wind turbines** (onshore and offshore)

- **Hydropower** from:
 - **Dams**
 - **Rivers**
 - **Tidal flow**

How it works:

- Wind or water flow **spins a turbine** connected to a **shaft**.
- The shaft rotates a **generator**, converting **kinetic energy** into **electrical energy**.
- Generator is often located in the **nacelle** (top part of a wind turbine).

2. Solar Power (Photovoltaics - PV)

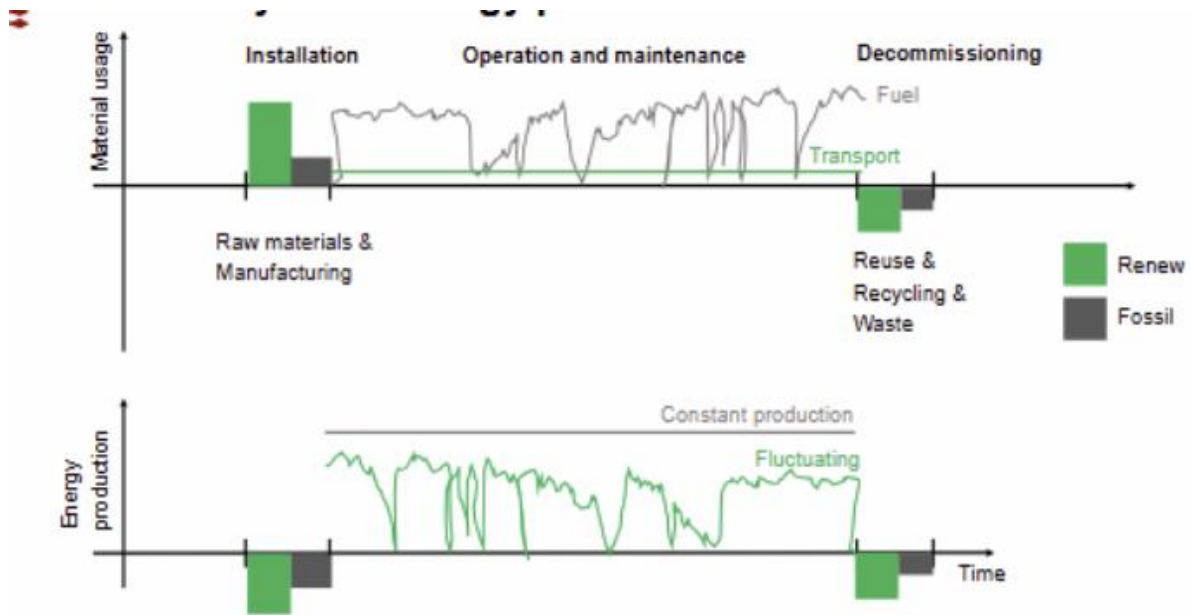
How it works:

- Solar panels are made of **silicon (Si) wafers**.
- **Sunlight (photons)** hits the panel → generates **electron-hole pairs** in the silicon.
- Electrons (-) and holes (+) move to opposite sides, creating **electric current**.
- A **resistor or load** can be connected to complete the circuit and draw power.
- **No moving parts** — direct **conversion of light to electricity**.

Material manufacturing causes CO2 emissions

Materials	Chemical process	Impact
Mining Iron ore and making metal	$\text{Fe}_2\text{O}_3 + 3 \text{CO} \rightarrow 2 \text{Fe} + 3 \text{CO}_2$	1.9 ton CO ₂ e /ton _{Fe}
Remelting recycled steel (NLMK Dansteel)	$\text{Fe} + \text{heat/electricity} \rightarrow \text{Fe}$	0.7 ton CO ₂ e /ton _{Fe}
Refining Iron ore using hydrogen ("Green steel")	$\text{Fe}_2\text{O}_3 + 3 \text{H}_2 \rightarrow 2 \text{Fe} + 3 \text{H}_2\text{O}$	? (Demo at Salzgitter)
Concrete (turning limestone to CaO)	$\text{CaCO}_3 + \text{Heat} \rightarrow \text{CaO} + \text{CO}_2$	0.1- 0.2 ton CO ₂ e /ton _{Concrete}
Green cement (Aalborg Portland)	As above but use minerals and heating with less CO ₂ foot-print	30 % lower than cement ~ 0.6 ton CO ₂ e /ton _{Cement}
Making sand into silicon for PV wafers	$\text{SiO}_2 + 2 \text{C} \rightarrow \text{Si} + 2 \text{CO}$	5-16 ton CO ₂ e /ton _{Copper}
Making Copper ore into copper for wires	$2 \text{Cu}_2\text{S} + 3 \text{O}_2 \rightarrow 2 \text{Cu}_2\text{O} + 2 \text{SO}_2$ $2 \text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6 \text{Cu} + 2 \text{SO}_2$	3-4 ton CO ₂ e /ton _{Copper} SO ₂ causes acid rain

Life cycle of energy plants



1. Material Usage Over Time

- **Renewable plants (green line):**
 - **High material usage during installation** (e.g. wind turbines, solar panels).
 - **Low material use** during operation (mainly for transport/maintenance).
 - **Higher material demand during decommissioning** due to larger infrastructure.
- **Fossil fuel plants (gray line):**
 - **Lower installation material use.**
 - **Continuous material use during operation** (mainly fuel input).
 - **Less material needed for decommissioning.**

2. Energy Production Over Time

- **Fossil fuels:**
 - Provide **constant energy output** (can be controlled based on demand).
 - **Renewables:**
 - **Fluctuating output** — depends on **external conditions** (sunlight, wind).
 - E.g., solar only produces when the sun shines; wind turbines only when wind blows.
-

3. Life Cycle Considerations

- To compare energy systems fairly, we must analyze:
 - **Material in/out** over the full lifecycle (installation → operation → decommissioning).
 - **Total energy produced** over that time.
 - This helps estimate **overall environmental impact** (e.g., emissions per MWh).
-

Key Takeaway:

- **Renewables have front-loaded material impact and intermittent energy output.**
- **Fossil fuels have steady energy output, but ongoing fuel/material needs and emissions.**
- A full **life cycle assessment** is crucial for evaluating sustainability.

Normalization of emissions [emission / kWh]

- The bill of material (BOM) of an energy plant accounts for the usage of the different materials m_i used to manufacture the energy plant
- The life time fuel consumption m_f is the sum of the annual fuel consumptions AFC used during the life time LT of the energy plant
- The recycling masses $m_{r,i}$ are the masses of the original bill of material recovered during the decommissioning phase, which can replace raw materials for the production of new energy plants. This means that emissions can be subtracted.
- The life time energy production E is then sum of the annual energy production AEP of the energy plant over the life time LT of the plant
- Total emission factors of the technology EF_j are now calculated from the specific emission factors $\varepsilon_{i,j}$ of the materials and normalized by the life time energy produced

Example:

Gram CO₂ / kWh

$$EF_j = \frac{\sum_{i=1}^N m_i \varepsilon_{i,j} + m_f \varepsilon_{f,j} + \sum_{i=1}^N m_{r,i} \varepsilon_{r,i,j}}{E}$$

Goal:

To compare the environmental impact of different energy technologies by calculating **emissions per unit of electricity produced** (e.g., grams of CO₂ per kWh).

Key Definitions:

- **Bill of Materials (BOM):**
Total materials m_i used to construct the plant.
- **Lifetime Fuel Consumption m_f :**
Total amount of fuel used during the **lifetime LT** of the plant.
$$m_f = \sum AFC$$
- **Recycling Masses $m_{r,i}$:**
Materials recovered at decommissioning, which can **offset raw material needs** in future plants → **emissions can be subtracted**.
- **Lifetime Energy Production E :**
Total electricity generated over the lifetime of the plant.
$$E = \sum AEP$$

Formula for Normalized Emissions:

$$EF_j = \frac{\sum_{i=1}^N m_i \varepsilon_{i,j} + m_f \varepsilon_{f,j} + \sum_{i=1}^N m_{r,i} \varepsilon_{r,i,j}}{E}$$

Where:

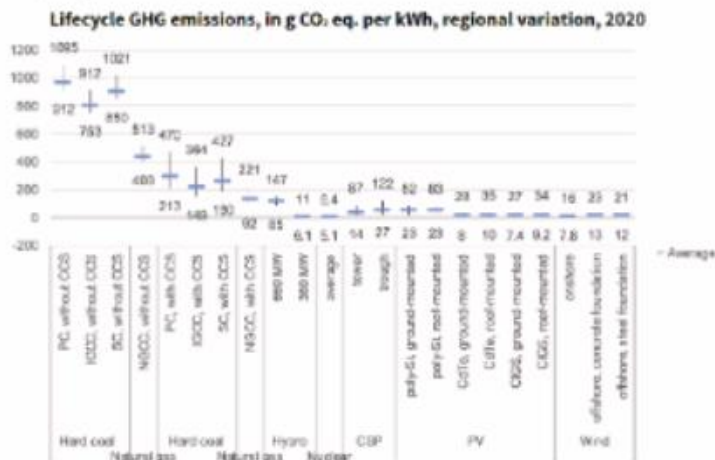
- EF_j : Total emission factor for technology j
- $\varepsilon_{i,j}$: Emission factor per material
- $\varepsilon_{f,j}$: Emission factor from fuel
- $\varepsilon_{r,i,j}$: Emission factor reduction from recycling
- E : Lifetime energy produced

Key Takeaways:

- Helps compare **renewables** (low fuel use, high material use) vs **fossil plants** (high fuel use).
- Accounts for:
 - **Material-related emissions**
 - **Fuel-related emissions**
 - **Recycling credits**
- Enables calculation of **CO₂ emissions per kWh**, a key metric for sustainability.

LCA overview of electricity producing technologies

Figure 1 Lifecycle greenhouse gas emission ranges for the assessed technologies



Carbon Neutrality in the UNECE Region:
Integrated Life-cycle Assessment
of Electricity Sources



Purpose of the Figure:

- Based on a **UN Life Cycle Assessment (LCA)** of electricity sources (UNECE, 2022).
- Compares **lifecycle greenhouse gas emissions** across technologies.
- Metric used: **g CO₂-equivalents per kWh** of electricity generated (2020 data).

Key Observations from the Graph:

High-emission technologies:

- **Coal-fired plants:**
 - Pulverized coal (PC): Up to **~1100 g CO₂/kWh**
 - Integrated coal gasification (IGCC): **~850–1000 g CO₂/kWh**
- **Natural gas:**
 - Combined cycle without CCS: **~400–500 g CO₂/kWh**
 - Lower than coal, but still significant.

Carbon Capture and Storage (CCS):

- Adding **CCS** reduces emissions for coal and gas plants.
- Coal with CCS: Drops emissions **down to ~400 g CO₂/kWh**
- Still not low enough to be considered "clean."

Low-emission technologies:

- **Hydropower, Nuclear, Concentrated Solar Power (CSP), Photovoltaics (PV), Wind:**
 - Emissions typically **below 100 g CO₂/kWh**
 - Wind and PV are often **as low as ~10–30 g CO₂/kWh**

Conclusion:

- **Coal and gas** (even with CCS) remain **high-emission sources**.
- **Renewables and nuclear** are the **lowest-emission electricity options**.
- This LCA helps justify why focus is shifting toward **low-emission technologies** for clean energy transitions.

Energy technologies

Table 1 Summary of life cycle inventories' scopes, per type of technology

TECHNOLOGY	INCLUDED	EXCLUDED	Recycling of materials in UN report is absent !	
Coal power	without CCS Energy carrier supply chain, from extraction to combustion, including methane leakage Infrastructure construction, operation, and dismantling (energy inputs and waste production) Connection to grid	Potential recycling of dismantled equipment	Concentrated solar power Infrastructure, site preparation and occupation, operation and maintenance (including 6-hour storage) Decommissioning (energy inputs and waste production) Connection to grid	Potential recycling of dismantled equipment
	with CCS Same as above, plus capture equipment and chemicals, transportation through and CO ₂ and storage infrastructure (well)	Same as above, plus Potential emissions leakage from captured CO ₂ transportation or from the storage site	Photovoltaics Infrastructure, site preparation and occupation, operation and maintenance Decommissioning (energy inputs and waste production) Connection to grid	Potential recycling of dismantled equipment
Natural gas power	without CCS Energy carrier supply chain, from extraction to combustion, including methane leakage Infrastructure construction, operation, and dismantling (energy inputs and waste production) Connection to grid	Potential recycling of dismantled equipment	Wind power Infrastructure, site preparation and occupation, operation and maintenance Decommissioning (energy inputs and waste production) Connection to grid	Potential recycling of dismantled equipment
	with CCS Same as above, plus capture equipment and chemicals, transportation through and CO ₂ and storage infrastructure (well)	Same as above, plus Potential emissions leakage from captured CO ₂ transportation or from the storage site		
Hydropower	Construction, site preparation, transportation of materials Connection to grid	Potential recycling of dismantled equipment Storage site leakage, emissions of CO ₂ and CH ₄		
Nuclear power	Full device supply chain (from extraction to fabrication to fuel) Core processes (construction and decommissioning of power plant), as well as operations Back-end process (spent fuel management, storage, and final repository)	Potential recycling of dismantled equipment Reprocessing of spent fuel (conservative assumption that all fuel is reprocessed)		

Energy Technologies Covered in the Report:

- **Coal power** (with and without CCS)
- **Natural gas power** (with and without CCS)
- **Hydropower**
- **Nuclear power**
- **Concentrated Solar Power (CSP)**

- **Photovoltaics (PV)**
 - **Wind power**
-

LCA Scope – What’s Included vs. Excluded:

Commonly Included:

- Fuel/energy carrier supply chains (e.g., mining, extraction, combustion).
- Infrastructure construction, operation, decommissioning.
- Transport of materials and connection to the grid.

Commonly Excluded:

- **Recycling of dismantled equipment** (not covered in the UN report — marked as absent).
 - Leakage from CCS transport/storage (for coal/gas with CCS).
 - Reprocessing of spent fuel (nuclear, limited scope).
-

Key Impact Categories (LCA Perspectives):

1. **Climate Change**
→ *Measured in kg CO₂-equivalents per kWh*
→ Main focus of this lecture and report.
 2. **Freshwater Eutrophication**
→ *Nutrient emissions* (e.g., phosphorus/nitrogen runoff into freshwater).
 3. **Ionizing Radiation**
→ Not just from **nuclear** power — also from **mining** and raw material extraction (e.g., radon release).
 4. **Human Toxicity**
→ Mentioned but **not deeply covered** in the lecture.
-

Key Observation:

- Despite focusing on CO₂/kWh, a **comprehensive LCA** also accounts for environmental and health-related impacts like **radiation** and **toxicity**, even for **non-nuclear** sources.

Impacts

Table 3 Selected environmental indicators for Life Cycle Impact Assessment

CATEGORY	UNIT	REFERENCE	DESCRIPTION
Climate change	kg CO ₂ eq.	IPCC (2013)	Radiative forcing as global warming potential, integrated over 100 years (GWP100), based on IPCC baseline model.
Freshwater eutrophication	kg P eq.	EUTREND, Struijs, Beusen [16]	Expression of the degree to which the emitted nutrients reach the freshwater end compartment. As the limiting nutrient in freshwater aquatic ecosystems, a surplus of phosphorus will lead to eutrophication.
Ionising radiation	kBq ²³⁵ U eq	Frischknecht, Braunschweig [17]	Human exposure efficiency relative to ²³⁵ U radiation. The original model is Dreicer, Tort [18] and follows the linear no-threshold paradigm to account for low dose radiation (details in Box 5).
Human toxicity	CTUh (comparative toxic units)	USEtox 2.1. model Rosenbaum, Bachmann [19]	The characterization factor for human toxicity impacts (human toxicity potential) is expressed in comparative toxic units (CTUh), the estimated increase in morbidity in the total human population, per unit mass of a chemical emitted, assuming equal weighting between cancer and non-cancer due to a lack of more precise insights into this issue. Unit: [CTUh per kg emitted] = [disease cases per kg emitted]

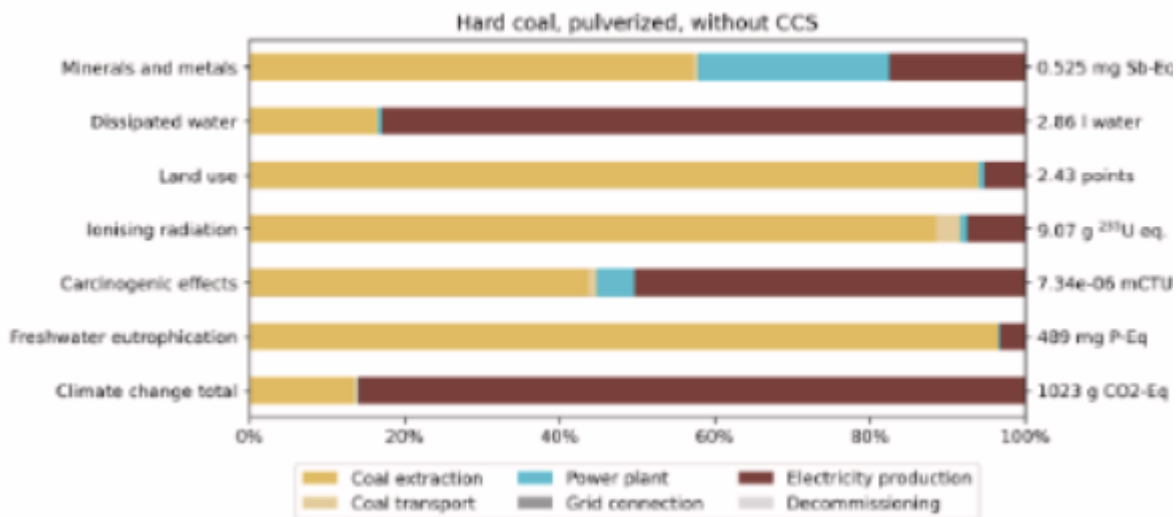
Land use	points	LANCA model, Bos, Horn [20]	The LANCA model provides five indicators for assessing the impacts due to the use of soil: 1. erosion resistance; 2. mechanical filtration; 3. physicochemical filtration; 4. groundwater regeneration and 5. biotic production.
Water resource depletion	m ³	Swiss Ecoscarcity Frischknecht, Steiner [21]	Water use related to local consumption of water. Note: only air emissions are accounted for. <i>In this method, all flows have an identical characterisation factor of 42.95 m³/m³ – we therefore choose to account for these flows uncharacterised, i.e. 1 m³/m³.</i>
Mineral, fossil and renewable resource depletion	kg Sb eq.	Van Oers, De Koning [22]	Scarcity of resource in relation to that of antimony. Scarcity is calculated as « reserve base ».

- These categories are important, but **climate change** (CO₂ eq./kWh) is the **primary focus**.
- More advanced categories like land use or mineral depletion are **complex** and not covered in detail here due to difficulty in defining **planetary boundaries**.

Technology examples : coal



Figure 4 Life cycle impacts from 1 kWh of coal power production, pulverised coal, Europe, 2020



Main Case:

Technology: Pulverized coal (PC), without CCS

Scope: Full life cycle impacts per 1 kWh of electricity generated

Key Environmental Impact Values:	
Category	Value
Climate change	1023 g CO ₂ -eq
Freshwater eutrophication	489 mg P-eq
Carcinogenic effects	7.34e-06 mCTUh
Ionizing radiation	9.07 g ²³⁵ U-eq
Land use	2.43 points
Dissipated water	2.86 liters
Minerals and metals	0.525 mg Sb-eq

Breakdown of Impact Contributions (by life cycle stage):

- **Electricity production** = Major contributor to most impact categories (especially climate change and water dissipation)

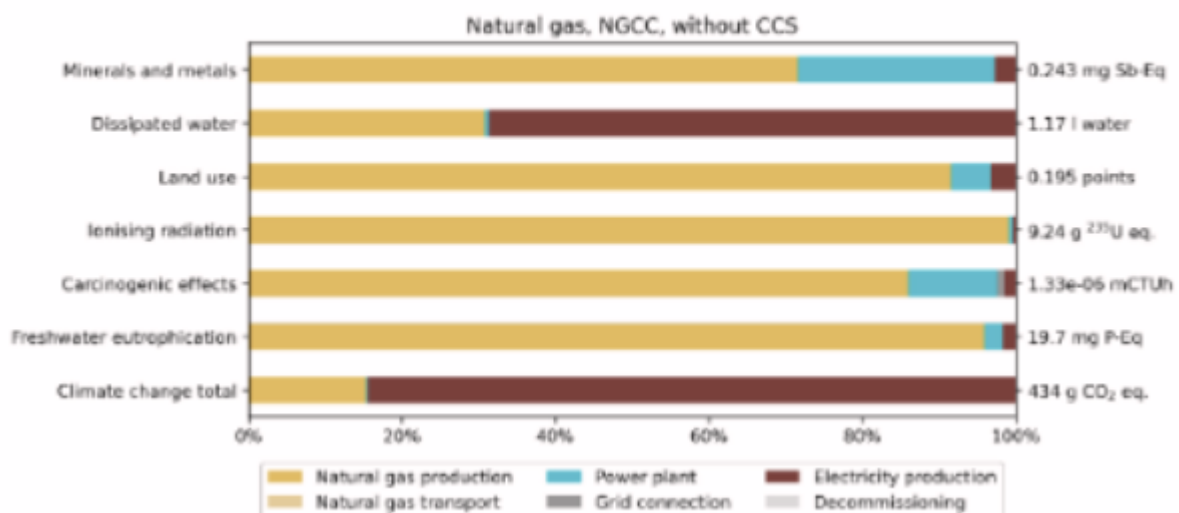
- **Coal extraction** = Significant contributor (~18% of climate impact), especially to water loss and radiation
- **Other stages (minor contributors):**
 - **Power plant construction**
 - **Coal transport**
 - **Grid connection**
 - **Decommissioning**

Notable Insights:

- The **largest share** of CO₂ emissions and water use is from **burning coal during electricity production**.
- Water loss (2.86 L/kWh) is mostly related to **cooling processes** in the plant.
- Even **non-emission factors** like **land use**, **mineral use**, and **radiation** are non-trivial and must be tracked in LCAs.

Technology examples : Natural gas

Figure 9 Life cycle impacts from 1 kWh of natural gas power production, NGCC without carbon dioxide capture and storage, Europe, 2020



Main Environmental Impact Values (per 1 kWh):

Impact Category	Value
Climate change	434 g CO ₂ -eq
Freshwater eutrophication	19.7 mg P-eq
Carcinogenic effects	1.33e-06 mCTUh
Ionizing radiation	9.24 g ²³⁵ U-eq
Land use	0.195 points
Dissipated water	1.17 liters
Minerals and metals	0.243 mg Sb-eq

Impact Breakdown by Life Cycle Stage:

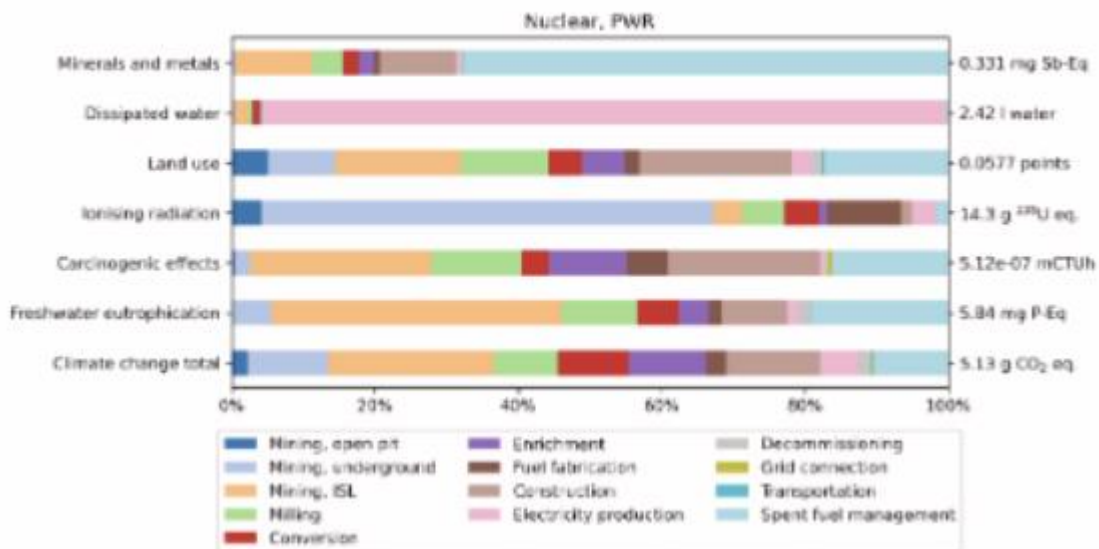
- **Natural Gas Production:**
Contributes ~**15–18%** of CO₂ emissions
Also responsible for a portion of **dissipated water** and **carcinogenic effects**
- **Electricity Production** (burning gas):
Major source of CO₂ emissions and **cooling-related water loss**
- Other stages (e.g., power plant construction, grid connection) are minor contributors

Key Takeaways:

- **CO₂ emissions** from NGCC are **lower than coal** (~434 g vs. ~1023 g CO₂/kWh), but still significant.
- **Water loss (1.17 L)** per kWh is also high—mainly due to **cooling systems**.
- **Ionizing radiation and toxicity** exist due to material extraction and processing.

Technology examples : Nuclear

Figure 35 Lifecycle impacts of nuclear power, global average reactor, per kWh and activity



Key Environmental Impact Values (per 1 kWh):

Impact Category	Value
Climate change	5.13 g CO ₂ -eq
Freshwater eutrophication	5.84 mg P-eq
Carcinogenic effects	5.12e-07 mCTUh
Ionizing radiation	14.3 g ²³⁵ U-eq
Land use	0.0577 points
Dissipated water	2.42 liters
Minerals and metals	0.331 mg Sb-eq

Key Observations from the Breakdown:

- **Extremely low CO₂ emissions** (only ~5 g CO₂/kWh) – among the **lowest of all technologies**.
- ~**65–70%** of climate impact is from **fuel production**, which includes:
 - **Mining** (open pit, underground, ISL)
 - **Milling**
 - **Conversion**
 - **Enrichment**
 - **Fuel fabrication**

- **Construction**

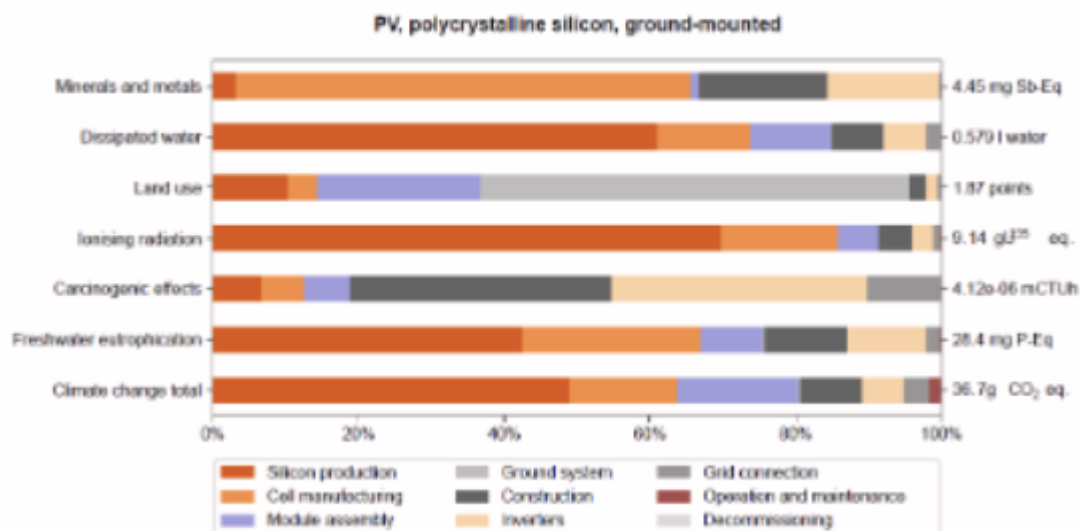
- **Electricity production** plays a smaller role in emissions but a larger role in **water use** (cooling).
- **Decommissioning, transportation, and spent fuel management** contribute noticeably—more than in other technologies.
- **Dissipated water:** 2.42 L/kWh—mainly from **cooling systems** during electricity generation.
- **Ionizing radiation and carcinogenic effects** are significant compared to other energy types, mostly due to **mining and fuel cycle steps**.

Conclusion:

- Nuclear has **very low climate impact** but **high complexity** and impacts in other categories (especially radiation and resource usage).
- Cooling water needs and fuel preparation are key contributors to overall life cycle impact.

Technology examples : Solar PV

Figure 25 Life cycle impacts from 1 kWh of poly-Si, ground-mounted, photovoltaic power production, Europe, 2020



Key Environmental Impact Values (per 1 kWh):	
Impact Category	Value
Climate change	36.7 g CO ₂ -eq
Freshwater eutrophication	28.4 mg P-eq
Carcinogenic effects	4.12e-06 mCTUh
Ionizing radiation	9.14 g ²³⁵ U-eq
Land use	1.87 points
Dissipated water	0.579 liters
Minerals and metals	4.45 mg Sb-eq

Breakdown of Emission Sources:

- **CO₂ emissions (36.7 g/kWh) are mostly (95%) from manufacturing**, including:
 - **Silicon production**
 - **Cell manufacturing**
 - **Module assembly**
 - **Construction and ground system**
- **Only ~5%** of emissions come from **operation and maintenance**.
- **Inverters, grid connection, and decommissioning** are minor contributors.

Other Environmental Insights:

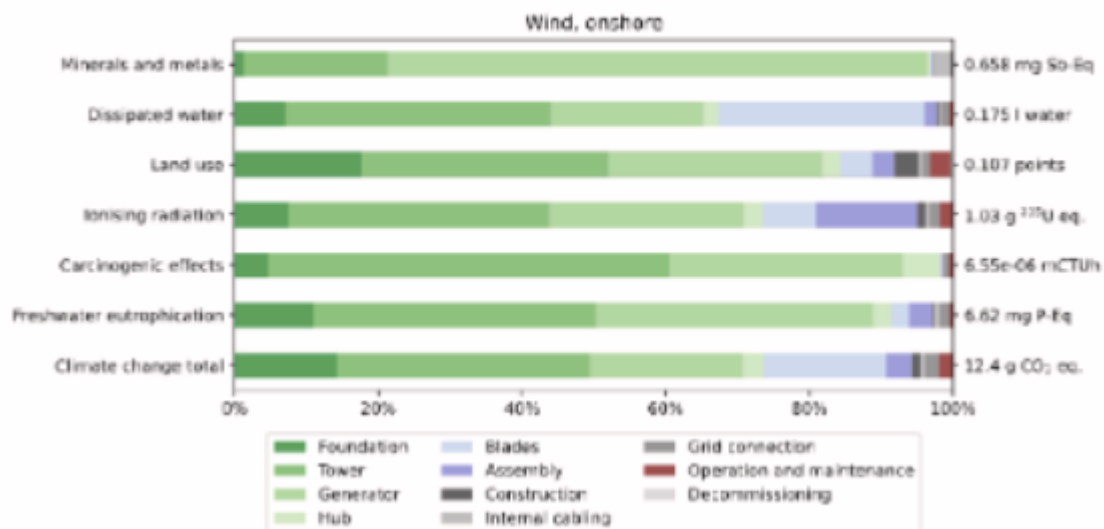
- **Water use (0.579 L/kWh)** is relatively low, with most from **silicon production and manufacturing**.
- **Ionizing radiation and toxicity** impacts exist, due to material sourcing and industrial processes.
- **Land use** impact is moderate (1.87 points), due to the physical footprint of ground-mounted systems.

Conclusion:

- Solar PV offers **very low CO₂ emissions** during operation.
- **Manufacturing stage** is the main contributor to lifecycle impacts.
- Still one of the **cleanest and most sustainable** technologies overall.

Technology examples : Onshore wind

Figure 12 Life cycle impacts from 1 kWh of onshore wind power production, Europe, 2020



Key Environmental Impact Values (per 1 kWh):

Impact Category	Value
Climate change	12.4 g CO ₂ -eq
Freshwater eutrophication	6.62 mg P-eq
Carcinogenic effects	6.55e-06 mCTUh
Ionizing radiation	1.03 g ²³⁵ U-eq
Land use	0.107 points
Dissipated water	0.175 liters
Minerals and metals	0.658 mg Sb-eq

Breakdown of CO₂ Emissions Sources:

- ~95% of emissions stem from **manufacturing and construction**, including:
 - **Foundation, tower, generator, hub, blades**
 - **Assembly, internal cabling, grid connection**
- **Operation & maintenance** contribute minimally.
- **Decommissioning** is almost **negligible**.

Water Use:

- **0.175 L of water per kWh**, mostly due to **manufacturing processes**.
- Very low compared to other technologies like nuclear or coal.

Conclusion:

- Onshore wind is one of the **lowest-emission technologies** (12.4 g CO₂/kWh).
- The **main environmental burden** lies in the **initial manufacturing**, not the operation.
- A highly sustainable energy source with minimal ongoing impacts

LCA impact summary based on main technologies

Technology and LCA Impact	Climate change total [g CO ₂ / kWh]	Freshwater eutrophication [mg P-Eq/kWh]	Carcinogenic effects [mCTUh/kWh]	Ionizing radiation [g ²³⁵ U Eq/kWh]	Land use [points/kWh]	Dissipated water [liter/kWh]	Minerals and metals [mg Sb-Eq / kWh]	Source
coal	1023	489	7,34E-06	9,07	2,43	2,86	0,525	Fig 4
Natural gas	434	19,7	1,33E-06	9,24	0,195	1,17	0,243	Fig 9
Nuclear	5,13	5,84	5,12E-07	14,3	0,0577	2,42	0,331	Fig 35
Solar PV	36,7	28,4	4,12E-06	9,14	1,87	0,579	4,45	Fig 25
Onshore wind	12,4	6,62	6,55E-06	1,03	0,107	0,175	0,658	Fig 12
Source :	Carbon Neutrality In the UNECE Region: Integrated Life-cycle Assessment of Electricity Sources, UN (2022)							

- Largest difference between technologies seen for climate change (CO₂e emission) ~ a factor of 200
- Smaller difference for other impacts such as Ionizing radiation (~ 14) and Dissipated water (~ 16)
- Table 13 and 14 on the next slides provide the full overview of the technology impacts

Key Takeaways:

Climate Change (CO₂ Emissions):

- **Coal is worst**, while **wind and nuclear** are best (factor ~200 difference).
- **PV** and **natural gas** are moderate, but far better than coal.

Freshwater Eutrophication:

- **Coal dominates** due to runoff and nutrient release.
- **Wind and nuclear** have the lowest freshwater impact.

Ionizing Radiation:

- **Nuclear has highest impact**, but others (e.g., gas, PV) also contribute due to **mining activities**.

- **Wind has the lowest radiation impact.**

Carcinogenic Effects:

- Roughly similar across most technologies.
- Still a factor to monitor, especially for coal and PV.

Land Use:

- **Solar PV requires most land** (due to ground-mounted installations).
- **Nuclear and wind** are much less land-intensive—wind even allows **co-use** (e.g., farming).

Dissipated Water:

- **Coal and nuclear** require most water (mainly for cooling).
- **Wind and PV** are **much lower**.

Minerals and Metals:

- All technologies require rare materials.
- **Solar PV** uses the most scarce materials, reflecting the intensive nature of **panel production**.

Final Note:

- **CO₂ emissions** vary widely by energy source and should be the primary focus for climate policy.
- However, **other impacts** (e.g., toxicity, water, land use, radiation) matter for **sustainable decision-making**.
- **Solar and wind** shine in most categories, but **manufacturing stages** are still impactful.
- **Nuclear** is extremely low-emission but requires careful handling of **radioactive impacts**.

Table 13 LCIA results for region EUR (Europe EU28), per kWh, in 2020, for select indicators, rounded to two significant figures.

PER kWh		CLIMATE CHANGE [g CO ₂ eq.]	FRESHWATER EUTROPHICATION [mg P eq.]	CARCINOGENIC EFFECTS [μCTUh]	IONISING RADIATION [g 1 ¹³¹ I eq.]	LAND USE [points]	DISSIPATED WATER [L]	MINERALS AND METALS [kg SO ₂ eq.]
Hard coal	PC, without CCS	1000	400	7.5	0.1	2.4	2.9	520
Hard coal	ISCC, without CCS	850	420	6.4	7.5	2.1	1.7	580
Hard coal	SC, without CCS	950	400	6.9	8.2	2.3	2.6	500
Natural gas	NGCC, without CCS	430	20	1.5	0.2	0.2	1.2	240
Hard coal	PC, with CCS	370	090	10	15	5.4	5.1	780
Hard coal	ISCC, with CCS	280	570	8.6	10	2.8	2.7	690
Hard coal	SC, with CCS	330	640	9.7	12	3.2	4.6	740
Natural gas	NGCC, with CCS	110	24	1.7	11	0.24	2.00	310
Hydro	660 MW	150	1.5	2.6	12	2.5	0.37	610
Hydro	360 MW	11	1.3	0.35	0.84	0.11	0.099	61
Nuclear	storage	5.1	5.8	0.51	14	0.058	2.4	310
CSP	tower	22	13	2.1	4.5	3.6	0.16	340
CSP	trough	42	14	6.3	5.1	3.5	0.34	650
PV	poly-Si, ground-mounted	37	28	4.1	0.1	1.9	0.58	4500
PV	poly-Si, roof-mounted	37	30	1.6	5.8	0.86	0.63	7700
PV	CdTe, ground-mounted	12	8.8	3.4	1.9	1.4	0.13	1500
PV	CdTe, roof-mounted	15	14	1.1	1.0	0.15	0.16	2600
PV	CIGS, ground-mounted	11	8.8	3.4	1.8	1.3	0.13	1700
PV	CIGS, roof-mounted	14	14	1.1	1.8	0.15	0.16	2800
Wind	onshore	12	6.2	6.6	1.0	0.11	0.18	680
Wind	offshore, concrete foundation	14	7.0	5.5	1.2	0.11	0.16	980
Wind	offshore, steel foundation	13	6.8	7	1.2	0.099	0.16	990

Full LCIA results

Table 14 is summarizing the resulting emission per kWh electricity produces in Europe

Table 14 LCIA results for region EUR (Europe EU 28), in 2020, all ILCD 2.0 indicators, three significant figures . Climate change

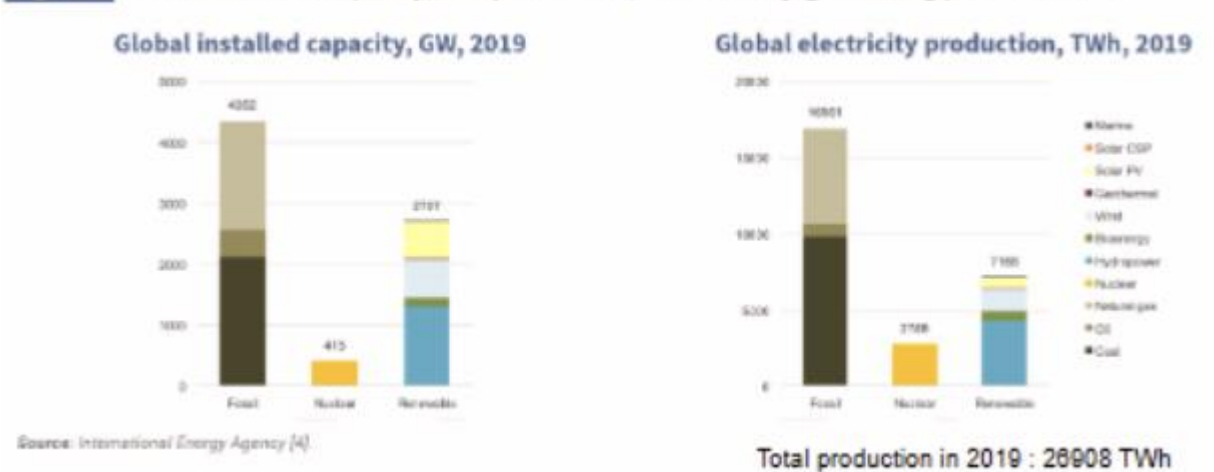
PRODUCT		INDICATORS							
		CLIMATE CHANGE: GLOBAL WARMING POTENTIAL	CLIMATE CHANGE: ACID EQUIVALENT POTENTIAL	CLIMATE CHANGE: OZONE DEPLETION POTENTIAL	CLIMATE CHANGE: HAPs	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND
		kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	
Material	PC, without CO ₂	6.87E-05	1.02E-08	1.07E-04	1.80E-08	1.78E-03	6.72E-01	6.80E-04	
Material	GGC, without CO ₂	1.38E-05	6.48E-01	1.48E-04	3.40E-02	1.89E-03	1.48E-01	4.19E-04	
Material	SC, without CO ₂	6.43E-05	9.53E-01	1.58E-04	9.53E-02	1.83E-03	4.23E-01	4.53E-04	
Natural gas	NGAL, without CO ₂	1.78E-05	6.34E-02	6.23E-05	4.34E-02	1.88E-04	1.38E-01	1.97E-05	
Material	PC, with CO ₂	1.86E-04	1.04E-01	2.47E-04	1.80E-08	1.89E-03	6.24E-01	7.28E-04	
Material	GGC, with CO ₂	7.26E-05	1.75E-04	1.98E-04	9.79E-04	1.93E-03	4.84E-01	1.94E-04	
Material	SC, with CO ₂	6.98E-05	1.88E-01	2.98E-04	1.30E-02	1.89E-03	1.52E-01	4.93E-04	
Natural gas	NGAL, with CO ₂	6.88E-05	1.29E-02	9.98E-05	1.24E-02	6.87E-04	2.94E-01	7.43E-05	
Hydro	H2H4H	6.33E-05	1.49E-01	1.08E-04	1.49E-02	4.23E-04	1.97E-01	1.28E-05	
Hydro	H2H4H	1.88E-05	1.07E-03	9.33E-05	1.97E-03	4.48E-05	1.73E-02	1.33E-05	
Number	average	1.58E-05	1.24E-03	1.24E-05	1.24E-03	4.28E-05	1.74E-02	6.43E-06	
CO ₂	lower	9.33E-05	2.14E-02	1.38E-05	1.37E-02	9.24E-05	1.83E-02	1.21E-05	
CO ₂	rough	6.57E-05	4.19E-02	6.68E-05	4.28E-02	1.83E-04	1.88E-01	2.88E-05	
PV	poly-PC, ground-mounted	6.43E-04	6.68E-01	1.63E-04	1.47E-02	9.68E-04	1.63E-01	6.43E-04	
PV	poly-PC, roof-mounted	6.43E-04	6.68E-01	1.63E-04	1.47E-02	9.68E-04	1.63E-01	6.43E-04	
PV	GFC, ground-mounted	6.43E-05	1.13E-02	2.98E-05	1.47E-02	6.83E-05	3.88E-02	1.43E-05	
PV	GFC, roof-mounted	1.58E-05	1.48E-02	4.58E-05	1.47E-02	6.83E-05	3.88E-02	1.43E-05	
PV	GFC, ground-mounted	6.58E-05	1.13E-02	2.58E-05	1.47E-02	6.83E-05	3.88E-02	1.23E-05	
PV	GFC, roof-mounted	1.47E-05	1.48E-02	4.58E-05	1.47E-02	6.83E-05	3.88E-02	1.23E-05	
Steel	crushed	1.87E-05	1.94E-03	1.08E-05	1.94E-03	1.38E-05	7.48E-03	1.94E-05	
Steel	crushed, concrete foundation	1.74E-05	1.43E-03	1.08E-05	1.43E-03	1.38E-05	6.83E-02	1.94E-05	
Steel	crushed, steel foundation	1.87E-05	1.94E-03	1.08E-05	1.94E-03	1.38E-05	6.83E-02	1.94E-05	

PRODUCT		INDICATORS									
		CLIMATE CHANGE: GLOBAL WARMING POTENTIAL	CLIMATE CHANGE: ACID EQUIVALENT POTENTIAL	CLIMATE CHANGE: OZONE DEPLETION POTENTIAL	CLIMATE CHANGE: HAPs	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND	CLIMATE CHANGE: RESOURCES AND ENERGY DEMAND
		kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq	kg CO ₂ -Eq
Material	PC, without CO ₂	6.87E-05	1.02E-08	1.07E-04	1.80E-08	1.78E-03	6.72E-01	6.80E-04	6.87E-05	1.02E-08	1.07E-04
Material	GGC, without CO ₂	1.38E-05	6.48E-01	1.48E-04	3.40E-02	1.89E-03	1.48E-01	4.19E-04	1.38E-05	6.48E-01	1.48E-04
Material	SC, without CO ₂	6.43E-05	9.53E-01	1.58E-04	9.53E-02	1.83E-03	4.23E-01	4.53E-04	6.43E-05	9.53E-01	1.58E-04
Natural gas	NGAL, without CO ₂	1.78E-05	6.34E-02	6.23E-05	4.34E-02	1.88E-04	1.38E-01	1.97E-05	1.78E-05	6.34E-02	6.23E-05
Material	PC, with CO ₂	1.86E-04	1.04E-01	2.47E-04	1.80E-08	1.89E-03	6.24E-01	7.28E-04	1.86E-04	1.04E-01	2.47E-04
Material	GGC, with CO ₂	7.26E-05	1.75E-04	1.98E-04	9.79E-04	1.93E-03	4.84E-01	1.94E-04	7.26E-05	1.75E-04	1.98E-04
Material	SC, with CO ₂	6.98E-05	1.88E-01	2.98E-04	1.30E-02	1.89E-03	1.52E-01	4.93E-04	6.98E-05	1.88E-01	2.98E-04
Natural gas	NGAL, with CO ₂	6.88E-05	1.29E-02	9.98E-05	1.24E-02	6.87E-04	2.94E-01	7.43E-05	6.88E-05	1.29E-02	9.98E-05
Hydro	H2H4H	6.33E-05	1.49E-01	1.08E-04	1.49E-02	4.23E-04	1.97E-01	1.28E-05	6.33E-05	1.49E-01	1.08E-04
Hydro	H2H4H	1.88E-05	1.07E-03	9.33E-05	1.97E-03	4.48E-05	1.73E-02	1.33E-05	1.88E-05	1.07E-03	9.33E-05
Number	average	1.58E-05	1.24E-03	1.24E-05	1.24E-03	4.28E-05	1.74E-02	6.43E-06	1.58E-05	1.24E-03	1.24E-05
CO ₂	lower	9.33E-05	2.14E-02	1.38E-05	1.37E-02	9.24E-05	1.83E-02	1.21E-05	9.33E-05	2.14E-02	1.38E-05
CO ₂	rough	6.57E-05	4.19E-02	6.68E-05	4.28E-02	1.83E-04	1.88E-01	2.88E-05	6.57E-05	4.19E-02	6.68E-05
PC	poly-PC, ground-mounted	6.43E-04	6.68E-01	1.63E-04	1.47E-02	9.68E-04	1.63E-01	6.43E-04	6.43E-04	6.68E-01	1.63E-04
PC	poly-PC, roof-mounted	6.43E-04	6.68E-01	1.63E-04	1.47E-02	9.68E-04	1.63E-01	6.43E-04	6.43E-04	6.68E-01	1.63E-04
PC	GFC, ground-mounted	6.43E-05	1.13E-02	2.98E-05	1.47E-02	6.83E-05	3.88E-02	1.43E-05	6.43E-05	1.13E-02	2.98E-05
PC	GFC, roof-mounted	1.58E-05	1.48E-02	4.58E-05	1.47E-02	6.83E-05	3.88E-02	1.43E-05	1.58E-05	1.48E-02	4.58E-05
PC	GFC, ground-mounted	6.58E-05	1.13E-02	2.58E-05	1.47E-02	6.83E-05	3.88E-02	1.23E-05	6.58E-05	1.13E-02	2.58E-05
PC	GFC, roof-mounted	1.47E-05	1.48E-02	4.58E-05	1.47E-02	6.83E-05	3.88E-02	1.23E-05	1.47E-05	1.48E-02	4.58E-05
Steel	crushed	1.87E-05	1.94E-03	1.08E-05	1.94E-03	1.38E-05	7.48E-03	1.94E-05	1.87E-05	1.94E-03	1.08E-05
Steel	crushed, concrete foundation	1.74E-05	1.43E-03	1.08E-05	1.43E-03	1.38E-05	6.83E-02	1.94E-05	1.74E-05	1.43E-03	1.08E-05
Steel	crushed, steel foundation	1.87E-05	1.94E-03	1.08E-05	1.94E-03	1.38E-05	6.83E-02	1.94E-05	1.87E-05	1.94E-03	1.08E-05

Estimating the consequence of an electricity mix

If we know the mix of the electricity producing technologies and the total production then we can estimate the impact and compare this with a defined limit

Figure 2 Global installed capacity, and production, of electricity-generating plants in 2019



🚧 Global Installed Capacity (GW):

- **Fossil fuels:** 4,352 GW
- **Renewables:** 2,707 GW
- **Nuclear:** 415 GW

⚡ Global Electricity Production (TWh):

- **Fossil fuels:** 16,951 TWh
- **Nuclear:** 2,789 TWh
- **Renewables:** 7,168 TWh

🌍 Key Takeaways:

1. **Fossil fuels dominate** global production:
 - Despite being ~60% of installed capacity, they generate ~63% of total electricity.
 - Reflects **high capacity factors** for coal and gas plants.
2. **Renewables have lower output relative to installed capacity:**
 - Although renewables make up ~38% of installed capacity, they contribute only ~27% of global production.

- This is due to **intermittency** (e.g. solar and wind are weather-dependent) and **lower capacity factors**.

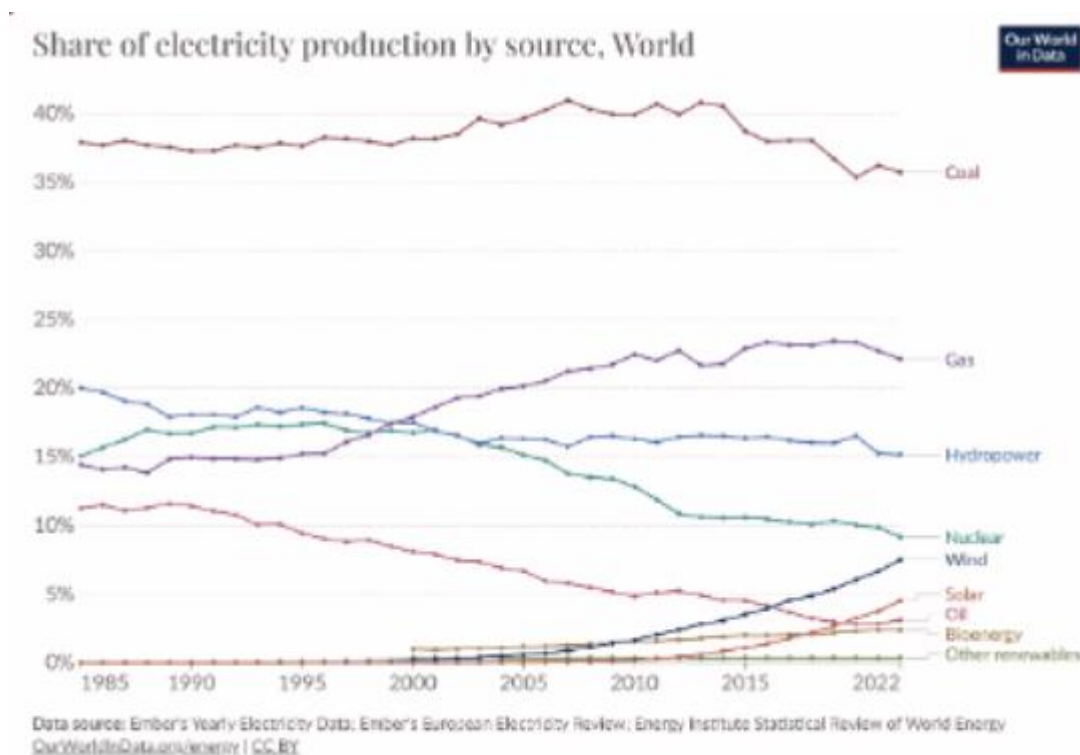
3. Nuclear is efficient but limited in scale:

- Small capacity (~6%) delivers ~10% of global electricity.
- Reflects high reliability and **high capacity factor**.

Implication:

- If we want to reduce global CO₂ emissions, shifting the **electricity mix** toward **low-emission sources (nuclear, wind, solar, hydro)** is crucial.
- Knowing the **share of each source** lets us **estimate total emissions** from electricity and **compare it to climate targets** (e.g., IPCC budgets).

Share of electricity production by source, world



Key Trends and Observations:

Coal

- Dominant source for global electricity (~35–40% share).

- **Stable from 1985 to ~2015, then gradual decline.**
- Still the **largest contributor** to electricity production and emissions.

● Gas

- **Increased steadily**, particularly after 2000.
- Helped **replace oil** as a source for electricity.
- Now at **~23% share**.

● Hydropower

- Long-established renewable.
- **Stable share (~15%)** since the 1990s.
- Limited growth due to **geographical constraints**.

● Nuclear

- Peaked in the **1990s (~17%)**, then **gradually declined** to below 10%.
- Decline driven by public concern, policy shifts, and aging plants.

🌀 Wind

- **Significant growth since 2005.**
- Now **above 7%** of global production.
- Expected to continue increasing due to falling costs and policy support.

☀️ Solar PV

- **Fastest-growing** technology.
- Near-zero share before 2005, now **over 3%**.
- Strong upward trend expected to continue.

● Oil

- Historically important, now **only ~2%** of electricity production.
- Declining due to cost, emissions, and better alternatives.

🌱 Bioenergy and Other Renewables

- Small but **growing segments**.

🧠 Contextual Notes:

- **1990** is used as the baseline for **global emission targets** (e.g. Kyoto Protocol, IPCC).
- The rise of **wind and solar** marks a shift toward **low-emission technologies**.
- Yet, fossil fuels (coal + gas) still account for **~60%** of global electricity.

Example of calculating world CO₂ emission from coal, gas, nuclear and hydro power

- The electricity production from coal, gas, nuclear and hydro power in 2019 is
 $E_{\text{Coal}} \sim 9500 \text{ TWh}$, $E_{\text{Gas}} \sim 6500 \text{ TWh}$, $E_{\text{Nuclear}} \sim 2800 \text{ TWh}$ & $E_{\text{Hydro}} \sim 4000 \text{ TWh}$ (Fig 2 in slide 26)
- And the climate change total emission factors with no Carbon Capture and Storage are

$EF_{\text{Climate change, Coal}}$	~ 1.02	kg CO ₂ e / kWh	
$EF_{\text{Climate change, Gas}}$	~ 0.434	kg CO ₂ e / kWh	
$EF_{\text{Climate change, Nuclear}}$	$\sim 5.29 \cdot 10^{-3}$	kg CO ₂ e / kWh	
$EF_{\text{Climate change, Hydro}}$	$\sim 1.07 \cdot 10^{-2}$	kg CO ₂ e / kWh	(Table 14 in slide 24)
- The resulting Climate Change (CO₂e) emission then becomes

$EM_{\text{Climate change, Coal}}$	$= EF_{\text{Climate change, Coal}} \times E_{\text{Coal}} = 1.02 \text{ kg CO}_2\text{e} / \text{kWh} \times 9500 \cdot 10^9 \text{ kWh} = 9.7 \cdot 10^{12} \text{ kg CO}_2\text{e}$	$= 9.7 \text{ Gt CO}_2\text{e}$
$EM_{\text{CC, gas}}$	$= EF_{\text{CC, gas}} \times E_{\text{Gas}} = 0.434 \text{ kg CO}_2\text{e} / \text{kWh} \times 6500 \cdot 10^9 \text{ kWh} = 2.8 \cdot 10^{12} \text{ kg CO}_2\text{e}$	$= 2.8 \text{ Gt CO}_2\text{e}$
$EM_{\text{CC, Nuclear}}$	$= EF_{\text{CC, Nuclear}} \times E_{\text{Nuclear}} = 5.29 \cdot 10^{-3} \text{ kg CO}_2\text{e} / \text{kWh} \times 2800 \cdot 10^9 \text{ kWh} = 1.5 \cdot 10^{10} \text{ kg CO}_2\text{e}$	$= 0.02 \text{ Gt CO}_2\text{e}$
$EM_{\text{CC, Hydro}}$	$= EF_{\text{CC, Hydro}} \times E_{\text{Hydro}} = 1.07 \cdot 10^{-2} \text{ kg CO}_2\text{e} / \text{kWh} \times 4000 \cdot 10^9 \text{ kWh} = 4.3 \cdot 10^{10} \text{ kg CO}_2\text{e}$	$= 0.04 \text{ Gt CO}_2\text{e}$
- Thus the CO₂ emission from nuclear and hydro is only about 0.6 % of the coal emission.

The unit Giga-ton(Gt)
is the same as 10⁹ ton

1 Gt = 10⁹ · 10³ kg
= 10¹² kg

⚡ Electricity Production (TWh) in 2019

- **Coal:** 9,500 TWh
- **Gas:** 6,500 TWh
- **Nuclear:** 2,800 TWh
- **Hydro:** 4,000 TWh

🌍 Emission Factors (EF) – kg CO₂e/kWh

- **Coal:** 1.02
- **Gas:** 0.434
- **Nuclear:** 0.00529
- **Hydro:** 0.0107



Calculated Climate Impact – CO₂e Emissions (2019)

(Multiply EF by electricity output)

Source	Emission (Gt CO ₂ e)
Coal	9.7 Gt CO ₂ e
Gas	2.8 Gt CO ₂ e
Nuclear	0.02 Gt CO ₂ e
Hydro	0.04 Gt CO ₂ e

■ Total from nuclear + hydro = ~0.06 Gt CO₂e

➡ Only ~0.6% of the CO₂ emissions from coal!



Key Takeaways:

- **Coal dominates CO₂ emissions** from electricity, contributing nearly 10 Gt CO₂e in 2019.
- **Gas is better than coal**, but still substantial.
- **Nuclear and hydro are extremely low-emission** energy sources.
- Replacing coal with low-carbon technologies (especially nuclear/hydro/renewables) would drastically reduce global emissions.



Quick Reference:

- 1 TWh = 10⁹ kWh
- 1 Gt = 10¹² kg = 1 billion metric tons

Global boundary on green house gas

The UN Emission Gap Report 2022 provides an estimate of the difference between the global emissions and the needed trajectory to fulfill the Paris agreement of a 1.5 °C, 1.8 °C and 2.0 °C global temperature increase by 2100. It is the accumulation of green house gasses in

the atmosphere that dictates the future allowed emission levels.

- Currently the global emissions are approximately 53 Gt CO₂e per year.

- Given the global electricity mix and resulting emissions one can estimate the fraction of the total permitted emission that is provided by electricity generation

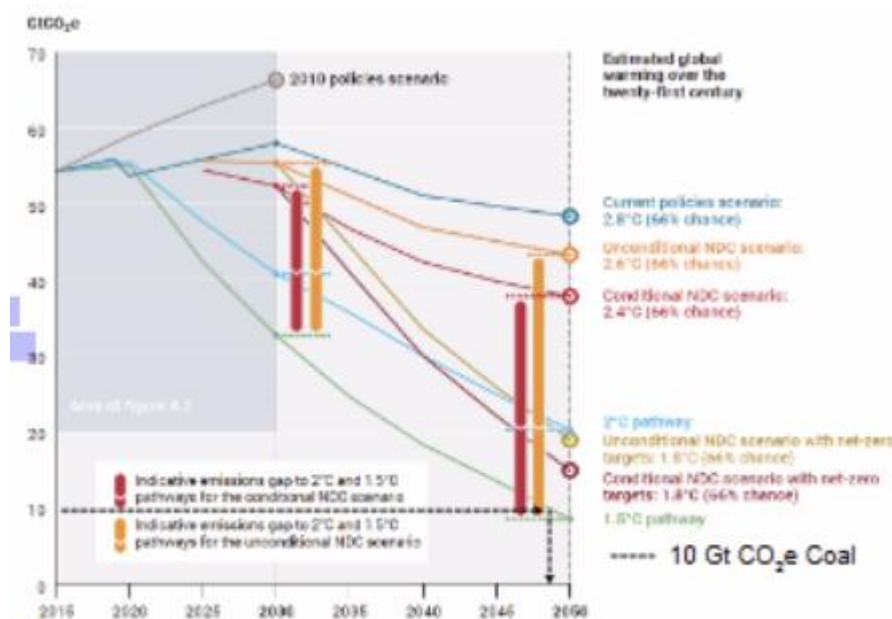


Figure 4.3 in the UN Report "The closing window – Emission gap report 2022"

🌍 Global Boundary on Greenhouse Gas Emissions

Source: UN Emissions Gap Report 2022

Focus: Tracking emissions vs. Paris Agreement targets (1.5°C, 1.8°C, 2.0°C)

📊 Current Global Emissions (2022–2023)

- Approximately **53 Gt CO₂e/year**

📈 Required Emission Trajectories to Meet Climate Targets

- **1.5°C pathway:** Rapid reduction needed to below **20 Gt CO₂e/year by 2045**
- **1.8°C pathway:** More gradual decline, but still significant
- **2.0°C pathway:** Slightly less ambitious, emissions must still fall notably
- **Current policies scenario:** Would result in **~2.8°C warming, insufficient reduction**
- **2010 policy scenario:** Even worse trajectory, **~60+ Gt CO₂e/year**

⚠️ Coal Alone Is a Major Contributor

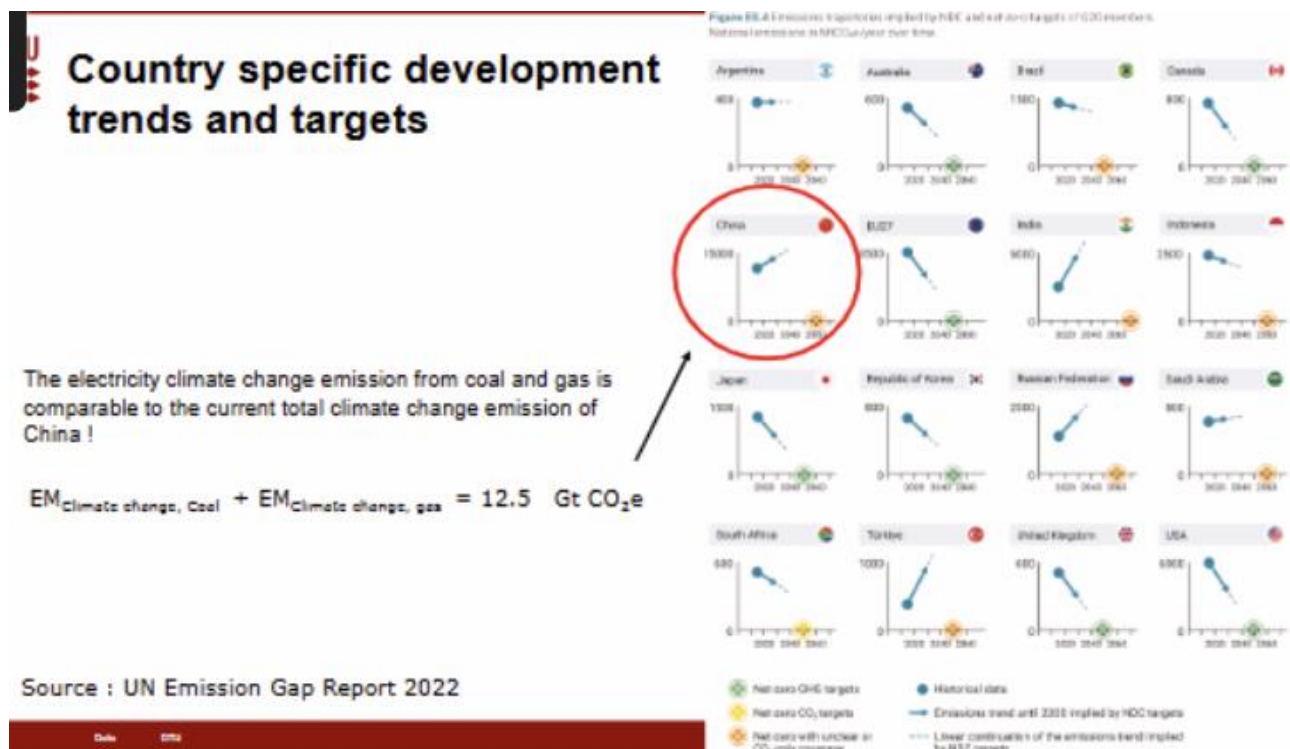
- Just coal-fired electricity emits:
 - ~10 Gt CO₂e/year
 - This alone places us well above the 1.5°C trajectory line
 - If coal isn't significantly reduced, the Paris Agreement goals are out of reach

💡 Key Insight

Reducing emissions from coal electricity is critical

Even a complete phase-out of coal could shift global emissions toward a 1.5–2.0°C compatible pathway.

Country specific development trends and target



🌍 Country-Specific Development Trends & Targets

Key Insight:

The combined electricity emissions from coal and gas (12.5 Gt CO₂e) are comparable to the entire annual emissions of China, the world's largest emitter.

Emissions Breakdown:

- **Coal (2019):** ~10 Gt CO_{2e}
 - **Gas (2019):** ~2.5 Gt CO_{2e}
 - **Total (electricity from fossil fuels):** ≈ 12.5 Gt CO_{2e}
-

Comparison with Countries:

- **China's total emissions (2020):** ≈ 12 Gt CO_{2e}
 - Coal + gas for electricity alone = **as much as China's total footprint**
 - Highlights **the immense weight of electricity generation** in global emissions
-

Why This Matters:

- It places electricity generation from **fossil fuels** into **global political context**
- Shows that **decarbonizing electricity** is just as important as national-scale reductions
- Urges transition to **low-emission technologies** like nuclear, hydro, wind, and sola

Comparison of electricity climate change emission to total global climate change emission limit

- The Coal climate change emission was found to be $EM_{CC,Coal} = 9.7 \text{ Gt CO}_2\text{e}$ in 2019
- The global emission limit was around $EM_{CC,Global} = 53 \text{ Gt CO}_2\text{e}$ (Fig ES.3 on slide 27)

Thus the fraction of the planetary limit from the electricity production based on coal is

$$f_{Coal} = \frac{EM_{CC,Coal}}{EM_{CC,Global}} = \frac{9.7 \text{ Gt CO}_2\text{e}}{53 \text{ Gt CO}_2\text{e}} = 18 \%$$

And if the emission from electricity production from natural gas is also included then one gets

$$f_{Coal+Gas} = \frac{EM_{CC,Coal} + EM_{CC,Gas}}{EM_{CC,Global}} = \frac{9.7 \text{ Gt CO}_2\text{e} + 2.8 \text{ Gt CO}_2\text{e}}{53 \text{ Gt CO}_2\text{e}} = 24 \%$$

If the 2030 climate emission target of 1.5 °C is used $EM_{CC,Global} = 33 \text{ Gt CO}_2\text{e}$ then $f_{Coal+Gas} = 38 \%$, which shows that the electricity production has a large impact on the climate change emissions. Unchanged coal and gas will violate limit around 2045.

Comparison of Electricity Emissions to Global Limits

Key Figures:

- **Global CO₂ emissions (2019):** ~53 Gt CO₂e
- **Coal-based electricity emissions:** ~9.7 Gt CO₂e
- **Gas-based electricity emissions:** ~2.8 Gt CO₂e
 - **Total (coal + gas):** 12.5 Gt CO₂e

Fraction of Global Limit (2019 Baseline):

- $f_{\text{coal}} = \frac{9.7}{53} \approx 18\%$
- $f_{\text{coal+gas}} = \frac{9.7+2.8}{53} = \frac{12.5}{53} \approx 24\%$

→ Just coal and gas-based electricity accounted for nearly 1/4 of all emissions allowed under 2019 levels.

Targeting 1.5 °C Scenario (2030 target: 33 Gt CO₂e):

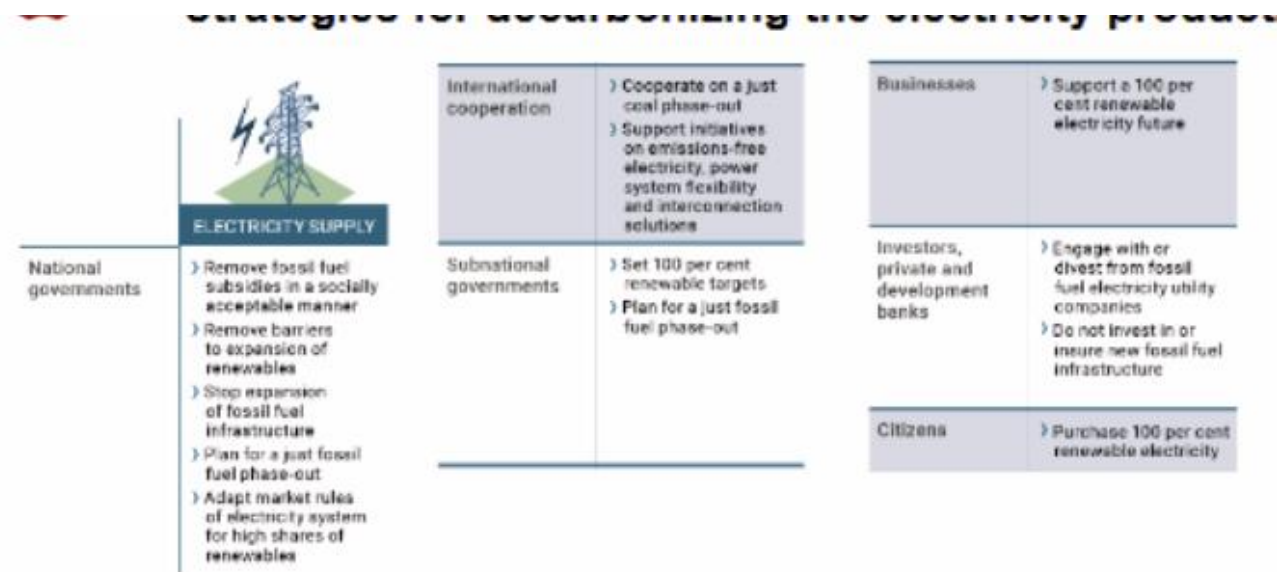
- $f_{\text{coal+gas}} = \frac{12.5}{33} \approx 38\%$

This means coal and gas power alone would take up more than a third of the entire emissions budget allowed to meet the 1.5 °C target.

Conclusion:

- If emissions from coal and gas **are not reduced**, they will **exceed the 1.5 °C-compatible planetary boundary by ~2045**.
- This underscores the **urgency** of a global shift toward **low-emission electricity sources** (renewables + nuclear).

Strategies for decarbonizing the electricity production



1. National Governments

- **End fossil fuel subsidies** in a **socially just** way.
 - **Remove legal/market barriers** to renewable energy expansion.
 - **Stop new fossil infrastructure** from being built.
 - **Plan for a just transition:** gradual fossil fuel phase-out with attention to equity.
 - **Adapt market rules** to accommodate **high shares of renewables** (e.g., grid management, pricing).
-

2. Subnational Governments

- **Set 100% renewable targets** at city, regional, or state level.
 - **Develop just transition plans** tailored to local contexts.
-

3. International Cooperation

- **Coordinate coal phase-out** across borders.
 - **Support emissions-free power systems**, interconnection, and system flexibility.
-

4. Businesses

- **Commit to 100% renewable electricity use.**
 - **Lead by example** in transitioning operations and supply chains.
-

5. Investors and Banks

- **Divest from fossil fuel utilities.**
 - **Avoid funding new fossil infrastructure.**
 - Redirect capital toward **renewable projects**.
-

6. Citizens

- **Choose 100% renewable electricity** when possible.
 - Support policies, utilities, and businesses aligned with the energy transition.
-

Takeaway

The transformation to a **zero-carbon electricity system** is feasible and requires:

- **Policy commitment**
- **Financial redirection**
- **Business innovation**
- **Individual action**

Conclusion

- The United Nations report "Carbon Neutrality in the UNECE Region: Integrated Life-cycle Assessment of Electricity Sources " (2022) provides a baseline reference on the emission expected from the technologies of the electricity mix of countries and regions.
- The electricity mix of different countries and regions can be found from statistical sources (Hannah Ritchie and Pablo Rosado (2020) - "Electricity Mix" Published online at [OurWorldInData.org](https://ourworldindata.org).)
- Impacts from the electricity mix can then be estimated using the reference values and compared to the UN emission gap report in order to understand how close the planetary climate change emission is to the planetary boundary.

The student of the course is encouraged to examine possible decarbonization strategies in the exercise in the following lectures.

Lecture 1: Conclusion – Life Cycle Impacts of Electricity Production

- The UN report "*Carbon Neutrality in the UNECE Region (2022)*" offers **standardized life-cycle emission values** for key electricity technologies (coal, gas, nuclear, hydro, wind, solar, etc.).
- These reference values allow us to:
 - **Quantify emissions** from an electricity mix.
 - **Compare electricity-related climate impacts** to global emission limits, including the **planetary boundary** for climate change.
- Country- and region-specific **electricity mix data** (e.g., from *OurWorldInData.org*) can be used in calculations to assess national and regional contributions to global emissions.
- This framework enables:
 - **Scenario analysis** (e.g., future mixes, decarbonization pathways).

- **Policy support** (e.g., understanding the urgency of coal/gas phase-outs).

Lecture 11 - Wind Energy as example

How sustainable is Wind Energy?

“Wind Energy is by definition green and once it supplies all production processed then the world will become purely green” (Wind sector around 2010)

There are however some challenges related to this point of view:

- The production of the materials will need energy that might not be green
- The production of materials might result in direct chemical emissions
- The installation and Operation and Maintenance might be fossil-based
- There are many emissions to the environment and how to define limits?
- Is the turbine fleet operated in a circular manner with zero waste, or are new materials needed to compensate for the waste?

• Life Cycle Analysis (LCA) is a method to quantify the impact on the environment (CO₂ emission per kWh produced energy as example, but 17 others exist)

Notes:

Reality check: Challenges to this claim

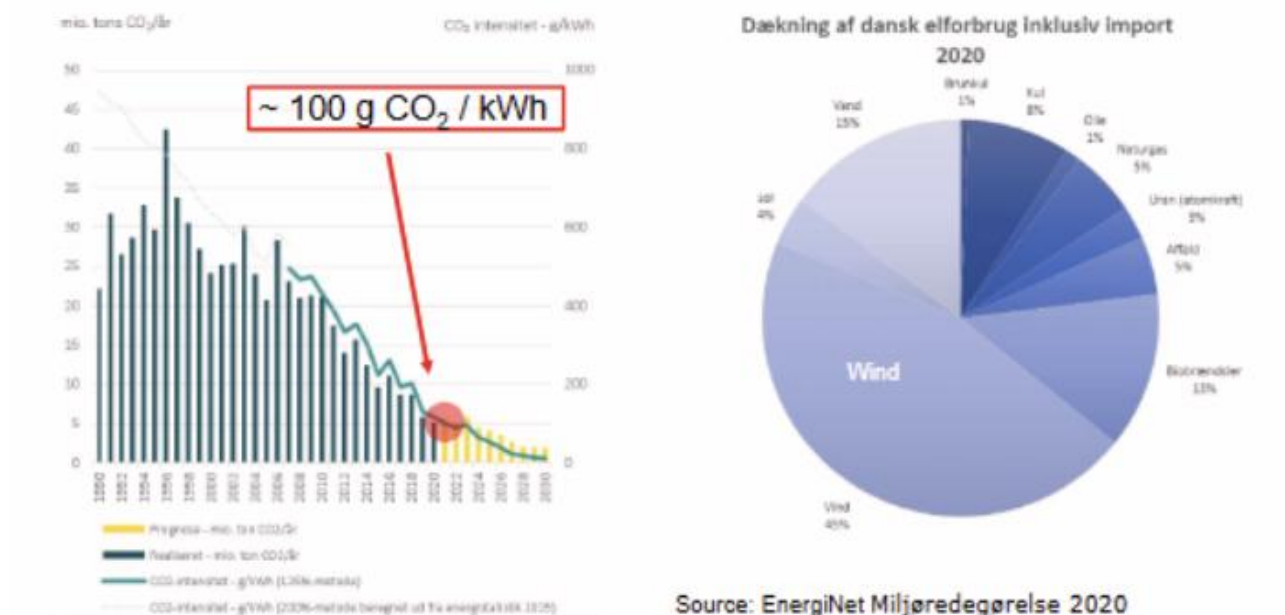
- **Material production** consumes energy, which might be fossil-based.
- **Manufacturing** could cause direct **chemical emissions**.
- **Installation, Operation & Maintenance (O&M)** often rely on fossil fuels.
- **Environmental emissions** (other than CO₂) must be defined and tracked.
- **End-of-life concerns:** Is the wind turbine fleet **circular** (i.e. zero-waste)? If not, virgin materials must compensate.

Using LCA to Quantify Sustainability

- **LCA (Life Cycle Analysis)** allows us to:
 - Calculate **CO₂ emissions per kWh** of electricity produced.
 - Examine other **17+ environmental impacts** (e.g., land use, water depletion, toxicity).
 - Assess true environmental cost—not just at operation, but across:

- Material extraction
- Manufacturing
- Transport
- Installation
- Operation & Maintenance
- End-of-life disposal/recycling

CO₂ emission from electricity of Denmark



DK Denmark: CO₂ Emission from Electricity Production

Trend in Emissions (Left graph)

- Shows a **consistent reduction** in Denmark's annual CO₂ emissions from electricity generation since 1990.
- By **2020**, CO₂ intensity reached approximately **100 g CO₂/kWh**.
- The decline is closely linked to changes in **fuel sources** and **renewable integration**.

Electricity Mix (Right pie chart, 2020)

- **Wind:** 45%
- **Biomass (Biofuels):** 13%
- **Hydro:** 15%

- **Solar:** 4%
- **Nuclear (imported):** 3%
- **Natural gas:** 5%
- **Waste:** 5%
- **Coal + Oil:** 9% combined (significantly reduced)

This mix helps explain the relatively low emission level of $\sim 100 \text{ g CO}_2/\text{kWh}$ — a value far below global fossil-based averages.

Next Analytical Focus

You noted a goal to **zoom in on wind energy** and assess **how much lower CO₂ emissions** can get. That sets up the next analysis:

- What is wind's **life cycle emission intensity**? (Already shown to be $\sim 12 \text{ g CO}_2/\text{kWh}$ for onshore in Lecture 1)
- Can Denmark **reach this minimum** if fossil fuels are entirely phased out?

Would you like to build a quick comparison between Denmark's current electricity CO₂ intensity and the theoretical minimum if it were powered 100% by wind?

The life cycle of offshore wind farm



Notes:

⚙️ Life Cycle Phases of Offshore Wind Farms and Their Environmental Impact

1. Manufacturing & Installation

- **Main materials:**
 - **Steel** (monopile, transition piece, tower)
 - **Glass fibre** (blades)
 - **Electronics, gear, generator** (nacelle)
- **Emissions source:** Extraction and processing of steel is CO₂ intensive.
- **Heavy transport and offshore piling** also contribute.

2. Operation & Maintenance (25 years)

- Typically **low emissions** during this phase.
- Emissions come from:
 - **Maintenance ships** (fossil-fueled)
 - Occasional **component replacement** (e.g., blades, nacelle parts)
- **Electricity produced** during this phase offsets emissions — key to calculating **g CO₂/kWh**.

3. Decommissioning

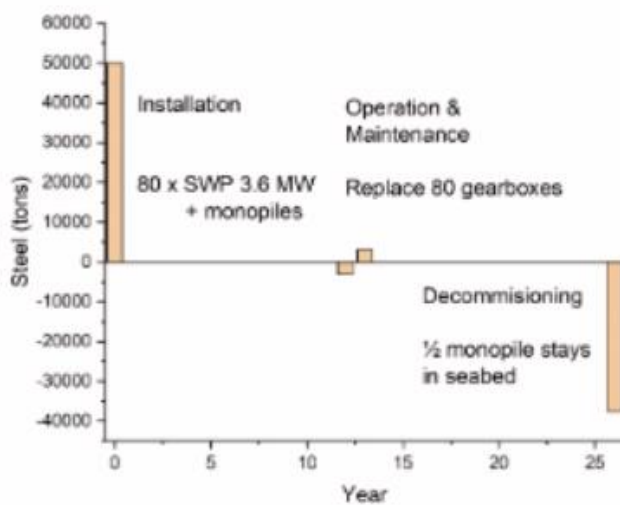
- Open question: do we leave a **clean seabed**, or is part of the monopile allowed to remain?
- This affects:
 - Environmental footprint
 - Marine ecosystem impact
 - **LCA boundary assumptions** (full removal vs. partial)

🔍 Why It Matters for LCA

This lifecycle framing is essential to:

- Calculate the **total embedded CO₂ emissions**
- Divide those emissions by total electricity output over 25 years
→ This gives the **CO₂ intensity (g CO₂/kWh)**

The steel life cycle balance of an offshore wind farm



CO₂ emission =

Bill of Material × emission per material +

Energy consumption × emission from energy +

Transport × emission per length +

- Recycled material (?)

Normalization:

Total energy produced in operational lifetime of the wind farm [kWh]

Notes:

Steel Life Cycle in Offshore Wind Farms (80 x 3.6 MW turbines)

- **Initial investment (installation):**
 - Requires approx. **50,000 tons of steel**
 - Includes monopiles, transition pieces, towers, nacelles
- **During lifetime:**
 - Small steel inflow: e.g., **80 gearboxes replaced** around year 12
- **End of life (decommissioning after 25 years):**
 - Only **~35,000 tons recovered**
 - **~15,000 tons remain** in seabed (1/2 of monopile)

CO₂ Emission Formula (LCA-based)

$$\text{CO}_2 \text{ emission (total)} = \underbrace{\text{Bill of Materials} \times \text{emission per material}}_{\text{e.g., tons steel} \times \text{kg CO}_2/\text{ton}} + \underbrace{\text{Energy use} \times \text{emission factor}}_{\text{e.g., diesel ships}} + \underbrace{\text{Transport distance} \times \text{emission rate}}_{\text{e.g., diesel ships}} + (- \text{recycled steel emission benefit})$$

Normalization:

To get g CO₂ per kWh, divide total emissions by energy output over 25 years:

$$\text{g CO}_2/\text{kWh} = \frac{\text{Total CO}_2 \text{ emissions [kg]}}{\text{Total energy produced [kWh]}}$$

For 80 turbines @ 3.6 MW:

- Total power = 288 MW
- Assume 45% capacity factor (typical offshore):

$$288 \text{ MW} \times 0.45 \times 8760 \text{ hrs/year} \times 25 \text{ years} = 28,400,000 \text{ MWh}$$

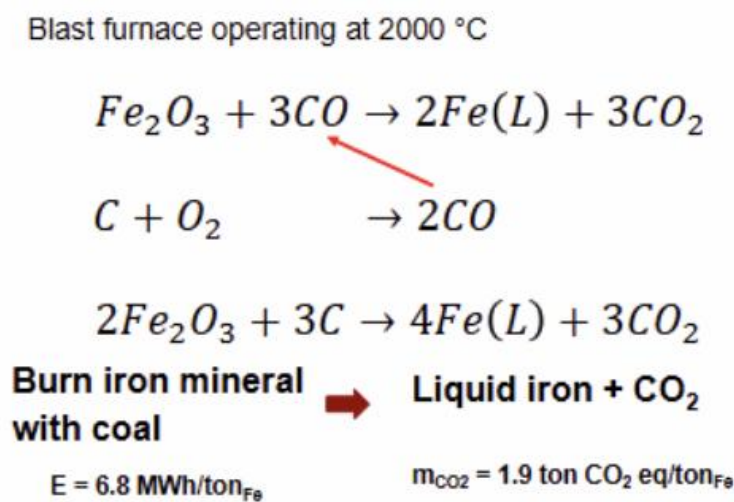
So you normalize by ~28.4 TWh of lifetime output.

💡 LCA Insight

This type of analysis:

- **Clarifies that wind energy is not emission-free**, especially due to steel and transport.
- But with proper recycling and large output, **CO₂ intensity remains very low** (in the range of 10–20 g CO₂/kWh).
- This is **10–100× lower** than fossil-based electricity.

Manufacturing of iron : From mine ore to metal



Notes:

🔥 **Steel Manufacturing = Embedded CO₂ Emissions**

🔪 **Chemical Process in a Blast Furnace**

- **Input:** Iron ore (Fe₂O₃), Coal (C), Oxygen (O₂)
- **Temperature:** ~2000 °C
- **Reaction pathway:**

- $Fe_2O_3 + 3CO \rightarrow 2Fe(L) + 3CO_2$
- $C + O_2 \rightarrow 2CO$
- **Combined:** $2Fe_2O_3 + 3C \rightarrow 4Fe(L) + 3CO_2$

📄 **Summary:**

- Iron ore is **reduced** to iron using **carbon monoxide**, releasing **CO₂ as a byproduct**
 - The **carbon** used comes from burning coal
 - This is **not combustion for energy**, but a **chemical reaction** that produces CO₂ directly
-

Environmental Impact per ton of steel:

- **Energy use:**
 $E = 6.8 \text{ MWh per ton Fe}$
- **CO₂ emission:**
 $m_{\text{CO}_2} = 1.9 \text{ tons CO}_2 \text{ eq. per ton Fe}$

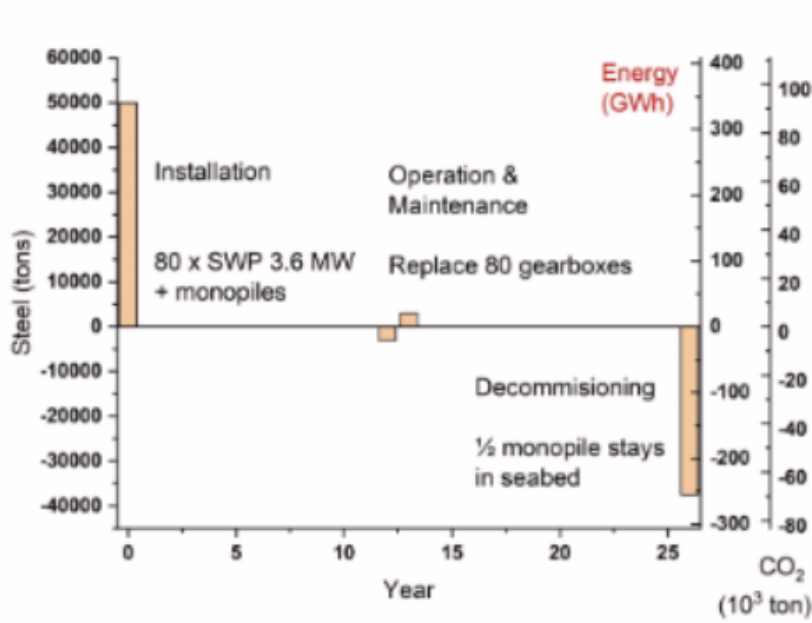
So for every **1 ton of iron** used in a turbine, there's **~1.9 tons of CO₂ emitted even before** transport, assembly, or operation.

Why this matters for wind energy LCA

When building wind turbines:

- **Steel is the largest material contributor**
- Especially for offshore turbines (monopiles, towers, nacelles)
- These emissions show up in **LCA-based CO₂ per kWh**

Energy usage and CO2 emissions due to steel balance



Emission

$$m_{CO_2} = 90 \cdot 10^3 \text{ tons } CO_2$$

Wind energy produced

$$E = 80 \text{ turbines} \times 3.6 \text{ MW/turbine} \times 365 \text{ days/year} \times 24 \text{ hours/day} \times 25 \text{ years} \times CF \\ = 25.2 \cdot 10^9 \text{ kWh (25 TWh)}$$

The capacity factor (CF) of the wind farms is assumed to be 0.4, which describes the average production due to the fluctuating wind.

Normalized emission

$$\frac{m_{CO_2}}{E} = \frac{90 \cdot 10^9 \text{ g } CO_2}{25 \cdot 10^9 \text{ kWh}} \sim 4 \frac{\text{g } CO_2}{\text{kWh}}$$

★ Steel Input → CO₂ Emissions

- **50,000 tons of steel** used for:
 - Monopiles
 - Towers
 - Nacelles
 - Gearboxes
- **Steel emission factor:**
 $1.8 \text{ ton } CO_2/\text{ton steel}$ → updated to **90,000 tons CO**

⚡ Wind Farm Production Estimate

📊 Formula:

$$E = 80 \text{ turbines} \times 3.6 \text{ MW} \times 365 \text{ days} \times 24 \text{ hours} \times 25 \text{ years} \times 0.4 \text{ (capacity factor)}$$

📌 Result:

$$E = 25.2 \cdot 10^9 \text{ kWh} = 25 \text{ TWh}$$

📊 Normalized Emission:

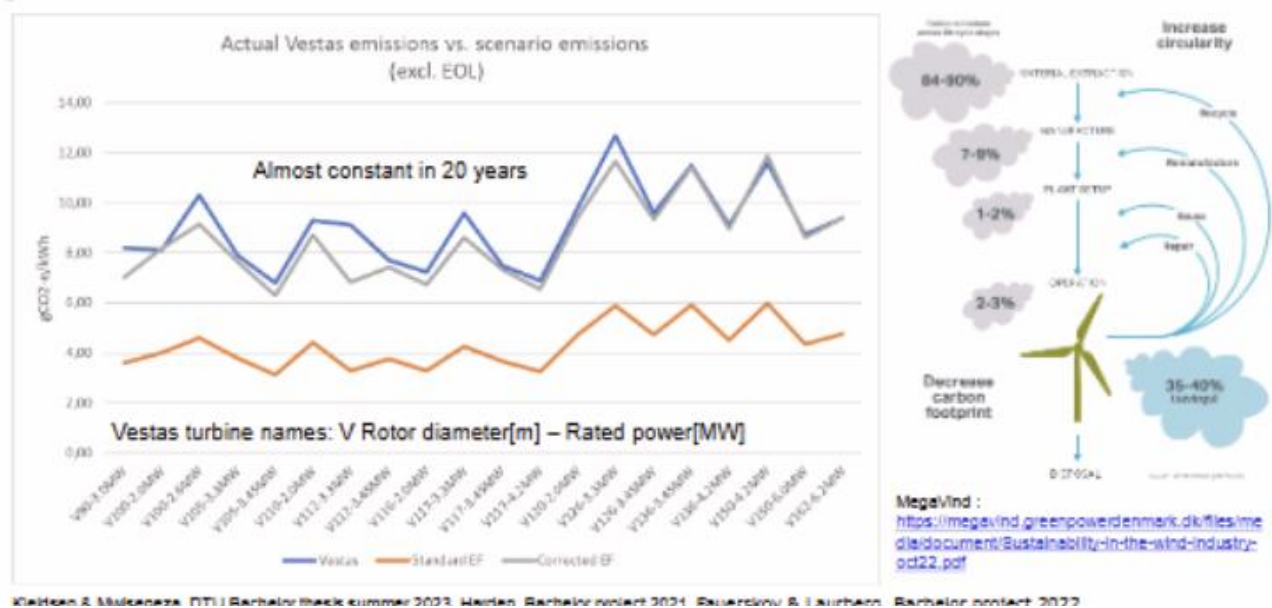
$$\frac{m_{\text{CO}_2}}{E} = \frac{90 \cdot 10^9 \text{ g CO}_2}{25 \cdot 10^9 \text{ kWh}} = 3.6 \text{ g CO}_2/\text{kWh} \approx 4 \text{ g CO}_2/\text{kWh}$$

💡 Key Insight:

Just considering **steel** in the wind turbine lifecycle (not transport, concrete, cables, or blades), the **CO₂ emission per kWh is only ~4 g**, which is:

- **Over 250× lower than natural gas** (~434 g CO₂/kWh)
- **Over 250× lower than coal** (~1023 g CO₂/kWh)

Onshore turbine emission from LCA studies of Vesta



Key Takeaways from the Left Graph

Graph title: *Actual Vestas emissions vs. scenario emissions (excl. End of Life)*

- **Turbine span:** From early 2000s models (e.g., V90-3.0MW) to modern turbines (e.g., V162-6.2MW)
- **Y-axis:** CO₂ emissions per kWh [g CO₂-eq/kWh]
- ◆ **Blue curve – Vestas Actual Emissions:**
 - Roughly **8–11 g CO₂/kWh**, almost constant over 20 years.
 - Reason: As turbines grow larger (producing more energy), material input grows too → ratio stays balanced.
- ◆ **Orange & Grey curves – Scenario and Corrected Emissions:**
 - Reflect either idealized or adjusted manufacturing footprints.
 - Lower values (~3.5–5 g CO₂/kWh) highlight potential with optimized practices or future improvements.

Lifecycle Breakdown (Right side graphic)

Where do wind turbine emissions come from?

- **84–90%:** *Material extraction & manufacture* (especially steel, concrete, composites).
- **7–9%:** *Plant setup* (transport, installation).

- 1–2%: *Operation*
- 2–3%: *Disposal*

🌀 Circular economy strategies (right side):

- Recycle, Remanufacture, Reuse, Repair
- **CO₂ savings potential: 35–40%**

📌 Core Insight

Despite dramatic growth in turbine size and performance:

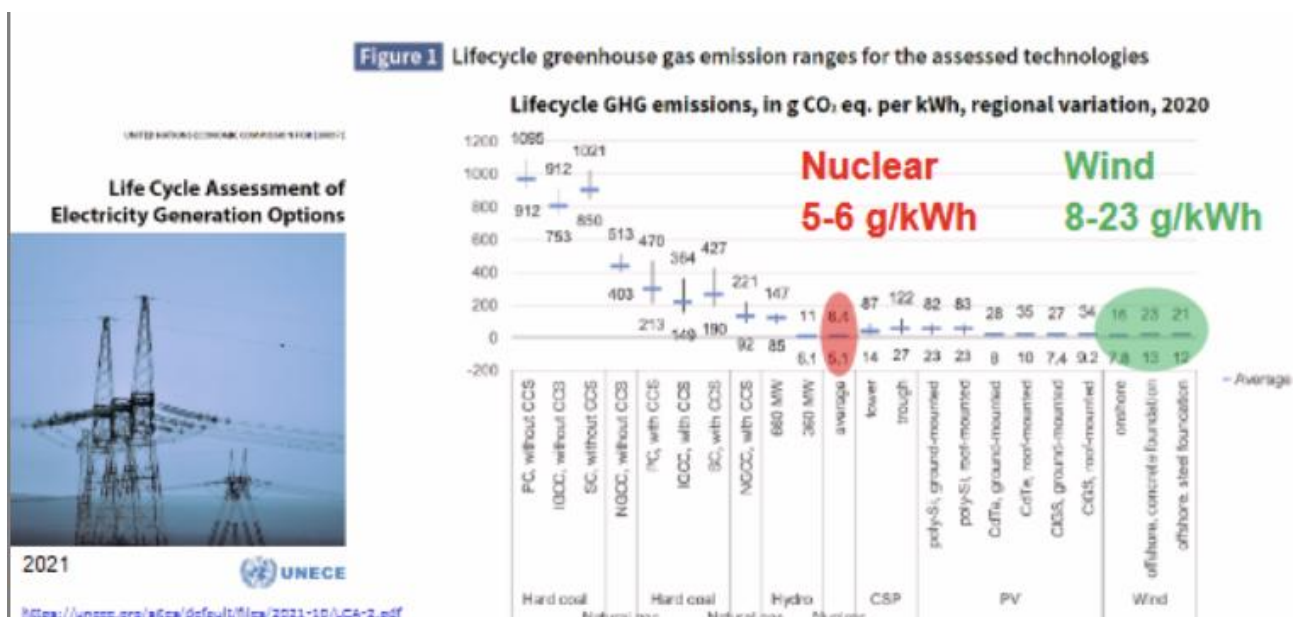
“Wind turbine CO₂ intensity per kWh remains steady due to proportional increases in material demand and energy output.”

This underscores the importance of:

- Decarbonizing **material production** (e.g., green steel)
- Designing for **reuse and recyclability**
- Continuing to improve **capacity factor and lifetime efficiency**

Let me know if you'd like a comparison chart summarizing CO₂ intensity of wind vs. solar, nuclear, etc., based on these LCA data.

UN ranking of electricity producing technologies



UN Electricity Production Emissions Comparison (UNECE, 2021)

Nuclear Energy

- **Lifecycle emissions:**
Average ~5–6 g CO₂/kWh
(Values shown: 6.4 and 5.1 for different scenarios)
- **Key point:** Despite the need for mining, enrichment, and waste management, **nuclear energy is among the lowest CO₂ emitters** per kWh produced.

Wind Energy

- **Lifecycle range:**
 - **Onshore wind:** ~7.8 to 16 g CO₂/kWh
 - **Offshore (concrete):** ~23 g CO₂/kWh
 - **Offshore (steel foundation):** ~13 g CO₂/kWh
 - **Key point:** Wind generally emits a little more CO₂ than nuclear, especially offshore systems due to heavy foundation material (concrete or steel).
-

Interpretation and Discussion Points

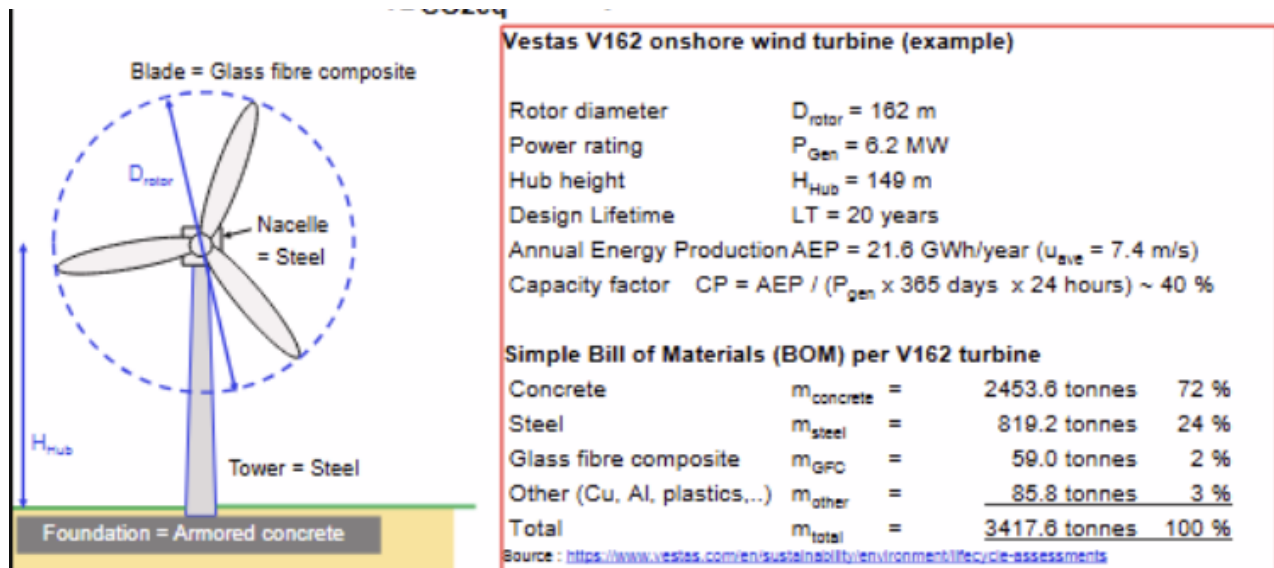
- **CO₂ Emissions Alone:**
If judging *only* by carbon emissions, **nuclear slightly outperforms wind**.
 - **Broader Sustainability Context:**
 - Nuclear faces *long-term waste storage, public perception challenges, and decommissioning complexity*.
 - Wind has *lower operational risks*, easier recycling potential (especially with steel), and minimal long-term waste.
 - **Offshore vs. Onshore:**
 - Offshore wind, especially with concrete foundations, has higher embodied carbon (materials and transport).
 - Onshore wind appears more climate-efficient per kWh over its lifetime.
-

Conclusion

Both nuclear and wind are **climate-friendly compared to fossil fuels**, but:

- **Nuclear** wins narrowly on **CO₂ intensity**.
- **Wind** wins on **renewability, modularity, and circularity potential**.

Simple model for estimating Global Warming Potential (g CO₂eq/kWh) of onshore wind turbine



⚙️ Vestas V162 Onshore Wind Turbine – Key Specifications

- **Rotor diameter:** 162 m
- **Power rating:** 6.2 MW
- **Hub height:** 149 m
- **Design lifetime:** 20 years
- **Annual Energy Production (AEP):** 21.6 GWh/year (at wind speed $u_{\text{ave}} = 7.4 \text{ m/s}$)
- **Capacity factor:** ~40%
This means the turbine produces energy equivalent to operating at full power for ~40% of the year.

Bill of Materials (BOM) per Turbine

Material	Mass (tonnes)	Share of Total
Concrete (foundation)	2453.6	72%
Steel (tower, nacelle)	819.2	24%
Glass fiber composite (blades)	59.0	2%
Other (Cu, Al, plastics, etc.)	85.8	3%
Total	3417.6	100%

 **Note:** The **concrete** (for the foundation) dominates by weight, while **steel** is the second-largest contributor and the **most impactful for CO₂ emissions** due to its energy-intensive production.

Why This Matters for CO₂e/kWh Calculation

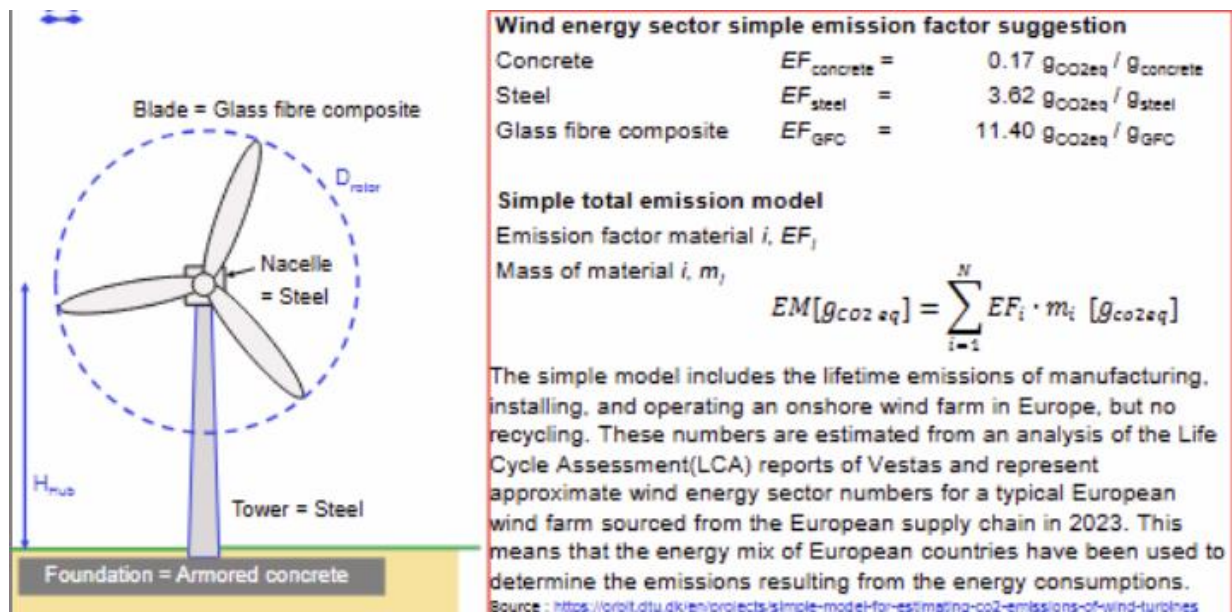
To calculate the **g CO₂e per kWh**, the method involves:

1. **CO₂ emissions per material** (e.g., ~1.9 t CO₂ per ton steel, ~0.13 t CO₂ per ton concrete, etc.).
2. **Total energy production over the lifetime** = 21.6 GWh/year × 20 years = **432 GWh**.
3. **Divide total emissions** (from production, materials, transport, etc.) **by total energy output** to get the **normalized GWP per kWh**.

Takeaway

This example shows that **even modern, efficient wind turbines involve substantial material use**, particularly concrete and steel. However, due to their high energy output over decades, **the lifecycle CO₂ emissions are still very low** — typically in the **single digits of g CO₂/kWh**, making them one of the **cleanest electricity sources** available.

Simple model for estimating Global Warming Potential



Simple Emission Calculation Model

The formula:

$$EM[g_{CO2eq}] = \sum_{i=1}^N EF_i \cdot m_i$$

Where:

- EF_i : Emission factor for material i (g CO_2e per gram of material)
- m_i : Mass of material i (in grams)

This model sums up emissions from each material used in the turbine over its **entire lifecycle**, assuming:

- **No recycling**
- Installation, transport, operation, and decommissioning are included
- European energy mix and manufacturing practices

Emission Factors (for Wind Energy Sector)

Material	Emission Factor EF_i
Concrete	$0.17 \text{ g } CO_2e / \text{g}$
Steel	$3.62 \text{ g } CO_2e / \text{g}$
Glass Fibre Composite	$11.40 \text{ g } CO_2e / \text{g}$

These values are based on **European supply chain conditions as of 2023** and extracted from Vestas LCA data.

Next Steps for Estimation

You can now calculate the **total CO₂ emissions** of a V162 wind turbine by inserting material weights (from Slide 10) into the formula:

- Concrete: 2,453,600 g
- Steel: 819,200 g
- Glass fibre: 59,000 g

The total emissions (in g CO₂e) are then **divided by the total energy output** over 20 years (e.g., 21.6 GWh/year × 20 years = 432 GWh = 4.32×10^{11} Wh) to get the **g CO₂e/kWh**.

Optional: Include Transport

For added precision, transport emissions could be included:

- Nacelle: 600 km by truck
- Tower: 425 km by truck
- Concrete: 50 km local delivery
- Others: 600 km
- Some parts also shipped (e.g., blades, hub)

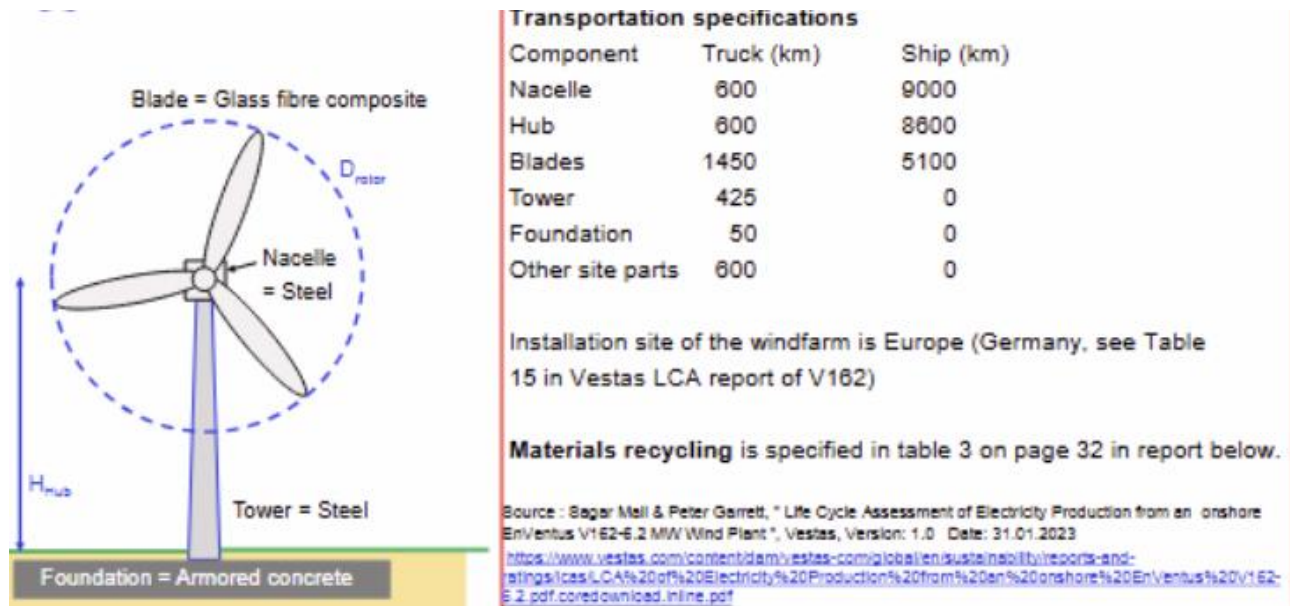
These values are found in the full Vestas LCA for the V162 turbine (Germany-specific installation context).

Simple model for estimating Global Warming Potential

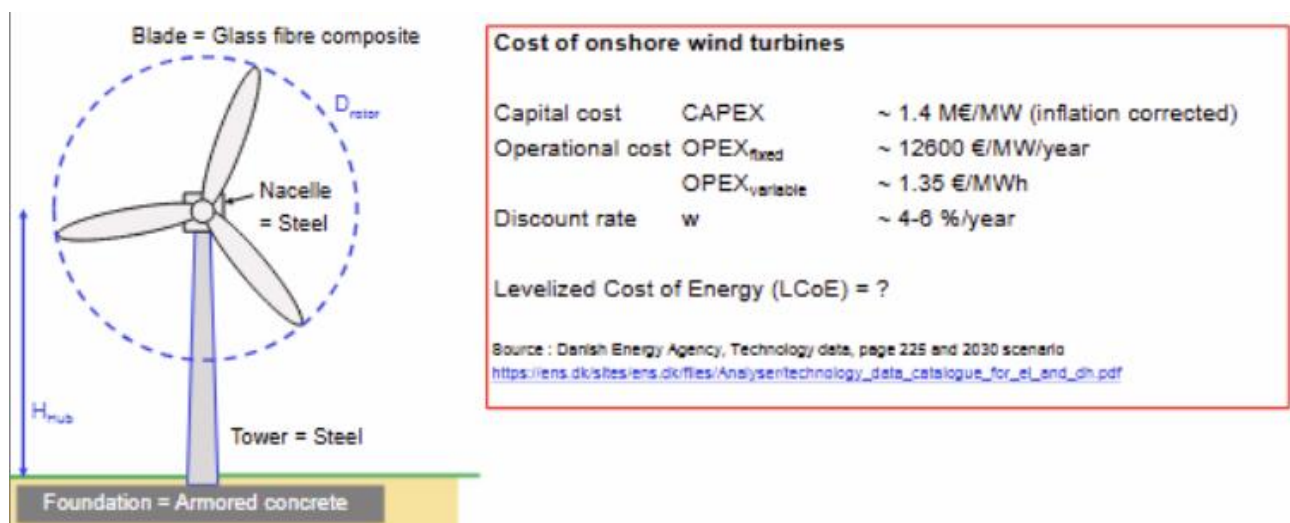
Bill of material example: Vestas V162 onshore turbine			Simple global warming potential emission factors		CO ₂ emissions	CO ₂ emissions	CO ₂ emission fraction
Materials	Mass [tonnes]	Mass fraction [%]	[g CO ₂ e/g material]		[g CO ₂ e]	[tonnes CO ₂ e]	[wt %]
Concrete	2453.6	71.8	0.17		417112000	417.1	10.3
Steel	819.2	24.0	3.62		2965504000	2965.5	73.1
Glass fibre composite	59.2	1.7	11.41		675472000	675.5	16.6
Other	55.8	2.5	0 Not included		0	0.0	0.0
Total	3417.8	100.0			4056088000	4056.1	100.0
Turbine Annual Energy Production							
Average wind speed installation site class	7.4 m/s						
Turbine rated power P _{rated} [MW]	6.2 MW						
Annual Energy Production AEP	21.6 GWh/year	Default	21.6 GWh/year				
Turbine design life time LT	20 years						
Hours per year	8760 Hours						
Capacity Factor CF	25.8 %						
Normalization of global warming potential by energy production							
Global warming potential per kWh	9.4 [gCO ₂ e/kWh]						

- Concrete constitutes 72 % of the mass used but the resulting CO₂ emission is only 10 % of the total CO₂ emission
- Steel constitutes 24 % of the mass used but results in 73 % of the CO₂ emission
- The glass fibre composite is only about 2 % of the total mass but results in 17 % of the CO₂ emission
- The resulting Global Warming Potential (GWP) is found to be 9.4 gCO₂e / kWh, which is considerably higher than shown on slide 8 since the recycling fraction of the bill of material after the end of life has not been subtracted.

Transport assumptions in simple GWP model



Simple model for estimating Levelized cost of Energy of onshore wind turbine electricity production



Notes:

■ LCoE Formula (Simplified)

The **Levelized Cost of Energy** is calculated as:

$$\text{LCoE} = \frac{\text{Total Lifetime Cost (CAPEX + OPEX)}}{\text{Total Lifetime Energy Production}}$$

For a more detailed financial model, use the **discounted cash flow formula**:

$$\text{LCoE} = \frac{\sum_{t=1}^T \frac{\text{CAPEX} + \text{OPEX}_{\text{fixed}} + \text{OPEX}_{\text{variable}} \cdot E_t}{(1+w)^t}}{\sum_{t=1}^T \frac{E_t}{(1+w)^t}}$$

Where:

- T = project lifetime (usually 20–25 years)
- w = discount rate (here 4–6%/year)
- E_t = energy produced in year t

📄 Input Values from the Slide

- **CAPEX** $\approx$ 1.4 million €/MW (inflation corrected)
- **OPEX (fixed)** $\approx$ 12,600 €/MW/year
- **OPEX (variable)** $\approx$ 1.35 €/MWh
- **Discount rate** $\approx$ 4–6%
- **Lifetime** $\approx$ 20–25 years
- **Capacity factor**: 40%
- **Turbine size**: 6.2 MW (from earlier slide)
- **Annual Energy Output** $\approx$ 21.6 GWh/year

📊 Back-of-the-envelope Estimate (Undiscounted)

If we simplify and ignore discounting for now (flat cash flows):

Annualized CAPEX:

$$\frac{1.4 \text{ M€}}{\text{MW}} \cdot 6.2 \text{ MW} = 8.68 \text{ M€}$$

$$\text{Annualized over 20 years: } \frac{8.68}{20} = 0.434 \text{ M€/year}$$

Fixed OPEX:

$$12,600 \cdot 6.2 = 78,120 \text{ €/year}$$

Variable OPEX:

$$1.35 \text{ €/MWh} \cdot 21,600 \text{ MWh/year} = 29,160 \text{ €/year}$$

Total Annual Cost:

$$434,000 + 78,120 + 29,160 = 541,280 \text{ €/year}$$

LCoE:

$$\text{LCoE} = \frac{541,280}{21,600} \approx 25.1 \text{ €/MWh}$$

↓

✅ Conclusion

- **Estimated LCoE:** ~25 €/MWh (undiscounted, ideal case)
- Including discounting and uncertainties, **realistic LCoE range:** likely 30–50 €/MWh

Levelized Cost of Energy estimate of onshore wind energy from Vestas V162-6.2 MW turbin

The levelized cost of energy can in a simple form be defined as

$$\text{LCoE} = \frac{\sum_{t=0}^{LT} \frac{C_t}{(1+w)^t}}{\sum_{t=0}^{LT} \frac{E_t}{(1+w)^t}} = \frac{C_{\text{CAPEX},0}}{E_{\text{AEP}}} \cdot \text{CRF} + \frac{C_{\text{OPEX,Annual}}}{E_{\text{AEP}}} + \text{LCoE}_{\text{OPEX,variable}} = 32.2 \frac{\text{€}}{\text{MWh}} + 3.6 \frac{\text{€}}{\text{MWh}} + 1.4 \frac{\text{€}}{\text{MWh}} = 37.2 \frac{\text{€}}{\text{MWh}}$$

Where

- The capital expenditure (CAPEX) is
 - $C_{\text{CAPEX},0} = 1.4 \text{ M€/MW} \times 6.2 \text{ MW} = 8.68 \text{ M€}$ (see slide 14)
- The annual operational expenditure (OPEX) is
 - $C_{\text{OPEX,Annual}} = 12600 \text{ €/MW/year} \times 6.2 \text{ MW} = 78120 \text{ €/year}$ (see slide 14)
- The variable operational expenditure is $\text{LCoE}_{\text{OPEX,variable}} = 1.35 \text{ €/MWh}$ (see slide 14)
- The Annual Energy Production $E_{\text{AEP}} = 21.6 \text{ GWh/year}$ (see slide 10)
- The Capital Return Factor (CRF) is given below using an interest rate $w = 5\%$ and a design life time $LT = 20$ years (see slide 14)

$$\text{CRF} = \frac{1}{\sum_{t=1}^{LT} \frac{1}{(1+w)^t}} = \frac{w}{1 - (1+w)^{-LT}} = \frac{0.05 \frac{1}{\text{year}}}{1 - (1 + 0.05)^{-20}} = 0.080 \frac{1}{\text{year}}$$

LCoE levels of electricity sources of Europe

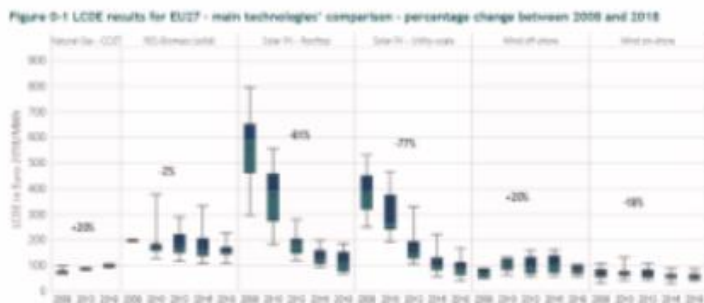
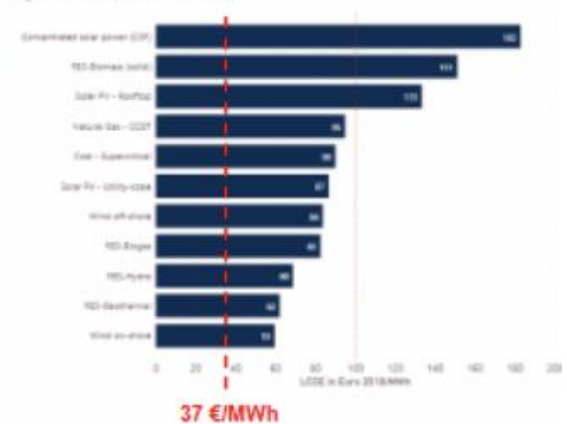
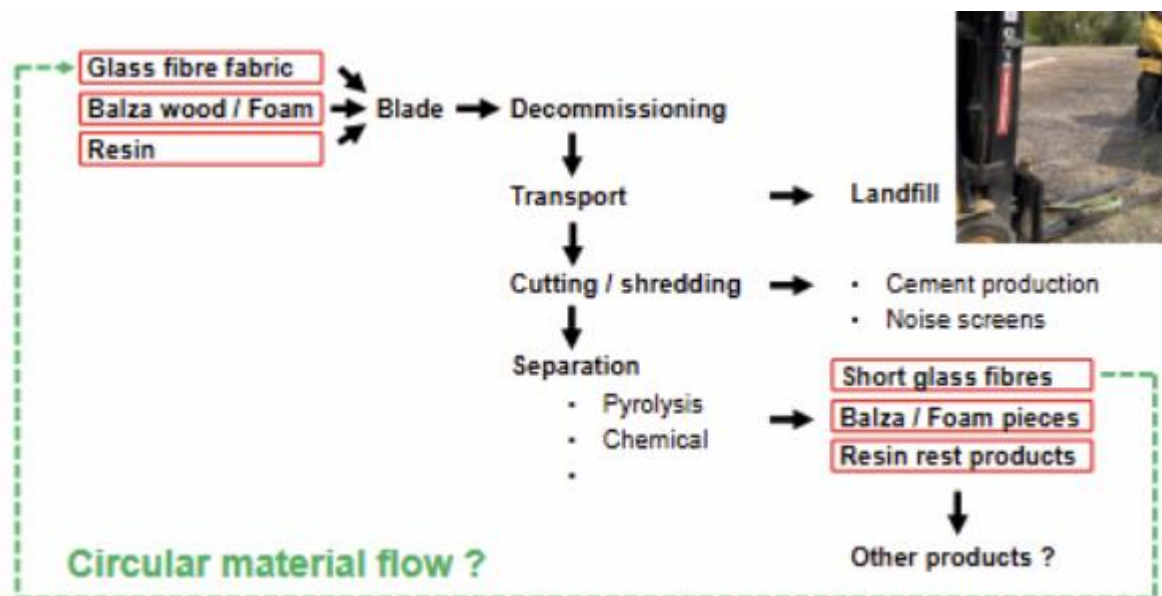


Figure 9-2 LCOE results for EU27 - in 2018



Thierry Badouard, Débora Moreira de Oliveira, Jessica Yearwood and Perla Torres, "Final Report Cost of Energy (LCOE). Energy costs, taxes and the impact of government interventions on investments", European Commission ENER/2018-A4/2018-471 (2020)

Circularity of wind turbine blades



Key Takeaways from the Slide on Blade Circularity

Main blade materials:

- **Glass fibre fabric:** provides structural strength
- **Balsa wood / foam:** lightweight core material
- **Resin (epoxy):** binds the materials together

End-of-life (EoL) Challenges and Pathways

1. Decommissioning and Transport

- Blades are bulky, requiring specialized logistics

2. Cutting/Shredding

- Shredded for downstream processing

3. Separation Processes

- **Pyrolysis** or **chemical treatments** to extract usable fractions
- Current efforts aim to break down the epoxy matrix

4. Output Fractions

- **Short glass fibres** (but with lower mechanical quality)
- **Balsa/Foam pieces**
- **Resin rest products** (often not reusable)

5. End Uses Today

- Cement co-processing (e.g. as fuel or additive)
- Noise barriers or construction filler (downcycling)

Is it Circular?

The dashed green arrow asks whether these recovered materials can **re-enter blade production** — at the moment, **not yet** at large scale.

What's Needed for True Circularity?

- New blade designs enabling easier **disassembly** or **recycling**
- **Thermoplastic composites** or bio-based resins
- More efficient **chemical separation technologies**
- Integration of **recovered fibres/resins** into new production chains

Great news on turbine blade recycling in 2023

DecomBlades and 3B-Fibreglass are ready to unlock circular recycling of glass fibre in wind turbine blades



Remelting recycled glass fibers

- 1) Cut turbine blades (5-10 tons)
- 2) Shredding
- 3) Grinding
- 4) Pyrolysis to remove epoxy
- 5) Milling recycled glass fiber

Mix 1-5 % recycled glass fibers into melt of production for new glass fibers by 3B (72 metric ton)

DTU Wind and Energy Systems showed mechanical properties of remelted glass fibers are as good as normal wind turbine grade fibers. This can enable a fully blade to blade circularity of the glass fibers of the wind industry.

Notes:

Collaborative effort by:

- **DecomBlades**
- **3B-Fibreglass**
- Supported by major players: Vestas, Ørsted, Siemens Gamesa, LM Wind Power, SDU, DTU, and more.

Remelting Recycled Glass Fibers – The Process

1. **Cut turbine blades** (5–10 tons)
2. **Shredding**
3. **Grinding**
4. **Pyrolysis** – removes epoxy resin matrix
5. **Milling** recycled fibres for reuse

These recycled fibres are **reintroduced** into the melt for **new glass fibre production**, enabling a **partial closed-loop** (1–5% mix in production so far).

Results from DTU Wind and Energy Systems

- **Mechanical properties** of recycled glass fibres are **comparable** to new fibres used in wind turbines.
 - Validates **technical feasibility** of full blade circularity.
-

Why This Is Important

- Wind turbines are already among the **lowest emitters** of lifecycle CO₂ (as seen in earlier slides).
- Solving **blade circularity** improves their **material sustainability** and avoids landfill or downcycling into cement or fillers.

Conclusion

- CO₂ emission of the Danish Energy mix is expected to reach ~ 10-15 g CO_{2eq}/ kWh using only wind turbines unless the production method of the materials used to build the turbines is changed.
- The main materials used in the wind turbines are: steel, concrete and glass fibre composite
- For offshore wind turbines the emissions will depend on the amount of recycled steel from monopiles
- Circularity in material recycling of wind turbines is improving by new blade recycling technologies
- New solutions
 - Turbine designs with less material usage (especially concrete in the foundation of onshore)
 - Usage of Green Steel and Green Cement?
 - Low CO₂ footprint steel in tower and monopile by utilizing remelted steel
 - Refining iron ore to iron using hydrogen route (water as chemical emission instead of CO₂)
 - Turbines based on completely different materials (wood in towers?)
- Turbine design standards dictate current material usage (IEC 61400-1 and IEC 61400-3) since the turbine manufacturers have to guarantee that the chance of major failures of the turbines is small during the design lifetime.

Notes:

Main Conclusions on Wind Energy Sustainability

CO₂ Emissions

- Denmark's future energy mix (based on 100% wind power) is expected to emit **~10–15 g CO_{2eq}/kWh**.
- This is **90% lower** than current average emissions from conventional fossil-based energy systems.

- However, **this limit is governed by the production methods** used for steel, concrete, and glass fibre composites — not the wind turbine operation itself.
-

Main Materials in Wind Turbines

- **Steel** – tower, nacelle, monopile (offshore)
 - **Concrete** – mainly onshore foundations
 - **Glass fibre composites** – rotor blades
-

Circularity & Material Recycling

- **Steel** is relatively easy to recycle (especially tower and nacelle).
 - **Glass fibre blade recycling** is now improving thanks to pyrolysis and remelting methods (e.g., via DecomBlades & 3B-Fibreglass).
 - Offshore recycling efforts will gain momentum as early wind farms reach decommissioning in 5–10 years.
-

Future Solutions to Lower Emissions Further

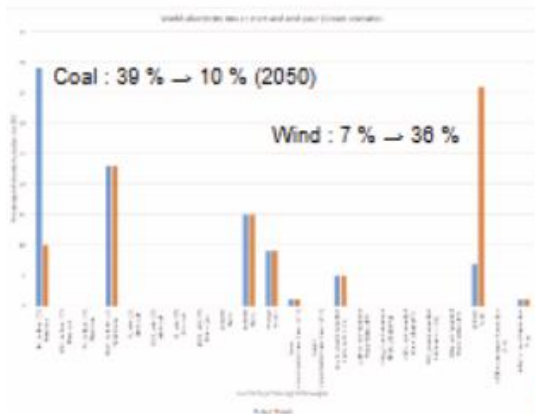
1. **Turbine designs using less material**, especially minimizing concrete in onshore foundations.
 2. **Green steel** (recycled or hydrogen-reduced iron) and **green cement** (low-emission formulations).
 3. **Use of recycled/remelted steel** in towers and monopiles.
 4. **Hydrogen-based iron production** – replacing CO₂ emissions with water ($\text{H}_2 + \text{Fe}_2\text{O}_3 \rightarrow \text{Fe} + \text{H}_2\text{O}$).
 5. **Innovative materials** – e.g., **wooden towers** (currently under research).
-

Design Standards and Constraints

- Wind turbines must comply with strict international safety and durability standards (IEC 61400-1 & 61400-3).
- These standards **dictate material requirements** to ensure that turbines can survive **20–30 years of operation** without catastrophic failure.

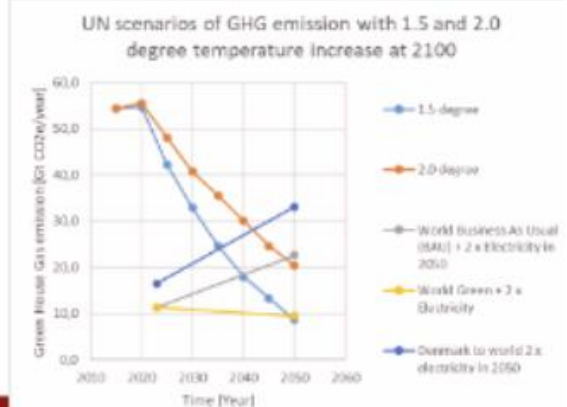
Lecture 11: Violation of planetary boundary of Global Warming Potential due to electricity mix

This exercise has been created to illustrate the Global Warming Potential or CO_{2e} emission resulting from a mix of electricity producing technologies on the scale of planet Earth and then to compare the emissions to the planetary boundary of 1.5 °C heating in 2100 as formulated in the Paris agreement. The students should suggest a future technology mix of 2050 and determine if the planetary boundary is violated. Secondly they should analyse a country scaled to planet scale.



Green scenario

- Double electricity demand in 2050
- Reduce coal
- Increase wind
- 6 million Danes
- 8 billion people on earth



Notes:

Green Scenario Highlights:

- Assumes **doubling of electricity demand** by 2050.
- Dramatic **reduction in coal usage**, replaced largely by wind and renewables.
- Goal: See if the **future mix still violates** the planetary boundary.

Graphs and Analysis Tools:

- UN projections of GHG emissions vs. Paris targets (1.5 °C and 2.0 °C pathways).
- World electricity mix transition (bar chart showing 2020 vs. 2050).
- Excel-based GWP calculator.

Final Assignment:

Students will:

- Propose a **future global electricity technology mix** for 2050.
- Test whether that mix **stays within the planetary boundary**.
- Analyze **their own region or country** (e.g., Denmark scaled globally).

- Validate if their proposed **decarbonization pathway** is truly sustainable.

Simple method of evaluating Global Warming Potential of the electricity mix of the world

A simple way to estimate the Global Warming Potential (and other impacts) of the electricity mix of the world is to use the emission factors EF_i provided by the UN report "UNITED NATIONS ECONOMIC COMMISSION FOR EUROPE : "Carbon Neutrality in the UNECE Region: Integrated Life-cycle Assessment of Electricity Sources" (2022)" and then determine the weighted average emission factor $EF_{average}$ of the electricity produced by the mix of the technologies with a fraction given as f_i :

$$EF_{average} = \sum_{i=1}^N EF_i \cdot f_i$$

Where i is the index of the N different electricity technologies each with a fraction f_i of the electricity mix. Thus the sum over all the fractions f_i must give 1.

The emission of the world E_m is now determined by multiplying the average Emission Factor $EF_{average}$ with the global electricity consumption E :

$$E_m = EF_{average} \cdot E$$

Notes:

The Simple Method for Evaluating GWP from Electricity Mix

To assess whether the **global warming potential (GWP)** from electricity production violates the **planetary boundary**, we follow a simple and data-driven method based on the **UNEP 2022 Life Cycle Assessment report** on electricity sources.

Step-by-Step Methodology:

1. Use UN Emission Factors

From the 2022 UNECE report, we take the **CO₂eq emission factors** for each electricity generation technology (e.g., coal, wind, nuclear, gas, solar, hydro).

2. Apply Regional or Global Electricity Mix

We combine those emission factors with the **actual electricity mix** of a country or the world (e.g., % of electricity from coal, wind, nuclear, etc.).

3. Calculate Weighted Average Emission Factor

Multiply the emission factor for each source by its share in the electricity mix and sum the values.

This gives the **average CO₂eq emission per kWh** for that mix.

4. Estimate Total Emissions

Multiply the average CO₂eq/kWh by the **total electricity demand** (in TWh or kWh) for the region or world.

This gives the **total emissions in Gt CO₂eq/year** from electricity production.

What Do We Compare It To?

We compare this result to the **planetary boundary for GWP** based on the **1.5 °C climate target** by 2100 (from the Paris Agreement):

- This boundary sets an **annual global emissions budget** (typically in Gt CO₂eq/year) that **must not be exceeded** if we are to stay below 1.5 °C warming.
- The GWP from electricity must **fit within this carbon budget** — if it exceeds the limit, we say the **boundary is violated**.

Scaling a country electricity production to global impact

The Impact of a country can be written by the "I = PAT" equation as

$$I_{country} = P_{country} \cdot A \cdot T$$

Where

- $I_{country}$ is the impact of the country in terms of CO₂ emission in Gt CO₂eq/year
- $P_{country}$ is the population of the country in the unit of [persons]
- A is the affluence in terms of energy consumption per citizen per year [kWh/ person / year]
- T is the emission of the technology mix in the unit of [g CO₂e / kWh]

This country impact can be scaled to the global impact I_{global} by multiplying with the ratio of citizens on Earth and in the country

$$I_{global} = I_{country} \frac{P_{global}}{P_{country}}$$

where P_{global} is the global population in the units of [persons].

Scaling National Electricity Impact to Global Level

Using the "I = P × A × T" Identity

To determine how a country's electricity consumption impacts the climate globally — especially in relation to the **planetary boundary for global warming** — we can use a structured formula:

$$I_{country} = P_{country} \times A \times T$$

Where:

- **I_country**: Total annual CO₂ emissions from electricity in the country (Gt CO₂e/year)
- **P_country**: Population of the country (persons)
- **A**: Affluence – electricity used per person annually (kWh/person/year)
- **T**: Emission intensity – CO₂ emissions per kWh (g CO₂e/kWh) from the country's electricity mix

This equation helps quantify how a country's electricity mix and consumption contributes to climate impact.

Scaling to Planetary Level

To project the **global impact** if **everyone on Earth lived like this country**, we scale the country emissions by population:

$$I_{Global} = I_{country} \cdot \frac{P_{Global}}{P_{country}}$$

Where:

- **P_global** is the total global population
- This gives a **hypothetical global emission scenario**:
 ► *What if everyone used electricity like the average Dane, German, American, etc.?*

Application to the Planetary Boundary

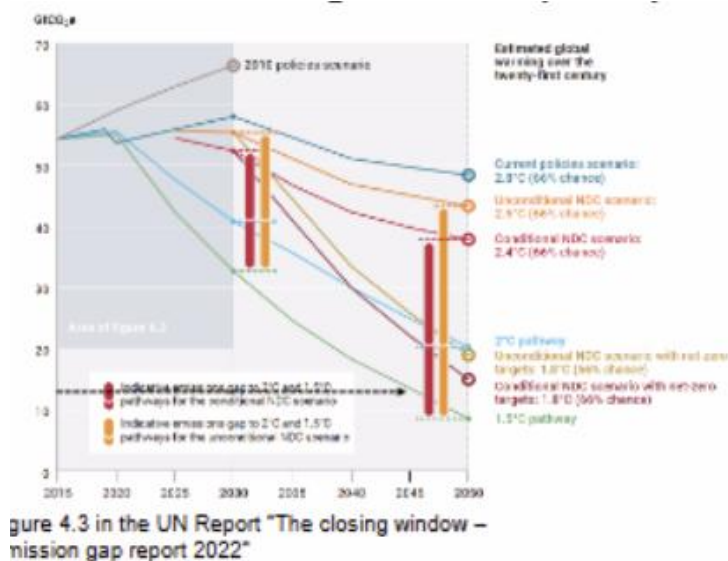
Now we compare **I_Global** with the **UN Emission Gap Report's GWP boundary**:

- ~33 Gt CO₂e/year in 2030 (to meet 1.5 °C)
- ~20 Gt CO₂e/year in 2050 (tightened further)

If **I_Global** exceeds the budget →  **Planetary boundary is violated**

If **I_Global** stays below the limit →  **Scenario is within climate-safe range**

Comparison to planetary boundary of Global Warming Potential(GWP)



- The estimated global emission can be compared directly to the UN report on the emission gap of 2022.
- Different formulations of the planetary boundary have been formulated, but the 1.5 °C pathway is used for the comparison.
- It should be noted that the 13 GtCO₂e of coal and gas emission from electricity production in 2022 from Lecture 1 will violate the 1.5 °C path around 2045.
- An excel spreadsheet has been created for calculating the impact of an electricity system and should be used to solve the exercise.

Comparison to Planetary Boundary of Global Warming Potential (GWP)

Based on UN Emission Gap Report 2022 (Fig. 4.3)

Key Messages:

- **The chart from the UN shows different future emission scenarios**, depending on whether current policies, unconditional NDCs, or conditional NDCs are followed.
 - The **1.5 °C pathway** is shown in **green** — a steep decline is needed.
 - Business-as-usual (**current policy**) leads to **2.8 °C** warming by 2100.
- A **dashed black line** was added showing **13 Gt CO₂/year**, which was calculated in Lecture 1 as the amount emitted **just from global electricity production by coal and natural gas** in 2022.
- If this level of emission **remains unchanged**, we will hit the **planetary CO₂ budget limit for 1.5 °C warming around 2045** — **just from power generation**, leaving no room for emissions from heating, transport, industry, agriculture, etc.

What this means:

- **Electricity production is already consuming a major portion** of the climate budget.
- Decarbonizing this sector is **non-negotiable** if the world is to stay within the 1.5 °C target.
- Students are asked to model different 2050 electricity mixes and determine:

1. Are they within the budget?
2. What happens if the world scales energy use like Denmark

Instruction of exercise

Purpose of the Tool

The goal is to evaluate whether a given electricity technology mix (for a country, region, or the whole world) in a specified year violates the planetary boundary for global warming potential (GWP), defined by the Paris Agreement's 1.5 °C target.

Excel Tool Structure (5 Sheets)

Front Page – Overview and metadata

World Business as Usual – Current mix + scenario (doubling electricity production by 2050)

World Green Scenario – A greener mix (less coal, more wind)

Denmark Case – Danish electricity production scaled to global level

Result Plot – GWP results compared to UN planetary boundaries

How It Works (Core Calculation Steps)

1. Define Electricity Mix

For both 2023 and 2050, define the % mix of electricity sources (e.g., coal, gas, wind, solar).

Change these values in the gray cells of the Excel sheets.

2. Use UN Emission Factors

Taken from the UN 2022 LCA Report: values in g CO₂eq per kWh for each technology.

Focus is on the "climate change total" impact category.

3. Input Global Data

Electricity production (e.g., 2.25×10^{13} kWh/year for 2023).

Population (8 billion people in 2023, scenario-adjustable for 2050).

4. Calculate Average Emissions

Tool calculates the average CO₂ emissions per kWh for the mix.

Then multiplies by electricity production to get total emissions in Gt CO₂eq/year.

Comparison to Planetary Boundary

From UN Emissions Gap Report (2022):

The world must stay below ~10 Gt CO₂eq/year in 2050 to stay on the 1.5 °C path.

You compare the scenario output to this number:

<100% → Within boundary

>100% → Violating boundary

Examples from the Lecture

Scenario	2050 GHG Emission	Result
World Business-as-Usual	22.6 Gt CO ₂ eq	263% of the 1.5°C limit → Violates
World Going Green (Wind↑, Coal↓)	~10.9 Gt CO ₂ eq	~110% of limit → Slight violation
Denmark Electricity Scaled Globally	~38.5 Gt CO ₂ eq	385% → Far above limit

Special Notes

- You can scale **any country's** mix to the global level using the **I = PAT equation**:
 - $I = P \cdot A \cdot T$
 - Where:
 - P = population
 - A = electricity per person
 - T = emissions per kWh
- Then adjust using:

$$I_{\text{global}} = I_{\text{country}} \cdot \frac{P_{\text{global}}}{P_{\text{country}}}$$

Student Exercise Instructions

You're encouraged to:

Design your own 2050 energy mix

Try to stay under the 1.5°C boundary

Test for a country or region using real data:

Technology mix (% share)

Electricity production

Population

Learning objectives of decarbonizing the energy system



Learning objectives of decarbonizing the energy system

The student should be able to

- **Describe** the main electricity generating technologies used around the world
- **Explain** the origin of CO₂ emissions from electricity producing units
- **Describe** the typical technology mix used for electricity production in different countries
- **Estimate** the resulting CO₂ emission of the electricity mix when producing 1 kWh of electricity
- **Discuss** the expected future CO₂ emission of electricity production and propose strategies of decarbonizing the electricity system
- **Discuss** which additional technologies that might be needed in the electricity system
- **Analyse** expected CO₂ emissions from a specific country or region and compare the Global Warming Potential with the UN CO₂ emission scenarios.
- **Evaluate** if the planetary CO₂ boundary is violated by the electricity generation ?

Date

DTU

Title

20

. Describe & Explain

- Know the **main electricity generation technologies**: coal, gas, nuclear, hydro, wind (on/offshore), solar (PV/CSP), biomass, etc.
- Understand **where emissions come from**: e.g. combustion of fossil fuels, production of steel and cement, resin for blades, etc.
- Be able to **name typical mixes** (e.g. Denmark = wind heavy, France = nuclear heavy, Poland = coal-heavy).

2. Estimate CO₂ Emissions per kWh

Using the UN LCA report data:

- Each electricity technology has a **CO₂ emission factor** in grams of CO₂ per kWh.
- Combine the emission factors with a region's electricity mix:

$$\text{Average emissions} = \sum (\text{share}_i \cdot \text{emission factor}_i)$$

- Result is **g CO₂/kWh** for that region or scenario.

3. Calculate Total Emissions

Use:

$$\text{Total CO}_2 \text{ emissions} = \text{Electricity produced (kWh)} \times \text{Average emissions (g CO}_2\text{/kWh)}$$

Convert to **gigatons (Gt CO₂)** to compare with global boundaries.

4. Compare to Planetary Boundary

UN 2022 Report:

- **1.5°C target** allows about **10 Gt CO₂/year** from all sectors by 2050.
- Electricity often makes up **~25% of global CO₂ emissions**.
- Use the result to check:
 - Are we **within** or **above** this 10 Gt/year limit?
 - Use % comparison:

$$\frac{\text{Scenario emissions}}{10 \text{ Gt CO}_2} \times 100\%$$

5. Scale a Country to Global Impact

Use the **IPAT formula**:

$$I = P \cdot A \cdot T$$

Where:

- P = population
- A = energy consumption per person (kWh/person/year)
- T = CO₂ intensity (g CO₂/kWh)

Then scale to global:

$$I_{\text{global}} = I_{\text{country}} \cdot \frac{P_{\text{global}}}{P_{\text{country}}}$$

6. Use the Excel Tool (Steps Recap)

1. **Download tool** from DTU Orbit or DTU Learn.
2. **Input data:**
 - Technology mix (2023 & 2050)
 - Electricity production (kWh/year)
 - Population
3. **Read results:**
 - g CO₂/kWh
 - Gt CO₂/year
 - % violation of planetary boundary
4. **Check result plot** for visual comparison with 1.5°C & 2.0°C paths.

7. Be Able to Discuss

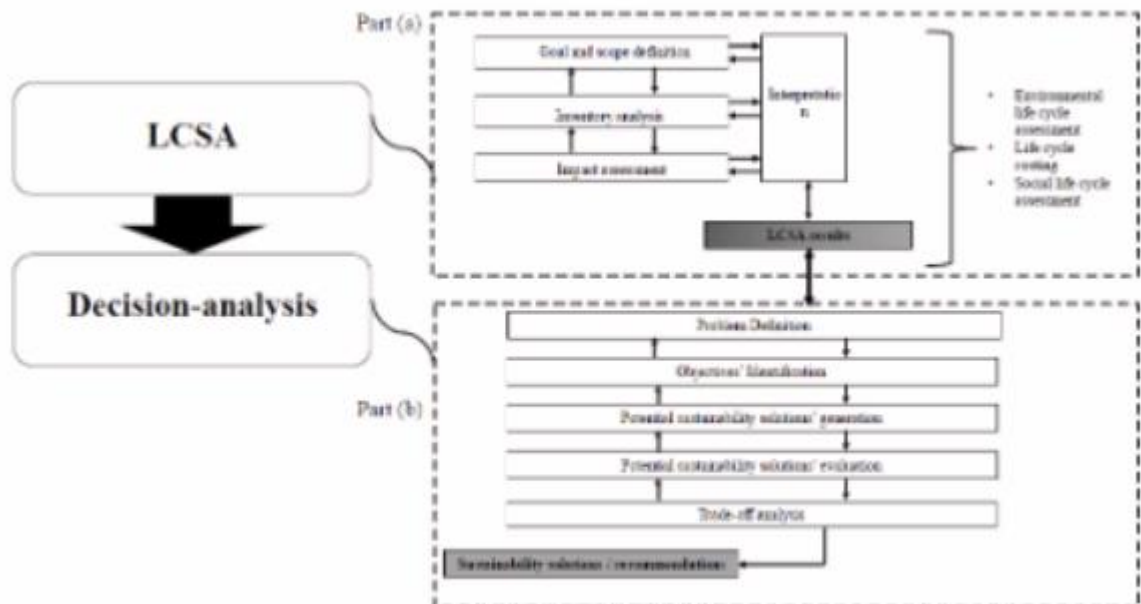
- Why the Danish system, if scaled, violates the boundary
 - How changes (e.g. more wind, less coal) affect the result
 - Which **new technologies** might be needed (e.g. batteries, hydrogen, PtX)
 - Suggest **strategies to decarbonize** (mix changes, CCS, energy efficiency)
-

Final Thoughts

You're now trained to **evaluate**, **calculate**, **analyze**, and **propose** electricity system scenarios and check if they are compatible with the Paris Agreement and planetary CO₂ boundary.

Let me know if you'd like a template or cheat sheet for use during exam prep or assignments!

Lecture 12 - Sustainability transitions (Interpretation and decision making)



Sustainability **2018**, *10*, 3863; doi:10.3390/su10113863

✓ Main Concepts from the Slide & Explanation

◆ LCSA (Life Cycle Sustainability Assessment)

- **Part (a)** in the figure shows LCSA includes:
 - **Goal and scope definition**
 - **Inventory analysis**
 - **Impact assessment**
 - **Interpretation**
- It integrates three types of assessments:
 - **Environmental LCA** (e.g., climate impact, resource use)
 - **Life Cycle Costing (LCC)** (economic aspect)
 - **Social LCA (S-LCA)** (e.g., labor rights, human health)

◆ Decision-Analysis

- **Part (b)** is about how results from LCSA are used in real decision-making processes:

1. **Problem Definition** – e.g., Should I choose electric or gasoline cars?
2. **Objectives Identification** – What matters: cost, CO₂ emissions, performance?
3. **Generation of Alternatives** – What options exist?
4. **Evaluation of Alternatives** – Using LCSA results to compare.
5. **Trade-off Analysis** – When objectives conflict (e.g., low cost vs. low emissions).
6. **Sustainability Recommendations** – Decision based on the best compromise.

💬 **Additional Insight (from narration)**

- The **goal of the assessment** greatly affects decisions.
- The **scope** determines:
 - Which system boundaries are used (e.g., full life cycle? cradle-to-gate?).
 - What quality of data is needed.
 - Which sustainability indicators are included

Quantitative Sustainability framework contain many different indicator

	Life Cycle Stage					
Sustainability Impact area	Extraction of raw materials	Manufacturing stage	Use stage	Disposal stage	Measured by:	Covered in module
Resources					Use of biotic and abiotic resources	4, DAVLU
					Circular economy Indicators	8, TMCA
Environment					Climate change, Carbon footprint	3, OJOLL
					Absolute boundaries	7, MZHA
Economic					Life Cycle Costs	5, KAMORR
Social/Health					Socioeconomic Impacts	6, KAMORR, OJOLL
					Health Impacts	
Transition					Interpretation System dynamics	9, ASAB 10, DAVLU, BIOL

◆ Sustainability Impact Areas and How They're Measured

Impact Area	Measured By	Covered in Module
Resources	Use of biotic/abiotic resources	Module 4, DAVLU
	Circular economy indicators	Module 8, TMcA
Environment	Climate change, carbon footprint	Module 3, OJOLL
	Absolute boundaries (planetary boundaries)	Module 7, MZHA
Economic	Life Cycle Costs (LCC)	Module 5, KAMORR
Social/Health	Socioeconomic impacts	Module 6, KAMORR & OJOLL
	Health impacts	<i>(covered, but no module listed)</i>
Transition	System dynamics and interpretation tools	Modules 9 & 10 (ASAB, SIOL)

🔍 Additional Key Points from the Lecture

- The framework simplifies complex LCSA practices into **digestible categories**.
- Multiple **indicators** can be derived per stage, but:
 - Too many can overwhelm decision makers.
 - So it's critical to **interpret and prioritize** what matters most.
- The **decision context** (as shown in the previous slide) helps determine which indicators are weighted more heavily.

Results must be interpreted and evaluated

- Consider your assumptions and main uncertainties
 - Which assumption are most sensitive and uncertain?
 - Which data do you consider being most uncertain? Can you improve?
- Completeness
 - Which parts of the life cycle did you not include? Potential influence?
 - Do you consider all dimensions and aspects? Or can you reasonably argue for the Omissions.

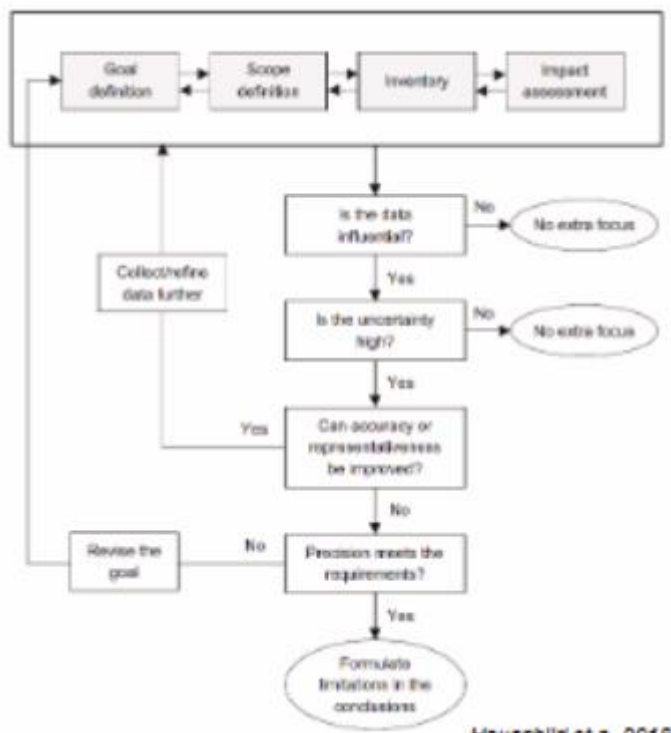
Notes:

1. Assumptions & Uncertainty

- **Identify key assumptions:** What simplifications or choices did you make that might affect results?
- **Check sensitivity:** Which assumptions have the **biggest influence** on your outcome?
- **Assess uncertainty** in the data:
 - Are certain datasets outdated or location-inappropriate?
 - Is there **high variance** between data sources?
 - Can any of these uncertainties be **reduced** with better data?

2. Completeness

- **Life cycle stages:** Did you include all relevant phases (e.g., use, disposal)? If not, how might that omission affect the result?
- **Sustainability dimensions:**
 - Did you cover environmental, economic, social, and transitional aspects?
 - If not, do you have a **justifiable reason** (e.g., negligible impact, lack of data, or study scope)?



◆ 1. Begin with LCSA Process

- Conduct:
 - Goal definition
 - Scope definition
 - Inventory analysis
 - Impact assessment

◆ 2. Ask: Is the data influential?

- **No** → No further action.
- **Yes** → Go to next step.

◆ 3. Is the uncertainty high?

- **No** → No further action.
- **Yes** → Proceed.

◆ 4. Can you improve accuracy or representativeness?

- **Yes** → Collect/refine more data → go back to start.
- **No** → Continue.

◆ 5. Does the precision meet your goal's requirements?

- Yes → Formulate **limitations in conclusions**.
- No → You may need to **revise your goal**.

💡 **Key Insights:**

- **Focus your effort** on influential and uncertain data.
- If better data isn't available, **transparency** about limitations becomes crucial.
- If uncertainty is too high to meet your original aim, it may be necessary to **revise the study goal** or **narrow the scope**.

Ecolabels provide visual decision support EPDs provide more in depth information




POTENTIAL ENVIRONMENTAL IMPACTS		UPSTREAM		CRUISE	DOWNSTREAM		TOTAL
data referred to 1 kg of product		Raw material production	Refining and auxiliary materials production	Production	Distribution up to shop	Private recycling and use	
GLOBAL WARMING POTENTIAL - GWP	kg CO ₂ eq	1.350E+01	2.550E+00	3.980E+01	1.440E+02	2.020E+01	2.070E+02
Acidification Potential	kg SO ₂ eq	6.390E+01	5.470E+01	3.800E+01	2.510E+01	4.100E+01	4.800E+02
Eutrophication Potential	kg PO ₄ eq	3.880E+01	4.540E+00	6.750E+00	1.440E+01	3.940E+00	3.800E+02
Photochemical Oxidant Formation Potential	kg O ₃ eq	2.140E+01	2.470E+00	3.660E+02	1.700E+02	2.810E+01	3.800E+05
Alkaline Degradation Potential	kg NaOH eq	3.880E+01	5.140E+01	6.750E+00	8.430E+01	3.430E+00	3.170E+01
Water scarcity potential	kg eq	1.450E+01	3.320E+01	7.370E+00	8.350E+00	7.280E+00	3.800E+01
Global Warming Potential - GWP	kg CO ₂ eq	7.330E+00	5.330E+01	5.370E+01	4.330E+01	3.730E+00	5.340E+02
Alkaline Degradation Potential	kg NaOH eq	1.540E+01	7.150E+01	5.330E+01	4.330E+01	4.090E+00	1.420E+01
Water scarcity potential	kg eq	1.050E+01	5.850E+00	4.920E+00	2.810E+00	2.270E+00	2.470E+01
Water scarcity potential	kg eq	7.640E+01	7.090E+01	5.800E+04	1.470E+01	3.790E+00	8.470E+01

WASTE PRODUCTION		UPSTREAM		CRUISE	DOWNSTREAM		TOTAL
data referred to 1 kg of product		Raw material production	Refining and auxiliary materials production	Production	Distribution up to shop	Private recycling and use	
Non-hazardous waste	kg	2.000E+01	3.110E+01	0.000E+00	0.000E+00	0.000E+00	5.110E+01
Hazardous waste	kg	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0.000E+00
Radioactive waste	kg	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0.000E+00

The figures contribute to Global Warming Potential refer only to Europe's nations.
The contribution given by Europe's CO₂ is equal to zero, since the demand amount is equal to the one that Europe's CO₂ within the reference 100 years period.

✅ **Eco-labels (e.g., Nordic Swan, EU Ecolabel)**

- **Purpose:** Help consumers make quick and sustainable choices.
- **Function:** Visual decision support.
- **Interpretation:** Simple – if a product has the label, it's among the best ~30% in its category regarding environmental criteria.
- **Scope:** Based on agreed indicators (resource use, emissions, durability, etc.).
- **Users:** Primarily for **consumers**.

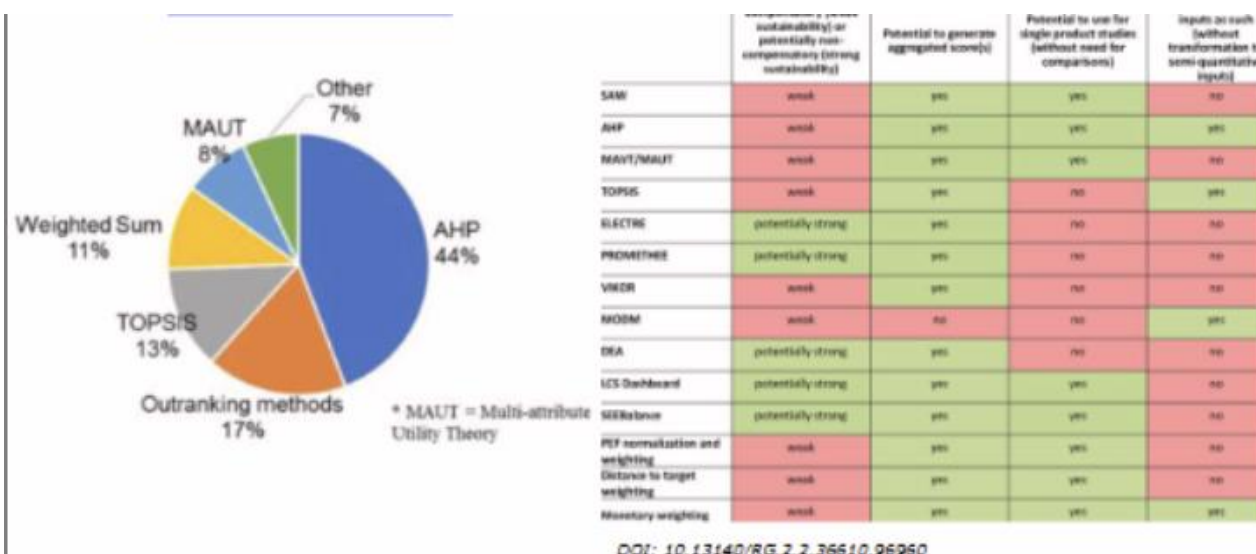
Environmental Product Declarations (EPDs)

- **Purpose:** Provide **detailed** life cycle data on product environmental performance.
- **Function:** In-depth, transparent reporting based on LCA.
- **Interpretation:** Complex – includes quantified impacts across all stages (upstream, core, downstream).
- **Scope:** Covers multiple impact categories like:
 - Global warming potential (fossil, biogenic)
 - Acidification
 - Water scarcity
 - Waste (hazardous, non-hazardous, radioactive)
- **Users:** Intended for **business-to-business (B2B)** and professional procurement, not general consumers.

Why This Matters:

- **Consumers** benefit from eco-labels for fast, trustworthy choices.
- **Businesses and decision-makers** rely on EPDs to compare detailed sustainability impacts when selecting products, suppliers, or strategies

Many different Multi criteria decision analysis methods exists



Notes:

💡 Why Use MCDA?

When evaluating sustainability trade-offs (e.g., cost vs. emissions vs. social impact), MCDA helps to:

- Prioritize among alternatives
- Structure complex decision problems
- Incorporate qualitative inputs (like expert judgment)
- Combine multiple indicators into a final decision or ranking

🌸 Common MCDA Methods (pie chart, left):

- **AHP (Analytical Hierarchy Process)** – 44%: Most widely used; allows pairwise comparisons of criteria; works with qualitative input.
- **TOPSIS** – 13%: Ranks alternatives based on distance from an ideal solution.
- **Weighted Sum** – 11%: Simple scoring approach with assigned weights.
- **MAUT (Multi-Attribute Utility Theory)** – 8%: Uses utility functions to reflect stakeholder preferences.
- **Outranking methods** – 17%: Includes ELECTRE and PROMETHEE, which rank options without full aggregation.
- **Other** – 7%: Includes advanced or customized approaches.

📊 Evaluation of MCDA Methods (table, right):

Feature	Meaning	Key Observations	📄
Compensatory (weak) vs non-compensatory (strong)	Whether good performance in one area can compensate for poor performance in another	Most widely used methods (like AHP, TOPSIS) are compensatory = weak sustainability	
Aggregate score potential	Can the method generate one final score?	Most methods can ✓	
Usable for single-product evaluations	Can it be used for just one system, not comparisons?	Only a few (AHP, MAUT, SEEBalance) work for single product cases ✓	
Qualitative input accepted	Can it use non-numeric inputs directly?	Only AHP, MAUT, MODM, and monetary weighting ✓	

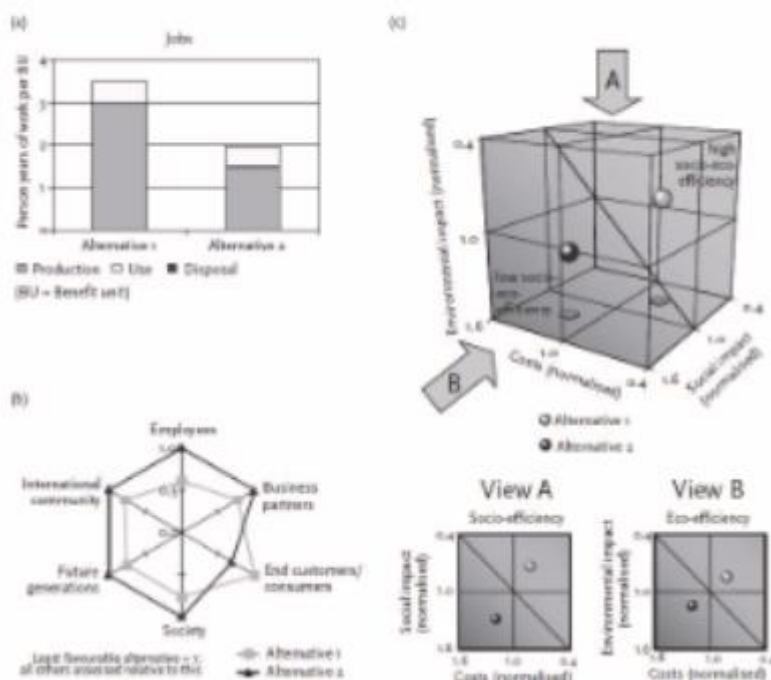
✓ When to Use AHP:

- You want to **involve expert opinions** early
- You lack precise data and need a **qualitative approach**
- You want **transparent and structured comparisons** of multiple sustainability criteria

✦ Takeaway:

MCDA is particularly helpful when your sustainability assessment includes trade-offs between different criteria or when you need to **turn complex results into actionable decisions**. AHP is often the go-to choice for sustainability decision-making due to its flexibility and user-friendliness.

Visualizations can provide overview



🖼️ Summary: How Visualizations Support Sustainability Decision-Making

🇮🇹 (a) Bar Chart – Job creation over life cycle stages

- **Y-axis:** Person-years of work per Benefit Unit (BU)
- Shows contributions from production, use, and disposal stages
- Useful for **single-indicator communication** (e.g., social impact: jobs)

🌟 (b) Radar Plot – Stakeholder-based comparison

- Plots indicators relative to various **stakeholders**: employees, society, customers, future generations, etc.
- Comparison of **Alternative 1 vs. 2**
- Can illustrate **social sustainability dimensions** clearly
- The **larger the area**, the better the overall stakeholder performance

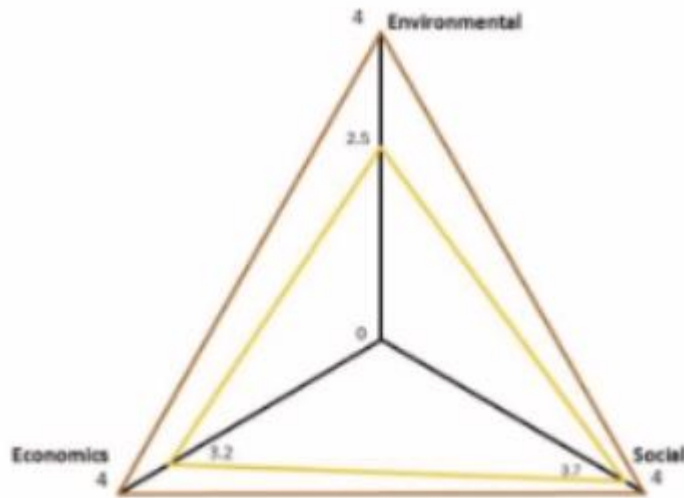
(c) 3D Plot – *Aggregated sustainability cube*

- Axes:
 - **X-axis**: Costs (normalized)
 - **Y-axis**: Social impacts (normalized)
 - **Z-axis**: Environmental impacts (normalized)
- Helps visualize **socio-eco-efficiency trade-offs**
- Top corner of the cube (high social, low cost, low environmental impact) = most desirable zone
- Projection views (**View A** = socio-efficiency, **View B** = eco-efficiency) simplify interpretation

Key Takeaways

- **1 indicator** → **bar/column graph**
- **2 indicators** → **2D scatter plots**
- **>2 indicators** → **radar plots or 3D cubes**
- Visualizations are especially useful when comparing alternatives, highlighting trade-offs, and making results understandable to stakeholders

Three sided triangle still requires aggregation within each dimension



Summary: Triangle Sustainability Visualization

What it shows:

- Each axis (from the center to a vertex) represents a **dimension**:
 - Top: **Environmental**
 - Bottom Left: **Economic**
 - Bottom Right: **Social**
- The value on each axis (from 0 to 4 in this case) is an **aggregated score** that combines multiple indicators into a single metric per dimension.
- The resulting triangle shape illustrates the **balance or imbalance** between the three dimensions for a given alternative.

Why it matters:

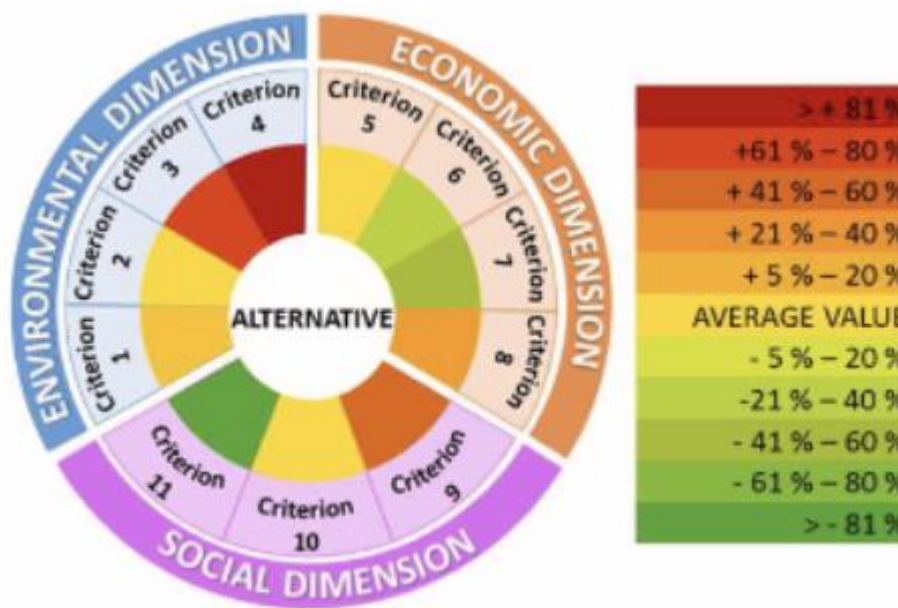
- This method enables a **quick visual comparison** of different solutions.
- Trade-offs become obvious—for example:
 - One alternative might score high environmentally but poorly economically.
 - Another might be well-balanced but not excellent in any one area.
- You can easily see where a solution needs improvement.

Key Takeaway:

While this triangle provides a **simple and powerful overview**, it **hides the internal complexity**—each point (e.g., 2.5 in environment) comes from multiple underlying indicators. So:

- It's helpful for **communicating with decision-makers**
- But you should also keep a **detailed view** available in your analysis to explain how each score was derived

Sustainability crowns compare several indicators within each dimension



Notes:

Purpose:

To **visually compare the performance of one alternative** (e.g., a product, technology, or policy) **against a reference or competing alternative**, across many individual indicators grouped by sustainability dimension.

How to Read It:

-  **Blue Sector – Environmental Dimension**
 - Includes indicators like CO₂ emissions, water usage, resource depletion.
-  **Orange Sector – Economic Dimension**
 - Includes cost indicators like capital cost, operational cost, lifecycle cost.
-  **Purple Sector – Social Dimension**

- Includes social indicators such as job creation, health impacts, and societal acceptance.
-  **Color Coding (based on legend at right):**
 - **Dark Green:** Much better than the reference (>81% improvement)
 - **Light Green to Yellow:** Slightly better or similar to the reference
 - **Orange to Red:** Worse than the reference (darker = much worse)

Each **criterion** (1 through 11) is individually color-coded depending on **how much better or worse it performs** compared to the baseline/alternative.

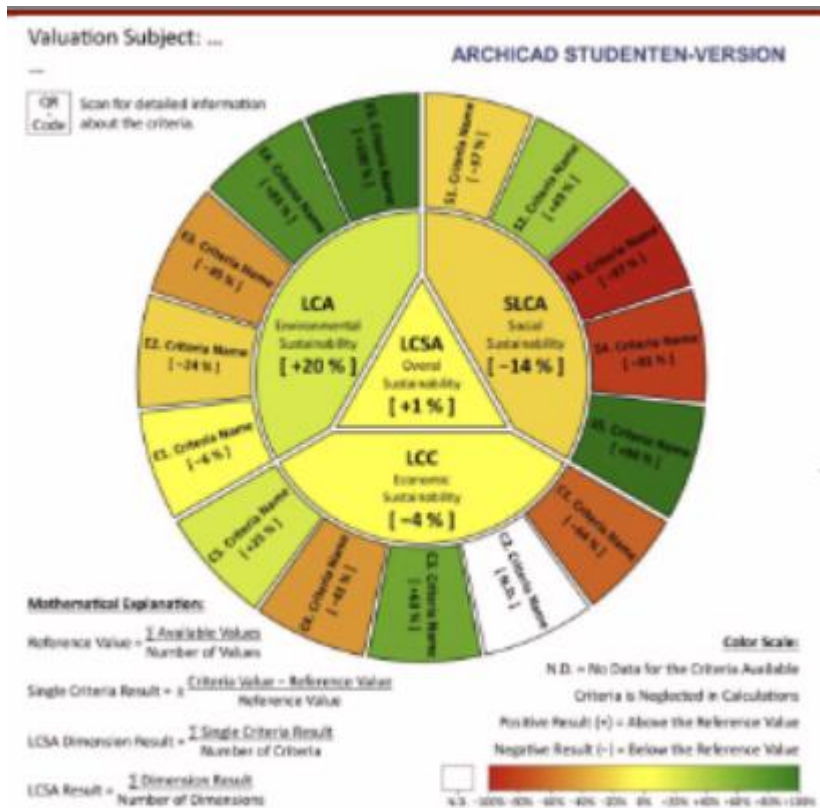
Advantages:

- Provides a **quick overview** across many criteria without heavy reading.
 - Easy to identify **which sustainability pillar** is driving improvement or underperformance.
 - Excellent for stakeholder presentations and communication.
-

Limitation:

As mentioned, this visualization **does not aggregate** the scores into a single metric. It's intentionally designed to **preserve the disaggregated view**—which is a strength when trade-offs need transparency, but a limitation if you want an overall “score.”

LCSA wheel fulfils all comparison criteria



Notes:

What the LCSA Wheel Does

Same core principles as the Sustainability Crown:

- Breaks down the three sustainability dimensions:
 - LCA (Environmental)
 - LCC (Economic)
 - SLCA (Social)
- Each of these contains multiple **individual criteria** (e.g., E1–E5 for environment).
- Color-coded scale shows **performance relative to a reference**:
 - Green = better than reference
 - Red = worse
 - Yellow = near reference
 - Grey = no data

💡 What's New in the LCSA Wheel?

- **Central Aggregation:**
 - Each dimension has a calculated **average deviation from the reference** (e.g., LCA = +20%, SLCA = -14%, LCC = -4%)
 - The **center circle** shows the **overall sustainability score** (here, LCSA = +1%)
 - **Quantified Percentage Improvements/Declines:**
 - Each criterion shows how much better/worse it is than the benchmark (e.g., "S1 = -97%")
 - **Mathematical Transparency:**
 - Calculation methods are shown at the bottom:
 - $\text{Single Criterion} = \text{Value} - \text{Reference}$
 - $\text{Dimension} = \text{Average of criteria}$
 - $\text{Overall} = \text{Average of dimensions}$
-

✅ Strengths of the LCSA Wheel

- **Combines granularity and overview:** You can still drill down into each indicator *and* get a high-level summary.
 - **Great for stakeholders:** It's both data-rich and intuitive to interpret.
 - **Self-explanatory format:** Mathematical definitions and color key make it reproducible without software.
-

🔧 Note:

This is not (yet) part of a software tool — but it can be **replicated in Excel or PowerPoint** by using conditional formatting, chart layering, or a circular dashboard layout

Summary

- Visualizations have strengths in providing overview
- MCDA methods have strengths in the guidance
- Evaluate assumptions, uncertainty and completeness
- Based on the types of data acquired and goal and scope of study, consider which types of decision support are adequate

notes:

1. Visualization tools:

- Help **communicate complex sustainability results**.
- Make it easier to spot **trade-offs** between alternatives.
- Examples include:
 - **Bar charts, radar plots, and 3D cubes** for single/multiple indicators.
 - **Sustainability crowns and LCSA wheels** for aggregated comparisons across environmental, economic, and social dimensions.

2. Multi-Criteria Decision Analysis (MCDA):

- Provides **systematic guidance** in choosing between alternatives.
- Methods like **AHP, TOPSIS, PROMETHEE**, etc., are used.
- MCDA is **useful when trade-offs must be made** between conflicting objectives (e.g. cost vs. environmental benefit).

3. Evaluate quality of your assessment:

- **Assumptions:** Are they sensitive or uncertain?
- **Data:** Which indicators are weak or missing? Can accuracy be improved?
- **Completeness:** Did you include all relevant life cycle stages and sustainability dimensions?

4. Matching method to situation:

- Choose decision support tools **based on data availability, goal/scope, and target audience** (e.g., public, business, policy).
 - **Simple visuals** (e.g., eco-labels) work for consumers.
 - **Detailed MCDA or LCSA tools** suit professional and policy contexts.

Lecture 12 - Sustainability transitions (Pathways and interventions)

Content

Interpretation

- Concepts
- Mathematical
- Visual

System dynamics

- Concepts
 - System states
 - Hysteresis and early warning signals
- Interventions
 - Economic, behavioural, and governance
- Tools to identify sustainability transitions
 - Focus on agent-based models

☐ Interpretation

- Concepts, mathematical logic, and visual tools used to make sense of LCSA results.

☐ System Dynamics

- Understanding system states (e.g. equilibrium, tipping points).
- Hysteresis and early warning signals — how systems resist and then suddenly change.
- Types of interventions: economic, behavioral, and governance-focused.
- Tools for identifying sustainability transitions — with an emphasis on **agent-based models** (ABMs).

Concepts

Notes:

☐ Why model?

When evaluating sustainability—whether of a product, activity, or lifestyle—we **develop models** to understand the process. These models can help us **predict outcomes** and assess **impacts** (e.g., environmental, social, or economic).

☐ Types of Models:

- **Mental models:**
Every human constantly builds mental models. Since we can't fully perceive or understand reality directly, our brains integrate sensory inputs to **build internal representations** of the world.
 - **Qualitative models:**
These can include **mind maps** or **causal loop diagrams** that show how elements are interconnected.
 - **Quantitative models:**
These might take the form of **life cycle assessments (LCAs)**, **mathematical functions**, or **differential equations** that describe change over time.
 - **System models:**
These describe **components** and **linkages** (e.g. feedback loops, delays, accumulations). They can be used to simulate behavior of the system under different conditions.
- **Purpose of Modeling:**
- Models help us **make inferences** and **test scenarios**.
 - They allow us to **explore dynamics** that would be hard to observe directly.
 - The aim is not to create a “perfect model” but one that is **useful**—that helps reveal insights.

What does sustainability transition mean?

- Systemic interventions to
 - Sustain acceptable outcomes on the three dimensions of sustainability
 - Change the system currently in unacceptable state
- Applies to all scales:
 - Manufacturing system
 - Resource exploitation system
 - Socioecological system

Notes:

What Does Sustainability Transition Mean?

Sustainability transition refers to systemic interventions aimed at:

1.  Sustaining acceptable outcomes across all three dimensions of sustainability:
 - Environmental

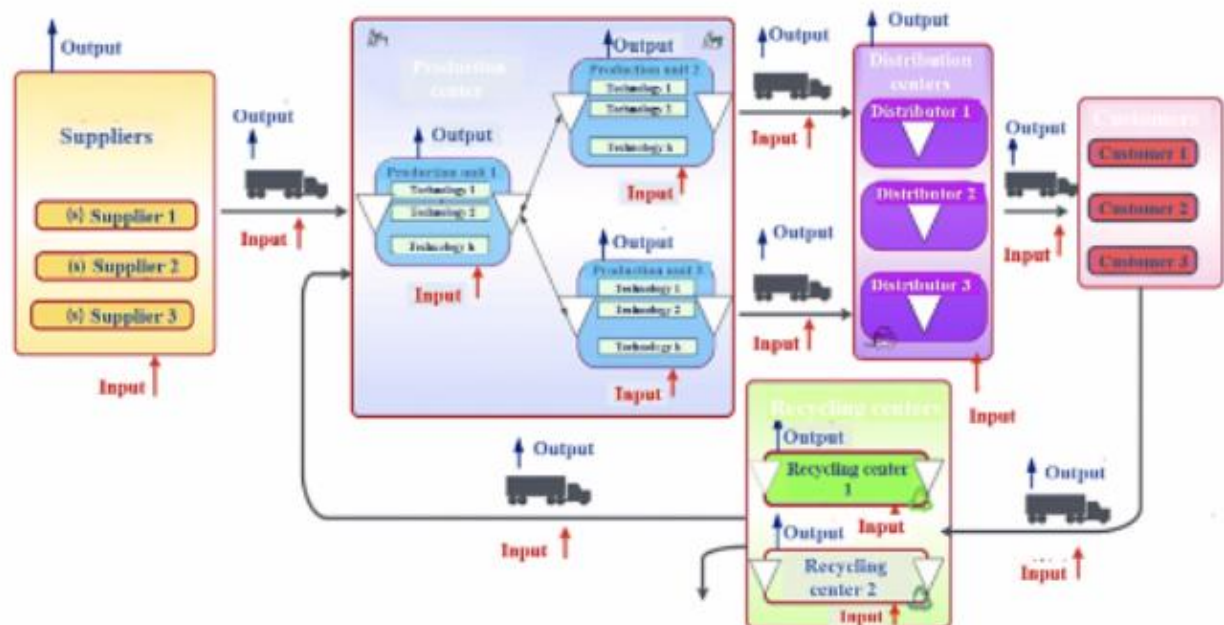
- Social
 - Economic
2.  Changing systems that are currently in unacceptable states:
- Systems where outcomes in one or more dimensions fall short of sustainability goals.
 - The aim is to realign them with sustainable trajectories.
-

 Applies at All Scales:

Sustainability transitions are relevant across many system types, for example:

-  Manufacturing systems
→ e.g. how factories produce goods, energy and material efficiency, emissions.
-  Resource exploitation systems
→ e.g. forestry, fisheries, or mining. A good example is your fishing exercise:
You modeled how overfishing disrupts ecological balance and identified limits for sustainable harvesting.
-  Socio-ecological systems
→ broader systems where human and environmental elements are tightly interlinked.
Examples include land use in agriculture, water basin management, or city planning.

System perspectives



Notes:

✿ Key System Components (from left to right):

1. Suppliers

- Provide **inputs (raw materials)** into the system.
- These inputs move to the **Production center**.

2. Production Center

- Contains multiple **production units** (e.g., Unit 1, 2, 3) each using different **technologies**.
- Transforms inputs from suppliers into outputs (intermediate or final products).
- Shows the **modular nature** of technological choices.

3. Distribution Centers

- Receive **output** from production.
- Serve as intermediaries to **deliver goods to customers**.
- Example: warehouses or logistics hubs.

4. Customers

- Final recipients or users of the products.
- Can represent households, businesses, or institutions.

5. Recycling Centers

- Handle **post-use materials**.
 - Close the loop by **recycling outputs** and feeding them back into the system.
 - Key for **circular economy modeling**.
-

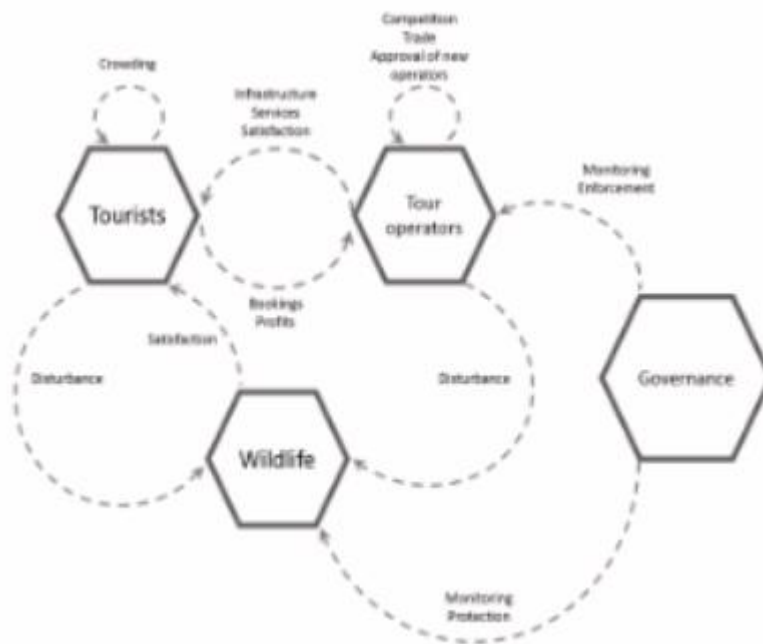
System Dynamics and Flow:

- **Inputs (red arrows):** Flow of raw materials or energy into a component.
 - **Outputs (blue arrows):** Flow of goods, products, or waste out of a component.
 - **Transport links (trucks):** Indicate **logistics/transport processes** between stages.
-

Interpretation:

- This is a **complex, multi-stage system** with **feedback loops** (e.g., recycling).
- Each block could be a **node** in a supply chain or life cycle model.
- It's ideal for **Life Cycle Assessment (LCA)**, **Agent-Based Modeling**, or **System Dynamics** modeling.
- Sustainability interventions could be analyzed at:
 - Supplier level (e.g., using greener inputs)
 - Technology level (e.g., switching to cleaner production)
 - Recycling efficiency (e.g., improving recovery rates)

Socioecological system perspectives



Multi-state dynamics



Key Concept: Multi-State Systems

The image shows a "landscape" with multiple **basins of attraction** and a **marble** (representing your system). Each basin represents a possible **system state** — like different configurations or modes the system can settle into.

Interpretation:

- The **marble** = your current system (e.g. a manufacturing system, resource management regime, city, or even a planet).
 - The **basins** = alternative stable states (e.g. a sustainable or unsustainable state).
 - The **contour lines** = gradients of stability or forces acting on the system (like policy, behavior, climate forces, etc.).
-

In the context of sustainability transitions:

There are two **strategic objectives**:

1. **Stay in a sustainable basin**

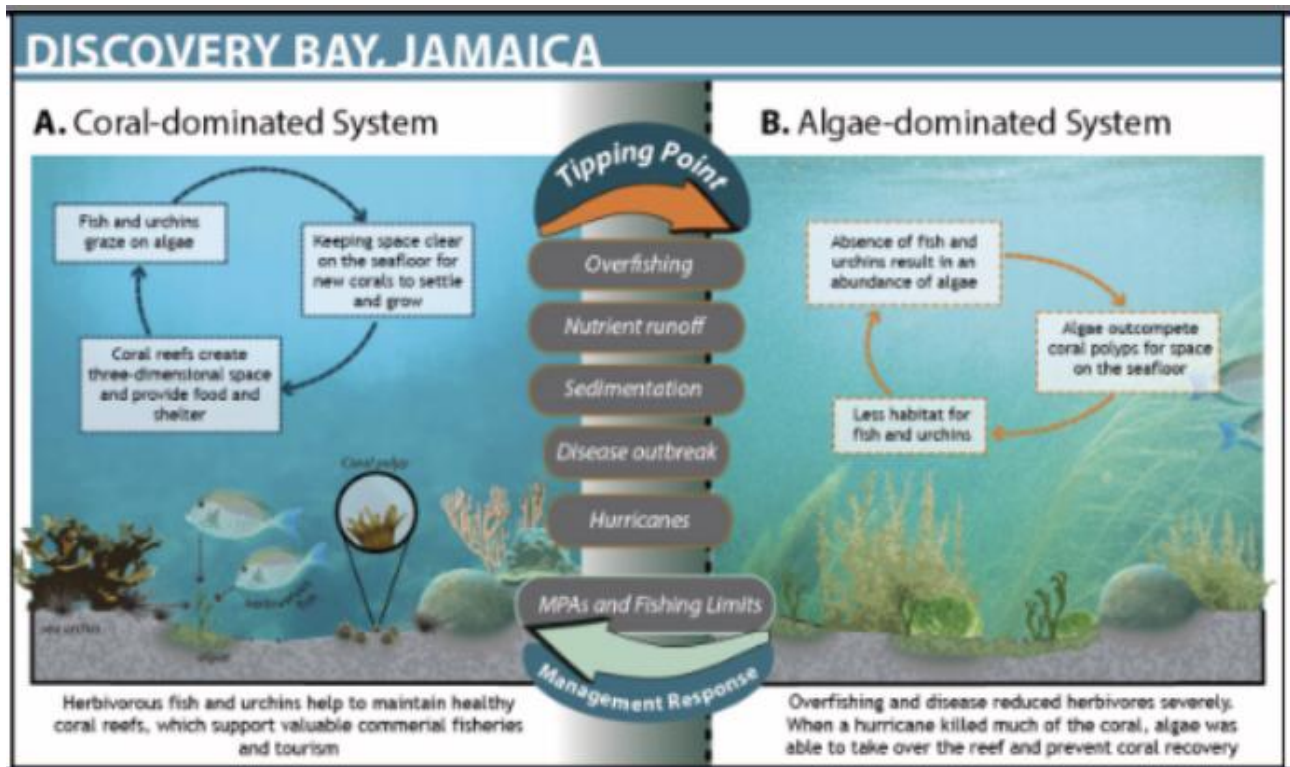
- If the system is already in a desirable, sustainable state, the goal is to maintain resilience.
- Interventions: increase system robustness, build adaptive capacity, monitor early warning signals.

2. **Shift out of an unsustainable basin**

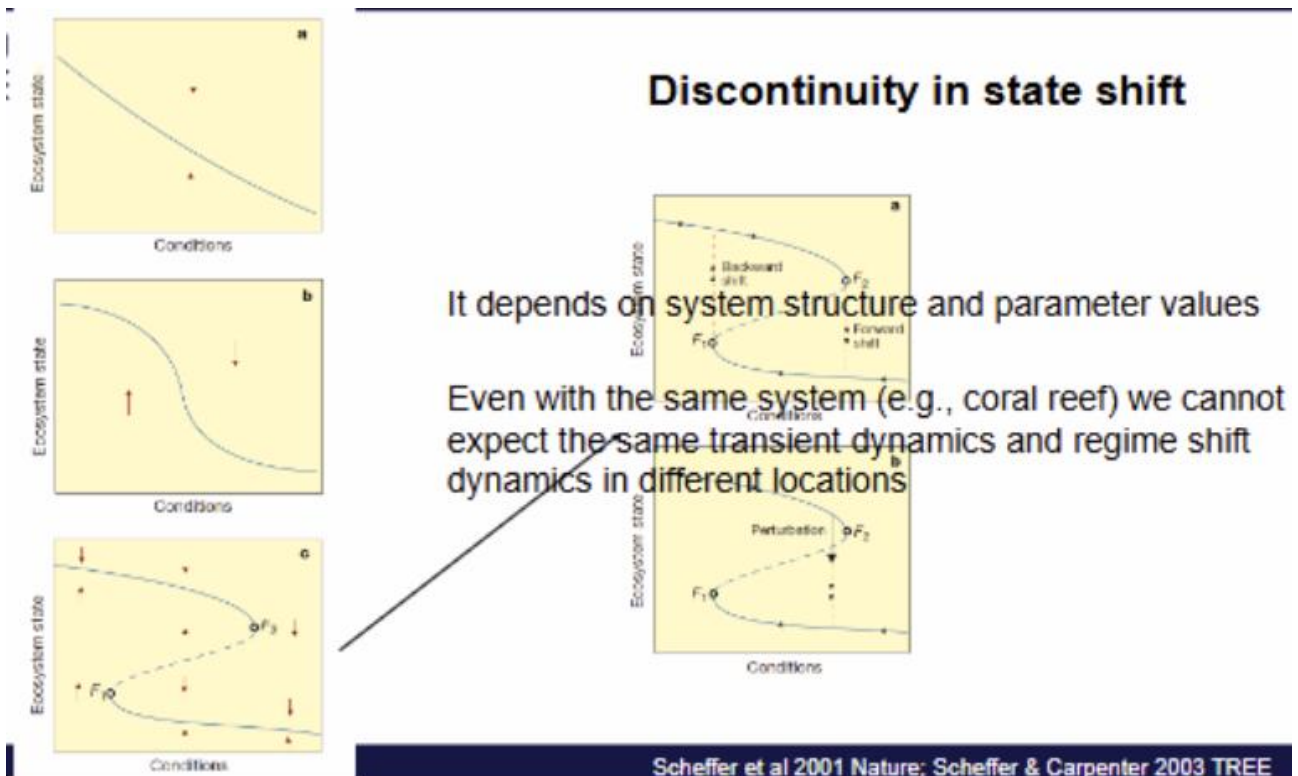
- If the system is in an undesirable (e.g. environmentally harmful or socially unjust) state, the goal is to push it out of that basin.
 - Interventions: redesign institutions, shift behavior, invest in clean technologies, etc.
-

This idea is tied to:

- **System dynamics** (feedbacks, delays, thresholds)
- **Hysteresis** (the effort to return after crossing a threshold is greater than leaving it)
- **Tipping points** and **resilience** theory
- **Agent-based modeling** to simulate interventions and behavior over time



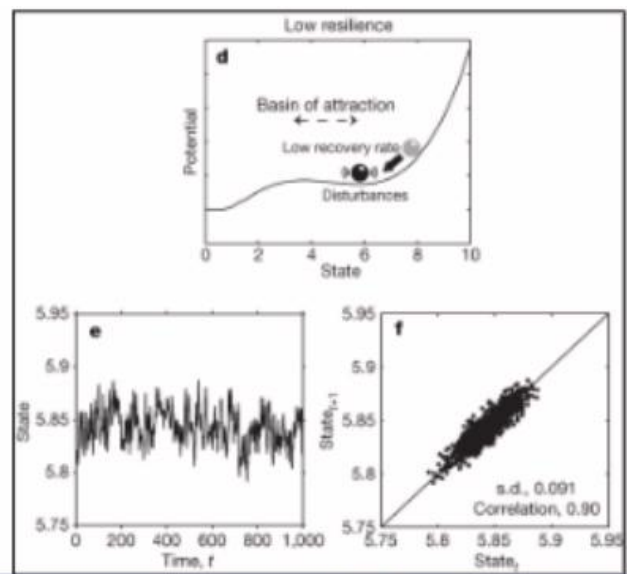
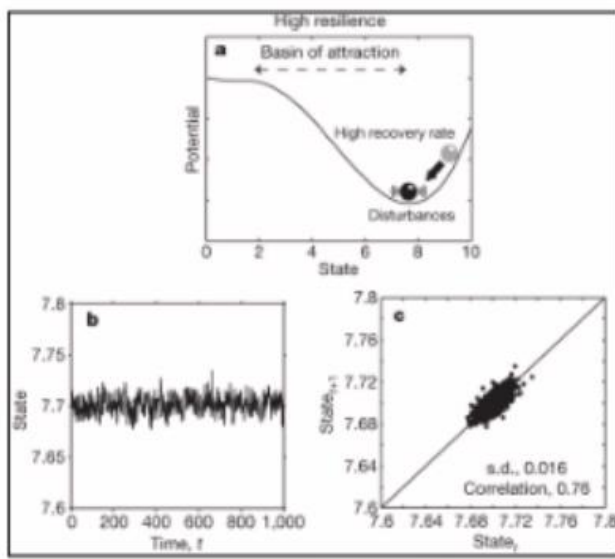
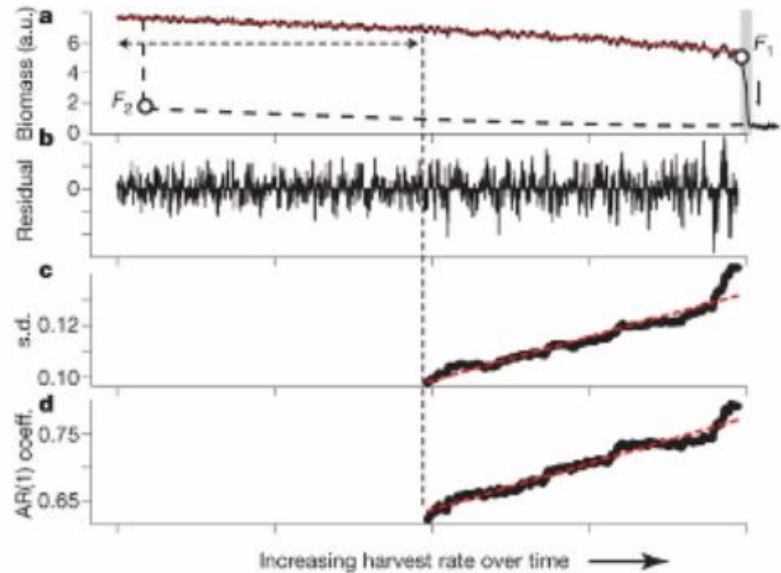
Hysteresis: Dependence of system state on its history



Early warning signals

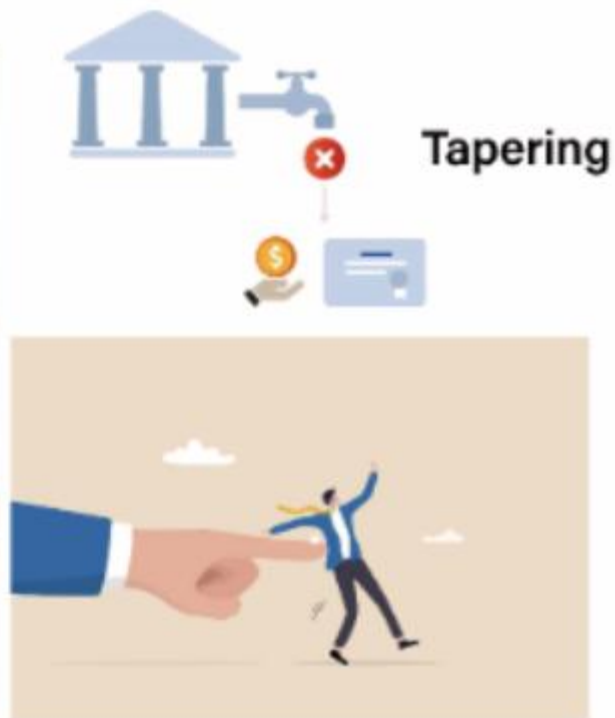
Behaviour as system approaches critical transition

- The system becomes less resilient to perturbations (a small change is more likely to tip it into another state)
- If it is less resilient the system is therefore more 'wobbly', do we see more variance?



Interventions

Systemic interventions



🔑 Types of Systemic Interventions (as suggested by the images):

1. Regulatory/Legal Interventions (Top Left)

- **Examples:** Laws, regulations, policies, bans.
- Often used by governments to mandate or prohibit behaviors (e.g. banning plastic bags, emissions caps).

2. Economic/Financial Interventions (Top Right - "Tapering")

- **Examples:** Subsidies, taxes, incentives, carbon pricing, grants.
- “Tapering” refers to **gradually removing subsidies or support**, encouraging a shift away from unsustainable practices.

3. Behavioral or Social Nudges (Bottom Right)

- **Examples:** Social pressure, education campaigns, behavioral nudges.
- Aimed at individuals or organizations to shift habits without enforcing hard rules.

🎯 Where Do Interventions Act?

Systemic interventions can target:

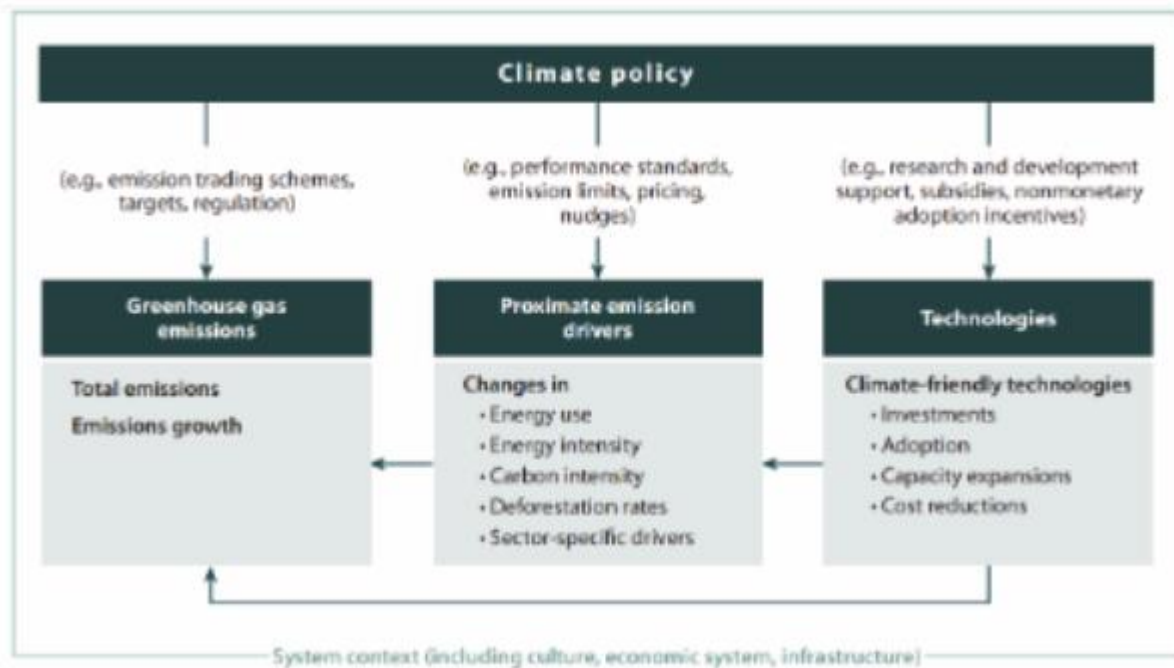
- **Individuals** (consumer behavior, awareness)
 - **Firms or producers** (technology choices, supply chain practices)
 - **Institutions** (policy frameworks, governance mechanisms)
 - **Socio-ecological interactions** (land use, conservation efforts)
-

✿ Why Systemic?

Because these interventions consider:

- Feedback loops
- Interdependencies
- Long-term effects
- The dynamic nature of systems (not static fixes)

Climate – is it all doom and gloom?



Key Takeaway: It's Not All Doom and Gloom—We Know What to Do

We already understand **what drives emissions, where to intervene, and which policies work**. The challenge lies in implementation and coordination, not in knowledge.

How the System Works

1. Climate Policy (Top Row)

Policymakers use a variety of tools:

- **Regulations & targets** (e.g. EU ETS)
- **Performance standards and pricing instruments** (e.g. CO₂ taxes)
- **Support for R&D and adoption incentives**

These shape the outcomes further down the system.

2. Technologies (Right Column)

Policies incentivize or enable:

- **Investment** in low-emission technologies
- **Wider adoption** (e.g. EVs, renewables)
- **Scaling capacity** (e.g. grid infrastructure)
- **Cost reduction** via innovation and economies of scale

These, in turn, affect how emissions are produced.

3. Proximate Emission Drivers (Middle Column)

These are **direct determinants** of emissions:

- Energy use and energy efficiency
 - Carbon intensity of energy
 - Deforestation
 - Sector-specific practices (e.g. agriculture, industry)
-

4. GHG Emissions (Left Column)

The **actual emissions** depend on the proximate drivers.

- Total emissions
- Growth or decline in emissions over time

🌍 System Context (Bottom)

All of this occurs **within a broader context**:

- Culture
- Economic systems
- Infrastructure

Context shapes how effective policies and technologies are across regions.

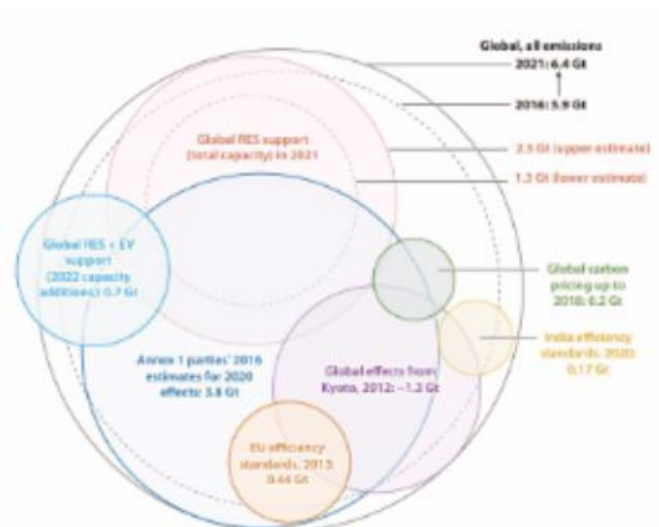
✅ Conclusion:

This model shows us where and how we can intervene. It provides optimism by outlining clear pathways to climate mitigation and demonstrates that:

- We are not blind.
- Tools exist.
- Interventions **can** work — with the right systemic coordination.

Climate - is it all doom and gloom

- Policy support for renewable development and uptake has been relatively most impactful to date



Notes:

● Key Point:

“Policy support for renewable development and uptake has been relatively most impactful to date.”

This is backed by **measurable emission reductions** from key interventions:

● Emission Reductions by Policy Type (Estimates in Gt CO₂ saved)

- **Global RES support (total capacity):**
 - ▲ 1.3–2.5 Gt (2021)
(RES = Renewable Energy Sources)
 - **Global RES + EV support (2022 additions):**
 - ◆ 0.7 Gt
 - **Annex I countries’ policy effects (2016 est. for 2020):**
 - 3.8 Gt
 - **Global effects from Kyoto Protocol:**
 - ~1.3 Gt
 - **EU efficiency standards (2013):**
 - 0.44 Gt
 - **India’s efficiency standards (2020):**
 - 0.17 Gt
 - **Global carbon pricing up to 2018:**
 - 0.2 Gt
-

🔄 Why Do Some Circles Overlap?

- Overlapping **indicates attribution overlap**.
For instance, some **Kyoto reductions** may also be counted under **national policies** or **renewable support schemes**.
-

🌍 Global Emissions Context:

- **2016 total:** 5.9 Gt
- **2021 total:** 6.4 Gt

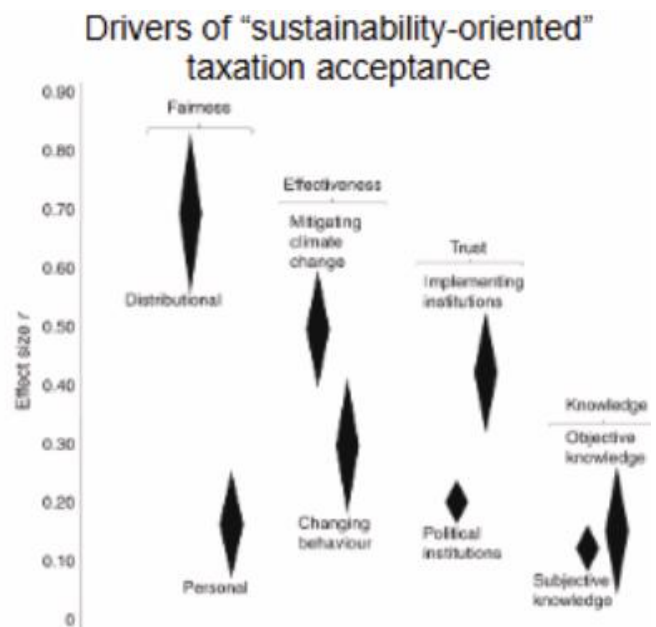
- While still growing, **emissions would be much worse** without these actions.

✓ Takeaway:

- We are **not paralyzed**. We **know what works**.
- Systemic public policies**—especially those supporting **renewables**—have been the **most impactful to date**.
- Further acceleration** is both possible and necessary.

The carrot and the stick

- Economic interventions: pricing, taxation, subsidies
- Internalise in price of goods & services their socioecological costs



Notes:

📌 Summary of Key Points

🏛️ Economic Interventions:

- Pricing** (e.g. carbon pricing)
- Taxation** (e.g. fuel/carbon/environmental taxes)
- Subsidies** (e.g. for renewable energy, EVs)

These aim to:

- 🔄 **Internalize the socioecological costs** (like carbon emissions) into **market prices** of goods and services.

Right Panel: Drivers of Public Acceptance for “Sustainability-Oriented” Taxes

(From Bergquist et al., 2022, *Nat Clim Change*)

This meta-analysis identified **which psychological and perceptual factors** most influence whether people **accept environmental taxes**:

Strongest Drivers:

- **Fairness (Distributional)**
→ If citizens perceive the tax as **fair** and **equitably distributed**, acceptance is high.
- **Effectiveness**
→ Especially if taxes are seen to **mitigate climate change**, not just change behavior.
- **Trust in Implementing Institutions**
→ Confidence that taxes will be used properly increases support.

▼ Weaker Drivers:

- **Personal Benefit**
- **Knowledge** (both subjective and objective)
- **Trust in Political Institutions**

Interpretation

This tells policymakers:

- It's not enough to say a tax is “green” —
You need to **show it's effective** and **fairly implemented**.
- **Public perception** and **trust** are as important as technical design.

Value creation

- There is public demand for ‘more sustainable’ goods & services
- A market creating opportunities (value creation)
- n.b. data showing the sustainable ‘value’ of goods & services is therefore valuable

Notes:

Public Demand → Market Opportunity

- There is increasing **consumer demand** for *more sustainable* goods and services.
 - This demand **creates new markets** for sustainable innovations.
 - Companies and public institutions can respond by **designing offerings** with sustainability benefits (e.g., carbon-labeled products, circular economy services).
-

Market Opportunity → Value Creation

- As new markets emerge, **value is created** not just economically, but also **socially and environmentally**.
 - Businesses that can demonstrate the **sustainability credentials** of their products gain a **competitive edge**.
-

Data as a Value Asset

- Data that proves or quantifies sustainability (e.g., life cycle assessments, carbon savings, biodiversity impact) becomes a **valuable asset**.
- When **consumers, producers, or regulators** can access and verify this data, it supports:
 - **Consumer trust**
 - **Regulatory compliance**
 - **Pricing premiums**
 - **Impact investing**
- There's growing discussion around **data ownership**, where individuals or organizations that generate sustainability data should **retain control and benefit** from it.

Behavioural interventions

- All of this works if people are interested in the new goods and services or can change habits
- The concept of nudging, as a public policy tool, has been proposed as a way to change the average behaviour of population to increase sustainability

notes:

All Interventions Depend on Human Action

- Regardless of how technically or economically sound an intervention is, it will only work **if people adopt it**.
 - This includes:
 - Choosing **sustainable products**
 - **Changing daily habits** (e.g., energy use, diet, transport)
 - **Supporting policies** that encourage sustainability
-

● Nudging as a Policy Tool

- **Nudging** means *gently steering people* toward better choices without restricting options.
 - It relies on **behavioural economics**, which studies how humans actually make decisions — often influenced by defaults, framing, or social norms.
 - Famous examples include:
 - Setting green energy as the **default option**
 - Placing vegetarian meals first in canteens
 - Comparing your energy use to your neighbors to promote **social competition**
-

🎯 Why It Matters for Sustainability

- Nudging helps **align personal behavior** with **collective environmental goals** without relying on coercion or large incentives.
- It can be **low-cost**, **scalable**, and **effective** when designed properly.
- However, its success **depends on context**, transparency, and **ethical design**.

Nudging and plastic bags

- 5 pence levy fee introduced in Scotland in 2015
 - About 80% reduction in bags used (replicated in England, Wales and Northern Ireland)
- 2021: levy increased to 10p (partly to ‘re-shock’ the system)
 - There is a dampening of the effect over time

Notes:

Initial Success of a Small Intervention

- **2015:** Scotland introduced a **5 pence levy** on single-use plastic bags.
- Result: **~80% reduction** in plastic bag use!
- The scheme was so effective it was **replicated in England, Wales, and Northern Ireland**.

 This is classic *nudge theory*: introducing a small friction (like a fee) makes people reconsider habitual behaviour — here, grabbing a plastic bag without thought.

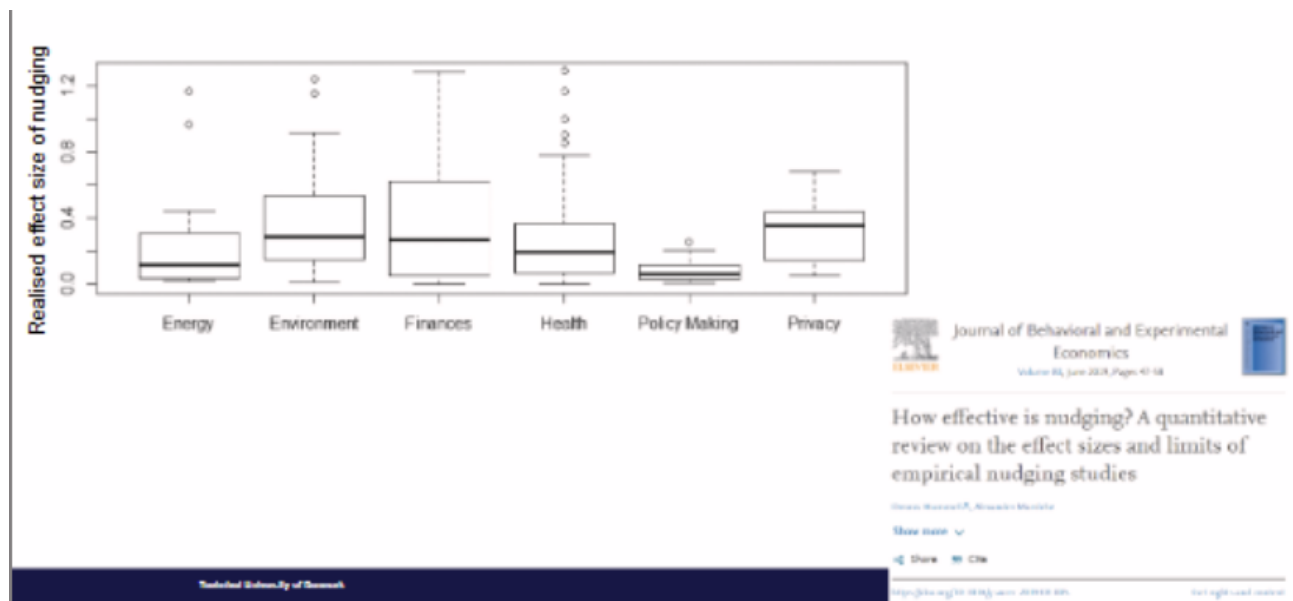
Re-shocking the System

- **2021:** The fee was **increased to 10p**.
 - Why? Because the initial **effectiveness began to decline**.
 - Over time, **habituation** reduces the behavioural impact of small nudges.
-

Key Insight from Research

- Nudging works best **initially** but tends to have a **dampening effect** over time.
- Periodic adjustments (like increasing the fee) are sometimes needed to *reinvigorate* behavioural change.
- The cited study (Adeyanju et al., 2021) is a **systematic review** showing that interventions like these can work — especially when part of a broader systemic change.

Meta-analysis of effects



Notes:



What's shown here:

- **Boxplots** of realized effect sizes from nudging interventions across six domains:
 - **Energy**
 - **Environment**
 - **Finances**
 - **Health**
 - **Policy Making**
 - **Privacy**



Key Takeaways:

- **Effect sizes vary widely** by domain.
 - **Finance** and **environment** show the highest median effects.
 - **Policy making** and **energy** show weaker results on average.
- **Outliers** (dots above the boxplots) indicate that *some interventions work extremely well*, even if the median effect is modest.
- There's substantial **variability and inconsistency**, suggesting that context matters a lot.

⚠ Important Caveat: Publication Bias

As mentioned in your notes:

Studies that find no effect are **less likely to be published**, which **inflates the perceived effectiveness** of nudging.

Recent critiques argue that if you **include unpublished or null-result studies**, the overall evidence base for nudging becomes far weaker.

🎯 Implication:

While **nudging** can be powerful, it's **not a silver bullet**. For policy or sustainability strategies, it should be:

- **Complemented** with regulatory, economic, or structural changes,
- **Carefully tested** for context-specific impact,
- And continuously **monitored and adapted** over time.



Notes:

- This high-profile **meta-analysis** reviewed **hundreds of studies** on nudging.

- After correcting for **publication bias** (the tendency to publish only positive results), the **average effect of nudging disappeared**.
 - Conclusion:
 - 👉 The evidence base for nudging may be **inflated**.
 - 👉 Nudging might **not be as robust or generalizable** as initially believed.
-

⚖️ So, Is Nudging Still Useful?

- **Yes, conditionally:**
 - In **specific, low-barrier decision contexts** (e.g., reminders, default options),
 - When **combined with structural or economic policies**,
 - And when **iteratively tested and adapted**.
 - But it **shouldn't be relied on** as the sole intervention for achieving sustainability transitions.
-

✿ Big Picture:

This reinforces the central message of the lecture:

Sustainability transitions require systemic interventions, combining:

- Regulatory levers 🏛️
- Economic instruments 💰
- Behavioral strategies 🧠
- Technological innovation ⚙️

Estimation

Methods – summary

- From a SES map to understanding its state, transient dynamics, and drivers of dynamics
- System dynamics (a branch of Ergodic theory)
- Deterministic approach:
 - Coupled ODEs (and PDEs)

- Qualitative approach: causal loop analysis / Bayesian Belief Network
- Stochastic approach:
- Simulations: agent-based models / multi-agent model

Notes:

Summary of Key Concepts

From SES Map to Dynamics

- Start with a **map of the system** (e.g., feedback loops, interactions).
- Use it to understand:
 - **System state**: current condition.
 - **Transient dynamics**: how the system evolves over time.
 - **Drivers of change**: what influences or pushes the system.

Methods in System Dynamics (Ergodic Theory)

Deterministic Approaches

- Focus on **predictable, rule-based changes** over time.
- **Mathematical tools**:
 - **Coupled ODEs/PDEs** (Ordinary/Partial Differential Equations) → used to model interactions quantitatively.
- **Qualitative methods**:
 - **Causal loop diagrams** → feedback-based visual models.
 - **Bayesian Belief Networks** → probabilistic reasoning about relationships.

Stochastic Approaches

- Use **probabilistic simulations** to explore uncertainty and interaction effects.
- **Agent-Based Models (ABMs)**:
 - Model **individual agents** (e.g., consumers, producers) with **simple rules**.
 - Observe **emergent patterns** from micro-level interactions.
 - Also called **Multi-Agent Models** (interchangeable term here).

Agent-based models – what does this mean?

- Systems of interest are often composed of many interacting components and can exist in multiple states
- Tractability challenge
- Their dynamics and emergent properties (sustainability is often an emergent property) are often near impossible to estimate deterministically or statistically
- Stochasticity plays a non-trivial role in system dynamics
- We can replicate a model of the interacting components to assess when to simulate their behaviour and estimate parameters of interest.

Notes:

Key Points from the Slide

1. Complex Systems with Many Interacting Parts

- Most sustainability-relevant systems (like food systems, energy markets, ecosystems) are composed of many **interdependent elements**.
- These systems can **exist in multiple stable or unstable states**, depending on how the parts interact.
- **Challenge:** It's often hard to *track* or *predict* the full system's behavior due to this complexity.

2. Emergence & Dynamics Are Hard to Estimate

- Sustainability is not always a fixed input but often an **emergent property** of system-wide interaction.
- This makes **deterministic** (fixed-rule) or **statistical** (pattern-based) methods insufficient or misleading.

3. Stochasticity Matters

- Real systems contain **randomness** or noise (e.g. climate variability, human behavior unpredictability).
- ABMs can incorporate **stochastic elements**, making them more realistic for simulating uncertainty.

4. Simulation Enables Learning

- With ABMs, you can simulate **individual agent behaviors** and see **how the system evolves**.
 - You can **test interventions**, observe **emergent outcomes**, and adjust model parameters to assess **policy effectiveness** or **thresholds for transitions**.
-

✅ Why Use ABMs for Sustainability Transitions?

- Suitable for **bottom-up modeling**: understanding how individual actions aggregate into system-level patterns.
- Allow for **exploration of “what-if” scenarios**: e.g. “What happens if we subsidize electric cars?” or “How do people respond to carbon pricing?”
- Handle **heterogeneity and local interactions**, which are key in socio-environmental systems.

Socioecological models of sustainable wildlife tourism

Notes:

🔗 Why Use Socioecological Models for Wildlife Tourism?

Wildlife tourism involves **human–nature interactions** that must be balanced:

- Too many tourists degrade the environment or disturb animals.
- Too few tourists may result in insufficient income for local communities or conservation efforts.

This makes it a **classic socioecological system (SES)**:

- **Social dimension**: tourists, local guides, policymakers, businesses.
 - **Ecological dimension**: animal populations, habitats, biodiversity.
 - **Economic dimension**: income from tourism, local livelihoods, infrastructure costs.
-

🧠 How ABMs Apply Here

1. Agents

- **Tourists**: with preferences (e.g. wildlife sightings, costs), behaviors (e.g. frequency of visits), and reactions to regulations (e.g. fees, caps).

- **Wildlife:** modeled as dynamic populations affected by human presence (e.g. stress, displacement).
- **Tour operators / park rangers:** may react to demand, follow rules, or enforce guidelines.

2. Interactions

- Tourist decisions impact wildlife behavior and stress.
- Wildlife abundance impacts tourist satisfaction.
- Park managers may implement rules (quotas, zones, fees) in response to ecological or social conditions.

3. Emergence

The model can show:

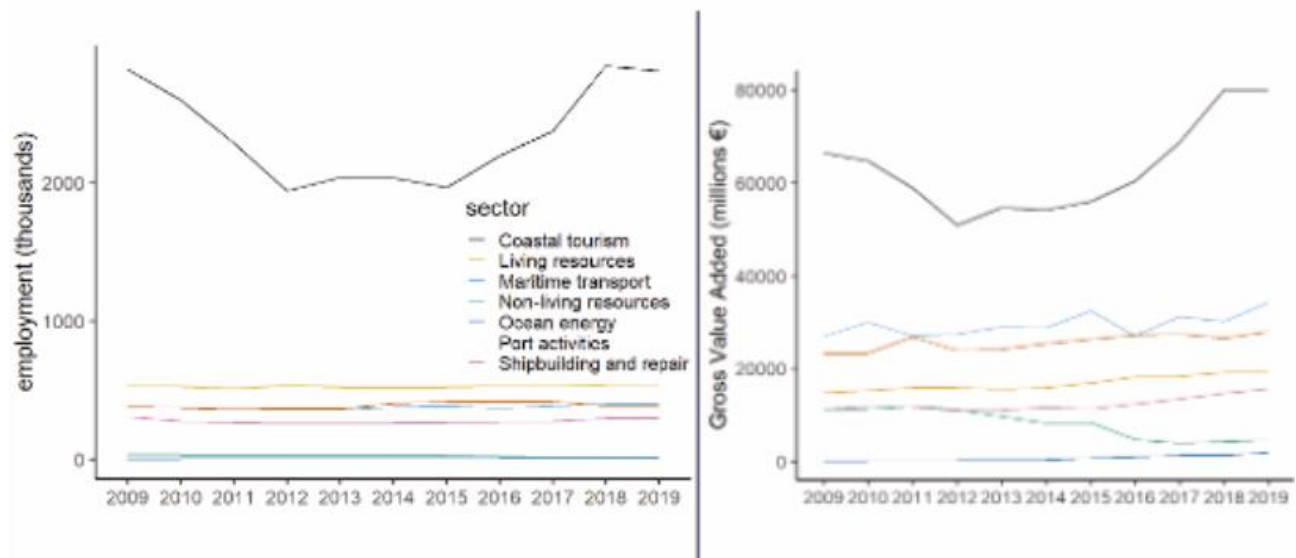
- **Tipping points** (e.g. where too many visits cause population collapse).
- **Sustainable states** (where tourism and conservation coexist).
- **Unintended consequences** of rules (e.g. displacement of tourists to unregulated areas).

Policy Testing with Simulations

Using this model, you can test interventions:

- **Introduce a daily cap on tourist visits** → See if wildlife recovers.
- **Subsidize off-season tourism** → Assess if pressure is distributed better.
- **Add eco-education nudges** → Explore if behavior changes without coercion.

EU Coastal tourism – dominant sector of the Blue Economy



Notes:

Why Coastal Tourism Matters for Sustainability

Sheer Size & Value

- **Employment** (left graph): Coastal tourism employs far more people than any other maritime sector — over 2 million by 2019.
- **Gross Value Added (GVA)** (right graph): The sector generates the most economic value in the Blue Economy, reaching nearly €80 billion by 2019.

Under-recognized Influence

- The dominance is often **underappreciated** because:
 - It is composed of **many small or medium-sized enterprises (SMEs)**.
 - These businesses may lack lobbying power or unified governance structures.
 - As a result, policy design may **under-prioritize** sustainability in tourism compared to more centralized industries like shipping or energy.

Why This Matters for Socioecological Modeling

- Coastal tourism's **scale and complexity** make it an ideal candidate for **agent-based models**:
 - Each business (or tourist) can be modeled as an **agent** with local behavior, preferences, or strategies.

- Environmental feedbacks (e.g. overcrowding, marine pollution) and economic consequences (e.g. declining visitor satisfaction) can emerge from these micro-level interactions.
- Understanding how **small decisions accumulate** to produce **large-scale impacts** is essential for policy.

Tourism & Recreation Biodiversity Footprint

Threats	all	marine
<i>Habitat modification</i>	6,491	1,569
<i>Disturbance</i>	4,291	1,097
Both	8,836	1,757
Total species assessed	147,517	17,081
Proportion of assessed	6%	10%

Notes:

Tourism & Recreation: Profitable but Impactful

Biodiversity Footprint of “Innocent” Leisure

- While tourism feels benign — just people enjoying nature — it **has measurable negative impacts on biodiversity**:
 -  **Habitat Modification** (e.g. building infrastructure like trails, marinas, viewing platforms).
 -  **Disturbance** (e.g. human presence stressing or displacing wildlife, altering feeding or nesting behavior)

Implication for Sustainable Tourism:

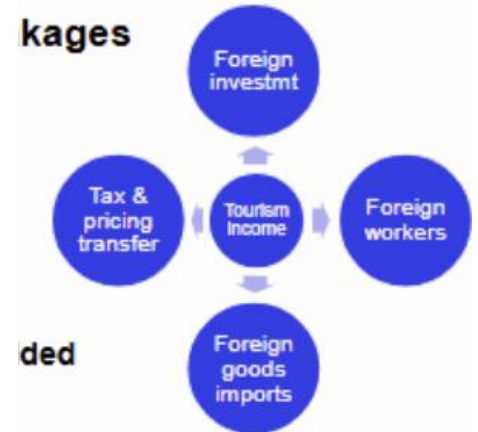
- These statistics are **not trivial**. Up to **10% of assessed marine species** are negatively affected by tourism activities.
- This demonstrates that **even non-consumptive uses of nature (like viewing dolphins or birds)** can lead to **population-level pressures** when done at scale or without regulations.

- It reinforces the need for **agent-based modeling or socioecological system thinking** — because cumulative, emergent impacts from many small “harmless” activities can be major.

Tourism & recreation economic leakages

1 € spent on tourism generate the following value added

country	Inbound tourism	Domestic tourism
Spain	0.97	1.22
Portugal	0.78	0.55
Italy	1.20	1.14
Germany	~0.63	~0.63
UK	0.67	0.51



Notes:

Tourism & Recreation Economic Leakages

Why doesn't all tourism income stay local?

Even when tourism brings money in, **that value often leaks out** due to:

-  **Foreign investment** in hotels or infrastructure
-  **Foreign labor** not spending income locally
-  **Imported goods** for tourist services (e.g. food, souvenirs)
-  **Taxation & pricing structures** benefiting entities outside the destination

Implication:

A **high tourism footprint** doesn't always translate into high **local benefit**. Sustainable tourism isn't just about environmental impact — it must also maximize **local economic retention**. That might mean:

- Supporting local ownership
- Reducing reliance on imported goods
- Promoting policies to retain tax and wage benefits in host regions

Navigating towards Sustainability?

Integrative modelling of resource management actions in coupled nature-human systems

Notes:

✿ The Problem:

- Tourism brings **economic benefits** but causes **biodiversity threats**, **social tension**, and **economic leakages**.
 - As shown by the graffiti “TOURIST GO HOME,” **community frustration** emerges when tourism growth **erodes local quality of life** or benefits outsiders disproportionately.
 - Nature and society are **coupled systems**: one affects the other constantly.
-

🔄 The Solution Direction:

► Integrative Modelling of Resource Management

That means combining:

- **Ecological dynamics** (wildlife disturbance, habitat change, etc.)
- **Economic structures** (value retention, leakages, pricing)
- **Social dimensions** (community well-being, acceptance, fairness)
- **Behavioral responses** (tourist and local reactions)

Through:

- 📊 **System dynamics** (e.g. feedback loops, thresholds, resilience)
 - 🤖 **Agent-based models** (simulating individual behaviors in systems)
 - 🇮🇹 **Multi-criteria decision tools** (balancing trade-offs in interventions)
-

💡 The Aim:

Create **sustainability transitions** that:

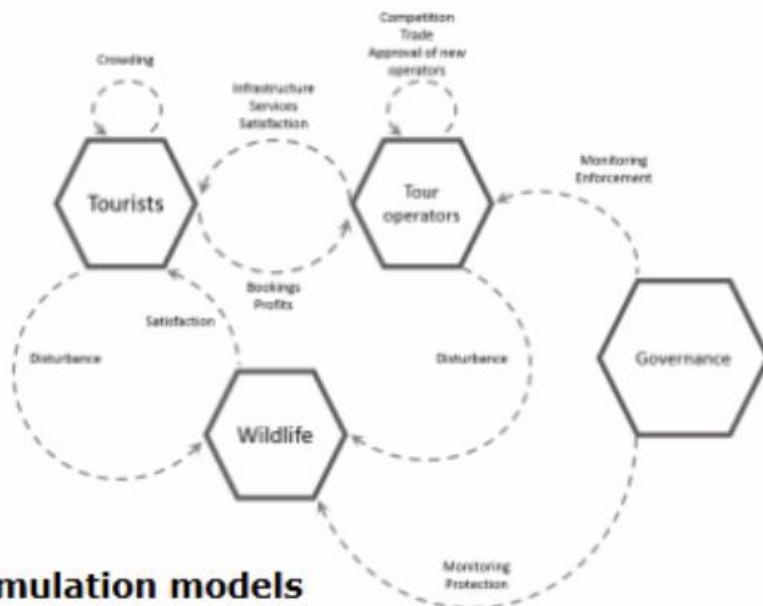
- Maintain ecological integrity
- Maximize **local economic benefit**
- Avoid social backlashes
- Foster **community ownership and acceptance**

Governance structure most likely to yield sustainability?

Triple win:

Economic prosperity
Environmental quality
Social justice
But still economic target
(lack of monopolisation)

Multi-agent simulation models



Notes:

🧠 What This Model Does:

It identifies **key agents** and **interactions**:

- **Tourists**, who contribute money but also cause **disturbance** to wildlife.
- **Tour operators**, who benefit economically but also mediate the disturbance.
- **Wildlife**, whose well-being underpins the sustainability of the system.
- **Governance**, which intervenes through **monitoring and enforcement**.

🎯 Sustainability Framed as a "Triple Win":

1. **Economic Prosperity**
→ Satisfied tourists and financially viable operators.
2. **Environmental Quality**
→ Protected wildlife and low long-term disturbance.
3. **Social Justice**
→ Avoid monopolisation; ensure fair access and opportunity across operators.

🔄 Feedback Loops in the Model:

This is key to **system dynamics** and agent-based modeling:

- Wildlife health affects tourist satisfaction.
 - Crowding reduces experience quality.
 - Governance enforcement shapes operator behavior.
 - Operator decisions affect both **profit** and **disturbance** levels.
-

From Model to Policy:

As you noted, this isn't just theory:

“... this system actually has been practically used to inform governance structures for tourism in some countries.”

That's the **real value** of system-based sustainability models:

- They **simulate alternative governance strategies**.
- Allow prediction of **emergent properties** like collapse or fairness.
- Help policymakers balance **trade-offs**.

Governance models

- Structures
 - Free market
 - Regulations
 - Hybrids (eg Cap and Trade/co-management)
- Accounting for
 - Responsibility assignment
 - Uncertainties (mngnt & monitoring)
 - Operator & customer behaviour

Notes:

Types of Governance Structures Simulated:

1. **Free Market**
 - No limits; operators and tourists act freely.
 - Often leads to overuse and unsustainability.
2. **Regulations**
 - Top-down restrictions like quotas or bans.
 - May be effective but can be rigid and costly to enforce.

3. Hybrids

→ **Cap and Trade** systems allow for flexible economic exchange under an ecological ceiling.

→ **Co-management** involves shared responsibility between state and stakeholders.

What Makes These Simulations Powerful:

You account for:

- **Responsibility assignment**
→ Who is accountable for environmental outcomes?
 - **Uncertainties**
→ Simulations reflect real-world limits in knowledge, monitoring, and management.
 - **Behavioral variation**
→ Different responses from **tour operators** and **tourists** are modeled, capturing diversity in real-world decision-making.
-

“Carrot and Stick” Approaches:

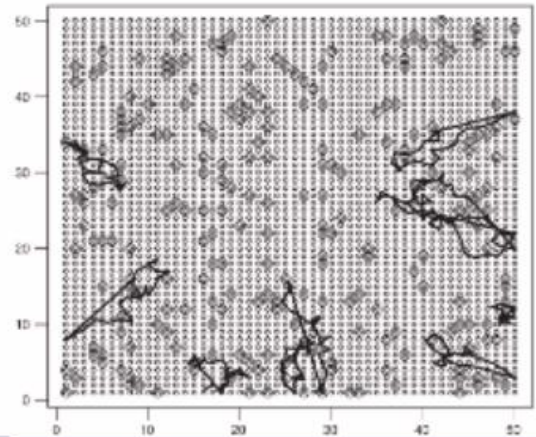
- **Carrot = Incentives** (e.g. tax benefits, reputation rewards, flexible trading)
- **Stick = Penalties** (e.g. fines, usage caps, revoking licenses)

Simulating these helps test which combination of rewards and punishments best leads to the **triple win**:

- Economic prosperity
- Environmental quality
- Social justice

Tour operator component

- At each tour, correlated random walk in the 'study area'
- Discrete grid: encounter overlap wildlife/operator
- Encounter duration: Weibull distribution
- Each year, operators are assigned a sustainable quota (delivered or not depending on management scheme)
- Each operator run one tour per day run if enough bookings are obtained
- Encounter dependent on $p_{\text{encounter}}$, duration dependent on $\max_y$ (/365 days). Decision: cooperate and adhere to maximum encounter duration or not
 - Phenotypes: optimist, pessimist, trustful or envious
- Rating dependent on tourist satisfaction



- Tradeable Wildlife Allowance – TWA: When a tour operator at the end of the day has spent all the time they were allowed with the animals, they will try to buy some extra TWA from tour operators that did not spend much of their allowed time.
- If a defecting tour operator is detected, 1000 money units are subtracted from their profits
 - A ticket is about 30 units).
- At the end of the year
 - the tour operators also decide on investments for the next year.
 - infrastructure or services
 - Update ticket price – function of demand:supply ratio.
 - If profits= for 3 years the operator retires.
 - New tour operators can start every year or every 6 years (scenarios)

Notes:

Tour Operator Component – Key Mechanics:

1. Spatial Movement & Encounters

- Each operator conducts a **correlated random walk** (i.e., non-random but not fully predictable) in a **discrete grid** representing the study area.
- An **encounter** occurs when an operator overlaps spatially with wildlife.
- Encounter **duration** is drawn from a **Weibull distribution**, allowing flexibility to model varying length encounters.

2. Sustainable Quotas

- Operators are assigned a **yearly sustainable quota**, which may be enforced or not, depending on governance style.
- This mimics policies like cap-and-trade systems or open-access markets.

3. Behavior-Based Decisions

- Operators can **cooperate or defect** from recommended encounter durations.
- Decisions are influenced by **phenotypes**:
 - **Optimist**: Believes others will cooperate.
 - **Pessimist**: Assumes others will defect.
 - **Trustful**: Cooperates even at personal risk.
 - **Envious**: Acts to outperform others regardless of ecological cost.

4. Income & Ratings

- Operators run **one tour/day** if they get enough bookings.
- Their **ratings**, and thus future bookings/income, depend on **tourist satisfaction**.
- Tourist satisfaction is linked to **time spent with wildlife**—a tension between profit maximization and conservation ethics.

5. Spatial Realism

- The visual (on the right) shows actual **simulated operator paths** interacting with **wildlife distribution points**.
- This spatial component is crucial for modeling **disturbance overlap** and **wildlife stress zones**.

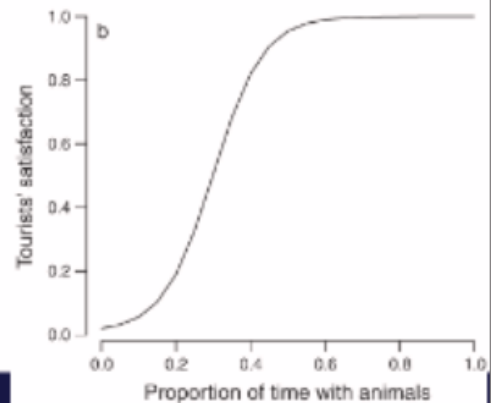
Why This Matters

This agent-level modeling allows you to:

- Examine **how individual decisions scale up** to system-level sustainability or collapse.
- Evaluate **trade-offs** between economics and ecology.
- Test which governance designs (free market, regulated, hybrid) work **under different behavioral assumptions**.

Tourists component

- A tourist population (1,000,000) is sampled every day and visitors try to book tours
- Scenario: specialists, generalists, mixed
- 3 income groups influencing willingness to pay
 - High income (10% of the total population) $N(60, 1.30)$
 - Middle income $N(45, 3.5)$
 - Low income $N(30, 1.5)$
- Satisfaction dependent on time spent with animals
- Seasonal and annual trends in daily numbers



Notes:

Tourist Component – Key Features:

1. Large-Scale Population Sampling

- 1,000,000 virtual tourists sampled daily.
- Each tries to **book tours** with available operators.

2. Tourist Typologies (Phenotypes)

- **Specialists:** want to see a specific species and spend significant time with them.
- **Generalists:** value the overall nature experience, less specific expectations.
- **Mixed:** a realistic combination of both.

These preferences **influence satisfaction and booking behavior**.

3. Economic Differentiation

Tourists are assigned to **income groups** that influence their **willingness to pay**:

- **High income:** $N(60, 1.3)$ → top 10% (most willing to pay).
- **Middle income:** $N(45, 3.5)$.
- **Low income:** $N(30, 1.5)$.

This affects **operator revenue potential** and pricing strategies.

4. Satisfaction Driven by Time with Wildlife

- Shown in the **satisfaction curve** (right): steep increase up to ~50% time with animals, then levels off.
- Operators optimizing satisfaction must balance **tour frequency** with **encounter quality**.

5. Seasonality & Trend Dynamics

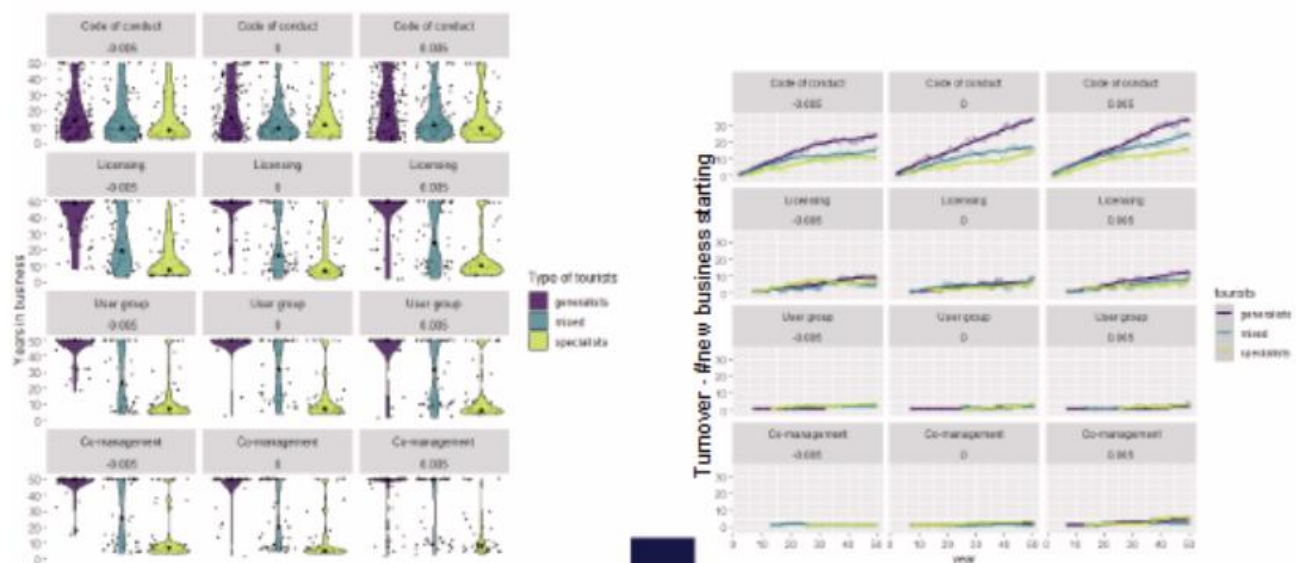
- Simulation includes **seasonal** and **annual fluctuations** in tourist numbers.
- Reflects real-world demand patterns (e.g., holidays, migrations, climate).

Systems-Level Insight

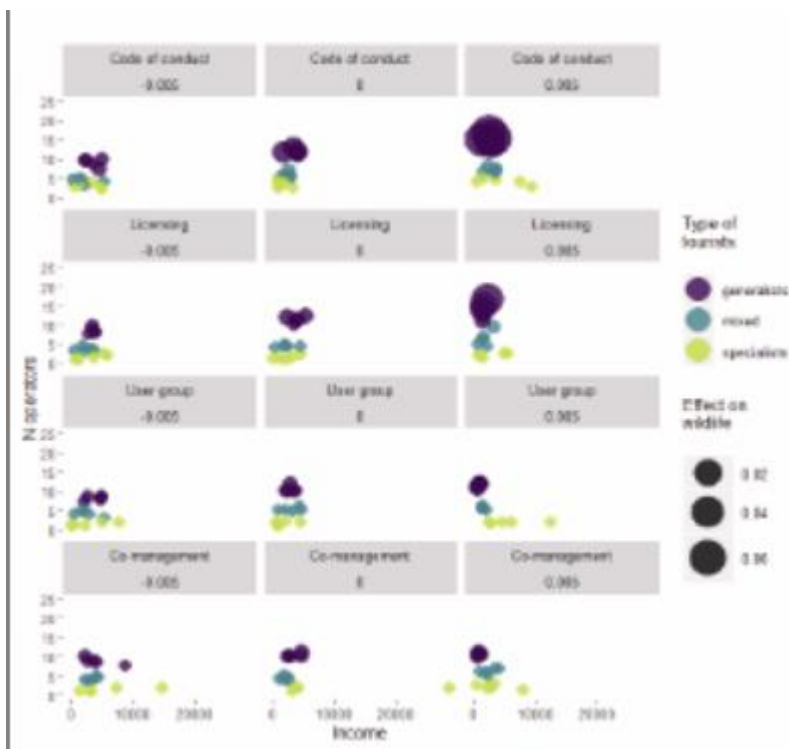
The tourist module feeds back into:

- **Operator performance** (via satisfaction-driven ratings).
- **System pressure** (more tourists = more disturbance).
- **Revenue flows and social equity** (depends on access by different income groups).

Companies lifespan



Monopolisation



Outcomes: sustainable governance space



Community-led public-private partnerships

offer the wider sustainable space
(over management error & defection rate)

Adaptive management

Changes with destination
maturation phases

Tourist typology

Tourist behaviour &
typology drives
sustainability too

Notes:

1. Community-led Public-Private Partnerships

- **Most robust governance model** across varying:
 - Tourist behaviors
 - Wildlife ecologies

- Operator strategies
 - **Why it works:** Broad sustainable space because it tolerates higher:
 - **Management error**
 - **Operator defection**
 - Emphasizes **local accountability** and **shared responsibility** (operators aren't passive rule followers—they co-manage).
-

2. Adaptive Management

- **Governance needs to evolve with the life cycle of a destination:**
 - **Early phases:** Fewer tourists, often specialists → different needs.
 - **Mature phases:** Generalists dominate → higher pressure and expectations.
 - Management tools must adapt accordingly (e.g., stricter caps, more nuanced incentives).
-

3. Tourist Typology

- **Tourist behavior directly influences sustainability** outcomes:
 - **Specialists:** Often higher satisfaction per encounter, willing to pay more.
 - **Generalists:** May tolerate lower-quality experiences but create crowding.
 - **Attraction strategy matters:** Sustainability is affected by the kind of tourists a site attracts.
-

Bottom Line from the Model

Even with strong governance and environmental intentions:

- If the **wrong type of tourists dominate**, operators' behavior changes (e.g., overuse).
- Sustainable outcomes require alignment between:
 - Governance model
 - Tourist typology
 - Business incentives

Summary

- We have a toolkit of systemic interventions that can change functional relationships in the way people use the planet
- We have the mathematical tools to understand whether and how system can change state and identify factors that can drive this change
- Systemic changes to improve 'sustainability' are possible but contextual (the same change will not necessarily provide the same systemic outcomes across systems)

Notes:

Toolkit of Systemic Interventions

- We have **diverse levers** to shift systems:
 - **Economic:** taxes, subsidies, markets
 - **Behavioral:** nudging, habit change
 - **Governance:** regulation, co-management, hybrid models
-

Mathematical and Modeling Tools

- System dynamics and agent-based models let us:
 - Explore **nonlinear dynamics**
 - Identify **thresholds, feedbacks, emergent behaviors**
 - Simulate and **evaluate intervention outcomes**
-

Context Matters

- Systemic interventions are **not one-size-fits-all**:
 - **Same policy** (e.g. tourist levy) yields **different effects** in different systems.
 - Outcomes depend on **ecology, behavior, governance, culture**.
- But: we **can predict** likely outcomes by understanding system structure and **running simulations** tailored to context.

Activities quiz

Self-assessment

Module 1:

Question 1 what are your world views about the role of humans in their environment?

- living from nature
- living with nature
- don't know
- living in nature
- living as nature

Question 2 What is sustainability about? Question 2 options:

- it is about saving the environment
- it is about finding a way for people to live a healthy and productive life in harmony with nature
- it is about maximising economic welfare
- it is about eradicating poverty
- Note: sustainability is about ensuring wellbeing equitability within planetary limits and ensuring that

this balance is maintained for future generations

Question 3 when did the current consensus definition of the sustainability concept emerge?

Question 3 options:

- in Stockholm in 1972
- in Rio in 1992
- in Paris in 1948

Question 4: what are the key dimensions to be considered when appraising sustainability?

Question 4 options:

- environmental and social
- environmental and economic
- judicial, economic and environmental
- futuristic, social and environmental
- environmental, economic and social

Question 5 an equitable solution to a sustainability challenge satisfies which dimensions?

Question 5 options:

- economic and environmental
- social and environmental
- economic and social
- none of the above

Question 6 the Sustainable Development Goals are the first framework from the United Nations to operationalise the aim of sustainability Question 6 options:

- True
- False

Question 7 we have clearly defined indicators for all agreed sustainability goals

- True
- False

Question 8 the goal for climate action is the SDG with the most data available options:

- True
- False

Question 9 what is a S.M.A.R.T. objective

- an objective which is specific, measurable, achievable, rationale and time-bound
- an objective which is special, motivated, achievable, relevant and time-bound
- an objective which is specific, measurable, achievable, relevant and time-bound
- an objective which is special, motivated, achievable, relevant and timely

Question 10 Many companies (in EU) must report on their sustainability performance

- True
- False

Question 11 There are a number of initiatives for companies to engage in sustainability.

- Science based targets initiative
- World economic forum
- DGNB
- IOC
- UN Global compact

Question 12 What is the latest reporting requirement from EU?

- NFRD (Non financial reporting directive)
- ENFRD (European non-financial reporting directive)
- CSRD (Corporate sustainability reporting directive)

Module 2:

Self assessment - Module 2, 1210X Quantitative sustainability

Question 1

What does the $I=PAT$ equation represent?

Question 1 options:

That the environmental impact (I) is a function of the power consumption (P), the area used for producing the power (A) and the efficiency of the technology used to produce it (T)

That the environmental impact (I) is a function of how much we produce (P), the age of our production equipment (A), and the amount of time we use our products (T)

That the overall environmental impact (I) is a function of how many people we are (P), how wealthy we are (A), and the environmental efficiency of our technologies (T)

Question 2

How much should the factor T decrease, in order to compensate for increasing P and A and an already large I?

Question 2 options:

It should decrease by a factor 2

It should decrease by a factor 4-20

It should decrease by a factor of at least 10

Question 3

The Rebound effect may counterbalance the technological improvements causing consumers to buy even more products.

Question 3 options:

True

False

Question 4

The object of assessment in sustainability assessment focus on the service that the product, system or technology provides?

Question 4 options:

True

False

Question 5

In the suggestion for a stepwise Life Cycle Sustainability Assessment by Neugebauer et al. there are some suggestion for which indicators should be included at each step. Below are the social indicators. Please order them according to which tier they belong starting with the lowest tier.

Question 5 options:

2 Health

1 Fair wages

3 Human rights

Question 6

How will environmental sustainability be addressed in this course?

Question 6 options:

You will assess the impact on the nine boundaries of the planetary boundaries framework

You will apply a life cycle impact assessment including several different environmental impact categories and relate them to their absolute boundaries

You will assess impact on climate change through a carbon footprint and relate this result to the absolute boundaries (earth's carrying capacity) for climate change

Question 7

Question 7 options:

The environmental aspects is represented by

LCA or life cycle assessment

The economic aspect is represented by

LCC or life cycle costing

The social aspect is represented by

SLCA or social life cycle assessment

Question 8

The functional unit quantifies the product/system/technology e.g. in terms of it's efficiency, for example as kg CO₂ emitted/km driven for a car?

Question 8 options:

True

False

Module 4:

Module 5:

Self assessment - Module 5, 1210X, Quantitative Sustainability

Question 1

Why are common goods easy to overexploit?

The benefits of taking more than your share of a common good are all received by the one user overexploiting, and the costs (consequences) of this overexploitation are shared among all users of the resource. So, the user taking more than its share always receive more net benefits than other users.

Question 2

If a resource exploitation extracts (yields) less of that resource than MSY (maximum sustainable yield); is that resource exploitation sustainable?

Not necessarily. while the yield is below MSY, it may relates to an economic yield which is too small to sustain the exploiter economically or to afford fair wages to employees. It is also possible that the resource exploitation causes biodiversity degradation, affecting resources (eg species) that are not the target resource of the exploitation. the yield may be below MSY, but the yield impact on biodiversity may be greater than a commonly agreed acceptable threshold.

Question 3

Can you sustain the exploitation of a non-renewable resource?

Question 3 options:

It depends, it is a matter of recyclability (While ultimately, we will always run out of a non-renewable resource if we keep extracting it, we may be able to extend its use to a 'quasi-sustainable' timescale if the recycling rate is very very high (eg one could argue this with aluminium))

Yes, indefinitely

No, you will run out of it at some point

Question 4

What is the problem with common goods?

Question 4 options:

Their access cannot be restricted and the use of the resource prevent others from using it

Their access cannot be restricted and the use of the resource does not prevent others from using it

People have to compete for them

They can be overexploited by one user alone

Question 5

Which of these resource is not a renewable resource?

Question 5 options:

Timber

Wind

Corn

Lithium

Question 6

What is the difference between Maximum Sustainable Yield (MSY) and Maximum Economic Yield (MEY)?

MSY only relate to the resource extraction: it is the maximum amount of a renewable resource you can extract theoretically without jeopardising the sustainability of the resource exploitation (ie without overexploiting it). MEY is the maximum economic gain an exploiter can make from extracting a renewable resource, it accounts for the amount of resource extracted and the price at which it can be sold as well as the total cost of extracting that amount of resource. MEY is typically achieved at a resource yield which is smaller than MSY.

Module 6:

Question 1

Various tools are available to support sustainability assessments at the company level. These tools should be used to:

- Only examine the social outcomes of their goods and services
- Be used to maximise profit, while ensuring minimum impact to the environment
- Support companies to make decisions that minimise the impact to the environment, whilst also enhancing the social and economic wellbeing of society
- Exclusively examine the environmental impacts of their goods and services

Question 2

Input-Output Models describe the interdependencies between:

- different branches of the economy

- different ecosystems
- different households groups (for example: low income groups and high income group households)
- different government investment decisions

Question 3

Cost Benefit Analysis can be used to calculate the direct and indirect impacts of the production and consumption of different products and services on key economic parameters

- True
- False

Question 4

Functions of different sustainability tools

[1] can be used to evaluate the economic cost of a good or a service over its lifecycle

[3] is an economic method that estimates the equivalent monetary value of the benefits and costs of a project

[2] is developed towards evaluating social impacts of a good or a service over its lifecycle

[4] can be used to calculate the direct and indirect impacts of the production and consumption of different products and services on key economic parameters

1. Lifecycle cost analysis
2. Social-Lifecycle cost analysis
3. Cost-Benefit Analysis
4. Input-output models

Question 5

Life cycle costing (LCC) is an approach that assesses the total cost of an asset over its life cycle, including

- initial capital costs
- operating costs
- overall social costs
- the asset's residual value at the end of its life

Question 6

The reference scale approach to S-LCA (RF S-LCA) is used to assess the consequential social impacts from a product by characterizing the cause-effect chain

- True
- False

Question 7

Important social conditions/outcomes to measure regarding the mining and resource extraction sector involved in the production of electric vehicles include:

- Income
- Employment
- Education
- Health